

Gaussian Natural Gradient Flows for Variational Inference

Abstract

We study Gaussian variational inference for a target $\rho_{\text{post}} \propto e^{-V}$ through the natural gradient flow of the Fisher–Rao metric and two discretizations, a Riemannian retraction scheme and a KL/Bregman resolvent scheme. Convergence proceeds in three stages, covariance burn-in, global localization, and a local phase, each with its own sharp rate. The burn-in is logarithmic in the initial covariance and removable by a warm start. The global phase is controlled by the condition number after whitening by the optimizing covariance; a logarithmic-spiral construction shows this cannot be improved, and a bump train shows the fixed-step factor $\kappa/\Delta t$ is sharp. The local phase runs at the spectral gap of a self-adjoint score operator, bounded two-sidedly, and a convex-ridge family rules out every dimension-free logarithmic local rate. The same three stages persist for stochastic algorithms built from a joint Price/Hessian oracle, with pathwise covariance bands and variance floors that vanish for Gaussian targets. Beyond log-concavity, a double well destabilizes both unprojected schemes, while a covariance-constrained step retains an $O(1/N)$ stationarity guarantee. Finally, we classify all affine-invariant metrics on the Gaussian manifold and show that Fisher–Rao is the unique member, up to scale, that balances the mean, trace, and traceless covariance modes.

Keywords. Gaussian variational inference; natural gradient flow; Fisher–Rao geometry; affine invariance; convergence rates; stochastic gradient.

1 Introduction

We consider the problem of sampling the target posterior distribution with density $\rho_{\text{post}}(\cdot)$, for the parameter $\theta \in \mathbb{R}^{N_\theta}$, given by

$$\rho_{\text{post}}(\theta) \propto \exp(-V(\theta)), \quad \theta \in \mathbb{R}^{N_\theta},$$

where $V(\theta) : \mathbb{R}^{N_\theta} \rightarrow \mathbb{R}$ is a known function while the normalizing constant $Z = \int \exp(-V(\theta)) d\theta$ is intractable. Markov chain Monte Carlo (MCMC) methods [4] aim to sample from the target distribution ρ_{post} asymptotically, but can be computationally expensive in high-dimensional or ill-conditioned settings [20]. Variational inference (VI) [3], on the other hand, replaces exact sampling by optimization over a tractable family of probability distributions on its Kullback–Leibler (KL) divergence to the true target posterior, which is often more scalable than MCMC [8].

We focus on the Gaussian variational family

$$\mathcal{P}_G := \left\{ \rho_a = \mathcal{N}(m, C) : a = (m, C) \in \mathcal{A} := \mathbb{R}^{N_\theta} \times \mathbb{S}_{++}(N_\theta) \right\},$$

which is the simplest nontrivial VI family, where the Gaussian variational inference problem is

$$a_\star = (m_\star, C_\star) \in \arg \min_{a \in \mathcal{A}} \mathcal{E}(a), \quad \mathcal{E}(a) := \text{KL}(\rho_a \parallel \rho_{\text{post}}).$$

Recent theoretical work has studied Gaussian variational inference through Wasserstein and Bures–Wasserstein geometry [7, 11]. This viewpoint connects Gaussian variational inference with optimal transport and constrained Wasserstein gradient flows, and yields a convergence

theory for both the Wasserstein gradient flow and its forward–backward discretizations. On the other side, the Fisher–Rao gradient flow [5] develops a complementary information-geometric viewpoint [2]. When the Fisher–Rao gradient flow is restricted to the Gaussian family, we obtain the natural gradient flow [1] for Gaussian variational inference [6],

$$\begin{aligned}\frac{dm_t}{dt} &= C_t \mathbb{E}_{\rho_{a_t}} [\nabla_{\theta} \log \rho_{\text{post}}(\theta)], \\ \frac{dC_t}{dt} &= C_t + C_t \mathbb{E}_{\rho_{a_t}} [\nabla_{\theta} \nabla_{\theta} \log \rho_{\text{post}}(\theta)] C_t.\end{aligned}\tag{1}$$

The Fisher–Rao metric restricted to the Gaussian family is affine invariant [6]: the flow (1) commutes with every invertible affine change of variables $\theta \mapsto A\theta + b$, so that its behavior is unchanged by an arbitrary linear reparameterization of the target. This is the structural property that distinguishes it from the Bures–Wasserstein flow, and it is why we expect natural gradient methods to be insensitive to the conditioning of the target. It also determines which quantities can appear in a sharp convergence rate. However, while the natural gradient flow is useful in practice, its convergence theory for Gaussian variational inference remains underexplored. We develop that theory here, for the flow (1) and for two discretizations of it, in both the deterministic and the stochastic setting.

The convergence of (1) and of its discretizations proceeds in three stages,

covariance burn-in + global localization + local convergence,

(2)

each governed by a different mechanism and each carrying its own sharp rate. The first stage is an artifact of the geometry rather than of the target. In the covariance equation of (1) the growth of a small covariance is multiplicative, and the mean equation carries the factor C_t , so an underdispersed initialization suppresses the mean motion until the covariance has reached the curvature scale of the target. The second stage is the global descent proper, and it is controlled not by the curvature ratio $\kappa = \beta/\alpha$ of the target but by the intrinsic ratio $\kappa_{\star}$ obtained after whitening by the optimizing covariance $C_{\star}$; the two agree for a poorly conditioned Gaussian target only in the sense that $\kappa_{\star} = 1$ there, however ill-conditioned κ may be. The third stage is local, and its rate is governed neither by κ nor by $\kappa_{\star}$ but by the spectral gap of a self-adjoint operator built from the first- and second-Hermite components of the whitened Hessian, which measures the non-Gaussian coupling between the mean and covariance modes. The three mechanisms are logically independent: a transport warm start removes the first without touching the other two.

The deterministic theory occupies Sections 2 and 3 and gives matching upper and lower bounds at every stage. We prove in Theorem 2.3 an affine-invariant continuous-time estimate that separates the initial covariance from the intrinsic conditioning, and we show in Theorem 2.5 that its $\kappa_{\star}$ dependence cannot be improved, by means of a two-dimensional logarithmic-spiral potential. For the two discretizations, a Riemannian retraction scheme and a KL/Bregman resolvent scheme, we prove in Theorems 2.9 and 2.11 global complexity bounds of a common form, and we show in Theorem 2.12 that their factor $\kappa/\Delta t$ is sharp at a fixed stepsize, by means of a one-dimensional bump-train potential that is even, strongly convex, and uniformly Hessian-Lipschitz. The covariance burn-in, and only the covariance burn-in, can be removed algorithmically: we give in Theorem 2.13 a Bures–Wasserstein transport warm start and in Theorem 2.16 a single gradient–Hessian query that lands the covariance on the curvature band, which for a Gaussian target recovers the exact optimizer in one query by Theorem 2.15.

The local theory keeps the nonlinear Gaussian covariance dynamics exactly and perturbs only the non-Gaussian remainder. We prove in Theorem 3.3 a two-sided spectral estimate for the linearized generator, and we show in Theorem 3.15 that its $\beta_{\star} - 1$ branch is attained: a smooth convex-ridge family in dimension $N_{\theta} = \Theta(\kappa_{\star}^2)$, with a dimension-free Hessian-Lipschitz constant, has local gap $\Theta(\kappa_{\star}^{-1/2})$, which rules out every dimension-free logarithmic local rate.

The Gaussian-core decomposition gives in [Theorems 3.8](#) and [3.12](#) certified local regions that are exact for Gaussian targets in every dimension and degrade continuously for general strongly log-concave targets, together with an optimizer-free entry test in [Theorem 3.14](#). [Theorem 3.16](#) assembles the three stages into a single complexity statement for the flow and for both schemes.

The stochastic theory occupies [Section 4](#) and reproduces the same three stages under a joint Price/Hessian oracle with a sticking-the-landing mean estimator [\[16\]](#). Because the sampled Hessian obeys an almost-sure spectral bound, the covariance bands hold pathwise with no clipping and no stopping time, and the global theorems [Theorems 4.4](#) and [4.5](#) therefore hold in plain expectation, with explicit residual floors in two variance regimes. The local statement is weaker than its deterministic counterpart: we prove in [Theorems 4.7](#) and [4.8](#) a mean-square contraction on the non-exit event together with a finite-horizon containment bound in [Theorem 4.9](#), and we do not claim infinite-horizon containment, since the mean-score noise has unbounded support. The additive floors vanish for Gaussian targets, and a decreasing stepsize removes them in general by [Theorem 4.11](#). [Theorem 4.10](#) is the stochastic mirror of [Theorem 3.16](#).

Two further results mark the boundaries of the theory. Beyond log-concavity, the unprojected schemes genuinely fail: a smooth double well with uniformly bounded Hessian produces the covariance cascade of [Theorem 2.18](#) and the resolvent pole of [Theorem 2.19](#), while a covariance-constrained KL step retains an $O(1/N)$ stationarity guarantee by [Theorem 2.21](#). Finally, in [Section 5](#) we classify, for $N_\theta \geq 2$, the complete three-parameter family of affine-invariant Riemannian metrics on the Gaussian manifold, compute in [Theorem 5.4](#) the exact rates at which each metric preconditions the mean, trace, and traceless covariance modes on a Gaussian target, and show in [Theorem 5.5](#) that Fisher–Rao is the unique member of the family, up to scale, that balances all three. This recovers the metric with which we began.

Related work

Natural gradients and information geometry provide the classical background for the Fisher–Rao metric [\[1, 2, 15\]](#), and Gaussian and stochastic variational inference are surveyed in Blei et al. [\[3\]](#) and Hoffman et al. [\[8\]](#). The Fisher–Rao gradient flow and its Gaussian restriction are studied in Carrillo et al. [\[5\]](#) and Chen et al. [\[6\]](#), the latter establishing the affine invariance we exploit throughout. Closest to our deterministic theory, Kumar et al. [\[10\]](#) prove convergence guarantees for Gaussian natural-gradient variational inference, in discrete time and for its continuous-time flow, through a square-root covariance parameterization; for stochastic natural-gradient variational inference, Wu and Gardner [\[21\]](#) obtain the first $O(1/T)$ non-asymptotic rate in the conjugate-likelihood case. Our analysis differs from these and from a generic natural-gradient convergence argument in that it uses the exact Gaussian covariance equation in the intrinsic Fisher–Rao geometry rather than a smoothness surrogate or a chosen parameterization, and keeps every constant in optimizer-whitened, affine-invariant form; this is what yields the three-stage decomposition, the optimizer-whitened condition number, and the matching lower bounds at every stage.

Wasserstein Gaussian variational methods supply a complementary geometry. Lambert et al. [\[11\]](#) study variational inference through Bures–Wasserstein gradient flows and analyze a stochastic scheme with non-asymptotic guarantees, and Diao et al. [\[7\]](#) analyze a forward–backward scheme with state-of-the-art rates and the first stationarity guarantees beyond log-concavity. Kim et al. [\[9\]](#) analyze stochastic Gaussian variational inference with Price’s Hessian-based gradient estimator across the Bures–Wasserstein and parameter-space geometries, an estimator-level overlap with our joint Price/Hessian oracle, and Luu et al. [\[13\]](#) add control-variate variance reduction on the Bures–Wasserstein manifold; neither treats the Fisher–Rao geometry or its variance floors. We use the Bures–Wasserstein flow in two limited roles, as a warm start that removes the covariance burn-in and as a point of comparison beyond log-concavity; we do not blend the two geometries into a Wasserstein–Fisher–Rao method here. Affine-invariant metrics on the cone of positive-definite matrices were classified in several forms by Thanwerdas and

Pennec [18, 19]; our classification includes the mean block of the Gaussian family and, for $N_\theta \geq 2$, uses it to expose the exact modal rates.

The stochastic construction rests on the sticking-the-landing principle of Roeder et al. [16], and natural-gradient variational inference with nonconjugate models and constrained covariance parameterizations is studied in Sun et al. [17]. Our stochastic results treat the joint geometry of the mean and the full covariance, separate constant-step residual levels from decreasing-step convergence, and keep local containment separate from local contraction.

Organization and notation

Section 2 develops the global deterministic theory, its two discretizations, the removal of the covariance burn-in, and the non-log-concave boundary. Section 3 gives the local spectral and nonlinear theory and the three-stage complexity theorem. Section 4 treats the Price/Hessian stochastic algorithms. Section 5 classifies the affine-invariant metrics, Section 6 presents numerical illustrations, and Section 7 collects open problems. Longer proofs and the three lower constructions are given in the appendices.

We write N_θ for the dimension, $\|\cdot\|_{\text{op}}$ and $\|\cdot\|_{\text{F}}$ for the operator and Frobenius norms, and $\Delta(a) = \mathcal{E}(a) - \mathcal{E}(a_\star)$ for the energy gap, with $\Delta_0 = \Delta(a_0)$. The symbol $\kappa = \beta/\alpha$ always denotes the curvature ratio in the original coordinates and $\kappa_\star$ its optimizer-whitened counterpart, and similarly for $\alpha_\star$ and $\beta_\star$. We write Δt for the stepsize, B for the batch size, and $\log_+ x := \max\{\log x, 0\}$. Constants hidden by $O(\cdot)$ are numerical unless their dependence is displayed.

2 Global convergence and discretization

We study the Gaussian natural gradient flow where the target posterior distribution ρ_{post} is logconcave, and then propose practical discretization algorithms.

2.1 Setup and the flow

Assumption 2.1. We assume that ρ_{post} is α -strongly logconcave and β -smooth, i.e.

$$\alpha I \preceq -\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta) \preceq \beta I, \quad \theta \in \mathbb{R}^{N_\theta},$$

which is equivalent to

$$\alpha I \preceq \nabla^2 V(\theta) \preceq \beta I.$$

We write $\kappa := \beta/\alpha \geq 1$ for the condition number.

A third of our results, all of them stated where they are used, need in addition that the Hessian varies slowly.

Assumption 2.2 (Hessian-Lipschitzness). There is $L_H < \infty$ with $\|\nabla^2 V(x) - \nabla^2 V(y)\|_{\text{op}} \leq L_H \|x - y\|$ for all $x, y \in \mathbb{R}^{N_\theta}$. We write the dimensionless parameter $\bar{L}_H := L_H/\alpha^{3/2}$; a Gaussian target has $L_H = \bar{L}_H = 0$.

For a Gaussian parameter $a = (m, C) \in \mathcal{A}$ we abbreviate the averaged score and the averaged curvature,

$$g(a) := \mathbb{E}_{\rho_a}[\nabla_\theta \log \rho_{\text{post}}(\theta)] = -\mathbb{E}_{\rho_a}[\nabla V(\theta)], \quad A(a) := \mathbb{E}_{\rho_a}[\nabla^2 V(\theta)], \quad (3)$$

and the precision residual $R(a) := C^{-1} - A(a)$. Under Theorem 2.1 we have $\alpha I \preceq A(a) \preceq \beta I$ for every $a \in \mathcal{A}$, and at the minimizer the first-order conditions read

$$g(a_\star) = 0, \quad A(a_\star) = C_\star^{-1}, \quad \text{that is} \quad R(a_\star) = 0. \quad (4)$$

In this notation the Gaussian natural gradient flow (1) reads

$$\frac{dm_t}{dt} = C_t g(a_t), \quad \frac{dC_t}{dt} = C_t - C_t A(a_t) C_t. \quad (5)$$

We also use the energy splitting $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$ with

$$\mathcal{E}_1(a) = \int \rho_a \log \rho_a = -\frac{1}{2} \log \det C + \text{const.}, \quad \mathcal{E}_2(a) = -\mathbb{E}_{\rho_a}[\log \rho_{\text{post}}] = \mathbb{E}_{\rho_a}[V] + \text{const.} \quad (6)$$

The Fisher–Rao metric induced on the Gaussian family, at $a = (m, C)$ and for tangent vectors $(u, U), (v, W) \in \mathbb{R}^{N_\theta} \times \mathbb{S}(N_\theta)$, is

$$\langle (u, U), (v, W) \rangle_a := u^\top C^{-1} v + \frac{1}{2} \text{Tr}(C^{-1} U C^{-1} W), \quad \|(u, U)\|_a^2 = u^\top C^{-1} u + \frac{1}{2} \|C^{-1/2} U C^{-1/2}\|_F^2, \quad (7)$$

which is the Fisher information inner product of the Gaussian family. We first identify the flow (1) as the gradient flow of $\mathcal{E}$ in this metric. Writing $\theta = m + C^{1/2} Z$ with $Z \sim \mathcal{N}(0, I)$ and differentiating under the integral sign, which is justified by dominated convergence since $\|\nabla^2 V\|_{\text{op}} \leq \beta$ forces $\|\nabla V(\theta)\| \leq \|\nabla V(0)\| + \beta \|\theta\|$, we obtain $\partial_m \mathcal{E}_2 = -g(a)$ and, by Price’s theorem, $\partial_C \mathcal{E}_2 = \frac{1}{2} A(a)$. Together with $\partial_C \mathcal{E}_1 = -\frac{1}{2} C^{-1}$ this gives the differential

$$D\mathcal{E}(a)[(u, U)] = -g(a)^\top u + \frac{1}{2} \text{Tr}((A(a) - C^{-1})U). \quad (8)$$

Matching (8) against (7) block by block, the Riemannian gradient is

$$\text{grad } \mathcal{E}(a) = (-Cg(a), CA(a)C - C), \quad (9)$$

so that $\dot{a}_t = -\text{grad } \mathcal{E}(a_t)$ is exactly (1). Its squared norm is

$$\|\text{grad } \mathcal{E}(a)\|_a^2 = G(a)^2 := g(a)^\top Cg(a) + \frac{1}{2} \|C^{1/2} R(a) C^{1/2}\|_F^2, \quad (10)$$

because $C^{-1}(CA(a)C - C) = A(a)C - I$ and cyclicity of the trace turn the covariance block into $\text{Tr}(CR(a)CR(a))$. The chain rule then gives the exact dissipation identity

$$\frac{d}{dt} \Delta(a_t) = D\mathcal{E}(a_t)[- \text{grad } \mathcal{E}(a_t)] = -G(a_t)^2. \quad (11)$$

Two inequalities convert (11) into a rate. The first is the Wasserstein Polyak–Łojasiewicz inequality induced by the α -strong convexity of V and restricted to the Gaussian family [11]: for every $a \in \mathcal{A}$,

$$\|\text{grad}^{W_2} \mathcal{E}(a)\|_{W_2}^2 = \|g(a)\|^2 + \text{Tr}(R(a)CR(a)) \geq 2\alpha \Delta(a). \quad (12)$$

The second compares the two gradient norms. If $C \succeq \ell I$, then $g(a)^\top Cg(a) \geq \ell \|g(a)\|^2 \geq \frac{\ell}{2} \|g(a)\|^2$, while $R(a)CR(a) \succeq 0$ gives $\frac{1}{2} \text{Tr}(CR(a)CR(a)) = \frac{1}{2} \text{Tr}(C[R(a)CR(a)]) \geq \frac{\ell}{2} \text{Tr}(R(a)CR(a))$; adding the two estimates and inserting (12),

$$G(a)^2 \geq \frac{\ell}{2} \|\text{grad}^{W_2} \mathcal{E}(a)\|_{W_2}^2 \geq \alpha \ell \Delta(a), \quad (13)$$

and the same computation with $\|C\|_{\text{op}}$ in place of ℓ gives the reverse comparison $G(a)^2 \leq \|C\|_{\text{op}} \|\text{grad}^{W_2} \mathcal{E}(a)\|_{W_2}^2$. We combine (11) and (13) with a lower bound on the covariance.

2.2 Continuous-time global convergence

The covariance lower bound is cleanest in the precision variable $P_t := C_t^{-1}$, along which (1) is affine: differentiating $P_t = C_t^{-1}$ and using (5),

$$\dot{P}_t = A(a_t) - P_t, \quad P_t = e^{-t}P_0 + \int_0^t e^{-(t-s)}A(a_s)ds. \quad (14)$$

The two-sided covariance envelope along the flow, [Theorem E.1](#), is proved in [Section E](#).

The rate produced by (13) and [Theorem E.1](#) ([Section E](#)) still carries the affine anisotropy of the target, which the flow itself does not see. We therefore whiten by the optimizing Gaussian covariance. Let $a_\star = (m_\star, C_\star)$ be the optimizing Gaussian and set

$$\hat{m} := C_\star^{-1/2}(m - m_\star), \quad \hat{C} := C_\star^{-1/2}CC_\star^{-1/2}, \quad \hat{V}(z) := V(m_\star + C_\star^{1/2}z), \quad (15)$$

so that the optimizer becomes $(\hat{m}_\star, \hat{C}_\star) = (0, I)$, and define the optimal whitened curvature constants and the whitened initial covariance floor,

$$\alpha_\star := \inf_z \lambda_{\min}(\nabla^2 \hat{V}(z)), \quad \beta_\star := \sup_z \lambda_{\max}(\nabla^2 \hat{V}(z)), \quad \kappa_\star := \frac{\beta_\star}{\alpha_\star}, \quad (16)$$

$$\lambda_{0,\star} := \lambda_{\min}(\hat{C}_0) = \lambda_{\min}(C_\star^{-1/2}C_0C_\star^{-1/2}). \quad (17)$$

Writing $S = C_\star^{1/2}$ we have $\nabla \hat{V}(z) = S \nabla V(m_\star + Sz)$ and $\nabla^2 \hat{V}(z) = S \nabla^2 V(m_\star + Sz) S$, hence $\hat{g} = Sg$ and $\hat{A} = SAS$, so the whitened parameters solve the same Fisher–Rao system $\hat{m} = \hat{C}\hat{g}$, $\hat{C} = \hat{C} - \hat{C}\hat{A}\hat{C}$. The first-order conditions (4) become $\mathbb{E}_{Z \sim \mathcal{N}(0,I)}[\nabla \hat{V}(Z)] = 0$ and $\mathbb{E}_{Z \sim \mathcal{N}(0,I)}[\nabla^2 \hat{V}(Z)] = I$, so that $\alpha_\star \leq 1 \leq \beta_\star$. Under an invertible affine change $y = Bx + b$ the transformed optimizer covariance is $BC_\star B^\top$, and $O := (BC_\star B^\top)^{-1/2}BC_\star^{1/2}$ is orthogonal; whitening before and after the change therefore differs by an orthogonal map only, so $\alpha_\star$, $\beta_\star$, $\kappa_\star$ and the spectrum of $\hat{C}_0$, hence also the product $\beta_\star \lambda_{0,\star}$, are affine invariants. Finally, $C_\star^{-1} = A(a_\star)$ and $\alpha I \preceq A(a_\star) \preceq \beta I$ give $\beta^{-1}I \preceq C_\star \preceq \alpha^{-1}I$, so $\kappa^{-1}I \preceq \nabla^2 \hat{V}(z) \preceq \kappa I$ and

$$1 \leq \kappa_\star \leq \kappa^2. \quad (18)$$

Theorem 2.3 (Affine-invariant continuous-time global convergence). *Under [Theorem 2.1](#), the Gaussian natural gradient flow (1) satisfies, for every $t \geq 0$,*

$$\Delta(a_t) \leq \Delta_0 [1 + \beta_\star \lambda_{0,\star} (e^t - 1)]^{-1/\kappa_\star}.$$

Consequently, for every $\delta \in (0, \Delta_0)$ the hitting time $T^{\text{glob}}(\delta) := \inf\{t \geq 0 : \Delta(a_t) \leq \delta\}$ obeys

$$T^{\text{glob}}(\delta) \leq \log \left(1 + \frac{(\Delta_0/\delta)^{\kappa_\star} - 1}{\beta_\star \lambda_{0,\star}} \right) \leq \log_+ \frac{1}{\beta_\star \lambda_{0,\star}} + \kappa_\star \log \frac{\Delta_0}{\delta}.$$

The original-coordinate estimate remains valid as well, so we may take the minimum of the two, replacing $(\beta_\star, \lambda_{0,\star}, \kappa_\star)$ by $(\beta, \lambda_0, \kappa)$.

Proof. The energy and the Fisher–Rao dissipation are invariant under the whitening (15), so we work in whitened coordinates and omit hats. Applying [Theorem E.1](#) with the whitened constants gives $C_t \succeq \ell_\star(t)I$ with $\ell_\star(t) = \lambda_{0,\star}e^t/(1 + \beta_\star \lambda_{0,\star}(e^t - 1))$, and (13) gives $G(a_t)^2 \geq \alpha_\star \ell_\star(t) \Delta(a_t)$. With (11) this yields $\dot{\Delta}_t \leq -\alpha_\star \ell_\star(t) \Delta_t$, and by Gronwall’s inequality

$$\Delta_t \leq \Delta_0 \exp \left(-\alpha_\star \int_0^t \ell_\star(s) ds \right).$$

The scalar integral is exact, $\int_0^t \ell_\star(s) ds = \beta_\star^{-1} \log[1 + \beta_\star \lambda_{0,\star}(e^t - 1)]$, and $\alpha_\star/\beta_\star = 1/\kappa_\star$, which gives the decay estimate. Solving the equality in it gives the exact hitting time. For the simplified form, set $R = (\Delta_0/\delta)^{\kappa_\star} \geq 1$ and $b = \beta_\star \lambda_{0,\star}$: if $b \leq 1$ then $1 + (R-1)/b \leq R/b$, and if $b \geq 1$ then $1 + (R-1)/b \leq R$; taking logarithms in the two cases gives the bound. $\square$

The intrinsic condition number removes anisotropy that can be eliminated by an affine reparametrization. It does not, by itself, encode the initialization size: the factor $\log(\Delta_0/\delta)$, or an equivalent radius term, is indispensable, since if arbitrary initial means are allowed then the worst-case time to reach a fixed energy level is infinite even for a quadratic target. The comparison in (18) can be very loose in the useful direction: for an anisotropic quadratic target $\kappa_\star = 1$ while κ is arbitrarily large.

Remark 2.4. From Corollary 3 of Lambert et al. [11], the convergence rate of Gaussian approximate Wasserstein gradient flow does not depend on the initial covariance, while in Theorem 2.3 the convergence rate of the Gaussian approximate natural gradient flow depends on the spectral bound of the initial covariance through Theorem E.1. We show by example that the dependence on the lower spectral bound of the initial covariance is not a proof artifact, instead, it is determined by the properties of the Fisher–Rao gradient flow. For the one-dimensional Gaussian target

$$\rho_{\text{post}} = \mathcal{N}(0, 1), \quad V(\theta) = \frac{1}{2}\theta^2,$$

with $a = (m, c)$, the Gaussian variational objective is

$$\mathcal{E}(m, c) - \mathcal{E}(a_\star) = \frac{1}{2}(m^2 + c - 1 - \log c), \quad a_\star = (0, 1).$$

Since

$$\nabla \log \rho_{\text{post}}(\theta) = -\theta, \quad \nabla^2 \log \rho_{\text{post}}(\theta) = -1,$$

the Gaussian natural gradient flow becomes

$$\dot{m}_t = -c_t m_t, \quad \dot{c}_t = c_t(1 - c_t).$$

Let $c_0 = \varepsilon \in (0, 1)$. Then the covariance equation is exactly solvable:

$$c_t = \frac{\varepsilon e^t}{1 - \varepsilon + \varepsilon e^t}.$$

Consequently,

$$m_t = m_0 \exp\left(-\int_0^t c_s \, ds\right) = \frac{m_0}{1 - \varepsilon + \varepsilon e^t}.$$

Therefore, for every fixed $t > 0$,

$$\lim_{m_0 \rightarrow \infty} \frac{\mathcal{E}(m_t, c_t) - \mathcal{E}(a_\star)}{\mathcal{E}(m_0, c_0) - \mathcal{E}(a_\star)} = \frac{1}{(1 - \varepsilon + \varepsilon e^t)^2}.$$

Taking $\varepsilon \downarrow 0$ gives

$$\lim_{\varepsilon \downarrow 0} \lim_{m_0 \rightarrow \infty} \frac{\mathcal{E}(m_t, c_t) - \mathcal{E}(a_\star)}{\mathcal{E}(m_0, c_0) - \mathcal{E}(a_\star)} = 1.$$

Hence no estimate of the form

$$\mathcal{E}(a_t) - \mathcal{E}(a_\star) \leq e^{-rt}(\mathcal{E}(a_0) - \mathcal{E}(a_\star)), \quad r > 0,$$

with r depending only on the log-concavity and smoothness constants, can hold uniformly over all initial covariances. In this example $\alpha = \beta = 1$, and the rate in the preceding result depends on

$$\lambda_{\min} = \min\{\varepsilon, 1\} = \varepsilon,$$

which correctly degenerates as $\varepsilon \downarrow 0$.

This should be contrasted with the Gaussian Bures–Wasserstein gradient flow for the same target. In one dimension it satisfies

$$\dot{\tilde{m}}_t = -\tilde{m}_t, \quad \dot{\tilde{c}}_t = 2(1 - \tilde{c}_t),$$

and hence

$$\tilde{m}_t = \tilde{m}_0 e^{-t}, \quad \tilde{c}_t = 1 + (\tilde{c}_0 - 1)e^{-2t}.$$

Thus the mean component contracts immediately at the target curvature rate, independently of the initial covariance. This illustrates the structural difference between the Fisher–Rao and Bures–Wasserstein geometries: in the Fisher–Rao Gaussian flow, small covariance preconditions and slows down the mean dynamics, while the Bures–Wasserstein flow does not contain this covariance prefactor in the mean equation. The same contrast is what makes a short transport warm start remove the covariance burn-in term altogether, as we show in [Section 2.6](#).

The linear dependence on κ_* in [Theorem 2.3](#) cannot be improved, up to logarithmic initialization and localization factors, under the general spectral initialization class.

Theorem 2.5 (Continuous-time logarithmic-spiral lower bound). *For every $K \geq 16$ there is an even potential $V_K \in C^3(\mathbb{R}^2)$, given explicitly in (86), and a scale r_0 with $r_0\sqrt{\alpha} \geq 10^{25}K^{3/2}\sqrt{\log(eK)}$, such that $\alpha I \preceq \nabla^2 V_K \preceq \beta I$ with $\kappa = 2K + 1$ and*

$$K \leq \kappa_*(V_K) \leq 1.03K.$$

There is an initial mean m_0 with $\|m_0\| = e^2 r_0$ and an anisotropic initial covariance C_0 , admissible under the general spectral condition $\beta^{-1}I \preceq C_0 \preceq \alpha^{-1}I$, such that for every local level $\delta_{\text{loc}} < \frac{\alpha}{2}e^{-0.08}\|m_0\|^2$ the exact Gaussian Fisher–Rao flow obeys

$$T_{\text{loc}} := \inf\{t \geq 0 : \Delta(a_t) \leq \delta_{\text{loc}}\} \geq \frac{K}{100} \geq \frac{\kappa_*(V_K)}{103}.$$

Moreover $\beta_\lambda_{0,*} \geq (1-\eta_K)/(1+\eta_K)$ with $\eta_K := Ke^{-\alpha r_0^2/2} \leq 10^{-3}$, so the covariance-initialization term $\log_+(1/(\beta_*\lambda_{0,*}))$ in [Theorem 2.3](#) is smaller than 0.003 for this family.*

The proof is given in [Section B](#).

A $\Theta(\kappa_*)$ packaging of this bound, [Theorem B.1](#), is stated in [Section B](#).

The matching construction uses an admissible anisotropic initial covariance. It does not establish an $\Omega(\kappa_*)$ lower bound under the stricter initialization $C_0 = \beta^{-1}I$. It also allows the initial radius to grow with K , and the logarithm of that radius is part of the input size charged by [Theorem 2.3](#). A Hessian-Lipschitz version of the construction is available for every prescribed $L_H > 0$ by enlarging r_0 , and is recorded in [Section B](#).

2.3 Discretization with Riemannian retraction

We decompose the objective as $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$ as in (6), and define the squared base-point penalty at the current iterate $a_n = (m_n, C_n)$ as

$$d(a, a_n)^2 = (m - m_n)^\top C_n^{-1}(m - m_n) + \frac{1}{2}\|Z\|_F^2, \quad Z = \log(C_n^{-1/2}CC_n^{-1/2}), \quad (19)$$

built blockwise from the metric (7) at the base point a_n : the mean block freezes the covariance at C_n , and the covariance block is the affine-invariant distance on $\mathbb{S}_{++}(N_\theta)$. It is not the geodesic distance of (7), whose geodesics couple the mean and the covariance ([Theorem 2.7](#)). As the entropy term $\mathcal{E}_1$ is often non-smooth, we apply the proximal algorithm [14] for optimization. In the forward step,

$$a_{n+1/2} = \arg \min_{a \in \mathcal{A}} \left\{ \mathcal{E}_2(a_n) + \left\langle \text{grad } \mathcal{E}_2(a_n), \text{Retr}_{a_n}^{-1}(a) \right\rangle + \frac{1}{2\Delta t} d(a, a_n)^2 \right\},$$

where $\text{Retr}_{a_n}^{-1}$ is the inverse of the retraction Retr_{a_n} defined in [Theorem 2.6](#) below, which gives

$$m_{n+1/2} = m_n + \Delta t C_n g(a_n), \quad C_{n+1/2} = C_n^{1/2} \exp(-\Delta t C_n^{1/2} A(a_n) C_n^{1/2}) C_n^{1/2}.$$

The backward step reads

$$a_{n+1} = \arg \min_{a \in \mathcal{A}} \left\{ \mathcal{E}_1(a) + \frac{1}{2\Delta t} d(a, a_{n+1/2})^2 \right\},$$

which gives

$$m_{n+1} = m_{n+1/2}, \quad C_{n+1} = e^{\Delta t} C_{n+1/2}.$$

These collectively give the update through Riemannian geometry:

$$\boxed{\begin{aligned} m_{n+1} &= m_n + \Delta t C_n g(a_n), \\ C_{n+1} &= C_n^{1/2} \exp(\Delta t (I - C_n^{1/2} A(a_n) C_n^{1/2})) C_n^{1/2}. \end{aligned}} \quad (20)$$

We proceed to show the convergence theory of (20) under [Theorem 2.1](#). We first show that the update for each step is exactly a gradient descent step along the exponential retraction of the Gaussian manifold.

Proposition 2.6. *For $a = (m, C) \in \mathcal{A}$ and $(u, U) \in T_a \mathcal{A}$ define*

$$\text{Retr}_{(m,C)}(u, U) := \left(m + u, C^{1/2} \exp(C^{-1/2} U C^{-1/2}) C^{1/2} \right).$$

Then the discrete algorithm with Riemannian retraction (20) gives the one-step update

$$a_{n+1} = \text{Retr}_{a_n}(-\Delta t \text{grad } \mathcal{E}(a_n)).$$

Proof. By (9), in the mean direction

$$m_{n+1} = m_n + \Delta t C_n g(a_n) = m_n - \Delta t \text{grad}_m \mathcal{E}(a_n).$$

In the covariance direction,

$$\begin{aligned} -\Delta t \text{grad}_C \mathcal{E}(a_n) &= \Delta t C_n - \Delta t C_n A(a_n) C_n \\ &= C_n^{1/2} (\Delta t I - \Delta t C_n^{1/2} A(a_n) C_n^{1/2}) C_n^{1/2}, \end{aligned}$$

so that $C_n^{-1/2} (-\Delta t \text{grad}_C \mathcal{E}(a_n)) C_n^{-1/2} = \Delta t (I - C_n^{1/2} A(a_n) C_n^{1/2})$, which is exactly the exponent appearing in the covariance component of (20). This completes the proof. $\square$

Remark 2.7. The covariance block of Retr is the exponential map of the affine-invariant metric on $\mathbb{S}_{++}(N_\theta)$, so the covariance update follows a genuine geodesic of the covariance factor. The joint mean–covariance map is nevertheless a retraction and not the exponential map of the Fisher–Rao metric on $\mathcal{A}$, whose geodesics couple the mean and the covariance. All estimates below are proved for the retraction directly, so no geodesic smoothness statement is used.

We then prove the spectral bounds of the covariance along (20).

Lemma 2.8 (Riemannian covariance bootstrap). *Under [Theorem 2.1](#), if $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$ and $0 < \Delta t \leq \min\{1, 1/(\beta \lambda_{\max})\}$, then along (20) one has $C_n \preceq \lambda_{\max} I$ for all $n \geq 0$, and $C_n \succeq L_n I$ where*

$$L_0 := \min \left\{ \lambda_0, \frac{1}{2\beta} \right\}, \quad L_{n+1} := \frac{e^{\Delta t} L_n}{1 + (e - 1) \Delta t \beta L_n}.$$

With $y_n := (\beta L_n)^{-1}$, $y_0 = 1/(\beta L_0)$ and $y_\infty := (e - 1) \Delta t / (e^{\Delta t} - 1) \in [1, e - 1)$, the envelope solves the affine recursion $y_n = y_\infty + (y_0 - y_\infty) e^{-n \Delta t}$. Hence $C_n \succeq \frac{1}{2\beta} I$ for all

$$n \geq N_{\text{cov},-}^{\text{R}} := \left\lceil \frac{1}{\Delta t} \log_+ \frac{y_0 - y_\infty}{2 - y_\infty} \right\rceil = O\left(\frac{1}{\Delta t} \log_+ \frac{1}{\beta \lambda_0} \right).$$

Moreover, with $z_0 := 1/(\alpha \lambda_{0,\max})$ and $z_\infty := \Delta t / (e^{\Delta t} - 1) \in [\frac{1}{e-1}, 1)$, one has $z_n := \lambda_{\min}(C_n^{-1}) / \alpha \geq z_\infty + (z_0 - z_\infty) e^{-n \Delta t}$, hence $C_n \preceq \frac{2}{\alpha} I$ for all $n \geq N_{\text{cov},+}^{\text{R}}$, where $N_{\text{cov},+}^{\text{R}} := 0$ if $z_0 \geq \frac{1}{2}$ and $N_{\text{cov},+}^{\text{R}} := \lceil \Delta t^{-1} \log \frac{z_\infty - z_0}{z_\infty - 1/2} \rceil = O(\Delta t^{-1} \log_+(\alpha \lambda_{0,\max}))$ otherwise.

Proof. Write $B_n := C_n^{1/2} A(a_n) C_n^{1/2}$, so that $C_{n+1}^{-1} = C_n^{-1/2} \exp(\Delta t (B_n - I)) C_n^{-1/2}$. Using $e^X \succeq I + X$ for symmetric X and $C_n^{-1/2} B_n C_n^{-1/2} = A(a_n)$,

$$C_{n+1}^{-1} \succeq (1 - \Delta t) C_n^{-1} + \Delta t A(a_n).$$

If $C_n \preceq \lambda_{\max} I$ then $C_n^{-1} \succeq \lambda_{\max}^{-1} I$ and $A(a_n) \succeq \alpha I \succeq \lambda_{\max}^{-1} I$, and since $\Delta t \leq 1$ this gives $C_{n+1}^{-1} \succeq \lambda_{\max}^{-1} I$, which proves the upper bound by induction. For the lower envelope, $0 \preceq \Delta t B_n \preceq I$ because $\Delta t \leq 1/(\beta \lambda_{\max})$, and the scalar inequality $e^x \leq 1 + (e - 1)x$ on $[0, 1]$ gives $\exp(\Delta t B_n) \preceq I + (e - 1)\Delta t B_n$ by functional calculus, hence

$$C_{n+1}^{-1} = e^{-\Delta t} C_n^{-1/2} \exp(\Delta t B_n) C_n^{-1/2} \preceq e^{-\Delta t} [C_n^{-1} + (e - 1)\Delta t A(a_n)] \preceq e^{-\Delta t} (L_n^{-1} + (e - 1)\Delta t \beta) I,$$

that is $C_{n+1} \succeq L_{n+1} I$. In the variable y_n the recursion is $y_{n+1} = e^{-\Delta t}(y_n + (e - 1)\Delta t)$, whose solution is the stated affine one, and $\Delta t \mapsto y_\infty$ is continuous and monotone on $(0, 1]$ with limit $e - 1$ at 0 and value 1 at $\Delta t = 1$, so $y_\infty \in [1, e - 1)$. The condition $C_n \succeq \frac{1}{2\beta} I$ is $y_n \leq 2$, which gives $N_{\text{cov}, -}^{\text{R}}$. For the upper envelope, $e^X \succeq I + X$ again gives $C_{n+1}^{-1} \succeq e^{-\Delta t} (C_n^{-1} + \Delta t \alpha I)$, so $z_{n+1} \geq e^{-\Delta t}(z_n + \Delta t)$ and z_n dominates the stated affine sequence. Since $z_\infty > \frac{1}{2}$, if $z_0 \geq \frac{1}{2}$ then $z_n \geq \frac{1}{2}$ for all n as a convex combination of two numbers at least $\frac{1}{2}$; if $z_0 < \frac{1}{2}$ then $z_0 < \frac{1}{2} < z_\infty$, the sequence increases to z_∞ , and $z_n \geq \frac{1}{2}$ first holds at the stated index. The case split is genuine, since the one-line form with $\log_+$ is undefined for $z_0 > z_\infty$. $\square$

The descent estimate is proved for the retraction itself, using only a Gaussian coupling, smoothness of the potential, Gaussian integration by parts, and a scalar exponential inequality.

The one-step descent estimate, [Theorem E.4](#), is stated and proved in [Section E](#).

We proceed to prove the global convergence rate of [\(20\)](#).

Theorem 2.9 (Global hitting time, Riemannian scheme). *Under [Theorem 2.1](#), with $\lambda_{0, \min} I \preceq C_0 \preceq \lambda_{0, \max} I$ and*

$$0 < \Delta t \leq \frac{1}{2\beta\lambda_{\max}},$$

the iterates of [\(20\)](#) satisfy $\Delta(a_{n+1}) \leq (1 - \frac{1}{2}\alpha L_n \Delta t) \Delta(a_n)$ with L_n the covariance envelope of [Theorem 2.8](#), and $\Delta(a_{n+1}) \leq (1 - \frac{\Delta t}{4\kappa}) \Delta(a_n)$ once $n \geq N_{\text{cov}, -}^{\text{R}}$. In particular, for every $\delta \in (0, \Delta_0)$ the hitting time $N_{\text{R}}^{\text{glob}}(\delta) := \inf\{n \geq 0 : \Delta(a_n) \leq \delta\}$ satisfies

$$N_{\text{R}}^{\text{glob}}(\delta) = O\left(\frac{1}{\Delta t} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t} \log_+ \frac{\Delta_0}{\delta}\right).$$

Proof. Apply [Theorem E.4](#) with $\Lambda = \lambda_{\max}$ and $K = \beta\lambda_{\max}$. Since $\Delta t \leq 1/(2K)$, $\Delta(a_{n+1}) \leq \Delta(a_n) - \frac{\Delta t}{2} \mathbf{G}(a_n)^2$. If $C_n \succeq L_n I$, then [\(13\)](#) gives $\mathbf{G}(a_n)^2 \geq \alpha L_n \Delta(a_n)$, and therefore $\Delta(a_{n+1}) \leq (1 - \frac{1}{2}\alpha L_n \Delta t) \Delta(a_n)$. After $N_{\text{cov}, -}^{\text{R}}$ iterations [Theorem 2.8](#) gives $L_n \geq 1/(2\beta)$, so the per-step factor is at most $1 - \Delta t/(4\kappa)$. The covariance burn-in costs $O(\Delta t^{-1} \log_+(1/(\beta\lambda_0)))$ iterations, and reducing the energy gap from Δ_0 to δ afterwards costs $O((\kappa/\Delta t) \log_+(\Delta_0/\delta))$ iterations, using $-\log(1 - x) \geq x$. $\square$

2.4 Discretization with KL divergence

The discrete algorithm with Riemannian retraction [\(20\)](#) is geometrically natural but computationally expensive, which requires to compute a matrix exponential at every step, for the ease in computation, we develop the following discretization scheme with KL divergence.

At the forward step, we compute

$$a_{n+1/2} = \arg \min_{a \in \mathcal{A}} \left\{ \mathcal{E}_2(a_n) + \langle \nabla \mathcal{E}_2(a_n), a - a_n \rangle + \frac{1}{\Delta t} \text{KL}(\rho_a \| \rho_{a_n}) \right\},$$

which gives

$$m_{n+1/2} = m_n + \Delta t C_n g(a_n), \quad C_{n+1/2} = (C_n^{-1} + \Delta t A(a_n))^{-1}.$$

For the backward step, the update reads

$$a_{n+1} = \arg \min_{a \in \mathcal{A}} \left\{ \mathcal{E}_1(a) + \frac{1}{\Delta t} \text{KL}(\rho_a \| \rho_{a_{n+1/2}}) \right\},$$

which gives

$$m_{n+1} = m_{n+1/2}, \quad C_{n+1} = (1 + \Delta t) C_{n+1/2}.$$

Hence, the discrete algorithm with KL divergence reads

$$\boxed{\begin{aligned} m_{n+1} &= m_n + \Delta t C_n g(a_n), \\ C_{n+1} &= (1 + \Delta t)(C_n^{-1} + \Delta t A(a_n))^{-1}, \end{aligned}} \quad (21)$$

or equivalently $C_{n+1}^{-1} = \frac{1}{1+\Delta t}(C_n^{-1} + \Delta t A(a_n))$.

We denote $D(a, a_n) = \text{KL}(\rho_a \| \rho_{a_n})$ and take note that $D(a, a_n)$ is a Bregman divergence because

$$D_\psi(\xi, \xi_n) = \psi(\xi) - \psi(\xi_n) - \langle \nabla \psi(\xi_n), \xi - \xi_n \rangle = \text{KL}(\rho_a \| \rho_{a_n}), \quad (22)$$

where we define

$$\psi(\xi) = -\frac{1}{2} \log \det(\xi_2 - \xi_1 \xi_1^\top), \quad \xi = (\xi_1, \xi_2) = (m, C + mm^\top). \quad (23)$$

Thus (21) is a forward-backward proximal step with the Bregman penalty D : it linearizes the potential part $\mathcal{E}_2$ explicitly in the coordinates (m, C) and treats the entropy $\mathcal{E}_1$ implicitly; only one linear solve is required per iteration. We note that the linearization is taken in (m, C) , not in the expectation parameter ξ , so (21) is not the exact mirror-descent step in expectation coordinates; the two maps agree to first order in Δt . The step is the natural one for this divergence because $\mathcal{E}_2$ is relatively smooth with respect to D with constant $\beta \lambda_{\max}$, which we record as [Theorem E.6](#) in [Section E](#).

We proceed to prove the convergence rate of the discretization algorithm with KL divergence (21), under the standard [Theorem 2.1](#). We first prove the spectral bounds of the covariance along (21).

Lemma 2.10 (KL covariance bootstrap). *Under [Theorem 2.1](#), if $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$, then along (21) one has $C_n \preceq \lambda_{\max} I$ for all $n \geq 0$, and $C_n \succeq L_n I$ with*

$$L_0 := \min \left\{ \lambda_0, \frac{1}{2\beta} \right\}, \quad L_{n+1} = \frac{(1 + \Delta t) L_n}{1 + \Delta t \beta L_n}, \quad L_n = \frac{1}{\beta [1 + (\frac{1}{\beta L_0} - 1)(1 + \Delta t)^{-n}]}.$$

Hence $C_n \succeq \frac{1}{2\beta} I$ for all

$$n \geq N_{\text{cov},-}^{\text{KL}} := \left\lceil \frac{\log_+(\frac{1}{\beta L_0} - 1)}{\log(1 + \Delta t)} \right\rceil = O\left(\frac{1}{\Delta t} \log_+ \frac{1}{\beta \lambda_0}\right),$$

and $C_n \preceq \frac{2}{\alpha} I$ for all $n \geq N_{\text{cov},+}^{\text{KL}} := \lceil \log 2 / \log(1 + \Delta t) \rceil = O(1/\Delta t)$.

Proof. The update is $C_{n+1}^{-1} = \frac{1}{1+\Delta t}(C_n^{-1} + \Delta t A(a_n))$ with $\alpha I \preceq A(a_n) \preceq \beta I$. If $C_n \preceq \lambda_{\max} I$ then $C_n^{-1} \succeq \lambda_{\max}^{-1} I$ and $A(a_n) \succeq \alpha I \succeq \lambda_{\max}^{-1} I$, so $C_{n+1}^{-1} \succeq \lambda_{\max}^{-1} I$. If $C_n \succeq L_n I$ then $C_{n+1}^{-1} \preceq \frac{1}{1+\Delta t}(L_n^{-1} + \Delta t \beta) I$, that is $C_{n+1} \succeq L_{n+1} I$; with $y_n = (\beta L_n)^{-1}$ the recursion is $y_{n+1} = 1 + (y_n - 1)/(1 + \Delta t)$, so $y_n = 1 + (y_0 - 1)(1 + \Delta t)^{-n}$, which is the stated closed form, and $y_n \leq 2$ gives $N_{\text{cov},-}^{\text{KL}}$. For the upper envelope, $A(a_n) \preceq \beta I$ gives $z_{n+1} \geq 1 + (z_n - 1)/(1 + \Delta t)$ for $z_n := \lambda_{\min}(C_n^{-1})/\alpha$; since the map $z \mapsto 1 + (z - 1)/(1 + \Delta t)$ is increasing, induction gives $z_n \geq \bar{z}_n := 1 + (z_0 - 1)(1 + \Delta t)^{-n}$, the comparison sequence solving the equality recursion. If $z_0 \geq 1$ then $z_n \geq \bar{z}_n \geq 1$ for all n , and if $z_0 < 1$ then $z_n \geq \frac{1}{2}$ as soon as $(1 + \Delta t)^{-n} \leq \frac{1}{2}$, since $1 - z_0 \leq 1$. $\square$

The covariance band of [Theorem 2.10](#) holds for every $\Delta t > 0$, in contrast with the Riemannian bootstrap of [Theorem 2.8](#), which needs $\Delta t \leq \min\{1, 1/(\beta\lambda_{\max})\}$; the rational resolvent is unconditionally stable in the covariance variable, and the stepsize restriction below is used only for the descent estimate.

The forward-KL descent and Wasserstein-gradient control, [Theorem E.7](#), is stated and proved in [Section E](#). The relevant descent divergence is the forward divergence $\text{KL}(\rho_{a_{n+1}}\|\rho_{a_n})$; the reverse divergence does not satisfy the corresponding descent inequality for arbitrary mean displacement.

Theorem 2.11 (Global hitting time, KL scheme). *Under [Theorem 2.1](#), with $\lambda_{0,\min}I \preceq C_0 \preceq \lambda_{0,\max}I$ and $0 < \Delta t \leq 1/(\beta\lambda_{\max})$, the iterates of (21) satisfy*

$$\Delta(a_{n+1}) \leq \left(1 - \frac{\alpha L_n \Delta t}{2(1 + \Delta t \beta \lambda_{\max})^2}\right) \Delta(a_n)$$

with L_n the covariance envelope of [Theorem 2.10](#), and $\Delta(a_{n+1}) \leq (1 - \frac{\Delta t}{16\kappa})\Delta(a_n)$ once $n \geq N_{\text{cov},-}^{\text{KL}}$. In particular, for every $\delta \in (0, \Delta_0)$ the hitting time $N_{\text{KL}}^{\text{glob}}(\delta) := \inf\{n \geq 0 : \Delta(a_n) \leq \delta\}$ satisfies

$$N_{\text{KL}}^{\text{glob}}(\delta) = O\left(\frac{1}{\Delta t} \log_+ \frac{1}{\beta \lambda_0} + \frac{\kappa}{\Delta t} \log_+ \frac{\Delta_0}{\delta}\right).$$

Proof. Apply [Theorem E.7](#) with $L = L_n$, $\Lambda = \lambda_{\max}$ and $K = \beta\lambda_{\max}$. Combining the two displays of [Theorem E.7](#) with the Wasserstein Polyak–Łojasiewicz inequality (12) gives the stated per-step contraction. After $N_{\text{cov},-}^{\text{KL}}$ iterations [Theorem 2.10](#) gives $L_n \geq 1/(2\beta)$, and $\Delta t \beta \lambda_{\max} \leq 1$ gives $(1 + \Delta t \beta \lambda_{\max})^2 \leq 4$, so the per-step factor is at most $1 - \Delta t/(16\kappa)$. The hitting-time bound follows by adding the covariance burn-in and using $-\log(1 - x) \geq x$. $\square$

The two schemes carry the same hitting-time bound and differ in cost and in stability range. The Riemannian update requires one symmetric matrix exponential per iteration, whereas the KL update requires only the rational resolvent $(C_n^{-1} + \Delta t A(a_n))^{-1}$, that is one linear solve. The covariance band of (21) is unconditional in Δt , while that of (20) requires $\Delta t \leq \min\{1, 1/(\beta\lambda_{\max})\}$; the certified descent ranges are $\Delta t \leq 1/(2\beta\lambda_{\max})$ and $\Delta t \leq 1/(\beta\lambda_{\max})$ respectively, so the KL scheme also admits the larger certified step.

2.5 Sharpness of fixed-step localization

Both hitting times carry the global-localization scale $(\kappa/\Delta t) \log(\Delta_0/\delta)$, which at the globally certified stepsize $\Delta t = \Theta(\kappa^{-1})$ becomes $\Theta(\kappa^2)$ iterations for a constant-factor reduction of the objective. This polynomial dependence is sharp for both schemes.

Theorem 2.12 (Sharp fixed-step global localization). *Fix $\gamma \in (0, 1/2]$ and $L_H > 0$. There exist constants $T, Y, q > 0$ and $\kappa_0 \geq 1$, depending only on γ , L_H and one fixed flat-top bump profile, such that for every $\kappa \geq \kappa_0$ there is an even potential $V_\kappa \in C^\infty(\mathbb{R})$ with*

$$1 \leq V_\kappa'' \leq \kappa, \quad \|V_\kappa'''\|_\infty \leq L_H,$$

and an initialization

$$m_0 = \frac{Y\kappa^4}{\gamma}, \quad c_0 = \frac{1}{\kappa},$$

such that, when either (20) or (21) is run with exact expectations and stepsize $\Delta t = \gamma/\kappa$, one has

$$\Delta(a_n) \geq q \Delta_0 \quad \text{for every } 0 \leq n \leq \left\lfloor \frac{T\kappa^2}{\gamma} \right\rfloor.$$

Consequently, for a constant-factor reduction of the objective both schemes require

$$\Omega\left(\frac{\kappa^2}{\gamma}\right) = \Omega\left(\frac{\kappa}{\Delta t}\right)$$

iterations. The lower bound holds with $\kappa_\star = \kappa$, with the covariance initially in the curvature-scale band, and with a dimension-free Hessian-Lipschitz constant.

The construction and the proof are given in [Section C](#).

The $\Theta(\kappa^2)$ complexity reading at $\Delta t = \Theta(1/\kappa)$, [Theorem C.1](#), is stated in [Section C](#).

Along the shadowed trajectory $c_n A(m_n, c_n) = 1 + o(1)$, so both covariance maps are almost stationary, while $c_n \simeq 1/\kappa$ and $b(m_n, c_n)/m_n = \Theta(1)$. Hence the relative mean displacement per step is $\Delta t c_n b(m_n, c_n)/m_n = \Theta(\gamma/\kappa^2)$, and a constant-factor motion of the mean requires $\Theta(\kappa^2/\gamma)$ steps. The lower bound is the product of the globally safe fixed-step scale $\Delta t = \Theta(\kappa^{-1})$ and a Fisher–Rao global mean scale of order κ^{-1} .

The theorem proves sharpness of the polynomial condition-number dependence for explicit methods committed to one globally fixed step. It does not rule out an $O(\kappa)$ global-localization result for an adaptive relative-curvature step, an Armijo or backtracking line search, or an implicit or Rosenbrock-stabilized mean update. Indeed the construction has $c_n A(m_n, c_n) \simeq 1$, so a method whose stability test is based on the current relative curvature could accept a step of order one and avoid this particular slowdown.

2.6 Removing the covariance burn-in

The remaining $\log_+(1/(\beta\lambda_0))$ term in [Theorems 2.3, 2.9 and 2.11](#) is a genuine Fisher–Rao warm-up cost when C_0 is severely underdispersed, and two optional deterministic initializers remove it. The reason a transport step helps is the one already visible in [Theorem 2.4](#): along the Fisher–Rao covariance equation an eigenvalue obeys $\dot{\mu}_i = \mu_i(1 - \mu_i h_i)$ with $h_i \in [\alpha, \beta]$, so growth from $\lambda_0 \ll 1/\beta$ is multiplicative and costs time of order $\log(1/(\beta\lambda_0))$, whereas along the Bures–Wasserstein flow

$$\dot{m}_t = g(a_t), \quad \dot{C}_t = 2I - C_t A(a_t) - A(a_t) C_t \quad (24)$$

the same eigenvalue obeys $\dot{\mu}_i = 2(1 - \mu_i h_i) \approx 2$, which is additive and scale-independent.

The Bures–Wasserstein forward–backward step reads, for $a = (m, C)$ and a transport stepsize $\eta > 0$,

$$m^+ := m + \eta g(a), \quad \tilde{C} := (I - \eta A(a))C(I - \eta A(a)), \quad C^+ := \frac{1}{2}(\tilde{C} + 2\eta I + [\tilde{C}(\tilde{C} + 4\eta I)]^{1/2}). \quad (25)$$

The covariance envelope along (25), [Theorem E.8](#), is proved in [Section E](#).

Theorem 2.13 (Transport-warm-started natural gradient). *Assume [Theorem 2.1](#) and fix $c \in (0, 1]$, with $c < 1$ for the continuous statement, since $T_{\text{BW}}(c)$ is finite only there.*

In continuous time, run (24) until $T_{\text{BW}}(c)$ and then switch permanently to (1). Then, for every $\delta \in (0, \Delta_0)$,

$$T_{\text{BW} \rightarrow \text{FR}}(\delta; c) \leq T_{\text{BW}}(c) + \min \left\{ \frac{\kappa}{c} \log_+ \frac{\Delta_0}{\delta}, \log_+ \frac{\kappa}{c\beta_\star} + \kappa_\star \log_+ \frac{\Delta_0}{\delta} \right\} = O\left(\frac{1}{\beta} + \frac{\kappa}{c} \log_+ \frac{\Delta_0}{\delta}\right)$$

suffices for $\Delta(a_t) \leq \delta$.

Discretely, set $\eta := c/\beta$, perform one step (25) if $\lambda_0 < c/\beta$ and skip it otherwise, and then run (20) with $0 < \Delta t \leq 1/(2\beta\lambda_{\max})$, or (21) with $0 < \Delta t \leq 1/(\beta\lambda_{\max})$. Then

$$N_{\text{BW} \rightarrow \bullet}(\delta; c) = \mathbf{1}_{\{\lambda_0 < c/\beta\}} + O\left(\frac{\kappa}{c\Delta t} \log_+ \frac{\Delta_0}{\delta}\right), \quad \bullet \in \{\text{R}, \text{KL}\},$$

iterations suffice for $\Delta(a_n) \leq \delta$. In either case the covariance burn-in term $\log_+(1/(\beta\lambda_0))$ is replaced by a single transport step, respectively by a warm-start time of order $1/\beta$.

Proof. The Bures–Wasserstein stage decreases $\mathcal{E}$ [11], so at the switching state a_b we have $\Delta(a_b) \leq \Delta_0$ and $C_b \succeq (c/\beta)I$ by Theorem E.8. In continuous time the lower envelope ℓ of Theorem E.1 started from $\lambda = c/\beta \leq 1/\beta$ is nondecreasing, since $\ell'(t) = \lambda e^t(1 - \beta\lambda)/(1 + \beta\lambda(e^t - 1))^2 \geq 0$, so $C_t \succeq (c/\beta)I$ for the whole Fisher–Rao stage; then (13) and (11) give $\frac{d}{dt}\Delta(a_t) \leq -\frac{c}{\kappa}\Delta(a_t)$, whence the first entry in the minimum. Alternatively $\hat{C}_b \succeq \frac{c}{\beta}C_\star^{-1} \succeq \frac{c}{\kappa}I$, so Theorem 2.3 restarted at a_b gives the second entry. The big- O form uses $T_{\text{BW}}(c) \leq (2\beta)^{-1} \log(1/(1 - c))$ for fixed c .

Discretely, Theorem E.8 gives $C_b \succeq (c/\beta)I$, $C_b \preceq \lambda_{\max}I$ and $\Delta(a_b) \leq \Delta_0$, and the covariance envelopes of Theorems 2.8 and 2.10 started from $L_0 = \min\{c/\beta, 1/(2\beta)\} \geq c/(2\beta)$ are nondecreasing in n , since their y -variables are nonincreasing. Freezing the floor at $c/(2\beta)$ in the per-step contractions of Theorems 2.9 and 2.11 gives $\Delta(a_{n+1}) \leq (1 - \frac{c\Delta t}{4\kappa})\Delta(a_n)$ for (20) and $\Delta(a_{n+1}) \leq (1 - \frac{c\Delta t}{16\kappa})\Delta(a_n)$ for (21), whence the stated count. $\square$

The second initializer is available whenever a pointwise Hessian can be queried at a single point, and it lands on the full curvature band rather than only on its lower edge.

Definition 2.14 (Quadratic rescue). Given $m_{\text{in}} \in \mathbb{R}^{N_\theta}$, query the oracle once at m_{in} ,

$$G_{\text{in}} := \nabla V(m_{\text{in}}), \quad H_{\text{in}} := \nabla^2 V(m_{\text{in}}),$$

and set

$$m_0 := m_{\text{in}} - H_{\text{in}}^{-1}G_{\text{in}}, \quad C_0 := H_{\text{in}}^{-1}.$$

The band antitonicity fact, Theorem E.9, is recorded in Section E.

Theorem 2.15 (Exact one-query recovery of Gaussian targets). Suppose $V(\theta) = \frac{1}{2}(\theta - \mu)^\top H(\theta - \mu) + \text{const.}$ with $H \succ 0$. Then Theorem 2.14 gives $m_0 = \mu$ and $C_0 = H^{-1}$, so $a_0 = a_\star$ and $\Delta(a_0) = 0$, and both (20) and (21) remain at $a_\star$. Over the class of unknown Gaussian targets the oracle complexity is exactly one query.

Proof. For a quadratic potential $\nabla V(m_{\text{in}}) = H(m_{\text{in}} - \mu)$ and $\nabla^2 V(m_{\text{in}}) = H$, so $m_0 = m_{\text{in}} - H^{-1}H(m_{\text{in}} - \mu) = \mu$ and $C_0 = H^{-1}$. The target belongs to the Gaussian family, so $a_\star = (\mu, H^{-1}) = a_0$; there $g(a_\star) = 0$ and $A(a_\star) = H = C_\star^{-1}$, so both covariance maps preserve H^{-1} and both mean maps preserve μ . For necessity, an algorithm making zero queries is independent of (μ, H) , while one query at any θ returns $H(\theta - \mu)$ and H , from which $\mu = \theta - H^{-1}G$ and $C_\star = H^{-1}$. $\square$

Theorem 2.16 (Burn-in-free complexity after quadratic rescue). Assume Theorem 2.1 and initialize by Theorem 2.14. Then $\lambda_0 \geq 1/\beta$, so in the covariance envelopes of Theorems 2.8 and 2.10 one has $L_0 = \min\{\lambda_0, \frac{1}{2\beta}\} = \frac{1}{2\beta}$ and $y_0 = 2$; the envelopes decrease from $y_0 = 2$ to their limits, so $C_n \succeq \frac{1}{2\beta}I$ for all $n \geq 0$ and $N_{\text{cov},-}^{\text{R}} = N_{\text{cov},-}^{\text{KL}} = 0$. Consequently, running (20) with $0 < \Delta t \leq 1/(2\beta\lambda_{\max})$, or (21) with $0 < \Delta t \leq 1/(\beta\lambda_{\max})$, gives for every $\delta \in (0, \Delta_0)$

$$N_{\text{rescue} \rightarrow \bullet}(\delta) = 1 + O\left(\frac{\kappa}{\Delta t} \log_+ \frac{\Delta_0}{\delta}\right), \quad \bullet \in \{\text{R}, \text{KL}\},$$

where the additive 1 counts the single deterministic Hessian query and $\Delta_0 = \Delta(a_0)$ is finite, and equals 0 for Gaussian targets by Theorem 2.15. The stage of length $\log_+(1/(\beta\lambda_0))$ is eliminated.

Proof. By Theorem E.9, $C_0 \succeq \frac{1}{\beta}I \succ \frac{1}{2\beta}I$, so $L_0 = \frac{1}{2\beta}$ and $y_0 = 2$. For (20) the envelope of Theorem 2.8 is $y_n = y_\infty + (2 - y_\infty)e^{-n\Delta t}$ with $y_\infty \in [1, e - 1) \subset [1, 2)$, which decreases from 2; for (21) the envelope of Theorem 2.10 is $y_n = 1 + (1 + \Delta t)^{-n}$, which decreases from 2 to 1. In both cases $y_n \leq 2$, that is $C_n \succeq \frac{1}{2\beta}I$, for every $n \geq 0$, and the counts $N_{\text{cov},-}^\bullet = \lceil \Delta t^{-1} \log_+ 1 \rceil = 0$. With the covariance floor $\frac{1}{2\beta}$ in force from $n = 0$, the per-step contractions of Theorems 2.9 and 2.11 apply from $n = 0$, which gives the stated count. $\square$

Both warm starts delete the $\log_+(1/(\beta\lambda_0))$ stage but sit in different oracle models and geometries. The transport warm start of [Theorem 2.13](#) is expectation-level: it uses only $g(a)$ and $A(a)$, never a pointwise Hessian, and inflates the covariance additively, so it guarantees only the floor $C_0 \geq \frac{\varepsilon}{\beta}I$ at the lower band edge without matching the pointwise curvature. The quadratic rescue of [Theorem 2.14](#) is a single pointwise-Hessian query and lands directly on the full curvature band $[\frac{1}{\beta}, \frac{1}{\alpha}]$, exactly recovering Gaussian targets. Thus transport trades a weaker oracle for a coarser landing scale, while the rescue trades one Hessian evaluation for exact Gaussian recovery; the ensuing Fisher–Rao iterations are identical.

Both devices remove only the covariance burn-in stage. They do not improve the fixed-step global-localization factor $\kappa/\Delta t$, since the lower construction of [Theorem 2.12](#) already starts at $c_0 = 1/\kappa$, after the covariance has reached the curvature scale; and they do not improve the local stiffness factor of [Section 3](#). Covariance warm-up, fixed-step global localization and local stiffness are three distinct issues.

2.7 Stationary guarantees for non-logconcave targets

The convergence theory above uses [Theorem 2.1](#) twice, once through the Wasserstein Polyak–Łojasiewicz inequality on Gaussians and once through the covariance bootstrap. When ρ_{post} is not logconcave neither ingredient is available, but the flow [\(1\)](#) still dissipates energy at the rate of its own squared gradient norm, and this alone yields a stationarity guarantee.

Corollary 2.17. *Suppose $\inf_{\mathcal{A}} \mathcal{E} > -\infty$. If the target distribution ρ_{post} is not logconcave, then along [\(1\)](#) we have the following stationary guarantee:*

$$\min_{0 \leq t \leq T} \|\text{grad}^{\text{FR}} \mathcal{E}(a_t)\|_{a_t}^2 \leq \frac{\mathcal{E}(a_0) - \inf_{\mathcal{A}} \mathcal{E}}{T}.$$

Proof. Since [\(1\)](#) is a Riemannian gradient flow over the Gaussian manifold, then

$$\frac{d\mathcal{E}(a_t)}{dt} = -\|\text{grad}^{\text{FR}} \mathcal{E}(a_t)\|_{a_t}^2$$

and hence

$$\int_0^T \|\text{grad}^{\text{FR}} \mathcal{E}(a_t)\|_{a_t}^2 dt = \mathcal{E}(a_0) - \mathcal{E}(a_T) \leq \mathcal{E}(a_0) - \inf_{\mathcal{A}} \mathcal{E},$$

this gives

$$\min_{0 \leq t \leq T} \|\text{grad}^{\text{FR}} \mathcal{E}(a_t)\|_{a_t}^2 \leq \frac{\mathcal{E}(a_0) - \inf_{\mathcal{A}} \mathcal{E}}{T}.$$

□

The discretizations [\(20\)](#) and [\(21\)](#) inherit no such statement. Without [Theorem 2.1](#) the covariance is no longer confined to the band $[L_n, \lambda_{\max}]$ of [Theorem 2.8](#) and [Theorem 2.10](#), and a two-sided bound on $\nabla^2 V$ alone does not restore it. We exhibit the mechanism on a smooth one-dimensional target. For $R > 0$ let

$$V_R(\theta) := \frac{\theta^2}{2} - 2R^2 \log \cosh(\theta/R), \quad \theta \in \mathbb{R}, \quad (26)$$

whose derivatives are

$$V'_R(\theta) = \theta - 2R \tanh(\theta/R), \quad V''_R(\theta) = 1 - 2 \text{sech}^2(\theta/R),$$

so that $-1 \leq V''_R \leq 1$ uniformly in R . For $X \sim \mathcal{N}(0, c)$ we write $A_R(c) := \mathbb{E}[V''_R(X)]$. Since $V''_R = -1 + 2 \tanh^2(\cdot/R)$ and $\tanh^2(y) \leq y^2$, we have the quantitative curvature bound

$$0 \leq A_R(c) + 1 = 2\mathbb{E} \tanh^2(\sqrt{c}Z/R) \leq \frac{2c}{R^2}, \quad Z \sim \mathcal{N}(0, 1), \quad (27)$$

so that $A_R(c) \rightarrow -1$ as $R \rightarrow \infty$, uniformly on compact intervals of c . The potential (26) is even, hence the symmetric mean $m = 0$ is invariant under all the covariance dynamics below and only the scalar covariance moves.

Proposition 2.18 (Riemannian covariance cascade). *Fix $\Delta t > 0$ and let*

$$c_{n+1}^{(R)} = c_n^{(R)} \exp \left\{ \Delta t \left(1 - c_n^{(R)} A_R(c_n^{(R)}) \right) \right\}, \quad c_0^{(R)} = 1, \quad (28)$$

be the covariance component of (20) for the target $\exp(-V_R)$. For every $M > 0$ there are R and a finite n such that $c_n^{(R)} \geq M$. Consequently no bound depending only on $\sup_\theta |V_R''(\theta)|$, Δt and c_0 can control all trajectories of (20) uniformly over this family.

Proof. Replacing A_R by its limit -1 in (28) gives the recursion

$$\bar{c}_{n+1} = \bar{c}_n e^{\Delta t(1+\bar{c}_n)}, \quad \bar{c}_0 = 1,$$

which is increasing and diverges, so there is a finite N with $\bar{c}_N > 2M$. By (27) the map $(R, c) \mapsto A_R(c)$ converges to -1 uniformly on every compact c -interval, and an induction over the first N steps gives $c_n^{(R)} \rightarrow \bar{c}_n$ for $0 \leq n \leq N$ as $R \rightarrow \infty$. For R sufficiently large we therefore have $c_N^{(R)} > M$. $\square$

Proposition 2.19 (Pole of the KL covariance resolvent). *For $\Delta t = 1$ the covariance component of (21) for the target $\exp(-V_R)$ reads*

$$c_1 = \frac{2c_0}{1 + c_0 A_R(c_0)}. \quad (29)$$

There are sequences $\varepsilon \downarrow 0$ and $R_\varepsilon \rightarrow \infty$ such that, with $c_0 = 1 - \varepsilon$,

$$c_1 = \Theta(\varepsilon^{-1}), \quad (1 - c_1 A_{R_\varepsilon}(c_1))^2 = \Theta(\varepsilon^{-2}).$$

Proof. Take $0 < \varepsilon \leq 1/4$ and choose $R_\varepsilon^2 \geq 2\varepsilon^{-2}$. By (27), $0 \leq A_{R_\varepsilon}(1 - \varepsilon) + 1 \leq \varepsilon^2$, so the denominator of (29) is

$$1 + (1 - \varepsilon) A_{R_\varepsilon}(1 - \varepsilon) = \varepsilon + (1 - \varepsilon) (A_{R_\varepsilon}(1 - \varepsilon) + 1) = \varepsilon + O(\varepsilon^2),$$

which is positive and gives $c_1 = \Theta(\varepsilon^{-1})$, in particular $c_1 \leq 2/\varepsilon$. Applying (27) once more at c_1 ,

$$0 \leq A_{R_\varepsilon}(c_1) + 1 \leq \frac{2c_1}{R_\varepsilon^2} \leq 2\varepsilon \leq \frac{1}{2},$$

whence $1 + c_1/2 \leq 1 - c_1 A_{R_\varepsilon}(c_1) = 1 + c_1 - c_1(A_{R_\varepsilon}(c_1) + 1) \leq 1 + c_1$, and the residual scaling follows. $\square$

The blow-up in (29) is caused by the approaching loss of positive definiteness of the rational resolvent, and it occurs despite the uniform Hessian bound $|V_R''| \leq 1$; taken together, the two propositions show that a global smoothness constant cannot replace covariance control in a trajectory-wise Fisher–Rao theorem.

The Bures–Wasserstein forward–backward method carries a different guarantee beyond log-concavity. Under the assumptions and stepsize η of Diao et al. [7, Theorem 5.5], its iterates satisfy the running-minimum bound

$$\min_{0 \leq n < N} \|\text{grad}^{\text{BW}} \mathcal{E}(a_n)\|_{\text{BW}}^2 \leq \frac{150\{\mathcal{E}(a_0) - \inf \mathcal{E}\}}{\eta N}.$$

This is a stationarity statement in the Bures–Wasserstein geometry and does not control the pointwise Fisher–Rao residual appearing in Theorems 2.18 and 2.19. We use it only as a comparison between the two geometries.

2.8 A covariance-constrained KL scheme

Theorems 2.18 and **2.19** suggest constraining the covariance directly, as in Sun et al. [17]. Fix constants $0 < \lambda_- \leq \lambda_+ < \infty$ and define the covariance-truncated Gaussian parameter space

$$\mathcal{A}' := \left\{ a = (m, C) \in \mathbb{R}^{N_\theta} \times \mathbb{S}_{++}(N_\theta) : \lambda_- I \preceq C \preceq \lambda_+ I \right\},$$

and assume the following global smoothness condition throughout this subsection.

Assumption 2.20. The potential $V \in C^2(\mathbb{R}^{N_\theta})$ satisfies

$$\|\nabla^2 V(\theta)\|_{\text{op}} \leq \beta, \quad \forall \theta \in \mathbb{R}^{N_\theta},$$

equivalently $-\beta I \preceq \nabla^2 V(\theta) \preceq \beta I$.

For $C \in \mathbb{S}_{++}(N_\theta)$ with spectral decomposition $C = Q \text{diag}(c_1, \dots, c_{N_\theta}) Q^\top$ we define the clipping operator

$$\text{Clip}_{[\lambda_-, \lambda_+]}(C) := Q \text{diag}(\min\{\lambda_+, \max\{\lambda_-, c_1\}\}, \dots, \min\{\lambda_+, \max\{\lambda_-, c_{N_\theta}\}\}) Q^\top.$$

Clipping the spectrum is exactly the Bregman projection onto the covariance interval for the log-determinant divergence $D_\Phi(\cdot, \cdot)$ generated by $\Phi(C) = -\frac{1}{2} \log \det C$, which is the mechanism that lets the projected step retain a descent certificate.

At $a_n = (m_n, C_n)$ we write

$$b_n := \mathbb{E}_{\mathcal{N}(m_n, C_n)}[\nabla_\theta V(\theta)], \quad A_n := \mathbb{E}_{\mathcal{N}(m_n, C_n)}[\nabla_\theta \nabla_\theta V(\theta)],$$

so that (21) followed by the projection reads

$$\begin{aligned} m_{n+1} &= m_n - \Delta t C_n b_n, \\ \tilde{C}_{n+1} &= (1 + \Delta t)(C_n^{-1} + \Delta t A_n)^{-1}, \quad C_{n+1} = \text{Clip}_{[\lambda_-, \lambda_+]}(\tilde{C}_{n+1}). \end{aligned} \tag{30}$$

The inverse is well defined at the stepsizes considered below. The one-step stationarity measure of this split scheme is

$$S_n := \frac{1}{2} \|m_{n+1} - m_n\|_{C_n^{-1}}^2 + \text{KL}(\mathcal{N}(0, C_n) \parallel \mathcal{N}(0, C_{n+1})), \tag{31}$$

where the orientation of the covariance divergence matters: the forward divergence, with C_{n+1} in the first slot, carries the smoothness remainder, while the reverse one appearing in (31) is what the descent certificate leaves behind.

Theorem 2.21. Under *Theorem 2.20*, if $a_0 \in \mathcal{A}'$ and the stepsize satisfies $0 < \Delta t \leq \frac{1}{2\beta\lambda_+}$, then the iterates of (30) are well defined, remain in $\mathcal{A}'$, and satisfy

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t} S_n. \tag{32}$$

Consequently, for every $N \geq 1$,

$$\min_{0 \leq n < N} S_n \leq \frac{\Delta t}{N} \{\mathcal{E}(a_0) - \mathcal{E}(a_N)\}. \tag{33}$$

Proof. Since $\lambda_- I \preceq C_n \preceq \lambda_+ I$ and $\|A_n\|_{\text{op}} \leq \beta$, the stepsize restriction gives

$$C_n^{-1} + \Delta t A_n \succeq (\lambda_+^{-1} - \Delta t \beta) I \succeq (2\lambda_+)^{-1} I,$$

so $\tilde{C}_{n+1}$ is well defined, and $C_{n+1} \in [\lambda_- I, \lambda_+ I]$ by construction. Writing $\mathcal{E} = \Phi + \mathcal{E}_2 + \text{const}$ with $\Phi(C) = -\frac{1}{2} \log \det C$ and $\mathcal{E}_2(a) = \mathbb{E}_{\rho_a}[V(\theta)]$, a square-root coupling of ρ_{a_n} and $\rho_{a_{n+1}}$ together

with the scalar inequality $x - 1 - \log x \geq (\sqrt{x} - 1)^2$ shows that $\mathcal{E}_2$ is relatively smooth with constant $L = 2\beta\lambda_+$ with respect to the split divergence built from $\frac{1}{2}\|\cdot\|_{C_n^{-1}}^2$ and $\text{KL}(\mathcal{N}(0, \cdot) \|\mathcal{N}(0, C_n))$. The pair (m_{n+1}, C_{n+1}) is the unique minimizer over $\mathcal{A}'$ of the associated proximal model, because the covariance part of that model is, up to an additive constant, a multiple of $D_\Phi(\cdot, \tilde{C}_{n+1})$, whose Bregman projection onto $[\lambda_- I, \lambda_+ I]$ is spectral clipping. Mean optimality then contributes $b_n^\top(m_{n+1} - m_n) = -\frac{2}{\Delta t} \cdot \frac{1}{2}\|m_{n+1} - m_n\|_{C_n^{-1}}^2$, while the constrained log-determinant three-point inequality contributes the reverse covariance divergence in (31). Since $L\Delta t \leq 1$, the forward remainder is absorbed and the two estimates combine to (32). Summing (32) over $0 \leq n < N$ and bounding the minimum by the average gives (33). The details are given in Section G. $\square$

The full successive-Gaussian KL bound with the λ_+/λ_- factor is Theorem G.2 in Section G.

For centered scalar trajectories the mean term vanishes, so the factor-one bound (33) is the quantity plotted in Section 6.5.

Theorem 2.21 certifies stationarity of the projected split map on the covariance-truncated set $\mathcal{A}'$. If an iterate is pinned at the upper clip $\lambda_+ I$, then S_n can vanish while the unconstrained Fisher–Rao gradient points outward. The theorem therefore does not imply that the unconstrained gradient is small, nor that the iterates converge to the unconstrained Gaussian optimizer $a_\star$.

3 Local convergence near equilibrium

The global theory of Section 2 delivers the iterate into a neighborhood of the optimizer at a rate proportional to κ^{-1} . Near the optimizer the dynamics is governed by a different mechanism, and the condition number disappears from the rate. We develop that mechanism here, and we organize it around a decomposition of the natural gradient field into a Gaussian core and a non-Gaussian remainder,

$$F = F_G + H, \quad F_G(m, C) := (-Cm, C - C^2),$$

in which F_G , the field of the standard Gaussian target, is retained exactly and only H is Taylor-bounded. Since $H \equiv 0$ whenever the target is Gaussian, everything the local theory certifies is exact on Gaussian targets.

3.1 Whitened coordinates and the linearized generator

Definition 3.1 (Whitened coordinates). Given the optimizer $a_\star = (m_\star, C_\star)$, set

$$\hat{m} := C_\star^{-1/2}(m - m_\star), \quad \hat{C} := C_\star^{-1/2}CC_\star^{-1/2}, \quad \hat{V}(z) := V(m_\star + C_\star^{1/2}z),$$

and let $0 < \alpha_\star \leq \beta_\star < \infty$ satisfy $\alpha_\star I \preceq \nabla^2 \hat{V}(z) \preceq \beta_\star I$, with $\kappa_\star := \beta_\star/\alpha_\star$.

The Kullback–Leibler divergence and the dynamics (1), (20), (21) are invariant under $\theta \mapsto C_\star^{-1/2}(\theta - m_\star)$, so throughout this section $a = (m, C)$ denotes the whitened parameter and $a_\star = (0, I)$. The stationarity conditions read $\mathbb{E}_{\mathcal{N}(0, I)}[\nabla \hat{V}] = 0$ and $\mathbb{E}_{\mathcal{N}(0, I)}[\nabla^2 \hat{V}] = I$, whence $\alpha_\star \leq 1 \leq \beta_\star$, and one may take $\alpha_\star \geq 1/\kappa$, $\beta_\star \leq \kappa$, $\kappa_\star \leq \kappa^2$. Statements transfer back through the fixed equilibrium norms

$$\|a - a_\star\|_\star^2 := \|\hat{m}\|^2 + \frac{1}{2}\|\hat{C} - I\|_F^2, \quad \|a - a_\star\|_{\star, \Delta t}^2 := \|\hat{m}\|^2 + \frac{1+\Delta t}{2}\|\hat{C} - I\|_F^2, \quad (34)$$

of which $\|\cdot\|_\star$ is the Fisher–Rao tangent norm frozen at $a_\star$. We write $\xi := a - a_\star$, $D(a) := \|\xi\|_\star^2$, and $F(a) := (Cg(a), C - CA(a)C)$, so that (1) reads $\dot{a} = F(a)$ with $F(a_\star) = 0$. Finally we set

$$t_0 := \max\{1, \log(4\beta_\star)\}, \quad \Gamma := \min\{\beta_\star - 1, \sqrt{N_\theta}(4\sqrt{2t_0} + 4t_0 + 4)\}, \quad (35)$$

so that $\Gamma = O(1 + \sqrt{N_\theta} \log \kappa_\star) = O(1 + \sqrt{N_\theta} \log \kappa)$.

The linearization of F at $a_\star$ is expressed through three Gaussian score operators, the first two of Hermite degree one and the third of Hermite degree two.

Definition 3.2 (Linearized score operators). Let $Z \sim \mathcal{N}(0, I)$ and $B(Z) := \nabla^2 \hat{V}(Z)$, so that $\mathbb{E}B(Z) = I$. For $u \in \mathbb{R}^{N_\theta}$ and $X \in \mathbb{S}(N_\theta)$ define

$$\mathcal{T}[X]_k := \mathbb{E}[\text{Tr}(XB(Z))Z_k], \quad \mathcal{T}^*[u]_{ij} := \mathbb{E}[(u^\top Z)B_{ij}(Z)], \quad \mathcal{H}[X]_{ij} := \mathbb{E}[B_{ij}(Z)(Z^\top XZ - \text{Tr } X)].$$

The Gaussian score identities of [Section A](#) give $u^\top \mathcal{T}[X] = \text{Tr}(\mathcal{T}^*[u]X)$ and $\text{Tr}(\mathcal{H}[X]Y) = \text{Tr}(X\mathcal{H}[Y])$, so $\mathcal{T}$ and $\mathcal{T}^*$ are mutually adjoint and $\mathcal{H}$ is self-adjoint.

We reduce the spectral analysis to the diagonal modes of $\mathcal{H}$, on which the two score operators are represented by a matrix pair whose Schur complement is the quantity of interest.

The exact diagonal-mode identity behind both branches, [Theorem A.4](#), is stated and proved in [Section A](#); there, in an orthogonal basis with $X = \text{diag}(\lambda)$, $\tilde{T}_{ki} := \mathbb{E}[B_{ii}(Z)Z_k]$ and $D := G - \mathbf{1}\mathbf{1}^\top$ with $G_{ij} := \mathbb{E}[B_{ii}(Z)Z_j^2]$ is the diagonal fourth-moment matrix.

The two branches of Γ have different origins. The branch $\beta_\star - 1$ is the pointwise curvature bound applied to a diagonal quadratic form. The branch $\sqrt{N_\theta}(4\sqrt{2t_0} + 4t_0 + 4)$ comes from the exact representation $\lambda^\top D\lambda = \mathbb{E}[\text{Tr}(XB(Z))(Z^\top XZ - \text{Tr } X)]$ together with a Laurent–Massart tail bound for the chi-square chaos $Z^\top XZ - \text{Tr } X$; it is a genuine $\sqrt{N_\theta}$ and not N_θ , because the pointwise bound $|\text{Tr}(XB)| \leq \text{Tr}(|X|B)$ costs only $\|\lambda\|_1$.

Proposition 3.3 (Spectral sandwich for the linearized generator). Let $J_\star := DF(a_\star)$ and $\mathcal{L}_\star := -J_\star$. Then

$$\mathcal{L}_\star(u, X) = \left(u + \frac{1}{2}\mathcal{T}[X], X + \mathcal{T}^*[u] + \frac{1}{2}\mathcal{H}[X]\right), \quad (36)$$

$\mathcal{L}_\star$ is self-adjoint in the Fisher–Rao inner product $\langle \cdot, \cdot \rangle_\star$ frozen at $a_\star$, and with $\gamma_\star := \lambda_{\min}(\mathcal{L}_\star)$ and $\Lambda_\star := \lambda_{\max}(\mathcal{L}_\star)$,

$$\boxed{\max\left\{\alpha_\star, \frac{1}{4 + \Gamma}\right\} \leq \gamma_\star \leq \Lambda_\star \leq \min\left\{\beta_\star, \frac{4 + \Gamma}{2}\right\}.} \quad (37)$$

For a Gaussian target, $\mathcal{T} = 0$, $\mathcal{H} = 0$, and $\gamma_\star = \Lambda_\star = 1$.

Proof. Bonnet–Price differentiation at $a_\star = (0, I)$, recorded in [Section A](#), gives $\mathbb{E}_{\mathcal{N}(u, I+X)}[\nabla \hat{V}] = u + \frac{1}{2}\mathcal{T}[X] + O(\|(u, X)\|_\star^2)$ and $\mathbb{E}_{\mathcal{N}(u, I+X)}[\nabla^2 \hat{V}] = I + \mathcal{T}^*[u] + \frac{1}{2}\mathcal{H}[X] + O(\|(u, X)\|_\star^2)$; substituting into $F(m, C) = (-C\mathbb{E}[\nabla \hat{V}], C - C\mathbb{E}[\nabla^2 \hat{V}]C)$ gives (36), and the adjoint identities of [Theorem 3.2](#) make $\mathcal{L}_\star$ self-adjoint. Its quadratic form, after diagonalizing $X = \text{diag}(\lambda)$, is

$$Q(u, \text{diag } \lambda) := \langle \mathcal{L}_\star(u, X), (u, X) \rangle_\star = \|u\|^2 + u^\top \tilde{T}\lambda + \frac{1}{2}\|\lambda\|^2 + \frac{1}{4}\lambda^\top D\lambda. \quad (38)$$

Set $\nu := 1/(4 + \Gamma)$. Completing the square in u , the coercivity $Q \geq \nu \|\cdot\|_\star^2$ reduces through (77) to $\frac{1-\nu}{2} - \frac{1+\nu\Gamma}{4(1-\nu)} \geq 0$, that is to $1 - \nu(4 + \Gamma) + 2\nu^2 \geq 0$, whose left side equals $2\nu^2 > 0$ at $\nu = 1/(4 + \Gamma)$. For the matching upper bound, [Theorem A.4](#) gives

$$Q(u, \text{diag } \lambda) \leq \|u\|^2 + \sqrt{1 + \Gamma} \|u\| \|\lambda\| + \left(\frac{1}{2} + \frac{\Gamma}{4}\right) \|\lambda\|^2 \leq \frac{4 + \Gamma}{2} \left(\|u\|^2 + \frac{1}{2} \|\lambda\|^2\right).$$

For the second branch we use the exact identity $\mathbb{E}[\eta(Z)^\top B(Z)\eta(Z)] = \|u\|^2 + 2u^\top \tilde{T}\lambda + \|\lambda\|^2 + \lambda^\top D\lambda$ of [Theorem A.4](#) with $\eta(Z) = 2u + XZ$, which turns (38) into

$$Q(u, \text{diag } \lambda) = \frac{1}{4}\mathbb{E}[(2u + XZ)^\top B(Z)(2u + XZ)] + \frac{1}{4}\|\lambda\|^2.$$

Sandwiching $\alpha_\star I \preceq B(Z) \preceq \beta_\star I$ against $\mathbb{E}\|2u + XZ\|^2 = 4\|u\|^2 + \|\lambda\|^2$ gives $\alpha_\star \|u\|^2 + \frac{1+\alpha_\star}{4} \|\lambda\|^2 \leq Q \leq \beta_\star \|u\|^2 + \frac{1+\beta_\star}{4} \|\lambda\|^2$, and $\alpha_\star \leq 1 \leq \beta_\star$ yields $\frac{1+\alpha_\star}{4} \geq \frac{\alpha_\star}{2}$ and $\frac{1+\beta_\star}{4} \leq \frac{\beta_\star}{2}$, so $\alpha_\star \|\xi\|_\star^2 \leq Q(\xi) \leq \beta_\star \|\xi\|_\star^2$. The spectral theorem gives (37). $\square$

The two branches of (37) are complementary: the Γ branch is universal and carries the dimension dependence, while the $(\alpha_\star, \beta_\star)$ branch is sharper for well-conditioned targets, where the universal estimate is pessimistic. For the rest of the local theory any certified lower bound $0 < \gamma_0 \leq \gamma_\star$ may be used; the universal choice is $\gamma_0 = 1/(4 + \Gamma)$, and the Gaussian target has the exact value $\gamma_\star = 1$.

We now record what (37) gives for the two discretizations. The Riemannian map is a straight Euler step in the Fisher–Rao geometry and inherits the spectrum of $\mathcal{L}_\star$ directly; the KL map does not, because its covariance block carries the extra factor $(1 + \Delta t)^{-1}$ from the backward entropy step, and it becomes self-adjoint only after the covariance block of the norm is reweighted by $1 + \Delta t$, which is what $\|\cdot\|_{\star, \Delta t}$ in (34) does.

Proposition 3.4 (Linearized contractions of the two schemes). *Let T_R and T_{KL} denote the maps (20) and (21) in whitened coordinates. If $0 < \Delta t \leq 2/(4 + \Gamma)$, then*

$$\|DT_R(a_\star)\xi\|_\star \leq \left(1 - \frac{\Delta t}{4 + \Gamma}\right) \|\xi\|_\star, \quad \|DT_{KL}(a_\star)\xi\|_{\star, \Delta t} \leq \left(1 - \frac{\Delta t}{4 + 2\Delta t + \Gamma}\right) \|\xi\|_{\star, \Delta t}.$$

Proof. For the Riemannian map $DT_R(a_\star) = I - \Delta t \mathcal{L}_\star$, and $\Delta t \Lambda_\star \leq 1$ by (37), so its spectrum lies in $[0, 1 - \Delta t/(4 + \Gamma)]$. For the KL map, differentiating (21) at $(0, I)$ gives

$$M_{KL}(u, X) := DT_{KL}(a_\star)(u, X) = \left((1 - \Delta t)u - \frac{\Delta t}{2}\mathcal{T}[X], \frac{X - \Delta t \mathcal{T}^*[u] - \frac{\Delta t}{2}\mathcal{H}[X]}{1 + \Delta t}\right),$$

which is self-adjoint in $\langle (u, X), (v, Y) \rangle_{\star, \Delta t} := u^\top v + \frac{1 + \Delta t}{2} \text{Tr}(XY)$ and satisfies the exact identity $\|\xi\|_{\star, \Delta t}^2 - \langle M_{KL}\xi, \xi \rangle_{\star, \Delta t} = \Delta t Q(\xi)$ with Q as in (38). Completing the square in u as before, the weighted coercivity $Q(\xi) \geq \nu \|\xi\|_{\star, \Delta t}^2$ reduces to $1 - \nu(4 + 2\Delta t + \Gamma) + 2(1 + \Delta t)\nu^2 \geq 0$, whose left side equals $2(1 + \Delta t)\nu^2 > 0$ at $\nu = 1/(4 + 2\Delta t + \Gamma)$. Since $Q(\xi) \leq \frac{4 + \Gamma}{2} \|\xi\|_{\star, \Delta t}^2$ and $\Delta t \leq 2/(4 + \Gamma)$, the spectrum of M_{KL} is nonnegative, so its operator norm is its largest eigenvalue, which is at most $1 - \Delta t/(4 + 2\Delta t + \Gamma)$. $\square$

3.2 The Gaussian core and the certified region

The energy gap controls the covariance spectrum, not merely the parameter distance, and the region on which the local theory is certified is the corresponding energy sublevel set. The mechanism is the entropy term, which contributes a logarithmic barrier at degenerate covariances on top of the quadratic coercivity supplied by strong convexity.

Coercivity of the energy in the equilibrium norm, Theorem E.10, is proved in Section E; it bounds $\Delta(m, C)$ from below by $\frac{\alpha_\star}{2} \|m\|^2$ plus the scalar function

$$\phi_{\alpha_\star}(s) := s - 1 - \log s + \frac{\alpha_\star}{2}(s - 1)^2$$

summed over the eigenvalues s_i of $C^{1/2}$, with equality for Gaussian targets, where $\Delta_G(m, C) = \frac{1}{2} \|m\|^2 + \sum_i \varphi(c_i)$ with $\varphi(c) := \frac{1}{2}(c - 1 - \log c)$ and c_i the eigenvalues of C .

Since $\phi_{\alpha_\star}$ is strictly decreasing on $(0, 1)$, increasing on $(1, \infty)$, and vanishes at 1, for each $\delta > 0$ the equation $\phi_{\alpha_\star}(s) = \delta$ has roots $0 < s_-(\delta) < 1 < s_+(\delta)$, and we set $\ell_\delta := s_-(\delta)^2$, $U_\delta := s_+(\delta)^2$. Then $\Delta(a) \leq \delta$ forces $\ell_\delta I \preceq C \preceq U_\delta I$, and, with

$$\vartheta(\delta) := \sup_{s \in [s_-, s_+], s \neq 1} \frac{(s^2 - 1)^2}{2\phi_{\alpha_\star}(s)} \leq \frac{(1 + s_+)^2}{\alpha_\star + s_+^{-2}}, \quad \mathfrak{b}(\delta)^2 := \max\left\{\frac{2}{\alpha_\star}, \vartheta(\delta)\right\}, \quad (39)$$

one has $\Delta(a) \leq \delta \Rightarrow \|a - a_\star\|_\star \leq \mathfrak{b}(\delta)\sqrt{\delta}$ and $\|a - a_\star\|_{\star, \Delta t} \leq \mathfrak{b}_{\Delta t}(\delta)\sqrt{\delta}$ with $\mathfrak{b}_{\Delta t}(\delta)^2 := \max\{2/\alpha_\star, (1 + \Delta t)\vartheta(\delta)\}$. For a Gaussian target ℓ_δ and U_δ are the two roots of $\varphi(c) = \delta$.

Definition 3.5 (Region, hull, and non-Gaussian moduli). For $\delta > 0$ set $\mathcal{U}_\delta := \{a : \Delta(a) \leq \delta\}$ and let $\mathcal{V}_\delta := \{a_\star + s(a - a_\star) : a \in \mathcal{U}_\delta, s \in [0, 1]\}$ be its star-shaped hull. Set $\bar{\Lambda}_\delta := \max\{U_\delta, 1/\alpha_\star\}$ and

$$\mathcal{B}_\delta := \sup_{a \in \mathcal{V}_\delta} \|A(a) - I\|_{\text{op}}, \quad \Sigma_\delta := \sup_{a \in \mathcal{V}_\delta} (\mathbb{E}_{\rho_a} \|\nabla^2 \hat{V}(X) - A(a)\|_{\text{F}}^2)^{1/2}, \quad \bar{\Sigma}_\delta := \min\{\Sigma_\delta, \beta_\star - \alpha_\star\}.$$

All three moduli vanish for Gaussian targets and are nondecreasing in δ .

Every $a \in \mathcal{U}_\delta$ satisfies $\ell_\delta I \preceq C \preceq U_\delta I$ and $\|a - a_\star\|_\star \leq \mathfrak{b}(\delta)\sqrt{\delta}$, and so does every point of $\mathcal{V}_\delta$, since $C' = (1 - s)I + sC$ has eigenvalues in $[\ell_\delta, U_\delta]$ and $\|a' - a_\star\|_\star = s\|\xi\|_\star$. The region is compact, and it is forward invariant for the flow, because $\frac{d}{dt}\Delta = -\mathbf{G}^2 \leq 0$, and for any scheme step satisfying $\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n)$.

We next show that the Gaussian core dissipates exactly against the equilibrium norm. The Gaussian core is nonlinear in the covariance, but its dissipation against the equilibrium norm is an exact algebraic identity, so no Taylor expansion is spent on it and the certified rate degrades only through the covariance floor ℓ_δ .

Lemma 3.6 (Exact Gaussian-core dissipation). *Let $F_G(m, C) := (-Cm, C - C^2)$ be the natural gradient field of the standard Gaussian target $\hat{V}_G(z) = \frac{1}{2}\|z\|^2$. Then $DF_G(a_\star) = -I$ on $\mathbb{R}^{N_\theta} \times \mathbb{S}(N_\theta)$, and for every $a = (m, C)$, with $\xi = a - a_\star$,*

$$\langle \xi, F_G(a) \rangle_\star = -m^\top C m - \frac{1}{2} \text{Tr}(C(C - I)^2) = -\frac{1}{2} q_G(a) D(a), \quad q_G(a) := \frac{2m^\top C m + \text{Tr}(C(C - I)^2)}{\|m\|^2 + \frac{1}{2} \|C - I\|_{\text{F}}^2}.$$

Moreover $q_G(a) \geq 2\lambda_{\min}(C)$, with equality when $m = 0$ and $C - I$ is supported on a minimal eigendirection of C ; consequently $\langle \xi, F_G(a) \rangle_\star \leq -\ell_\delta D(a)$ on $\mathcal{U}_\delta$.

Proof. At $a_\star = (0, I)$ with $\zeta = (u, U)$ we have $D(-Cm)[\zeta] = -u$ and $D(C - C^2)[\zeta] = U - (UI + IU) = -U$. Writing $X = C - I$, we have $C - C^2 = -C(C - I) = -(X + X^2)$, so $\langle \xi, F_G \rangle_\star = -m^\top C m + \frac{1}{2} \text{Tr}(X \cdot (-(X + X^2))) = -m^\top C m - \frac{1}{2} \text{Tr}(CX^2)$, using $I + X = C$. Finally $2m^\top C m \geq 2\lambda_{\min}(C)\|m\|^2$ and, since C and X commute, $\text{Tr}(CX^2) \geq \lambda_{\min}(C)\|X\|_{\text{F}}^2$, so the numerator of q_G is at least $2\lambda_{\min}(C)D(a)$, with equality as stated; on $\mathcal{U}_\delta$, $\lambda_{\min}(C) \geq \ell_\delta$. $\square$

The remainder $H := F - F_G$ is, since $g_G(a) = -m$ and $A_G(a) = I$,

$$H(a) = (C(g(a) + m), -C(A(a) - I)C), \quad H(a_\star) = 0, \quad (40)$$

both components being built from the non-Gaussian discrepancies $g(a) + m = -\mathbb{E}_{\rho_a}[\nabla \hat{V}(X) - X]$ and $A(a) - I = \mathbb{E}_{\rho_a}[\nabla^2 \hat{V}(X) - I]$. Its second differential is controlled by the following modulus.

Lemma 3.7 (Non-Gaussian second-differential modulus). *Define*

$$K_{\text{ng}}(\delta) := \sup_{a \in \mathcal{V}_\delta} \sup_{\zeta \neq 0} \frac{\|D^2 H(a)[\zeta, \zeta]\|_\star}{\|\zeta\|_\star^2}.$$

Then $K_{\text{ng}}(\delta) \leq (K_{\text{ng},m}(\delta)^2 + \frac{1}{2} K_{\text{ng},C}(\delta)^2)^{1/2}$ with

$$K_{\text{ng},m}(\delta) := \frac{U_\delta^{3/2}}{\ell_\delta^2} \left(\sqrt{3} \bar{\Sigma}_\delta + \Sigma_\delta + \sqrt{\Sigma_\delta^2 + 2\mathcal{B}_\delta^2} \right), \quad K_{\text{ng},C}(\delta) := \frac{U_\delta^2}{\ell_\delta^2} ((4\sqrt{2} + \sqrt{6})\Sigma_\delta + 4\mathcal{B}_\delta).$$

In particular $K_{\text{ng}}(\delta) = 0$ for every Gaussian target, and, with the band factors U_δ, ℓ_δ held fixed, $K_{\text{ng}}(\delta) = O(\frac{U_\delta^2}{\ell_\delta^2}(\mathcal{B}_\delta + \Sigma_\delta))$. The proof is given in [Section A](#).

Since $H(a_\star) = 0$, the integral Taylor formula $R_{\text{ng}}(\xi) := H(a) - DH(a_\star)\xi = \int_0^1 (1-s) D^2 H(a_\star + s\xi)[\xi, \xi] ds$ has every segment point in $\mathcal{V}_\delta$, so $\|R_{\text{ng}}(\xi)\|_\star \leq \frac{1}{2} K_{\text{ng}}(\delta) \|\xi\|_\star^2$ for $a = a_\star + \xi \in \mathcal{U}_\delta$. Combining this with [Theorem 3.6](#) gives the local convergence of the flow.

Theorem 3.8 (Gaussian-core local convergence of the flow). *Let $0 < \gamma_0 \leq \gamma_\star$ and $\delta > 0$, and set*

$$\rho_{\text{core}}(\delta) := 2\gamma_0 - 2 + 2\ell_\delta - K_{\text{ng}}(\delta) \mathfrak{b}(\delta) \sqrt{\delta}. \quad (41)$$

If $\rho_{\text{core}}(\delta) > 0$ and $\Delta(a_0) \leq \delta$, then the flow (1) remains in $\mathcal{U}_\delta$ and

$$\|a_t - a_\star\|_\star^2 \leq e^{-\rho_{\text{core}}(\delta)t} \|a_0 - a_\star\|_\star^2, \quad t \geq 0.$$

Proof. The region $\mathcal{U}_\delta$ is compact and forward invariant, so the flow exists in it for all $t \geq 0$. Since $\|\cdot\|_\star$ is a fixed inner-product norm, $\frac{1}{2} \frac{d}{dt} D = \langle \xi_t, F_G(a_t) \rangle_\star + \langle \xi_t, H(a_t) \rangle_\star$, and the first term is at most $-\ell_\delta D(a_t)$ by [Theorem 3.6](#). Next $H(a_t) = B_\star \xi_t + R_{\text{ng}}(\xi_t)$ with $B_\star = DH(a_\star) = J_\star + I$ self-adjoint, so $\langle \xi_t, B_\star \xi_t \rangle_\star = \langle \xi_t, J_\star \xi_t \rangle_\star + \|\xi_t\|_\star^2 \leq (1 - \gamma_0) D(a_t)$ by [Theorem 3.3](#). Finally $\|\xi_t\|_\star \leq \mathfrak{b}(\delta) \sqrt{\delta}$ gives $\langle \xi_t, R_{\text{ng}}(\xi_t) \rangle_\star \leq \frac{1}{2} K_{\text{ng}}(\delta) \mathfrak{b}(\delta) \sqrt{\delta} D(a_t)$. Adding, $\frac{d}{dt} D \leq -\rho_{\text{core}}(\delta) D$, and Grönwall's inequality completes the proof. $\square$

Corollary 3.9 (Universal certified level). *With $\gamma_0 = 1/(4 + \Gamma)$ one has $\rho_{\text{core}}(\delta) \rightarrow 2/(4 + \Gamma)$ as $\delta \downarrow 0$, since $\ell_\delta \uparrow 1$ and $\mathfrak{b}(\delta) \sqrt{\delta} \downarrow 0$. Hence for every $\rho < 2/(4 + \Gamma)$ there is $\delta_c(\rho) > 0$ with $\rho_{\text{core}}(\delta) \geq \rho$ on $(0, \delta_c(\rho)]$, and every such level is admissible in [Theorem 3.8](#).*

Corollary 3.10 (Exact sharpness for Gaussian targets). *Let the target be Gaussian. Then $\gamma_\star = 1$, $K_{\text{ng}} \equiv 0$, and [Theorem 3.8](#) holds with the exact rate $\rho_{\text{core}}(\delta) = 2\ell_\delta$. Consequently, for every $\rho \in (0, 2)$ the largest energy sublevel on which rate ρ is certified is $\{\Delta \leq \Delta^\#(\rho)\}$ with*

$$\Delta^\#(\rho) = \varphi\left(\frac{\rho}{2}\right) = \frac{1}{2} \left(\frac{\rho}{2} - 1 - \log \frac{\rho}{2} \right), \quad (42)$$

independent of the dimension, and this level is exactly the supremal threshold $\inf\{\Delta(a) : a \neq a_\star, q_c(a) < \rho\}$ for the instantaneous rate $q_c(a) := -2 \langle a - a_\star, F(a) \rangle_\star / \|a - a_\star\|_\star^2$.

Proof. For a Gaussian target $F = F_G$ and $H \equiv 0$, so $\gamma_\star = 1$ by [Theorem 3.3](#) and $K_{\text{ng}} \equiv 0$ by [Theorem 3.7](#), whence $\rho_{\text{core}} = 2\ell_\delta$; rate ρ is certified precisely when $\ell_\delta \geq \rho/2$, that is, since ℓ_δ is the lower root of $\varphi(c) = \delta$ and φ decreases on $(0, 1)$, when $\delta \leq \varphi(\rho/2)$. For the benchmark, $q_c = q_G \geq 2\lambda_{\min}(C)$ by [Theorem 3.6](#); if $q_c(a) < \rho$ then some eigenvalue $c_j < \rho/2$, so $\Delta_G(a) \geq \varphi(c_j) > \varphi(\rho/2)$, while $m = 0$ and $C = \text{diag}(c, 1, \dots, 1)$ with $c \uparrow \rho/2$ has $q_G = 2c < \rho$ and $\Delta_G \rightarrow \varphi(\rho/2)$. $\square$

For general targets the certified rate degrades from $2\gamma_\star$ continuously, through ℓ_δ and through the moduli $\mathcal{B}_\delta, \Sigma_\delta$ that enter K_{ng} . As $\delta \downarrow 0$, $\rho_{\text{core}}(\delta) \rightarrow 2\gamma_\star$, which is the exact asymptotic squared-norm rate of the flow.

3.3 Discrete local convergence

For Gaussian targets both schemes are scalar in the eigenbasis of C_n and admit exact contraction intervals. We record the KL map first, since its interval is closed-form.

Proposition 3.11 (Exact Gaussian KL map). *Let the target be Gaussian, so that (21) in whitened coordinates becomes $m_{n+1} = (I - \Delta t C_n) m_n$ and $C_{n+1} = (1 + \Delta t) C_n (I + \Delta t C_n)^{-1}$. In the eigenbasis of C_n , each covariance eigenvalue c and the aligned mean component transform by*

$$c^+ = \frac{(1 + \Delta t)c}{1 + \Delta t c}, \quad r_m(c) = (1 - \Delta t c)^2, \quad r_C(c) = \frac{1}{(1 + \Delta t c)^2}.$$

Fix $0 < \Delta t < 2$ and $q \in (0, 1)$ with $q \geq q_0(\Delta t) := \max\{(1 + \Delta t)^{-2}, (1 - \Delta t)^2\}$, and set

$$I_{\Delta t}^{\text{KL}}(q) := \left[\frac{q^{-1/2} - 1}{\Delta t}, \frac{1 + \sqrt{q}}{\Delta t} \right] \ni 1. \quad (43)$$

Then $r_m(c), r_C(c) \leq q$ on $I_{\Delta t}^{\text{KL}}(q)$, the scalar map $c \mapsto c^+$ maps $I_{\Delta t}^{\text{KL}}(q)$ into itself, and if $\text{spec}(C_n) \subseteq I_{\Delta t}^{\text{KL}}(q)$ then $\|a_{n+k} - a_\star\|_{\star, \Delta t}^2 \leq q^k \|a_n - a_\star\|_{\star, \Delta t}^2$ for all $k \geq 0$. For $\Delta t \leq \sqrt{2}$, and in particular for $\Delta t \leq 2/(4 + \Gamma)$, one has $q_0(\Delta t) = (1 + \Delta t)^{-2}$.

Proof. Both updates share the eigenbasis of C_n ; there $c^+ - 1 = \frac{c-1}{1+\Delta t c}$ gives $r_C(c) = (1 + \Delta t c)^{-2}$ and the aligned mean scales by $1 - \Delta t c$. Now $r_m(c) \leq q$ is equivalent to $c \in [\frac{1-\sqrt{q}}{\Delta t}, \frac{1+\sqrt{q}}{\Delta t}]$ and $r_C(c) \leq q$ to $c \geq \frac{q^{-1/2}-1}{\Delta t}$; since $q^{-1/2} - 1 \geq 1 - \sqrt{q}$, the covariance lower bound dominates and gives (43), and 1 lies in the interval exactly when $q \geq q_0(\Delta t)$. Since $c^+ - 1$ has the sign of $c - 1$ with smaller magnitude and $c^+ - c = \frac{\Delta t c(1-c)}{1+\Delta t c}$ has the opposite sign, c^+ lies strictly between c and 1, so an interval containing 1 is preserved; each scalar factor being $\leq q$, the norm (34) contracts by q per step. $\square$

The parallel Riemannian statement, [Theorem E.11](#), is recorded in [Section E](#).

The Riemannian interval endpoints solve scalar transcendental equations, while the KL interval (43) is closed-form, so [Theorem 3.11](#) is the primary discrete Gaussian statement and [Theorem E.11](#) its parallel ([Section E](#)). For general targets we record the clean per-step rate used in the complexity statements, writing $\gamma_R := 1/(4 + \Gamma)$ and $\gamma_{\text{KL}} := 1/(4 + 2\Delta t + \Gamma)$ for the certified linearized gaps of [Theorem 3.4](#) in the norms $\|\cdot\|_\star$ and $\|\cdot\|_{\star, \Delta t}$, and $\mathbf{b}_{\star, R}(\delta) := \mathbf{b}(\delta)$, $\mathbf{b}_{\star, \text{KL}}(\delta) := \mathbf{b}_{\Delta t}(\delta)$ for the corresponding norm-conversion constants, which satisfy $\mathbf{b}_{\Delta t}(\delta) \leq \sqrt{1 + \Delta t} \mathbf{b}(\delta)$. For a scheme map T we write

$$K_{\text{disc}}(\delta) := \Delta t^{-1} \sup_{a \in \mathcal{V}_\delta} \|D^2 T(a)\|_{\star, \bullet}, \quad \|D^2 T(a)\|_{\star, \bullet} := \sup_{\|\xi\|_{\star, \bullet}=1} \|D^2 T(a)[\xi, \xi]\|_{\star, \bullet}, \quad (44)$$

the normalization by Δt reflecting that both scheme maps are $I + O(\Delta t)$.

Theorem 3.12 (General-target discrete local convergence). *Fix a scheme $\bullet \in \{R, \text{KL}\}$ with gap $\gamma_\bullet$ and norm $\|\cdot\|_{\star, \bullet}$, and suppose*

$$0 < \Delta t \leq \min\left\{\frac{2}{4 + \Gamma}, \frac{1}{\beta_\star \bar{\Lambda}_\delta}\right\}, \quad K_{\text{disc}}(\delta) \mathbf{b}_{\star, \bullet}(\delta) \sqrt{\delta} \leq \gamma_\bullet.$$

If $\Delta(a_0) \leq \delta$, then the iterates remain in $\mathcal{U}_\delta$ and

$$\|a_n - a_\star\|_{\star, \bullet} \leq \left(1 - \frac{1}{2} \Delta t \gamma_\bullet\right)^n \|a_0 - a_\star\|_{\star, \bullet},$$

so the Riemannian rate is $(1 - \frac{\Delta t}{2(4+\Gamma)})^n$ and the KL rate is $(1 - \frac{\Delta t}{2(4+2\Delta t+\Gamma)})^n$. Since $K_{\text{disc}}(\delta) \rightarrow K_{\text{disc}}(0) < \infty$ and $\mathbf{b}_{\star, \bullet}(\delta) \sqrt{\delta} \rightarrow 0$ as $\delta \downarrow 0$, the radius condition holds for all sufficiently small δ .

Proof. The descent estimates of [Theorem E.4](#) and [Theorem E.7](#) give $\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n)$ under $\Delta t \leq 1/(\beta_\star \bar{\Lambda}_\delta)$, so $a_{n+1} \in \mathcal{U}_\delta$ and $\|\xi_n\|_{\star, \bullet} \leq \mathbf{b}_{\star, \bullet}(\delta) \sqrt{\delta}$. Writing $\xi_n := a_n - a_\star$ and $M_\bullet := DT(a_\star)$, the integral Taylor identity

$$a_{n+1} - a_\star = T(a_\star + \xi_n) - T(a_\star) = M_\bullet \xi_n + \int_0^1 (1-s) D^2 T(a_\star + s\xi_n) [\xi_n, \xi_n] ds$$

holds, its linear part contracting by $1 - \Delta t \gamma_\bullet$ by [Theorem 3.4](#). Since $a_\star + s\xi_n \in \mathcal{V}_\delta$ for $s \in [0, 1]$, the remainder is bounded by $\frac{1}{2} \Delta t K_{\text{disc}}(\delta) \|\xi_n\|_{\star, \bullet}^2 \leq \frac{1}{2} \Delta t \gamma_\bullet \|\xi_n\|_{\star, \bullet}$ by (44) and the radius condition. Adding gives the per-step factor $1 - \frac{1}{2} \Delta t \gamma_\bullet$, and iterating completes the proof. $\square$

The one-step local contraction estimate isolated for the stochastic analysis, [Theorem E.12](#), is stated in [Section E](#).

Remark 3.13 (Status of K_{disc}). The constant $K_{\text{disc}}(\delta)$ is finite and continuous in δ , with $K_{\text{disc}}(\delta) \rightarrow K_{\text{disc}}(0) < \infty$, but it is not available in closed form. Under [Theorem 2.1](#) the maps $g(a) = -\mathbb{E}_{\rho_a}[\nabla V]$ and $A(a) = \mathbb{E}_{\rho_a}[\nabla^2 V]$ are C^∞ on every spectrally bounded compact subset of $\mathcal{A}$, since each parameter derivative brings down a Gaussian score polynomial, and both scheme maps are smooth functions of g, A, C on $\mathcal{V}_\delta$, so D^2T is continuous there and its supremum over the compact hull is finite. The radius condition of [Theorem 3.12](#) is therefore constructive rather than a practical formula.

3.4 Online entry criteria

The test $\Delta(a_n) \leq \delta$ is not directly computable, since $\mathcal{E}(a_\star)$ is unknown, and the local analysis is carried out in whitened coordinates while the algorithm runs in the original ones. The following sufficient tests use only the current iterate and known target quantities, in the original coordinates.

Proposition 3.14 (Computable residual gate). *Let $\alpha > 0$ be the strong-convexity constant of the target, and at $a = (m, C)$ in original coordinates set $g(a) = -\mathbb{E}_{\rho_a}[\nabla V(\theta)]$, $A(a) = \mathbb{E}_{\rho_a}[\nabla^2 V(\theta)]$, and the Bures–Wasserstein gradient residual*

$$\mathcal{R}_{\text{BW}}(a)^2 := \|g(a)\|^2 + \text{Tr}[(C^{-1} - A(a))C(C^{-1} - A(a))].$$

Then $\mathcal{R}_{\text{BW}}(a)^2 \geq 2\alpha \Delta(a)$, so for any a priori certified local level δ_{loc} ,

$$\mathcal{R}_{\text{BW}}(a_n)^2 \leq 2\alpha \delta_{\text{loc}} \implies \Delta(a_n) \leq \delta_{\text{loc}}.$$

The left-hand side is evaluated from $g(a_n)$, $A(a_n)$, C_n , and α , none of which requires $C_\star$, $m_\star$, or $\mathcal{E}(a_\star)$. When a covariance floor $C_n \succeq \ell_n I$ is available, the Fisher–Rao gradient norm gives the alternative intrinsic test $\mathcal{G}(a_n)^2 \leq \alpha \ell_n \delta_{\text{loc}} \implies \Delta(a_n) \leq \delta_{\text{loc}}$, which is more restrictive than the Bures–Wasserstein gate. Once either test triggers, for the flow and for either deterministic scheme under all hypotheses of its global theorem and its certified stepsize window ([Theorems 2.9](#) and [2.11](#)), energy descent keeps $\Delta(a_k) \leq \delta_{\text{loc}}$ for all $k \geq n$, so the local theorems apply permanently. The proof is given in [Section E](#).

Exact spectral entry gates for Gaussian targets, [Theorem E.13](#), are recorded in [Section E](#).

These spectral gates can be substantially less restrictive than the residual gate, since they impose no condition on the mean and depend only on the target-relative covariance spectrum. The level δ_{loc} itself is target-adaptive through $\alpha_\star, \beta_\star, \Sigma_\delta, \mathcal{B}_\delta, K_{\text{ng}}$; a precomputable conservative choice follows from the a priori bounds $\alpha_\star \geq \kappa^{-1}$, $\beta_\star \leq \kappa$, $\Sigma_\delta \leq \sqrt{N_\theta}(\beta_\star - \alpha_\star)$, $\mathcal{B}_\delta \leq \max\{\beta_\star - 1, 1 - \alpha_\star\}$, which reduce the continuous-time admissibility condition $\rho_{\text{core}}(\delta) > 0$ of [Theorem 3.8](#) to a scalar inequality in α, β, N_θ . For the discrete schemes the admissible level remains implicit, by [Theorem 3.13](#).

For the stochastic schemes of [Section 4](#) neither pathwise nor expected containment follows from these gates alone. The one-step recursion $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq q\Delta_n + \eta$ with a variance floor $\eta > 0$ gives only $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq q\delta_{\text{loc}} + \eta$, which is at most δ_{loc} only if $\delta_{\text{loc}} \geq \eta/(1 - q)$. Entry must therefore be combined with the containment analysis of [Section 4](#).

3.5 Sharpness: no dimension-free logarithmic local rate

The bound $\Gamma \leq \beta_\star - 1$ makes the certified local rate $1/(4 + \Gamma)$ deteriorate linearly in the whitened smoothness constant. We show that the true gap $\gamma_\star$ obeys no logarithmic bound uniformly in the dimension, even under the additional hypothesis of a dimension-free Hessian-Lipschitz constant.

Theorem 3.15 (Convex-ridge lower bound on the local gap). *There are numerical constants $K_0, c_\kappa, C_\kappa, C_\tau, C_\gamma > 0$, and $C'_\gamma := C_\gamma \sqrt{C_\kappa}$, with the following property. For every integer m with $K := (8m)^{1/4} \geq K_0$ and $N_\theta := m + 1$, there is a potential $V \in C^\infty(\mathbb{R}^{N_\theta})$ satisfying [Theorem 2.1](#) and the whitening conditions $\mathbb{E}_{\mathcal{N}(0,I)}[\nabla V] = 0$, $\mathbb{E}_{\mathcal{N}(0,I)}[\nabla^2 V] = I$ of [Theorem 3.1](#), and satisfying [Theorem 2.2](#) with the dimension-free constant $L_H = 64$, such that*

$$c_\kappa K^2 \leq \kappa_\star \leq C_\kappa K^2, \quad \tau_H^2 \geq \frac{K}{2} - C_\tau, \quad \gamma_\star \leq \frac{C_\gamma}{K} \leq \frac{C'_\gamma}{\sqrt{\kappa_\star}},$$

where $\tau_H := \sup_{\|w\|=1} \left\| \mathbb{E}[(w^\top Z) \nabla^2 V(Z)] \right\|_F = \|\mathcal{T}\|$ is the first-Hermite coupling norm of [Theorem 3.2](#). For this family the Schur bound (77) of [Theorem A.4](#) (Section A) gives $\tau_H^2 \leq 1 + \Gamma$ with $\Gamma = \beta_\star - 1 = \Theta(K)$, so that

$$\tau_H^2 = \Theta(K) = \Theta(\kappa_\star^{1/2}), \quad \gamma_\star = \Theta(K^{-1}) = \Theta(\kappa_\star^{-1/2}).$$

Consequently the $\beta_\star - 1$ branch of Γ in (35) is saturated, and there is no positive function $c(L_H)$ for which $\gamma_\star \geq c(L_H)/(1 + \log \kappa_\star)$ holds in every dimension under [Theorem 2.1](#) and [Theorem 2.2](#). The construction and the proof are given in [Section D](#).

The family uses $N_\theta = m + 1 = K^4/8 + 1 = \Theta(\kappa_\star^2)$. It therefore rules out estimates uniform over the dimension only. It does not settle whether an analogous obstruction exists at fixed N_θ , or for $N_\theta = o(\kappa_\star^2)$, and, since the branch it saturates is $\beta_\star - 1$, it says nothing about the sharpness of the $\sqrt{N_\theta} \log \kappa_\star$ branch of Γ .

3.6 Deterministic three-stage complexity

The complete run separates into three stages: a covariance burn-in in which an underdispersed initial covariance is lifted to the curvature scale, a global localization in which the energy gap is driven into a certified level, and a local convergence in which the parameter error contracts at the spectral rate of [Theorem 3.3](#). The total cost is their sum,

$$\boxed{\underbrace{\log_+ \frac{1}{\beta_\star \lambda_{0,\star}}}_{\text{covariance burn-in}} + \underbrace{\kappa_\star \log_+ \frac{\Delta_0}{\delta_c}}_{\text{global localization}} + \underbrace{(4 + \Gamma) \log \frac{1}{\varepsilon}}_{\text{local convergence}}} \quad (45)$$

with the error measured by the energy gap in the first two stages and by the equilibrium parameter norm in the third, the transition between the two metrics being effected by the conversion $\Delta(a) \leq \delta \Rightarrow \|a - a_\star\|_\star \leq \mathfrak{b}(\delta) \sqrt{\delta}$ of (39).

Theorem 3.16 (Deterministic three-stage complexity). *Assume [Theorem 2.1](#) and $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$. Let $\delta_c > 0$ be a level admissible for [Theorem 3.8](#) with $\rho_{\text{core}}(\delta_c) \geq 1/(4 + \Gamma)$, and let $\delta_R, \delta_{\text{KL}} > 0$ be levels admissible for [Theorem 3.12](#) for the two schemes at the stepsizes below; such levels exist by [Theorem 3.9](#) and [Theorem 3.12](#). For sufficiently small $\varepsilon > 0$, specifically $\varepsilon < \mathfrak{b}(\delta_c)^2 \delta_c$ for the flow and $\varepsilon < \mathfrak{b}_{\star,\bullet}(\delta_\bullet)^2 \delta_\bullet$ for the scheme $\bullet \in \{\text{R}, \text{KL}\}$:*

1. For the flow (1), the time

$$T_c(\varepsilon) = O\left(\log_+ \frac{1}{\beta_\star \lambda_{0,\star}} + \kappa_\star \log_+ \frac{\Delta_0}{\delta_c} + (4 + \Gamma) \log \frac{1}{\varepsilon}\right)$$

suffices for $\|a_T - a_\star\|_\star^2 \leq \varepsilon$. The first two terms may be replaced by the smaller of this affine-invariant expression and $\log_+(1/(\beta \lambda_0)) + \kappa \log_+(\Delta_0/\delta_c)$.

2. For (20) with $0 < \Delta t \leq \min\{1/(2\beta\lambda_{\max}), 2/(4 + \Gamma), 1/(\beta_\star \bar{\Lambda}_{\delta_R})\}$,

$$N_R(\varepsilon) = O\left(\frac{1}{\Delta t} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t} \log_+ \frac{\Delta_0}{\delta_R} + \frac{4 + \Gamma}{\Delta t} \log \frac{1}{\varepsilon}\right)$$

iterations suffice for $\|a_N - a_\star\|_\star^2 \leq \varepsilon$.

3. For (21) with $0 < \Delta t \leq \min\{1/(\beta\lambda_{\max}), 2/(4 + \Gamma), 1/(\beta_\star \bar{\Lambda}_{\delta_{KL}})\}$,

$$N_{KL}(\varepsilon) = O\left(\frac{1}{\Delta t} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t} \log_+ \frac{\Delta_0}{\delta_{KL}} + \frac{4 + 2\Delta t + \Gamma}{\Delta t} \log \frac{1}{\varepsilon}\right)$$

iterations suffice for $\|a_N - a_\star\|_{\star, \Delta t}^2 \leq \varepsilon$.

Proof. For the flow, Theorem 2.3 with $\delta = \delta_c$ gives $\Delta(a_t) \leq \delta_c$ after time $O(\log_+ \frac{1}{\beta_\star \lambda_{0,\star}} + \kappa_\star \log_+ \frac{\Delta_0}{\delta_c})$, so $a_t \in \mathcal{U}_{\delta_c}$ and $\|a_t - a_\star\|_\star \leq \mathfrak{b}(\delta_c) \sqrt{\delta_c}$ by (39). Restarting Theorem 3.8 at that time gives $\|a_{t+s} - a_\star\|_\star^2 \leq e^{-s/(4+\Gamma)} \mathfrak{b}(\delta_c)^2 \delta_c$, so a further time $s = (4 + \Gamma) \log(\mathfrak{b}(\delta_c)^2 \delta_c / \varepsilon)$ suffices, which is $O((4 + \Gamma) \log \frac{1}{\varepsilon})$ since δ_c does not depend on ε . For the two schemes, Theorem 2.9 and Theorem 2.11 with $\delta = \delta_\bullet$ give $\Delta(a_n) \leq \delta_\bullet$ after $O(\frac{1}{\Delta t} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t} \log_+ \frac{\Delta_0}{\delta_\bullet})$ iterations, and $\|a_n - a_\star\|_{\star, \bullet} \leq \mathfrak{b}_{\star, \bullet}(\delta_\bullet) \sqrt{\delta_\bullet}$. The stepsize windows satisfy the hypotheses of Theorem 3.12, whose per-step factors are $1 - \frac{\Delta t}{2(4+\Gamma)}$ and $1 - \frac{\Delta t}{2(4+2\Delta t+\Gamma)}$, so a further $O(\frac{4+\Gamma}{\Delta t} \log \frac{1}{\varepsilon})$, respectively $O(\frac{4+2\Delta t+\Gamma}{\Delta t} \log \frac{1}{\varepsilon})$, iterations suffice, using $-\log(1 - x) \geq x$. $\square$

Corollary 3.17 (Burn-in-free three-stage complexity). *Under Theorem 2.1, initialize by either the quadratic rescue of Theorem 2.14 or the Bures–Wasserstein transport warm-start of Theorem 2.13 with target $c = 1$, that is with $\eta = 1/\beta$ and at most one forward–backward step. Both land $C_0 \succeq \frac{1}{\beta} I$, so $\lambda_0 \geq \frac{1}{\beta}$ and the first stage of Theorem 3.16 is removed: for both discretizations*

$$N_\bullet(\varepsilon) = n_{\text{init}} + O\left(\frac{\kappa}{\Delta t} \log_+ \frac{\Delta_0}{\delta_\bullet} + \frac{c_\bullet}{\Delta t} \log \frac{1}{\varepsilon}\right), \quad c_R = 4 + \Gamma, \quad c_{KL} = 4 + 2\Delta t + \Gamma,$$

where $n_{\text{init}} \in \{0, 1\}$ is the warm-start, one Hessian query for the rescue or $\mathbf{1}_{\{\lambda_0 < 1/\beta\}} \leq 1$ forward–backward steps for transport, and $\Delta_0 = \Delta(a_0)$ is the energy of the warm-started initializer. For a Gaussian target the rescue gives $\Delta_0 = 0$, so the localization stage vanishes as well and the whole run collapses to the single initializing query.

Proof. For the rescue, Theorem E.9 gives $C_0 \succeq \frac{1}{\beta} I$; for transport with $c = 1$ the one-step Bures–Wasserstein floor of Theorem 2.13 with $\eta = 1/\beta$ gives $C^+ \succeq \frac{1}{\beta} I$ in one step, skipped when $\lambda_0 \geq 1/\beta$ already. In both cases $\lambda_0 \geq \frac{1}{\beta} > \frac{1}{2\beta}$, so the covariance envelopes of Theorem 2.8 and Theorem 2.10 keep $C_n \succeq \frac{1}{2\beta} I$ for all $n \geq 0$ and the burn-in stage of Theorem 3.16(2)–(3) is empty. The Gaussian case is Theorem 2.15. $\square$

Each of the three stages is sharp. The covariance burn-in cannot be removed within Fisher–Rao alone, since the underdispersed eigenvalue obeys $\mu \approx \mu$ and reaching the curvature scale from λ_0 costs time $\asymp \log(1/(\beta\lambda_0))$, which is why Theorem 3.17 must change geometry or oracle to delete it. The κ -dependence of the global localization stage is attained by the logarithmic spiral of Theorem 2.5 in continuous time and by the bump train of Theorem 2.12 for either scheme at a globally fixed stepsize. The $(4 + \Gamma)$ -dependence of the local stage is attained by the convex ridge of Theorem 3.15, which saturates the $\beta_\star - 1$ branch of Γ and rules out any dimension-free logarithmic replacement.

4 Stochastic algorithms

The deterministic schemes (20) and (21) both require the exact Gaussian expectations $g(a) = -\mathbb{E}_{\rho_a}[\nabla V(\theta)]$ and $A(a) = \mathbb{E}_{\rho_a}[\nabla^2 V(\theta)]$, which are unavailable in applications. We replace them by sample estimators built from a pointwise oracle for V , and we develop the convergence theory of the resulting algorithms in the three stages of Sections 2 and 3.

4.1 The oracle, the estimators, and the algorithm

The two expectations behave very differently under naive sampling. If we draw a single $\tilde{\theta} \sim \rho_{a_n}$ and take $b_n = -\nabla V(\tilde{\theta})$, $S_n = \nabla^2 V(\tilde{\theta})$, then β -smoothness of V gives, on a covariance band $C_n \preceq \lambda_{\max} I$,

$$\mathbb{E}[\|b_n - g(a_n)\|^2 \mid \mathcal{F}_n] \leq \beta^2 \lambda_{\max} N_\theta, \quad \mathbb{E}[\|S_n - A(a_n)\|_F^2 \mid \mathcal{F}_n] \leq N_\theta (\beta - \alpha)^2,$$

the first because ∇V is β -Lipschitz and $\mathbb{E}[\|\tilde{\theta} - m_n\|^2] = \text{Tr } C_n$, the second because $\alpha I \preceq \nabla^2 V(\tilde{\theta}) \preceq \beta I$ holds almost surely, so the sampled Hessian error is bounded pathwise in operator norm by $\beta - \alpha$. The mean estimator carries the whole burden: its variance grows with the curvature scale β^2 and with the spread $\lambda_{\max} N_\theta$ of the current variational density, and it does not vanish at the optimizer. Sticking the landing [16] removes exactly this defect by subtracting the score of the variational density itself, which has zero mean and cancels the leading fluctuation. We therefore use a sticking-the-landing score together with the sampled Hessian, and we keep the sampled Hessian rather than a gradient-only representation of $A(a)$ because it is the representation whose spectrum is controlled sample by sample.

Definition 4.1 (Price/Hessian oracle). Given $X \in \mathbb{R}^{N_\theta}$, the oracle returns the pair $(\nabla V(X), \nabla^2 V(X))$.

Fix a batch size $B \geq 1$, let $\mathcal{F}_n := \sigma(a_0, \dots, a_n)$, and draw $\xi_{n,s} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I)$, $X_{n,s} := m_n + C_n^{1/2} \xi_{n,s}$, $H_{n,s} := \nabla^2 V(X_{n,s})$ for $s = 1, \dots, B$. The sticking-the-landing mean estimator is

$$\bar{b}_n := \frac{1}{B} \sum_{s=1}^B (-\nabla V(X_{n,s}) + C_n^{-1}(X_{n,s} - m_n)) = \frac{1}{B} \sum_{s=1}^B (-\nabla V(X_{n,s}) + C_n^{-1/2} \xi_{n,s}), \quad (46)$$

and the covariance estimator is the sampled Hessian average $\bar{H}_n := \frac{1}{B} \sum_s H_{n,s}$ with associated precision residual $\bar{K}_n := C_n^{-1} - \bar{H}_n$. Both are conditionally unbiased,

$$\mathbb{E}[\bar{b}_n \mid \mathcal{F}_n] = g(a_n), \quad \mathbb{E}[\bar{K}_n \mid \mathcal{F}_n] = R(a_n), \quad (47)$$

the first because $\mathbb{E}_{\rho_{a_n}}[C_n^{-1}(\theta - m_n)] = 0$. The minibatch Price/Hessian Riemannian scheme is

$$m_{n+1} = m_n + \Delta t C_n \bar{b}_n, \quad C_{n+1} = C_n^{1/2} \exp(\Delta t (I - C_n^{1/2} \bar{H}_n C_n^{1/2})) C_n^{1/2}, \quad (48)$$

and the minibatch Price/Hessian KL scheme is

$$m_{n+1} = m_n + \Delta t C_n \bar{b}_n, \quad C_{n+1} = (1 + \Delta t)(C_n^{-1} + \Delta t \bar{H}_n)^{-1}. \quad (49)$$

Since each $H_{n,s}$ has spectrum in $[\alpha, \beta]$ almost surely under Theorem 2.1, so does the average,

$$\alpha I \preceq \bar{H}_n \preceq \beta I \quad \text{almost surely}, \quad (50)$$

and this almost-sure spectral band is the structural reason the schemes admit an unconditional global theory.

By Gaussian integration by parts, for $X = m + C^{1/2} \xi$ one has $\mathbb{E}[\xi \nabla V(X)^\top] = C^{1/2} A(a)$, so $A(a)$ admits both a sampled-Hessian and a gradient-only representation. We use the sampled Hessian, which is the representation carrying (50); the gradient-only representation has unbounded samplewise spectrum, and none of the pathwise statements below apply to it.

The stochastic pipeline is initialized by the single deterministic query of the quadratic rescue of [Theorem 2.14](#), which sets $m_0 = m_{\text{in}} - H_{\text{in}}^{-1}G_{\text{in}}$ and $C_0 = H_{\text{in}}^{-1}$ from $G_{\text{in}} = \nabla V(m_{\text{in}})$, $H_{\text{in}} = \nabla^2 V(m_{\text{in}})$. Because $\alpha I \preceq H_{\text{in}} \preceq \beta I$ and the matrix inverse is operator antitone, this initializer lands on the curvature band $\frac{1}{\beta}I \preceq C_0 \preceq \frac{1}{\alpha}I$ ([Theorem E.9](#)), which is exactly the band the covariance bootstraps of [Section 2](#) would otherwise have to reach by a burn-in of $O(\Delta t^{-1} \log_+ \frac{1}{\beta \lambda_0})$ steps; and for a quadratic potential it returns the optimizer itself, $a_0 = a_\star$ and $\Delta(a_0) = 0$ ([Theorem 2.15](#)). For nonquadratic targets we record the energy quality of the rescued initializer, which is what the global complexity bounds below consume.

The energy cost of the sampled rescue, [Theorem F.1](#), is bounded in [Section F](#). We may also retain the first-order covariance-segment estimate $\Delta(a_0) \leq \frac{\beta}{2} \|m_0 - m_\star\|^2 + \frac{\beta}{2} \sqrt{N_\theta} \|C_0 - C_\star\|_F$ and take the minimum of the two; the quadratic form is the one with the near-Gaussian $O(L_H^2)$ scaling that matches the stochastic local floors of [Section 4.4](#).

We now show that the rescued schemes keep their covariance on an explicit band along every sample path.

Lemma 4.2 (Covariance bands for the rescued schemes). *Assume [Theorem 2.1](#) and initialize with the quadratic rescue of [Theorem 2.14](#), so that $\frac{1}{\beta}I \preceq C_0 \preceq \frac{1}{\alpha}I$.*

1. *If $0 < \Delta t \leq 1/\kappa$, then along (48), almost surely $\frac{1}{2\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ for all $n \geq 0$.*

2. *For every $\Delta t > 0$, along (49), almost surely $\frac{1}{\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ for all $n \geq 0$.*

No stopping time, clipping, or containment hypothesis is used.

Proof. By (50), $\alpha I \preceq \bar{H}_n \preceq \beta I$ almost surely, so the deterministic covariance envelopes of [Theorems 2.8](#) and [2.10](#) apply sample by sample.

For the KL scheme, if $\frac{1}{\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ then $\alpha I \preceq C_n^{-1} \preceq \beta I$, so $(1 + \Delta t)\alpha I \preceq C_n^{-1} + \Delta t \bar{H}_n \preceq (1 + \Delta t)\beta I$; inverting and multiplying by $(1 + \Delta t)$ gives $\frac{1}{\beta}I \preceq C_{n+1} \preceq \frac{1}{\alpha}I$, with no stepsize restriction, by operator antitonicity of the inverse.

For the Riemannian scheme, $C_{n+1}^{-1} \succeq (1 - \Delta t)C_n^{-1} + \Delta t \bar{H}_n \succeq \alpha I$ when $C_n \preceq \frac{1}{\alpha}I$ and $\Delta t \leq 1$, which gives the upper bound. With that upper bound, $\bar{B}_n := C_n^{1/2} \bar{H}_n C_n^{1/2} \preceq \kappa I$, so $\Delta t \bar{B}_n \preceq I$ requires $\Delta t \leq 1/\kappa$; the lower envelope of [Theorem 2.8](#), started at $L_0 = \min\{\lambda_0, \frac{1}{2\beta}\} = \frac{1}{2\beta}$ because the rescue gives $\lambda_0 \geq \frac{1}{\beta}$, has $y_0 = (\beta L_0)^{-1} = 2$; since $y_\infty < 2$ and the recursion $y_{n+1} = e^{-\Delta t}(y_n + (e - 1)\Delta t)$ moves y_n monotonically toward y_∞ , we obtain $y_n \leq 2$, that is $C_n \succeq \frac{1}{2\beta}I$. $\square$

These bands hold pathwise and unconditionally, which is why every global theorem of [Section 4.3](#) is a plain expectation bound. [Algorithm 1](#) collects the pipeline in the implementable coordinates (m, C) .

4.2 Intrinsic sticking-the-landing variance

For a single sample $X \sim \rho_a$ the tangent noise is $\xi(a; X) := (-C\varepsilon_b, -C\varepsilon_R C)$ with $\varepsilon_b := \hat{b} - g(a)$ and $\varepsilon_R := -(\nabla^2 V(X) - A(a))$, and the minibatch noise is $\bar{\xi}_n := \widehat{\text{grad}} \mathcal{E}(a_n) - \text{grad} \mathcal{E}(a_n) = \frac{1}{B} \sum_s \xi(a_n; X_{n,s})$. Both blocks of the noise are governed by a single scalar, the fluctuation of the sampled Hessian measured in the metric of the current variational density.

Definition 4.3 (Intrinsic Hessian fluctuation). $\Psi(a) := \mathbb{E}_{X \sim \rho_a} [\|C^{1/2}(\nabla^2 V(X) - A(a))C^{1/2}\|_F^2]$.

Under [Theorem 2.1](#) the sticking-the-landing noises are conditionally centered and, with $\|\xi(a; X)\|_a^2 = \varepsilon_b^\top C \varepsilon_b + \frac{1}{2} \|C^{1/2} \varepsilon_R C^{1/2}\|_F^2$, the Gaussian Poincaré inequality applied to $F(X) := -\nabla V(X) + C^{-1}(X - m)$, whose derivative is the symmetric matrix $C^{-1} - \nabla^2 V(X)$, gives

$$\mathbb{E}[\varepsilon_b^\top C \varepsilon_b] \leq \|C^{1/2} R(a) C^{1/2}\|_F^2 + \Psi(a), \quad \mathbb{E}[\|\xi(a; X)\|_a^2] \leq 2G(a)^2 + \frac{3}{2}\Psi(a), \quad (51)$$

Algorithm 1 Quadratic-rescued minibatch Price/Hessian STL Fisher–Rao scheme

Require: initial mean m_{in} ; scheme selector $S \in \{\text{R}, \text{KL}\}$; batch size B ; stepsizes $\Delta t_n > 0$; iteration count N

```

1:                                     ▷ Phase 0: quadratic rescue, one deterministic oracle query
2: Query the oracle at  $m_{\text{in}}$ :  $G_{\text{in}} \leftarrow \nabla V(m_{\text{in}})$ ,  $H_{\text{in}} \leftarrow \nabla^2 V(m_{\text{in}})$ 
3:  $m_0 \leftarrow m_{\text{in}} - H_{\text{in}}^{-1} G_{\text{in}}$ ,  $C_0 \leftarrow H_{\text{in}}^{-1}$ 
4:                                     ▷ Phase I: minibatch Price/Hessian STL iterations
5: for  $n = 0, \dots, N - 1$  do
6:   Draw  $\xi_{n,s} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I)$  and set  $X_{n,s} \leftarrow m_n + C_n^{1/2} \xi_{n,s}$ ,  $s = 1, \dots, B$ ; query the oracle at
   each  $X_{n,s}$ 
7:    $\bar{b}_n \leftarrow \frac{1}{B} \sum_s (-\nabla V(X_{n,s}) + C_n^{-1/2} \xi_{n,s})$                                      ▷ sticking-the-landing score
8:    $\bar{H}_n \leftarrow \frac{1}{B} \sum_s \nabla^2 V(X_{n,s})$                                      ▷ Price/Hessian covariance
9:    $m_{n+1} \leftarrow m_n + \Delta t_n C_n \bar{b}_n$ 
10:  if  $S = \text{R}$  then
11:     $C_{n+1} \leftarrow C_n^{1/2} \exp(\Delta t_n (I - C_n^{1/2} \bar{H}_n C_n^{1/2})) C_n^{1/2}$ 
12:  else
13:     $C_{n+1} \leftarrow (1 + \Delta t_n) (C_n^{-1} + \Delta t_n \bar{H}_n)^{-1}$ 
14:  end if
15: end for

```

using $\|C^{1/2} R(a) C^{1/2}\|_{\text{F}}^2 \leq 2G(a)^2$; by conditional independence the minibatch noise satisfies $\mathbb{E}[\|\bar{\xi}_n\|_{a_n}^2 \mid \mathcal{F}_n] \leq B^{-1}(2G(a_n)^2 + \frac{3}{2}\Psi(a_n))$ for every $B \geq 1$. No third or fourth derivative of V enters: when $\nabla^2 V$ is only Lipschitz the Poincaré step is applied to the Lipschitz map F through its weak derivative. Two estimates of Ψ are available, and they separate the two regimes of the theory.

The self-bounding estimate for Ψ , [Theorem F.2](#), is proved in [Section F](#). The estimate (191) ([Section F](#)) charges the fluctuation against the gradient norm itself, which is what forces a smaller stepsize in the assumption-minimal regime below. Charging it instead against the nonquadraticity of the target decouples the two, at the price of one further regularity assumption.

Its Hessian-Lipschitz sharpening, [Theorem F.3](#), is also in [Section F](#). The anisotropic form is the sharper statement: it replaces N_θ^2 by the effective dimension $\text{Tr}(C^2) \text{Tr} C / \|C\|_{\text{op}}^3$ and is strictly smaller whenever the covariance spectrum decays, the worst case over the band being N_θ^2 . For a Gaussian target $\nabla^2 V(X) = A(a)$ pathwise, so $\Psi \equiv 0$ identically on $\mathcal{A}$; all the floors below are proportional to Ψ and vanish with it.

4.3 Global convergence and the variance floors

On the rescued bands the Fisher–Rao gradient dominates the energy gap. The comparison of [Section 2](#) states that whenever $C \succeq \ell I$, $G(a)^2 \geq \frac{\ell}{2} \|\text{grad}^{W_2} \mathcal{E}(a)\|_{W_2}^2 \geq \alpha \ell \Delta(a)$; applied to the two bands of [Theorem 4.2](#) this gives

$$\begin{aligned}
G(a_n)^2 &\geq \frac{1}{2\kappa} \Delta_n \quad \text{on the Riemannian band } C_n \succeq \frac{1}{2\beta} I, \\
G(a_n)^2 &\geq \frac{1}{\kappa} \Delta_n \quad \text{on the KL band } C_n \succeq \frac{1}{\beta} I.
\end{aligned} \tag{52}$$

We restate the two deterministic descent structures before perturbing them. The retraction-smoothness estimate of [Section 2](#) says that if $C \preceq \Lambda I$ and the tangent vector $\zeta = (u, U)$ satisfies the scope condition $\|C^{-1/2} U C^{-1/2}\|_{\text{op}} \leq 1$, then

$$\mathcal{E}(\text{Retr}_a(\zeta)) \leq \mathcal{E}(a) + \langle \text{grad} \mathcal{E}(a), \zeta \rangle_a + \beta \Lambda \|\zeta\|_a^2, \tag{53}$$

that is, the constant is $L/2 = \beta\Lambda$ and the estimate holds for an arbitrary tangent direction, not only for the deterministic gradient direction. The KL scheme, on the other hand, descends in the forward divergence: for $\Delta t \leq 1/(\beta\Lambda)$ and $L_n I \preceq C_n \preceq \Lambda I$,

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t} \text{KL}(\rho_{a_{n+1}} \| \rho_{a_n}), \quad \text{KL}(\rho_{a_{n+1}} \| \rho_{a_n}) \geq \frac{L_n \Delta t^2}{4(1 + \Delta t \beta \Lambda)^2} \left\| \text{grad}^{W_2} \mathcal{E}(a_n) \right\|_{W_2}^2, \quad (54)$$

which is [Theorem E.7](#). The forward-divergence margin (54) is exact for the deterministic map but is not stable under an unbiased perturbation of $A(a_n)$, since the divergence is evaluated at the perturbed endpoint. An identity exhibiting the KL resolvent update as a retraction removes the difficulty, after which (53) governs both schemes.

This identity is what lets the two schemes share one perturbation analysis: the implicit resolvent update is the exponential retraction $C^{1/2} e^{W(P)} C^{1/2}$ with $W(P)$ a spectral function of the sampled relative Hessian P , so its first-order stochastic part is unbiased and only a second-order fluctuation survives, and we may therefore perturb (53) for both schemes; it is recorded as [Theorem F.4](#) in [Section F](#).

The two one-step inequalities with their additive stochastic terms, [Theorems F.5](#) and [F.6](#), are stated and proved in [Section F](#). Substituting the two estimates of Ψ into these one-step inequalities and unrolling gives the global rates for each scheme.

Theorem 4.4 (Riemannian stochastic global convergence). *Assume [Theorem 2.1](#), initialize with the quadratic rescue of [Theorem 2.14](#), and run (48) with batch $B \geq 1$.*

1. (Hessian-Lipschitz regime.) *Under the additional [Theorem 2.2](#), with $\Delta t = \frac{1}{2\kappa(1+2/B)}$, for every $N \geq 0$*

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{1}{8\kappa^2(1+2/B)}\right)^N \Delta(a_0) + \frac{3\kappa}{B+2} N_\theta^2 \bar{L}_H^2.$$

For $B = 1$ this reads $\Delta t = \frac{1}{6\kappa}$, $\mathbb{E}\Delta_N \leq (1 - \frac{1}{24\kappa^2})^N \Delta(a_0) + \kappa N_\theta^2 \bar{L}_H^2$, so reaching accuracy ϵ above the floor costs $N = \tilde{O}(\kappa^2)$ iterations and $1 + \tilde{O}(\kappa^2)$ oracle pairs.

2. (Assumption-minimal regime.) *With $\rho_B := 1 + \frac{2+3\kappa}{B}$ and $\Delta t = \frac{1}{2\kappa\rho_B}$, for every $N \geq 0$*

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{1}{8\kappa^2\rho_B}\right)^N \Delta(a_0) + \frac{6\kappa^2 N_\theta}{B+2+3\kappa}.$$

The floor is $O(\kappa N_\theta)$ for every $B \lesssim \kappa$ and at most $2\kappa N_\theta$ for every $B \geq 1$. For $B = 1$, $\rho_1 = 3 + 3\kappa \asymp \kappa$ and $N = \tilde{O}(\kappa^3)$; for $B = \Theta(\kappa)$, $\rho_B = O(1)$ and $N = \tilde{O}(\kappa^2)$ rounds at $BN = \tilde{O}(\kappa^3)$ oracle pairs. No Hessian-Lipschitz assumption is used.

The KL scheme obeys the same pair of estimates on its own band.

Theorem 4.5 (KL stochastic global convergence). *Assume [Theorem 2.1](#), initialize with the quadratic rescue of [Theorem 2.14](#), and run (49) with batch $B \geq 1$.*

1. (Hessian-Lipschitz regime.) *Under the additional [Theorem 2.2](#), with $\Delta t = \frac{1}{8\kappa}$, for every $N \geq 0$*

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{1}{16\kappa^2}\right)^N \Delta(a_0) + \frac{7\kappa}{16B} N_\theta^2 \bar{L}_H^2,$$

so for $B = 1$ reaching accuracy ϵ above the floor costs $N = \tilde{O}(\kappa^2)$ iterations and $1 + \tilde{O}(\kappa^2)$ oracle pairs.

2. (Assumption-minimal regime.) *With $\Delta t = \min\{\frac{1}{8\kappa}, \frac{B}{14\kappa^2}\}$, for every $N \geq 0$*

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{\Delta t}{4\kappa}\right)^N \Delta(a_0) + 14\kappa^3 \Delta t \frac{N_\theta}{B}.$$

For $B \leq \frac{7}{4}\kappa$ one has $\Delta t = \frac{B}{14\kappa^2}$, the floor is exactly κN_θ and the contraction is $1 - \frac{B}{56\kappa^3}$; for $B = 1$ this gives $N = \tilde{O}(\kappa^3)$, and for $B = \Theta(\kappa)$, $N = \tilde{O}(\kappa^2)$ rounds at $\tilde{O}(\kappa^3)$ oracle pairs. No Hessian-Lipschitz assumption is used.

The proofs are given in [Section F](#).

The proof of [Theorem F.6](#) holds verbatim under the weaker batch-dependent condition $\Delta t \leq \frac{1}{4\kappa(1+1/B)}$, of which $\Delta t \leq \frac{1}{8\kappa}$ is the $B = 1$ case. Before the step is fixed, the Hessian-Lipschitz recursion unrolls to

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{\Delta t}{2\kappa}\right)^N \Delta(a_0) + \frac{7\kappa^2\Delta t}{2B}\Psi_H, \quad 0 < \Delta t \leq \frac{1}{4\kappa(1+1/B)},$$

which displays the rate–floor tradeoff: for large B the certified step nearly doubles, halving the outer-round constant at the cost of a proportionally larger fixed-step floor. The analogous self-bounding window is $\Delta t \leq \min\{\frac{1}{4\kappa(1+1/B)}, \frac{B}{14\kappa^2}\}$. The theorem statements retain the conservative choices.

The two schemes attain the same leading complexity in each regime, $\tilde{O}(\kappa^2)$ single-sample iterations under [Theorem 2.2](#) and $\tilde{O}(\kappa^3)$ under [Theorem 2.1](#) alone, both unconditionally. They differ in their bands and hence in their gradient-domination constants: the KL band $\frac{1}{\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ holds for every $\Delta t > 0$ and gives the constant $\frac{1}{\kappa}$ in (52), whereas the Riemannian band $\frac{1}{2\beta}I \preceq C_n$ requires $\Delta t \leq 1/\kappa$ and gives $\frac{1}{2\kappa}$. Both floors are proportional to $\Psi_H = N_\theta^2 \bar{L}_H^2$ in the Hessian-Lipschitz regime and to κN_θ in the assumption-minimal regime, and both vanish at the estimator level for Gaussian targets by [Theorem 4.12](#).

4.4 Local convergence

Once an iterate has entered a certified local region, the deterministic maps contract at the local rate of [Section 3](#) rather than the global κ -dependent one, and the stochastic analysis should reproduce that rate. We work in the whitened coordinates of [Theorem 3.1](#), in which $a_\star = (0, I)$ and V is replaced by $\hat{V}(z) = V(m_\star + C_\star^{1/2}z)$; the schemes (48)–(49) are equivariant under this reparametrization and take the same form with $\nabla^2 V \rightarrow \nabla^2 \hat{V}$, so $\alpha_\star I \preceq \bar{H}_n \preceq \beta_\star I$ almost surely. By the chain rule $\nabla^2 \hat{V}(z) = C_\star^{1/2} \nabla^2 V(m_\star + C_\star^{1/2}z) C_\star^{1/2}$, so under [Theorem 2.2](#) the whitened potential is Hessian-Lipschitz with modulus

$$L_{H,\star} \leq L_H \|C_\star\|_{\text{op}}^{3/2} \leq L_H \alpha^{-3/2} = \bar{L}_H, \quad (55)$$

using $C_\star = A(a_\star)^{-1} \preceq \alpha^{-1}I$. We write $\bar{\varepsilon}_{b,n} := \bar{b}_n - g(a_n)$ and $\bar{\varepsilon}_{H,n} := \bar{H}_n - A(a_n)$ for the whitened minibatch noises, conditionally centered by (47).

Definition 4.6 (Stochastically admissible level). For a scheme $\bullet \in \{\text{R}, \text{KL}\}$ and a fixed stepsize Δt , a level $\delta_\bullet > 0$ is *stochastically admissible* if the discrete radius condition $K_{\text{disc}}(\delta_\bullet) \mathfrak{b}_{\star,\bullet}(\delta_\bullet) \sqrt{\delta_\bullet} \leq \gamma_\bullet$ of [Theorem 3.12](#) holds and in addition $\ell_{\delta_\bullet} \geq \frac{1}{2}$, $U_{\delta_\bullet} \leq \frac{3}{2}$, and $\mathfrak{b}_{\star,\bullet}(\delta_\bullet) \sqrt{\delta_\bullet} \leq 1$.

This is a joint condition on the pair $(\Delta t, \delta_\bullet)$, and it is understood that Δt is fixed first and $\delta_\bullet$ chosen afterward. Such levels exist for every fixed admissible stepsize: as $\delta \downarrow 0$ the band roots of [Theorem E.10](#) satisfy $\ell_\delta \uparrow 1$ and $U_\delta \downarrow 1$, the norm conversion satisfies $\mathfrak{b}_{\star,\bullet}(\delta) \sqrt{\delta} \rightarrow 0$, and $K_{\text{disc}}(\delta) \rightarrow K_{\text{disc}}(0) < \infty$. On $\mathcal{U}_{\delta_\bullet}$ one then has $\frac{1}{2}I \preceq C \preceq \frac{3}{2}I$, which is the band hypothesis of the variance estimate. This is an existence statement; no closed-form value of $\delta_\bullet$ in terms of $\alpha, \beta, N_\theta, L_H$ is available, because K_{disc} is defined through a supremum of second derivatives of the scheme map over the local hull.

The oracle variance constants V_0, V_1 , [Theorem F.7](#), are derived in [Section F](#). We compare one stochastic step to the deterministic local step. The one-step contraction of [Theorem E.12](#) holds pointwise on $\mathcal{U}_{\delta_\bullet}$: every $a \in \mathcal{U}_{\delta_\bullet}$ satisfies $\|T_\bullet(a) - a_\star\|_{\star,\bullet} \leq (1 - \frac{1}{2}\Delta t \gamma_\bullet) \|a - a_\star\|_{\star,\bullet}$ with

$\gamma_R = \frac{1}{4+\Gamma}$ and $\gamma_{KL} = \frac{1}{4+2\Delta t+\Gamma}$, and it requires only $\Delta t \leq 2/(4+\Gamma)$, with no energy-descent restriction $\Delta t \leq 1/(\beta_\star \bar{\Lambda}_\delta)$ on the stepsize and no forward-invariance assertion. We perturb that estimate by the Price/Hessian noise. The energy-descent restriction would force $\Delta t = O(\kappa^{-2})$ and add a polynomial factor in κ to the local rate.

The two perturbation lemmas for the local maps, [Theorems F.9](#) and [F.10](#), are in [Section F](#). Combining the pointwise contraction with these perturbation estimates gives the local theorems.

Theorem 4.7 (Stochastic local convergence, Riemannian). *Assume [Theorems 2.1](#) and [2.2](#), work in whitened coordinates, and set $A_R := 4 + \Gamma$, $\mu_R := 1/(2A_R)$, $V_1 = 24 + 6N_\theta L_{H,\star}^2$. Fix a stepsize*

$$0 < \Delta t \leq \min\left\{\frac{2}{4+\Gamma}, \frac{B}{256 A_R V_1}\right\},$$

let δ_R be stochastically admissible for this Δt , run [\(48\)](#) with batch B from $a_0 \in \mathcal{U}_{\delta_R}$, and put $\tau_R := \inf\{n \geq 0 : a_n \notin \mathcal{U}_{\delta_R}\}$. Then for all $N \geq 0$,

$$\mathbb{E}[\|a_N - a_\star\|_\star^2 \mathbf{1}_{\{\tau_R > N\}}] \leq (1 - \mu_R \Delta t)^N \|a_0 - a_\star\|_\star^2 + \frac{384 \Delta t (4 + \Gamma) N_\theta^2 L_{H,\star}^2}{B}.$$

The KL scheme contracts at the same local rate in its own weighted norm.

Theorem 4.8 (Stochastic local convergence, KL). *Under the analogous hypotheses, set $A_{KL} := 4 + 2\Delta t + \Gamma$ and $\mu_{KL} := 1/(2A_{KL})$, fix*

$$0 < \Delta t \leq \min\left\{\frac{2}{4+\Gamma}, \frac{B}{272 A_{KL} V_1}\right\},$$

let δ_{KL} be stochastically admissible for this Δt , and put $\tau_{KL} := \inf\{n \geq 0 : a_n \notin \mathcal{U}_{\delta_{KL}}\}$. Running [\(49\)](#) with batch B from $a_0 \in \mathcal{U}_{\delta_{KL}}$, for all $N \geq 0$,

$$\mathbb{E}[\|a_N - a_\star\|_{\star, \Delta t}^2 \mathbf{1}_{\{\tau_{KL} > N\}}] \leq (1 - \mu_{KL} \Delta t)^N \|a_0 - a_\star\|_{\star, \Delta t}^2 + \frac{408 \Delta t (4 + 2\Delta t + \Gamma) N_\theta^2 L_{H,\star}^2}{B}.$$

The proofs are given in [Section F](#). The left-hand sides are non-exit second moments: trajectories that have left the certified region by time N contribute zero. The discarded mass is controlled separately.

Proposition 4.9 (Exit probability over a finite horizon). *In the setting of [Theorem 4.7](#) (resp. [Theorem 4.8](#)), fix a radius $r_\bullet \leq \frac{1}{2\sqrt{2}}$ with $\{a : \|a - a_\star\|_{\star, \bullet} \leq r_\bullet\} \subseteq \mathcal{U}_\bullet$ and set $\tau_{r_\bullet} := \inf\{n \geq 0 : \|a_n - a_\star\|_{\star, \bullet} > r_\bullet\}$. Write $\eta_R = 192\Delta t^2 N_\theta^2 L_{H,\star}^2/B$ and $\eta_{KL} = 204\Delta t^2 N_\theta^2 L_{H,\star}^2/B$. If $\|a_0 - a_\star\|_{\star, \bullet} \leq r_\bullet$, then for every $N \geq 1$,*

$$\mathbb{P}(\tau_{r_\bullet} \leq N) \leq \frac{\|a_0 - a_\star\|_{\star, \bullet}^2 + N \eta_\bullet}{r_\bullet^2}.$$

In particular $\mathbb{P}(\tau_{r_\bullet} \leq N) \leq \epsilon'$ holds whenever $\|a_0 - a_\star\|_{\star, \bullet}^2 \leq \frac{\epsilon'}{2} r_\bullet^2$ and

$$B \geq \frac{2c'_\bullet N \Delta t^2 N_\theta^2 L_{H,\star}^2}{\epsilon' r_\bullet^2}, \quad 2c'_R = 384, \quad 2c'_{KL} = 408.$$

The proof is given in [Section F](#). The two terms are controlled by different resources: the batch B enters only through $\eta_\bullet$ and not through the initial-condition term, so high-probability containment over a horizon N requires both a deep-entry condition on a_0 and a batch proportional to N . This motivates two nested regions, entry being certified on the smaller and the exit time measured against the larger.

[Theorems 4.7](#) and [4.8](#) are killed-process statements, not stopped-process bounds: they control the mean square on the survival event. The standing assumptions do not give the sticking-the-landing mean-score noise bounded support, since even for a Gaussian target it equals $(C^{-1/2} - C^{1/2})Z$ with $Z \sim \mathcal{N}(0, I)$ away from $C = I$. The sampled-Hessian covariance noise is by contrast bounded almost surely, since $\alpha_\star I \preceq \bar{H}_n \preceq \beta_\star I$, so it is the mean component that obstructs pathwise invariance of a bounded local sublevel at fixed stepsize, and we claim no infinite-horizon containment.

By $\beta_\star \leq \kappa$ the spectral scale of [Theorem 3.3](#) satisfies $\Gamma = O(1 + \sqrt{N_\theta} \log \kappa)$, so at the certified stepsizes $\Delta t_R^\star = \min\{\frac{2}{A_0}, \frac{B}{256A_0V_1}\}$ and $\Delta t_{\text{KL}}^\star = \min\{\frac{2}{A_0}, \frac{B}{272(A_0+1)V_1}\}$ with $A_0 := 4 + \Gamma$ the per-step factor is $1 - \Theta(A_0^{-2} \min\{1, B/V_1\})$ for both schemes and the floor is $O(N_\theta^2 L_{H,\star}^2 \min\{\frac{1}{B}, \frac{1}{V_1}\})$, the simplified form $O(N_\theta^2 L_{H,\star}^2/B)$ holding in the large-batch branch $B \gtrsim V_1$. This gives

$$N_{\text{loc}} = O\left((4 + \Gamma)^2 \max\{1, V_1/B\} \log \frac{r_0^2}{\epsilon}\right),$$

which under the universal spectral certificate is $O((1 + \sqrt{N_\theta} \log \kappa)^2 \log \frac{1}{\epsilon})$ in the large-batch branch, and which in the small- Γ regime $\Gamma = O(1)$, $B \gtrsim V_1$ is $O(\log \frac{1}{\epsilon})$, independent of κ . With any certified pair $\gamma_0 \leq \gamma_\star$, $\Lambda_0 \geq \Lambda_\star$ in place of the universal $(1/(4 + \Gamma), (4 + \Gamma)/2)$ the same proofs give the target-adaptive count $N_{\text{loc}} = O(\max\{\Lambda_\star/\gamma_\star, V_1/(B\gamma_\star^2)\} \log \frac{r_0^2}{\epsilon})$. We do not claim the intermediate $O(\sqrt{N_\theta} \log \kappa \cdot \log \frac{1}{\epsilon})$ bound, linear rather than quadratic in the spectral scale $4 + \Gamma$.

The deterministic three-stage theorem of [Section 3](#) spends $O(\Delta t^{-1} \log_+ \frac{1}{\beta\lambda_0})$ iterations on the covariance burn-in, $O(\kappa \Delta t^{-1} \log_+ \frac{\Delta_0}{\delta_\bullet})$ on the global phase, and $O((4 + \Gamma) \Delta t^{-1} \log \frac{1}{\epsilon})$ on the local phase; the following corollary is its stochastic mirror, with the first stage removed by the rescue, the second stage supplied by [Theorems 4.4](#) and [4.5](#) and Markov's inequality, and the third stage supplied by [Theorems 4.7](#) and [4.8](#) on the survival event.

Corollary 4.10 (Stochastic global-to-local pipeline). *Fix a scheme $\bullet$, a stochastically admissible outer level δ_{out} with radius $r_\bullet \leq \frac{1}{2\sqrt{2}}$ satisfying $\{a : \|a - a_\star\|_{\star,\bullet} \leq r_\bullet\} \subseteq \mathcal{U}_{\delta_{\text{out}}}$, and a pair $(\Delta t, B)$ satisfying the hypotheses of [Theorem 4.7](#) (resp. [Theorem 4.8](#)) at level δ_{out} . Fix failure budgets $\epsilon_{\text{ent}}, \epsilon' \in (0, 1)$ and an inner level $\delta_{\text{in}} \leq \delta_{\text{out}}$ small enough that $\mathfrak{b}_{\star,\bullet}(\delta_{\text{in}})^2 \delta_{\text{in}} \leq \frac{\epsilon'}{2} r_\bullet^2$. Run a global scheme of [Theorems 4.4](#) and [4.5](#) at constant step, or [Theorem 4.11](#), for N_1 steps until $\mathbb{E} \Delta_{N_1} \leq \epsilon_{\text{ent}} \delta_{\text{in}}$, then switch to the local scheme with $B \geq c_\bullet N_2 \Delta t^2 N_\theta^2 L_{H,\star}^2 / (\epsilon' r_\bullet^2)$, $c_R = 384$, $c_{\text{KL}} = 408$. Writing $E_{\text{in}} := \{a_{N_1} \in \mathcal{U}_{\delta_{\text{in}}}\}$ and $\tau^{(N_1)} := \inf\{n \geq 0 : \|a_{N_1+n} - a_\star\|_{\star,\bullet} > r_\bullet\}$, for every local horizon N_2 ,*

$$\mathbb{P}(E_{\text{in}} \cap \{\tau^{(N_1)} > N_2\}) \geq 1 - \epsilon_{\text{ent}} - \epsilon',$$

and

$$\mathbb{E}[\|a_{N_1+N_2} - a_\star\|_{\star,\bullet}^2; E_{\text{in}}, \tau^{(N_1)} > N_2] \leq (1 - \mu_\bullet \Delta t)^{N_2} \mathfrak{b}_{\star,\bullet}(\delta_{\text{in}})^2 \delta_{\text{in}} + F_\bullet,$$

with $F_R = 384 \Delta t (4 + \Gamma) N_\theta^2 L_{H,\star}^2 / B$ and $F_{\text{KL}} = 408 \Delta t (4 + 2\Delta t + \Gamma) N_\theta^2 L_{H,\star}^2 / B$.

The proof is given in [Section F](#). Both displays are restricted to E_{in} because on its complement the iterate carries only the outer bound $r_\bullet$, which the local contraction does not improve.

4.5 Removing the floors

At constant stepsize the certified bounds retain an additive residual. Under Hessian-Lipschitzness a standard decreasing schedule removes it for both schemes.

Theorem 4.11 (Decreasing-stepsize floor-free convergence). *Assume [Theorems 2.1](#) and [2.2](#), initialize with the quadratic rescue of [Theorem 2.14](#), and run either (48) or (49) with the schedule*

$$\Delta t_n = \frac{8\kappa}{n + n_0}, \quad n_0 \geq 64\kappa^2. \quad (56)$$

Then $\Delta t_n \leq 1/(8\kappa)$ for all $n \geq 0$, and with $c_S := \frac{7}{4}$ and $\Psi_H = N_\theta^2 \bar{L}_H^2$,

$$\mathbb{E}\Delta_N \leq \frac{K}{N + n_0}, \quad K := \max\left\{\frac{64 c_S \kappa^3 \Psi_H}{B}, n_0 \Delta(a_0)\right\}. \quad (57)$$

In particular, with the canonical offset $n_0 = \lceil 64\kappa^2 \rceil$, $K = \tilde{O}(\kappa^3 N_\theta^2 \bar{L}_H^2 / B + \kappa^2 \Delta(a_0))$; for a larger offset $n_0 \gg \kappa^2$ the initialization contribution $n_0 \Delta(a_0)$ scales with the chosen n_0 rather than with κ^2 .

The proof is given in [Section F](#).

The bound (57) is an $O(1/N)$ decay to zero, not to a floor: the Ψ_H -dependent term is carried inside K and divided by $N + n_0$. For a Gaussian target $\Psi_H = 0$ and, by [Theorem 2.15](#), $\Delta(a_0) = 0$, so $K = 0$. The schedule requires only $\bar{L}_H < \infty$ to define K ; the iteration itself does not use L_H . The theorem is stated only in the Hessian-Lipschitz regime, because in the assumption-minimal regime the stability threshold is $\Delta t = \Theta(\kappa^{-2})$ rather than $1/(8\kappa)$ and (56) does not apply as written.

Every stochastic floor above originates in the intrinsic fluctuation Ψ entering the additive stochastic terms of the one-step inequalities (196) and (197) of [Theorems F.5](#) and [F.6](#) ([Section F](#)), and Ψ vanishes identically for Gaussian targets. We record what this gives globally and locally.

Proposition 4.12 (Zero intrinsic fluctuation for Gaussian targets). *If $V(\theta) = \frac{1}{2}(\theta - \mu)^\top H(\theta - \mu) + c$ with $H \succ 0$, then $\nabla^2 V(X) = H$ for all X , so $A(a) = H$ and $\Psi(a) \equiv 0$ for every $a \in \mathcal{A}$, and the additive stochastic term in the one-step inequalities (196) and (197) of [Theorems F.5](#) and [F.6](#) ([Section F](#)) vanishes unconditionally. Consequently, under the covariance-band and stepsize hypotheses of the corresponding global theorem, that is the band $\frac{1}{2\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ for the Riemannian scheme and $\frac{1}{\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ for the KL scheme, both persisting pathwise from $n = 0$ under the rescued initialization by [Theorem 4.2](#), both schemes contract with no floor,*

$$\mathbb{E}\Delta_N \leq (1 - c\kappa^{-2})^N \Delta(a_0),$$

with $c = \frac{1}{24}$ for the Riemannian scheme at $B = 1$ and $c = \frac{1}{16}$ for the KL scheme. Under quadratic rescue $a_0 = a_\star$, so $\Delta(a_0) = 0$ and the iterates remain exactly at $a_\star$.

Proof. $\nabla^2 V \equiv H$ gives $\nabla^2 V(X) - A(a) = 0$ pathwise, so the integrand of [Theorem 4.3](#) vanishes. Setting $\Psi = 0$ in (196) and (197) leaves the pure descent $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \frac{\Delta t}{2} G(a_n)^2$ in both cases. The gradient-domination constant is scheme-specific through the covariance floor: on the Riemannian band (52) gives $G(a_n)^2 \geq \frac{1}{2\kappa} \Delta_n$, so at $\Delta t = \frac{1}{6\kappa}$ we get $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq (1 - \frac{\Delta t}{4\kappa}) \Delta_n = (1 - \frac{1}{24\kappa^2}) \Delta_n$; on the KL band it gives $G(a_n)^2 \geq \frac{1}{\kappa} \Delta_n$, so at $\Delta t = \frac{1}{8\kappa}$ we get $(1 - \frac{\Delta t}{2\kappa}) \Delta_n = (1 - \frac{1}{16\kappa^2}) \Delta_n$. Equivalently $\Psi_H = N_\theta^2 \bar{L}_H^2 = 0$ removes the Hessian-Lipschitz floors of [Theorem 4.4\(1\)](#) and [Theorem 4.5\(1\)](#) directly. Finally [Theorem 2.15](#) gives $a_0 = a_\star$ and $\Delta(a_0) = 0$. $\square$

Substituting the self-bounding estimate (191) for Ψ hides the vanishing in the assumption-minimal branches; the intrinsic one-step inequalities show it directly. The property is specific to Gaussian targets, since it requires the sampled Hessian to equal its mean pathwise, and the resulting upper bound is not a matching lower bound on the asymptotic error of the algorithm. The three-case Gaussian local calibration, [Theorem F.11](#), is recorded in [Section F](#).

4.6 Complexity summary

[Table 1](#) collects the stochastic complexities. Throughout, the rescue contributes one deterministic query and the reported counts are those of Phase I of [Algorithm 1](#); $\tilde{O}$ suppresses factors logarithmic in κ , N_θ , $\Delta(a_0)/\epsilon$, and $\bar{L}_H$.

Scheme and regime	quantity	floor	contraction	N to ϵ above floor
Riem., assumption-minimal	$\mathbb{E}\Delta_N$	$O(\kappa N_\theta)$	$1 - \Theta(\kappa^{-3})$	$\tilde{O}(\kappa^3)$
Riem., Hessian-Lipschitz	$\mathbb{E}\Delta_N$	$O(\kappa N_\theta^2 \bar{L}_H^2)$	$1 - \Theta(\kappa^{-2})$	$\tilde{O}(\kappa^2)$
KL, assumption-minimal	$\mathbb{E}\Delta_N$	$O(\kappa N_\theta)$	$1 - \Theta(\kappa^{-3})$	$\tilde{O}(\kappa^3)$
KL, Hessian-Lipschitz	$\mathbb{E}\Delta_N$	$O(\kappa N_\theta^2 \bar{L}_H^2)$	$1 - \Theta(\kappa^{-2})$	$\tilde{O}(\kappa^2)$
Riem., local, non-exit	$\mathbb{E}[r_N^2; \tau > N]$	$O(N_\theta^2 L_{H,*}^2/B)$	$1 - \Theta((4+\Gamma)^{-2})$	$\tilde{O}((4+\Gamma)^2)$
KL, local, non-exit	$\mathbb{E}[r_N^2; \tau > N]$	$O(N_\theta^2 L_{H,*}^2/B)$	$1 - \Theta((4+\Gamma)^{-2})$	$\tilde{O}((4+\Gamma)^2)$
Decreasing stepsize	$\mathbb{E}\Delta_N$	0	—	$\tilde{O}\left(\frac{\kappa^3 N_\theta^2 \bar{L}_H^2 + \kappa^2 \Delta(a_0)}{\epsilon}\right)$
Gaussian target	$\mathbb{E}\Delta_N$	0	—	0

Table 1: Stochastic complexity across regimes. The four global rows and the decreasing-stepsize row are unconditional expectation bounds displayed at $B = 1$; the two local rows retain the batch B and are non-exit bounds conditional on certified entry into $\mathcal{U}_{\delta_\bullet}$, controlling $\mathbb{E}[\|a_N - a_\star\|_{\star,\bullet}^2; \tau > N]$ rather than $\mathbb{E}\Delta_N$, paired with [Theorem 4.9](#) and [Theorem 4.10](#). The assumption-minimal rows use [Theorem 2.1](#); the Hessian-Lipschitz, local, and decreasing-step rows additionally use [Theorem 2.2](#). Minibatching $B = \Theta(\kappa)$ trades the assumption-minimal regime to $\tilde{O}(\kappa^2)$ rounds at $\tilde{O}(\kappa^3)$ total oracle pairs. The decreasing-step row uses the canonical offset $n_0 = \lceil 64\kappa^2 \rceil$, and the Gaussian row is exact after the one rescue query. The displayed local contraction and count are the large-batch branch $B \gtrsim V_1$; for general B they read $1 - \Theta((4 + \Gamma)^{-2} \min\{1, B/V_1\})$ and $\tilde{O}((4 + \Gamma)^2 \max\{1, V_1/B\})$. Enforcing the finite-horizon survival probability of [Theorem 4.9](#) requires the additional horizon-dependent batch $B = \Omega(N\Delta t^2 N_\theta^2 L_{H,*}^2/(\epsilon' r_\bullet^2))$, so the local rows are not high-probability oracle-complexity results.

The single-sample gap between the two constant-stepsize global regimes is exactly one factor of κ . Under [Theorem 2.1](#) alone the self-bounding estimate (191) charges the fluctuation against the gradient norm, forcing $\Delta t = \Theta(\kappa^{-2})$ through $\rho_B \asymp \kappa$ at $B = 1$ and giving contraction $1 - \Theta(\kappa^{-3})$; under [Theorem 2.2](#) the fluctuation is controlled by the nonquadraticity of the target directly, the stepsize rises to $\Delta t = \Theta(\kappa^{-1})$, and the contraction improves to $1 - \Theta(\kappa^{-2})$. The floor changes correspondingly from the dimension-linear, target-agnostic $O(\kappa N_\theta)$ to the dimension-quadratic $O(\kappa N_\theta^2 \bar{L}_H^2)$, which is proportional to the squared non-Gaussianity and hence far smaller for near-Gaussian targets. The table also shows that the three deterministic stages survive the noise: the covariance burn-in is spent by the single rescue query, the global phase costs $\tilde{O}(\kappa^2)$ or $\tilde{O}(\kappa^3)$ iterations to an explicit floor, and the local phase costs $\tilde{O}((4 + \Gamma)^2)$ iterations at a rate governed by the whitened spectral scale alone.

5 General affine invariant Gaussian flows

The Fisher–Rao metric restricted to the Gaussian family is affine invariant, and every rate obtained in [Sections 2 to 4](#) is expressed in affine invariant quantities. We aim to classify all affine invariant Riemannian metrics on the Gaussian manifold, study the gradient flows they induce for the KL divergence, and locate Fisher–Rao inside that family.

5.1 Classification of affine invariant metrics

Let $\varphi(\theta) = A\theta + b$ be an invertible affine transformation, and denote the map it induces on parameters by $\Phi_{(A,b)} : \mathcal{A} \rightarrow \mathcal{A}$, $\Phi_{(A,b)}(m, C) = (Am + b, ACA^\top)$. Its differential at $a = (m, C)$ is

$$D\Phi_{(A,b)}(m, C)[(u, X)] = \lim_{\epsilon \rightarrow 0} \frac{\Phi_{(A,b)}(m + \epsilon u, C + \epsilon X) - \Phi_{(A,b)}(m, C)}{\epsilon} = (Au, AXA^\top), \quad (58)$$

where we identify $T_a\mathcal{A}$ with $\mathbb{R}^{N_\theta} \times \mathbb{S}(N_\theta)$. A Riemannian metric g on $\mathcal{A}$ is affine invariant when every $\Phi_{(A,b)}$ is an isometry, that is, when

$$g_{(m,C)}((u, X), (v, Y)) = g_{\Phi_{(A,b)}(m,C)}((Au, AXA^\top), (Av, AY A^\top)) \quad (59)$$

for all $(u, X), (v, Y) \in T_a\mathcal{A}$.

Theorem 5.1 (Classification of affine invariant metrics on the Gaussian manifold). *Let $N_\theta \geq 2$. A Riemannian metric g on the Gaussian manifold $\mathcal{A} = \mathbb{R}^{N_\theta} \times \mathbb{S}_{++}(N_\theta)$ is affine invariant if and only if*

$$g_a((u, X), (v, Y)) = \eta u^\top C^{-1} v + \omega \operatorname{Tr}(C^{-1} X C^{-1} Y) + \tau \operatorname{Tr}(C^{-1} X) \operatorname{Tr}(C^{-1} Y) \quad (60)$$

for a triple of real parameters satisfying

$$\eta > 0, \quad \omega > 0, \quad \tau > -\frac{\omega}{N_\theta}, \quad (61)$$

and the triple (η, ω, τ) is uniquely determined by g . Replacing (η, ω, τ) by $c(\eta, \omega, \tau)$ with $c > 0$ multiplies g by c , so the affine invariant metrics form a two-parameter family up to an overall positive multiple, normalized below by $\eta = 1$.

Proof. Since $\Phi_{(C^{1/2}, m)}(0, I) = (m, C)$, the affine group acts transitively on $\mathcal{A}$, and by (59) the metric is determined by its value at $(0, I)$ together with the requirement of invariance under the isotropy group of $(0, I)$, which is the orthogonal group $O(N_\theta)$ acting by $(u, X) \mapsto (Qu, QXQ^\top)$.

We first state the established classification of affine invariant Riemannian metrics on $\mathbb{S}_{++}(N_\theta)$ [18, 19]. For $C \in \mathbb{S}_{++}(N_\theta)$ and $X, Y \in \mathbb{S}(N_\theta)$, every such metric is of the form

$$g_C(X, Y) = \omega \operatorname{Tr}(C^{-1} X C^{-1} Y) + \tau \operatorname{Tr}(C^{-1} X) \operatorname{Tr}(C^{-1} Y), \quad \omega > 0, \quad \tau > -\frac{\omega}{N_\theta}, \quad (62)$$

where $\omega > 0$ is positivity on the traceless directions and, taking $X = Y = sC$, positivity on the trace direction reads $N_\theta(\omega + N_\theta\tau)s^2 > 0$.

We then show that there are no cross terms between u and Y . We define $L(u, Y) := g_{(0,I)}((u, 0), (0, Y))$; for any orthogonal $Q \in O(N_\theta)$, affine invariance requires $L(u, Y) = L(Qu, QYQ^\top)$. Taking the sign element $Q = -I$, which fixes every Y and reverses every u , gives $L(u, Y) = L(-u, Y) = -L(u, Y)$ by linearity in the first argument, so $L(u, Y) = 0$ for all $u \in \mathbb{R}^{N_\theta}$ and all $Y \in \mathbb{S}(N_\theta)$, and affine invariant metrics on the Gaussian manifold carry no mean-covariance coupling.

We proceed to the mean part, and first consider the point $(0, I)$. Affine invariance requires $g_{(0,I)}((Qu, 0), (Qv, 0)) = g_{(0,I)}((u, 0), (v, 0))$ for every $Q \in O(N_\theta)$, hence $g_{(0,I)}((u, 0), (v, 0)) = \eta u^\top v$ for a scalar η , and $\eta > 0$ by positive definiteness. For an arbitrary covariance $C \in \mathbb{S}_{++}(N_\theta)$ there is A with $C = AA^\top$, and $\Phi_{(A,m)}(0, I) = (m, C)$ gives $g_{(m,C)}((Au, 0), (Av, 0)) = \eta u^\top v$; writing $u' = Au$ and $v' = Av$ yields $g_{(m,C)}((u', 0), (v', 0)) = \eta u'^\top C^{-1} v'$. Transporting (62) in the same way gives (60) with the positivity conditions (61).

Conversely, the right-hand side of (60) is invariant under (58) by direct substitution, and it is positive definite exactly under (61). Uniqueness of the triple follows by evaluating $g_{(0,I)}$ on $(u, 0)$ with $\|u\| = 1$, on a traceless X with $\operatorname{Tr}(X^2) = 1$, and on $X = I$, which returns η , ω and $N_\theta(\omega + N_\theta\tau)$ respectively. $\square$

The three blocks appearing in this proof, the mean, the traceless covariance and the trace covariance, are pairwise inequivalent irreducible representations of $O(N_\theta)$, and Schur's lemma gives (60) with one weight per block; this structural form of Theorem 5.1 is carried out in Section H.

Remark 5.2. For $N_\theta = 1$ there is no traceless block, the two covariance terms in (60) coincide, and only the sum $\omega + \tau$ is identifiable. The metric is then $g_a((u, X), (v, Y)) = \eta uv/C + (\omega + \tau)XY/C^2$, and positive definiteness requires $\eta > 0$ and $\omega + \tau > 0$. The uniqueness statement of Theorem 5.1 and the balancing statement of Theorem 5.5 concern the three-mode geometry and therefore require $N_\theta \geq 2$.

5.2 The general flow and its exact modal decomposition

We take the KL divergence $\mathcal{E}$ as the energy and the metrics of [Theorem 5.1](#) as the Riemannian structure, following Section 3 of [\[6\]](#). We abbreviate the precision residual and its whitened trace–traceless split by

$$\begin{aligned} R(a) &:= C^{-1} + \mathbb{E}_{\rho_a}[\nabla_{\theta} \nabla_{\theta} \log \rho_{\text{post}}(\theta)], & S(a) &:= C^{1/2} R(a) C^{1/2}, \\ s(a) &:= \text{Tr } S(a), & S_0(a) &:= S(a) - \frac{s(a)}{N_{\theta}} I, \end{aligned} \quad (63)$$

so that $CR(a)C = C + C\mathbb{E}_{\rho_a}[\nabla_{\theta} \nabla_{\theta} \log \rho_{\text{post}}(\theta)]C$ and $s(a) = \text{Tr}(CR(a))$. The gradient flow of $\mathcal{E}$ for the metric [\(60\)](#) is then characterized by the ODEs

$$\begin{aligned} \frac{dm_t}{dt} &= \frac{1}{\eta} C_t \mathbb{E}_{\rho_{a_t}}[\nabla_{\theta} \log \rho_{\text{post}}(\theta)], \\ \frac{dC_t}{dt} &= \frac{1}{2\omega} \left(C_t + C_t \mathbb{E}_{\rho_{a_t}}[\nabla_{\theta} \nabla_{\theta} \log \rho_{\text{post}}(\theta)] C_t \right) \\ &\quad - \frac{\tau}{2\omega(\omega + N_{\theta}\tau)} \text{Tr} \left(I + C_t \mathbb{E}_{\rho_{a_t}}[\nabla_{\theta} \nabla_{\theta} \log \rho_{\text{post}}(\theta)] \right) C_t. \end{aligned} \quad (64)$$

The mean equation involves only η and the covariance equation only (ω, τ) , so under the normalization $\eta = 1$ the mean drift is the same for every affine invariant metric and coincides with the mean drift of [\(1\)](#). The covariance equation, on the other hand, splits exactly, and not only to first order, along the whitened trace and traceless directions.

Theorem 5.3 (Exact modal decomposition of the general flow). *Let η, ω, τ satisfy [\(61\)](#). The Riemannian gradient of $\mathcal{E}$ for the metric [\(60\)](#) is*

$$\text{grad } \mathcal{E}(a) = \left(-\frac{1}{\eta} C \mathbb{E}_{\rho_a}[\nabla_{\theta} \log \rho_{\text{post}}(\theta)], -\frac{1}{2\omega} CR(a)C + \frac{\tau}{2\omega(\omega + N_{\theta}\tau)} \text{Tr}(CR(a))C \right), \quad (65)$$

its negative gradient flow is [\(64\)](#), and the covariance equation of [\(64\)](#) has the exact decomposition

$$C_t^{-1/2} \frac{dC_t}{dt} C_t^{-1/2} = \frac{1}{2\omega} S_0(a_t) + \frac{s(a_t)}{2N_{\theta}(\omega + N_{\theta}\tau)} I. \quad (66)$$

Along [\(64\)](#) the energy dissipates in three blocks,

$$\frac{d}{dt} \mathcal{E}(a_t) = -\frac{1}{\eta} \mathbb{E}_{\rho_{a_t}}[\nabla_{\theta} \log \rho_{\text{post}}(\theta)]^{\top} C_t \mathbb{E}_{\rho_{a_t}}[\nabla_{\theta} \log \rho_{\text{post}}(\theta)] - \frac{1}{4\omega} \|S_0(a_t)\|_{\text{F}}^2 - \frac{s(a_t)^2}{4N_{\theta}(\omega + N_{\theta}\tau)}. \quad (67)$$

Proof. The differential of $\mathcal{E}$ reads $D\mathcal{E}(a)[(u, X)] = -\mathbb{E}_{\rho_a}[\nabla_{\theta} \log \rho_{\text{post}}(\theta)]^{\top} u - \frac{1}{2} \text{Tr}(R(a)X)$. Matching the mean term against $\eta u^{\top} C^{-1} v$ gives the first component of [\(65\)](#). For the covariance block we write $W = C^{-1/2}(\text{grad } \mathcal{E})_C C^{-1/2}$ and $Z = C^{-1/2} X C^{-1/2}$, so that gradient duality requires

$$\omega \text{Tr}(WZ) + \tau \text{Tr}(W) \text{Tr}(Z) = -\frac{1}{2} \text{Tr}(S(a)Z) \quad \text{for every } Z \in \mathbb{S}(N_{\theta}).$$

Testing against traceless Z and against $Z = I$ separately gives $W_0 = -S_0(a)/(2\omega)$ and $\text{Tr } W = -s(a)/(2(\omega + N_{\theta}\tau))$, and recombining $W = W_0 + \frac{\text{Tr } W}{N_{\theta}} I$ gives the second component of [\(65\)](#) and, after conjugating by $C^{1/2}$, the decomposition [\(66\)](#). Since $\frac{d}{dt} \mathcal{E}(a_t) = -\|\text{grad } \mathcal{E}(a_t)\|_{a_t}^2$ and the metric [\(60\)](#) evaluates on W as $\omega \text{Tr}(W^2) + \tau(\text{Tr } W)^2 = \omega \|W_0\|_{\text{F}}^2 + (\omega + N_{\theta}\tau)(\text{Tr } W)^2/N_{\theta}$, substituting the two components gives [\(67\)](#). $\square$

On a Gaussian target the decomposition (66) becomes a set of three decoupled scalar rates, which is the sense in which (η, ω, τ) preconditions the flow.

Theorem 5.4 (Exact Gaussian modal rates). *Let $\rho_{\text{post}} = \mathcal{N}(m_\star, C_\star)$, and place the flow in optimizer-whitened coordinates, so that $\rho_{\text{post}} = \mathcal{N}(0, I)$. Then (64) reads*

$$\frac{dm_t}{dt} = -\frac{1}{\eta}C_t m_t, \quad \frac{dC_t}{dt} = \frac{1}{2\omega}(C_t - C_t^2) - \frac{\tau}{2\omega(\omega + N_\theta\tau)} \text{Tr}(I - C_t)C_t, \quad (68)$$

and writing $C_t = I + X_t$ with $X_t = X_{0,t} + \sigma_t I$, $\sigma_t = \text{Tr}(X_t)/N_\theta$ and $\text{Tr } X_{0,t} = 0$,

$$\frac{dX_{0,t}}{dt} = -\frac{1}{2\omega}X_{0,t} + O(\|X_t\|^2), \quad \frac{d\sigma_t}{dt} = -\frac{1}{2(\omega + N_\theta\tau)}\sigma_t + O(\|X_t\|^2). \quad (69)$$

Consequently the linearization of (64) at $a_\star = (0, I)$ has exactly the three decay rates

$$\gamma_{\text{mean}} = \frac{1}{\eta}, \quad \gamma_{\text{traceless}} = \frac{1}{2\omega}, \quad \gamma_{\text{trace}} = \frac{1}{2(\omega + N_\theta\tau)}. \quad (70)$$

Proof. By affine invariance we may assume $\rho_{\text{post}} = \mathcal{N}(0, I)$, for which $\mathbb{E}_{\rho_a}[\nabla_\theta \log \rho_{\text{post}}(\theta)] = -m$ and $\mathbb{E}_{\rho_a}[\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] = -I$, so that $R(a) = C^{-1} - I$ and $S(a) = I - C$. Substitution in (64) gives (68). With $C = I + X$ we have $S = -X$, $s = -\text{Tr } X$ and $S_0 = -X_0$, so (66) gives

$$\frac{dX_t}{dt} = -\frac{1}{2\omega}X_{0,t} - \frac{\text{Tr}(X_t)}{2N_\theta(\omega + N_\theta\tau)}I + O(\|X_t\|^2),$$

where the quadratic remainder comes from $C^{1/2}(\cdot)C^{1/2} = (\cdot) + O(\|X\|)$ applied to the right-hand side of (66). Projecting onto the traceless symmetric matrices and onto $\mathbb{R}I$ gives (69), and the mean equation of (68) gives $\dot{m}_t = -m_t/\eta + O(\|(m_t, X_t)\|^2)$. The three linear operators act on the mutually orthogonal blocks $\mathbb{R}^{N_\theta}$, $\mathbb{S}_0(N_\theta)$ and $\mathbb{R}I$, each as a negative multiple of the identity, which gives (70). $\square$

5.3 Fisher–Rao is the unique balanced metric

We then show that Fisher–Rao is the only member of the family that treats the three modes at equal speed.

Theorem 5.5 (Fisher–Rao is the unique balanced affine invariant metric). *Let $N_\theta \geq 2$ and let η, ω, τ satisfy (61). The three Gaussian modal rates (70) coincide if and only if*

$$(\eta, \omega, \tau) = c \left(1, \frac{1}{2}, 0\right) \quad \text{for some } c > 0. \quad (71)$$

Up to an overall metric scale, the Fisher–Rao metric is therefore the unique affine invariant metric on the Gaussian manifold that treats the mean, the traceless covariance and the trace covariance modes at equal speed, and under the normalization $\eta = 1$ its parameters are $\omega = 1/2$ and $\tau = 0$.

Proof. Equality of $\gamma_{\text{traceless}}$ and γ_{trace} gives $\omega = \omega + N_\theta\tau$, hence $\tau = 0$, and equality of γ_{mean} and $\gamma_{\text{traceless}}$ then gives $\eta = 2\omega$. Setting $c = \eta$ gives (71), and conversely (71) makes all three rates equal to $1/c$. Substituting $(\eta, \omega, \tau) = (1, 1/2, 0)$ in (60) returns the Fisher–Rao metric $g_a^{\text{FR}}((u, X), (v, Y)) = u^\top C^{-1}v + \frac{1}{2} \text{Tr}(C^{-1}XC^{-1}Y)$ and reduces (64) to (1). $\square$

Remark 5.6. The rates (70) are the exact eigenvalues of the linearization of (64) at a Gaussian optimizer. They are not global rates for the nonlinear covariance equation, and they are not rates for a general target. An unbalanced metric can outperform Fisher–Rao on a task dominated by a single mode. On a volume-dominated problem, where the initial error is carried by the trace $\text{Tr}(X_0)$, fixing η and ω and letting τ decrease to $-\omega/N_\theta$ makes γ_{trace} arbitrarily large while leaving γ_{mean} and $\gamma_{\text{traceless}}$ unchanged, and such a metric converges faster than Fisher–Rao on that task. The convergence theory of Sections 2 to 4 is established for $(\eta, \omega, \tau) = (1, 1/2, 0)$ and is not claimed for the remaining members of the family.

5.4 Discretizations of the general flow

The two schemes of [Section 2](#) extend to the whole family. The Riemannian scheme (20) advances the covariance by the exponential map of $\mathbb{S}_{++}(N_\theta)$ applied to $-\Delta t \text{grad } \mathcal{E}(a_n)$; carrying out the same construction with the gradient (65) in place of the Fisher–Rao gradient gives

$$\begin{aligned} m_{n+1} &= m_n + \frac{\Delta t}{\eta} C_n \mathbb{E}_{\rho_{a_n}} [\nabla_\theta \log \rho_{\text{post}}(\theta)], \\ C_{n+1} &= C_n^{1/2} \exp \left(\frac{\Delta t}{2\omega} \left[C_n^{1/2} \mathbb{E}_{\rho_{a_n}} [\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n^{1/2} \right. \right. \\ &\quad \left. \left. + \frac{\omega - \tau \text{Tr}(C_n \mathbb{E}_{\rho_{a_n}} [\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)])}{\omega + N_\theta \tau} I \right] \right) C_n^{1/2}. \end{aligned} \quad (72)$$

At $(\eta, \omega, \tau) = (1, \frac{1}{2}, 0)$ the bracket collapses to $C_n^{1/2} \mathbb{E}_{\rho_{a_n}} [\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n^{1/2} + I$ and the prefactor $\Delta t/(2\omega)$ to Δt , so (72) recovers

$$C_{n+1} = e^{\Delta t} C_n^{1/2} \exp \left(\Delta t C_n^{1/2} \mathbb{E}_{\rho_{a_n}} [\nabla_\theta \nabla_\theta \log \rho_{\text{post}}(\theta)] C_n^{1/2} \right) C_n^{1/2},$$

which is the covariance update of (20).

The Riemannian scheme requires a matrix exponential at every step. The KL resolvent scheme (21) of [Section 2](#) avoids it, but its rational form does not extend safely to the whole family: the natural general-metric analogue multiplies the resolvent by a scalar prefactor that is not sign-definite for $\tau < 0$, so the resulting update need not remain in the SPD cone for admissible metrics. We therefore state the general-metric extension only for the retraction scheme (72), whose covariance update is SPD by construction; at the Fisher–Rao point $(\eta, \omega, \tau) = (1, \frac{1}{2}, 0)$ the resolvent scheme (21) of [Section 2](#) remains available with its full convergence theory.

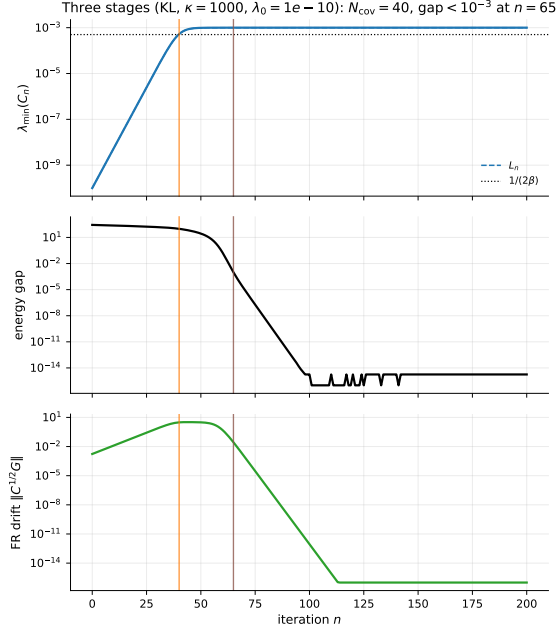
6 Numerical illustrations

We illustrate the mechanisms established above on a small number of diagnostic examples, each tied to a specific statement of the preceding sections. The deterministic Gaussian and separable log-concave expectations are evaluated analytically or by fixed Gauss–Hermite quadrature, and the nonseparable affine-mode target uses a fixed scrambled Sobol design. The stochastic runs use float64 arithmetic with explicit seeds, and each baseline and sticking-the-landing pair is driven by the same random-number stream, so that the comparison is paired. The target formulas, grids, seeds, quadrature rules, and stopping conditions are recorded in [Section I](#).

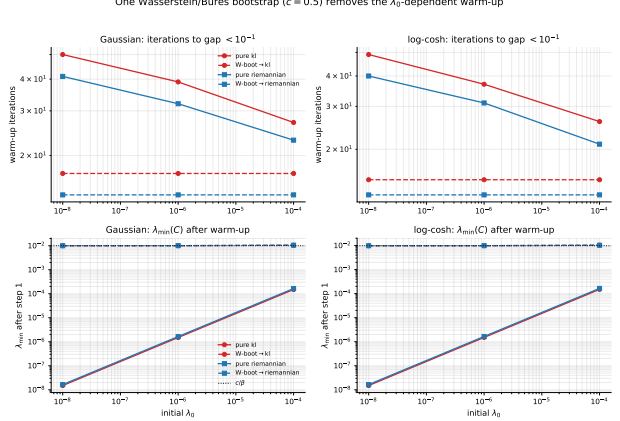
6.1 The three stages

The covariance-scaling sweep uses diagonal Gaussian targets with initial covariances as small as $\lambda_{0,\min} = 10^{-10}$, and we record as the burn-in endpoint the first iterate with $\lambda_{\min}(C_n) \geq 1/(2\beta)$. Across the sweep this time is linear in $\log_+(1/(\beta\lambda_{0,\min}))$, which is the burn-in count of the two-sided covariance recursions in [Theorems 2.8](#) and [2.10](#). [Figure 1](#) shows one hard KL run in dimension $N_\theta = 20$ with $\kappa = 1000$, where the three stages of [Theorem 3.16](#) are visible in sequence: the covariance calibrates, the energy then decays rapidly, and the Fisher–Rao mean-block drift $\|C_n^{1/2} g(a_n)\|$ decays last.

The transport panel supports one claim only. One Bures–Wasserstein step removes the small-covariance cost and hands the trajectory back to a Fisher–Rao scheme; it does not change the $\kappa_\star$ -scale obstruction of [Theorem 2.5](#), and it is not a blended Wasserstein–Fisher–Rao flow.



(a) A hard KL trajectory.



(b) One transport step, then a Fisher-Rao tail.

Figure 1: Covariance initialization and the three stages. On the left, a Gaussian run with $N_\theta = 20$, $\kappa = 1000$, and $\lambda_{0,\min} = 10^{-10}$ separates covariance burn-in, global energy localization, and the local regime. On the right, one Bures–Wasserstein step with $c = 1/2$, hence transport stepsize $\eta = c/\beta$, lifts the smallest covariance eigenvalue to the curvature scale irrespective of the initial underdispersion, and the subsequent Fisher–Rao warm-up is essentially flat in $\lambda_{0,\min}$, as in [Theorem 2.13](#).

6.2 The two discretizations

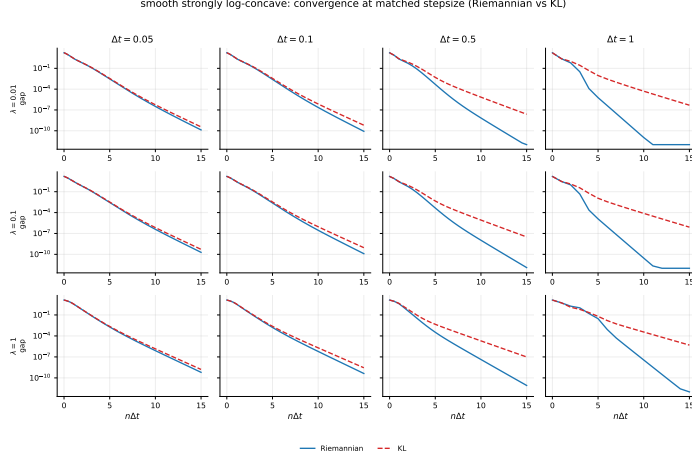
The deterministic comparison uses two-dimensional Gaussian and smooth log-concave targets together with a convex quartic target that satisfies neither half of [Theorem 2.1](#), its Hessian being unbounded above and degenerate along one direction. Both schemes use the same explicit mean update and evaluate the same population score and Hessian. We report the iteration count and the elapsed flow time $n\Delta t$ separately, the latter being a discretization diagnostic rather than a timing measurement.

The robustness recorded here is a property of the covariance map alone. On a log-concave target one has $C_n^{-1} + \Delta t A_n \succ 0$ for every $\Delta t > 0$, so the KL covariance update cannot leave the positive-definite cone, which is the content of [Theorem 2.10](#). The full algorithm can still fail at a large stepsize through the explicit mean update, which carries the factor $\Delta t C_n$.

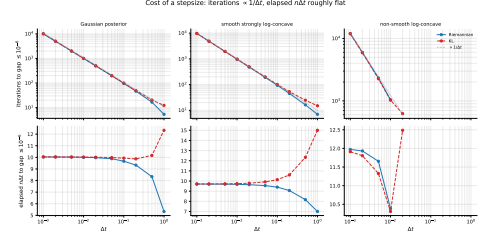
6.3 Sticking-the-landing variance and the constant-step floors

The stochastic grid uses Gaussian targets with $N_\theta \in \{2, 8, 32\}$ and $\kappa \in \{10, 100, 10^4\}$, together with smooth log cosh targets with $N_\theta \in \{2, 8, 32\}$ and $\kappa \in \{10, 100\}$, each cell using twenty seeds. The energy plotted is evaluated independently of the update sample.

These panels support the variance decomposition of [Section 4](#). They do not test the finite-horizon containment bound of [Theorem 4.9](#), whose horizon-dependent batch requirement is a worst-case statement and not a prediction for the median trajectory.



(a) Matched-step trajectories on a smooth target.



(b) Iterations to tolerance.

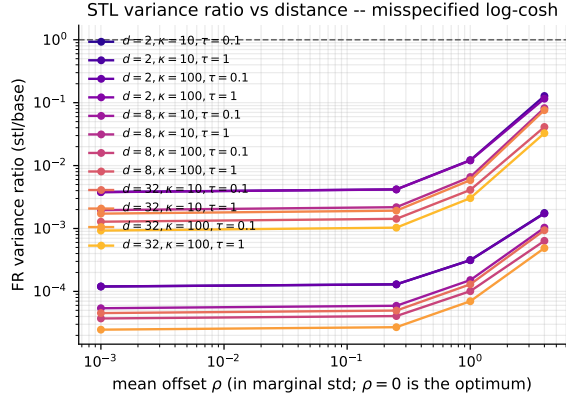
Figure 2: The Riemannian and KL discretizations. At matched stable stepsizes the two trajectories have comparable first-order behavior, and the Riemannian map often contracts faster per step in the tested range, while the KL covariance resolvent remains well defined at every stepsize and avoids a matrix exponential. The stepsize windows of [Theorems 2.9](#) and [2.11](#) are sufficient and conservative in these low-dimensional examples.

6.4 Affine-invariant modes

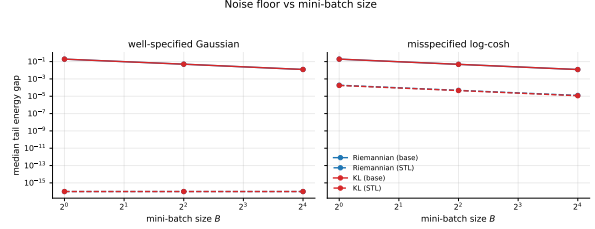
The affine-metric experiment varies $\tau \in \{-\omega/(2N_\theta), 0, \omega/(2N_\theta)\}$ for several values of ω , with five initializations isolating mean, volume-high, volume-low, traceless shape, and mixed perturbations. [Figure 4](#) reports the time-to-tolerance ratio $T(\tau)/T(0)$. At $\tau = -\omega/(2N_\theta)$ the trace rate $1/(2(\omega + N_\theta\tau))$ of [Theorem 5.4](#) doubles, and the volume-dominated runs take about half as long; at $\tau = \omega/(2N_\theta)$ their time scale grows by about $3/2$. The mean- and traceless-dominated runs remain near one, as the corresponding modal rates $1/\eta$ and $1/(2\omega)$ do not involve τ .

6.5 Beyond log-concavity

For the double well of [Section 2](#), all expectations are deterministic Gauss–Hermite quadratures and all trajectories are centered, so that the mean term of the split stationarity measure vanishes by symmetry and the plotted quantity is its covariance term.

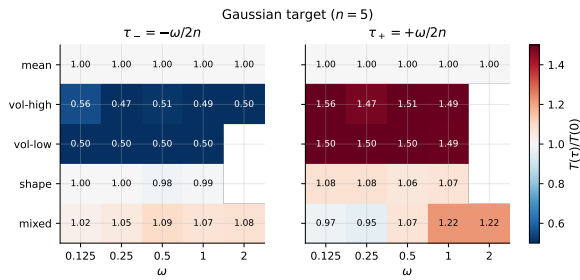


(a) Variance ratio versus mean displacement.

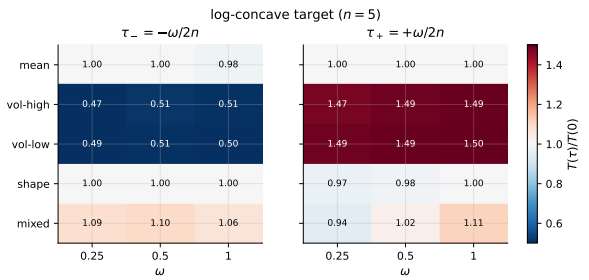


(b) Tail energy versus batch size.

Figure 3: Price/Hessian sticking-the-landing diagnostics. On a non-Gaussian log cosh target, the mean-block Fisher–Rao variance ratio is positive at the optimizer and increases with the mean displacement in every plotted configuration, which is the displacement-dependent part $V_1 \|a - a_\star\|^2$ of the local variance decomposition in [Theorem F.7](#). Mini-batching reduces the median tail energy gap approximately as $1/B$, which is the batch scaling of the Hessian-Lipschitz residual floors in [Theorems 4.4](#) and [4.5](#); the assumption-minimal floors of the same theorems do not scale in B at their prescribed stepsizes. The runs start from a generic initialization at a fixed stepsize rather than from the quadratic rescue those theorems assume, so the panels illustrate the scaling rather than instantiate the bounds.



(a) Gaussian target.



(b) Strongly log-concave target.

Figure 4: Mode preconditioning in the affine-invariant family. The Gaussian panel calibrates the exact modal computation of [Theorem 5.4](#); the non-Gaussian panel is an illustration of the same preconditioning effect and not a convergence theorem for the full metric family, in the sense of [Theorem 5.6](#).

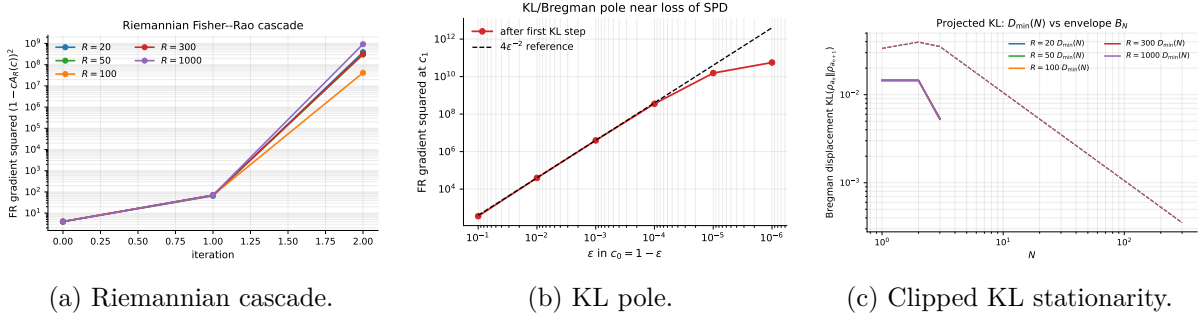


Figure 5: The non-log-concave boundary. Although the Hessian satisfies $|V_R''| \leq 1$, negative curvature drives the covariance cascade of [Theorem 2.18](#) and the rational pole of [Theorem 2.19](#). In panel (b) the witness condition $R_\varepsilon^2 \geq 2\varepsilon^{-2}$ of the proof of [Theorem 2.19](#) holds at $R = 10^3$ only for the two largest ε ; the next two points continue the ε^{-2} reference empirically, and the two smallest saturate at the fixed truncation radius. With the covariance constrained to $[\lambda_-, \lambda_+]$ and a stepsize in the window of [Theorem 2.21](#), the running minimum of the centered covariance KL stays below the energy-drop envelope. Once the upper constraint binds, the iterate remains pinned and the next displacement vanishes, which is constrained stationarity and not a small unconstrained Fisher–Rao gradient.

7 Discussion

The convergence theory of Gaussian natural gradient flows separates into three stages, and the separation persists from the continuous dynamics to the deterministic discretizations and to the stochastic algorithms. An underdispersed initial covariance costs a burn-in time logarithmic in $\beta\lambda_{0,\min}$; the global phase localizes the iterate at the intrinsic condition number obtained after whitening by the optimizing covariance; and the local phase runs at a rate set by the spectral gap of a self-adjoint score operator rather than by the curvature ratio of the target. The three stages have different sharpness certificates. The logarithmic spiral shows that the $\kappa_\star$ dependence of the global phase cannot be improved under the general spectral initialization, the bump train shows that the factor $\kappa/\Delta t$ of the fixed-step schemes is sharp for a constant-factor reduction, and the convex ridge shows that the $\beta_\star - 1$ branch of the local spectral scale is attained, so that no dimension-free logarithmic local rate can hold.

The stochastic theory preserves this structure with the sticking-the-landing and Price/Hessian estimators, at the cost of explicit variance floors: the Hessian-Lipschitz floors scale as $1/B$ in the batch size, while the assumption-minimal global floor plateaus at $O(\kappa N_\theta)$ for $B \lesssim \kappa$ and the local floor at $O(N_\theta^2 L_{H,\star}^2/V_1)$ for $B \lesssim V_1$; all of these floors vanish for Gaussian targets. The stochastic local result is a killed-process contraction together with a finite-horizon exit-probability bound; because the mean-score noise has unbounded support, we do not claim pathwise invariance of a bounded local region at a fixed stepsize, and we do not claim infinite-horizon containment. The discrete local radius for a general target is constructive through a supremum of second derivatives of the update map, and is not a closed-form quantity that a practitioner can evaluate.

Several questions remain open. The local spectral bound degrades with dimension through the factor $\sqrt{N_\theta} \log \kappa_\star$, and the convex-ridge family lives in dimension $N_\theta = \Theta(\kappa_\star^2)$. We do not know whether a stronger bound holds in fixed dimension, nor whether the $\sqrt{N_\theta} \log \kappa_\star$ branch is itself sharp. The fixed-step lower bound is stated for one globally fixed stepsize, and it leaves open whether an adaptive, line-search, or implicit scheme can evade the factor $\kappa/\Delta t$. A closed-form discrete local radius, and sharper stepsize windows for the stochastic local phase, would make the local theory usable as a stopping rule rather than as an existence statement. We prove a stationarity guarantee for the covariance-constrained KL scheme in the deterministic setting only; its stochastic counterpart requires a separate treatment of the interaction between the clipping step and the noise, which we do not treat. Finally, the classification of [Section 5](#) exhibits a three-parameter family of affine-invariant metrics whose modal rates we compute exactly on Gaussian targets, but the sharp global and local theory of [Sections 2](#) and [3](#) is developed for Fisher–Rao alone; extending it to unbalanced members of the family, and developing the Wasserstein–Fisher–Rao dynamics, which the transport warm start of [Section 2](#) does not develop, are natural directions for subsequent work.

A Gaussian differential identities and the score calculus

We collect the differential identities on which the Fisher–Rao gradient, the linearized generator of [Section 3.1](#), and the non-Gaussian modulus of [Theorem 3.7](#) rest. Throughout, $Z \sim \mathcal{N}(0, I)$, $a = (m, C) \in \mathcal{A}$, and $X = m + C^{1/2}Z$ has law ρ_a .

A.1 Price and Bonnet directional derivatives

Lemma A.1 (Directional derivatives of a Gaussian expectation). *Let $V \in C^2(\mathbb{R}^{N_\theta})$ satisfy $\|\nabla^2 V\|_{\text{op}} \leq \beta$, let $v \in \mathbb{R}^{N_\theta}$ and $Y \in \mathbb{S}(N_\theta)$, and write $a_\epsilon := (m + \epsilon v, C + \epsilon Y)$. Then*

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathbb{E}_{\rho_{(m+\epsilon v, C)}}[V] = \mathbb{E}_{\rho_a}[\nabla V]^\top v, \quad \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathbb{E}_{\rho_{(m, C+\epsilon Y)}}[V] = \frac{1}{2} \text{Tr}(Y A(a)), \quad (73)$$

where $A(a) = \mathbb{E}_{\rho_a}[\nabla^2 V]$, and

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathcal{E}_1(a_\epsilon) = -\frac{1}{2} \text{Tr}(Y C^{-1}), \quad \mathcal{E}_1(a) = -\frac{1}{2} \log \det C + \text{const.} \quad (74)$$

Proof. The mean identity is Bonnet's formula and follows from differentiating under the integral, which is licensed because $\|\nabla^2 V\|_{\text{op}} \leq \beta$ forces $\|\nabla V(\theta)\| \leq \|\nabla V(0)\| + \beta \|\theta\|$, so every integrand is bounded by a polynomial times a Gaussian density. For the covariance identity, let p_ϵ be the density of $\mathcal{N}(m, C + \epsilon Y)$, which is Schwartz for small $|\epsilon|$, with Fourier transform $\hat{p}_\epsilon(w) = \exp(i m^\top w - \frac{1}{2} w^\top (C + \epsilon Y) w)$, so that $\partial_\epsilon \hat{p}_\epsilon(w)|_{\epsilon=0} = -\frac{1}{2} (w^\top Y w) \hat{p}_0(w)$. The Fourier transform of $\partial_i \partial_j p_0$ is $-w_i w_j \hat{p}_0$, so the Fourier transform of $\frac{1}{2} \text{Tr}(Y \nabla_\theta^2 p_0)$ is exactly $-\frac{1}{2} (w^\top Y w) \hat{p}_0(w)$, and Fourier inversion gives the heat-type identity

$$\partial_\epsilon p_\epsilon|_{\epsilon=0} = \frac{1}{2} \text{Tr}(Y \nabla_\theta^2 p_0). \quad (75)$$

Since p_ϵ and its first two derivatives decay faster than any polynomial uniformly near $\epsilon = 0$ while V grows at most quadratically, we may differentiate under the integral and integrate by parts twice with vanishing boundary terms, obtaining

$$\left. \frac{d}{d\epsilon} \right|_0 \int V p_\epsilon d\theta = \int V \cdot \frac{1}{2} \text{Tr}(Y \nabla_\theta^2 p_0) d\theta = \frac{1}{2} \int \text{Tr}(Y \nabla^2 V) p_0 d\theta = \frac{1}{2} \text{Tr}(Y A(a)).$$

Finally (74) is Jacobi's formula $\left. \frac{d}{d\epsilon} \log \det(C + \epsilon Y) \right|_0 = \text{Tr}(C^{-1} Y)$. $\square$

A.2 The Fisher–Rao gradient

Lemma A.2 (Fisher–Rao gradient, flow, and gradient norm). *With $R(a) := C^{-1} - A(a)$ and the Fisher–Rao metric $g_a^{\text{FR}}((u, U), (v, Y)) = u^\top C^{-1} v + \frac{1}{2} \text{Tr}(C^{-1} U C^{-1} Y)$,*

$$\text{grad } \mathcal{E}(a) = (-Cg(a), -CR(a)C), \quad (76)$$

so that $\dot{a} = -\text{grad } \mathcal{E}(a)$ is exactly (1). Moreover, with $G(a)^2 := g(a)^\top Cg(a) + \frac{1}{2} \|C^{1/2} R(a) C^{1/2}\|_{\text{F}}^2$, one has $\|\text{grad } \mathcal{E}(a)\|_a^2 = G(a)^2$ and, along (1), $\frac{d}{dt} \Delta(a_t) = -G(a_t)^2$.

Proof. By Theorem A.1, the differential of $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$ at a in the direction (v, Y) is $d\mathcal{E}(a)[(v, Y)] = -g(a)^\top v - \frac{1}{2} \text{Tr}(R(a)Y)$, since $\mathbb{E}_{\rho_a}[\nabla V] = -g(a)$. Writing $(u^*, U^*) := \text{grad } \mathcal{E}(a)$ and matching blocks in $g_a^{\text{FR}}((u^*, U^*), (v, Y)) = d\mathcal{E}(a)[(v, Y)]$ gives $u^{*\top} C^{-1} v = -g(a)^\top v$ for all v , hence $u^* = -Cg(a)$, and $\frac{1}{2} \text{Tr}(C^{-1} U^* C^{-1} Y) = -\frac{1}{2} \text{Tr}(R(a)Y)$ for all $Y \in \mathbb{S}(N_\theta)$; since both $C^{-1} U^* C^{-1}$ and $R(a)$ are symmetric and $(S, Y) \mapsto \text{Tr}(SY)$ is nondegenerate on $\mathbb{S}(N_\theta)$, this forces $U^* = -CR(a)C$. Then $\dot{m} = Cg(a)$ and $\dot{C} = CR(a)C = C - CA(a)C$. Finally

$$\|\text{grad } \mathcal{E}(a)\|_a^2 = (Cg)^\top C^{-1} (Cg) + \frac{1}{2} \|C^{-1/2} (CRC) C^{-1/2}\|_{\text{F}}^2 = g^\top Cg + \frac{1}{2} \|C^{1/2} R C^{1/2}\|_{\text{F}}^2,$$

and the dissipation identity is the chain rule along the gradient flow. $\square$

A.3 The linearized score operators

We now work in the whitened coordinates of Theorem 3.1, so $a_\star = (0, I)$, $B(Z) := \nabla^2 \hat{V}(Z)$ and $\mathbb{E}B(Z) = I$.

Lemma A.3 (Adjoint identities and the linearization at $a_\star$). *For the operators of Theorem 3.2, $u^\top \mathcal{T}[X] = \text{Tr}(\mathcal{T}^*[u]X)$ and $\text{Tr}(\mathcal{H}[X]Y) = \text{Tr}(X\mathcal{H}[Y])$. Moreover*

$$\mathbb{E}_{\mathcal{N}(u, I+X)}[\nabla \hat{V}] = u + \frac{1}{2} \mathcal{T}[X] + O(\|(u, X)\|_\star^2), \quad \mathbb{E}_{\mathcal{N}(u, I+X)}[\nabla^2 \hat{V}] = I + \mathcal{T}^*[u] + \frac{1}{2} \mathcal{H}[X] + O(\|(u, X)\|_\star^2).$$

Proof. The first adjoint identity is immediate from the definitions, $u^\top \mathcal{T}[X] = \mathbb{E}[(u^\top Z) \text{Tr}(XB(Z))] = \text{Tr}(\mathcal{T}^*[u]X)$. For the second, write φ_{N_θ} for the standard Gaussian density and note that $\partial_{cd}\varphi_{N_\theta}(z) = (z_c z_d - \delta_{cd})\varphi_{N_\theta}(z)$, while $B = \nabla^2 \hat{V}$ gives $\partial_{cd}B_{ab} = \partial_{bc}B_{ad}$ as distributions. Pairing,

$$\mathbb{E}[(Z_c Z_d - \delta_{cd})B_{ab}(Z)] = \langle \partial_{cd}B_{ab}, \varphi_{N_\theta} \rangle = \langle \partial_{bc}B_{ad}, \varphi_{N_\theta} \rangle = \mathbb{E}[(Z_b Z_c - \delta_{bc})B_{ad}(Z)],$$

the pairings being justified by boundedness of B and by inserting compactly supported cutoffs of φ_{N_θ} . Contracting against $Y_{ab}X_{cd}$ and using $\mathbb{E}B = I$ gives $\text{Tr}(\mathcal{H}[X]Y) = \mathbb{E}[(YZ)^\top (B(Z) - I)(XZ)]$, which is symmetric in (X, Y) . The two expansions are [Theorem A.1](#) applied at $a_\star$: the first derivative of $a \mapsto \mathbb{E}_{\rho_a}[\nabla \hat{V}]$ in the mean direction u is $\mathbb{E}[B]u = u$ and in the covariance direction X is $\frac{1}{2}\mathbb{E}[\text{Tr}(X\nabla^3 \hat{V})] = \frac{1}{2}\mathcal{T}[X]$ by (73) and Gaussian integration by parts; the first derivative of $a \mapsto \mathbb{E}_{\rho_a}[\nabla^2 \hat{V}]$ in the mean direction is $\mathbb{E}[(u^\top Z)B(Z)] = \mathcal{T}^*[u]$ by Stein's identity and in the covariance direction is $\frac{1}{2}\mathcal{H}[X]$ by (73) applied entrywise. $\square$

Lemma A.4 (Diagonal-mode bounds). *Fix an orthogonal basis in which $X = \text{diag}(\lambda)$, and set $\tilde{T}_{ki} := \mathbb{E}[B_{ii}(Z)Z_k]$, $G_{ij} := \mathbb{E}[B_{ii}(Z)Z_j^2]$, and $D := G - \mathbf{1}\mathbf{1}^\top$. Then D is symmetric,*

$$\tilde{T}^\top \tilde{T} \preceq I + D, \quad \text{and} \quad \lambda_{\max}(D) \leq \Gamma. \quad (77)$$

Proof of Theorem A.4. Since $\mathbb{E}B = I$, two Gaussian integrations by parts give, for $i \neq j$,

$$D_{ij} = \mathbb{E}[B_{ii}(Z)(Z_j^2 - 1)] = \mathbb{E}[\hat{V}(Z)(Z_i^2 - 1)(Z_j^2 - 1)] = D_{ji},$$

and integrating one i - and one j -derivative gives $\mathbb{E}[B_{ij}(Z)Z_i Z_j] = D_{ij}$; for $i = j$, $\mathbb{E}[B_{ii}(Z)Z_i^2] = 1 + D_{ii}$ is immediate. Hence D is symmetric and $\mathbb{E}[B_{ij}(Z)Z_i Z_j] = \delta_{ij} + G_{ij} - 1$ for all i, j . For $\eta(Z) := u + \text{diag}(\lambda)Z$, expanding with this identity gives the exact quadratic form

$$\mathbb{E}[\eta(Z)^\top B(Z)\eta(Z)] = \|u\|^2 + 2u^\top \tilde{T}\lambda + \|\lambda\|^2 + \lambda^\top D\lambda. \quad (78)$$

Because $B(Z) \succeq 0$, the block quadratic form on the right is nonnegative for all (u, λ) , and its Schur complement with respect to the identity mean block is (77). Taking $u = 0$ and $B(Z) \preceq \beta_\star I$ in (78) gives $\|\lambda\|^2 + \lambda^\top D\lambda \leq \beta_\star \|\lambda\|^2$, so $\lambda_{\max}(D) \leq \beta_\star - 1$.

For the dimension-dependent branch, normalize $\|\lambda\| = 1$, set $X := \text{diag}(\lambda)$ in the fixed basis, and put $Q_X(Z) := Z^\top X Z - \text{Tr} X = \sum_j \lambda_j (Z_j^2 - 1)$. Since $D_{ij} = \mathbb{E}[B_{ii}(Z)(Z_j^2 - 1)]$, the quadratic form has the basis-free representation

$$\lambda^\top D\lambda = \mathbb{E}[\text{Tr}(XB(Z)) Q_X(Z)].$$

Because $B(Z) \succeq 0$, splitting $X = X_+ - X_-$ gives $|\text{Tr}(XB)| \leq \text{Tr}(|X|B)$ pointwise, with $\mathbb{E}\text{Tr}(|X|B) = \text{Tr}|X| = \|\lambda\|_1 \leq \sqrt{N_\theta}$ and, almost surely, $\text{Tr}(|X|B) \leq \beta_\star \text{Tr}|X| \leq \beta_\star \sqrt{N_\theta}$. The Laurent–Massart bound [12] for $\sum_j \mathbf{a}_j (Z_j^2 - 1)$ with $\mathbf{a} \succeq 0$, applied to the positive and negative parts of λ so that $\|\lambda_+\| + \|\lambda_-\| \leq \sqrt{2}$ and $\|\lambda\|_\infty \leq 1$, gives the two-sided tail $\mathbb{P}(|Q_X| \geq 2\sqrt{2t} + 2t) \leq 4e^{-t}$ for all $t \geq 0$. With t_0 as in (35) and $T := 2\sqrt{2t_0} + 2t_0$, split the expectation. On $\{|Q_X| \leq T\}$, $\mathbb{E}[\text{Tr}(|X|B) |Q_X|] \leq T \mathbb{E}\text{Tr}(|X|B) \leq T\sqrt{N_\theta}$. On the complement, the pointwise bound and the tail integral

$$\mathbb{E}[|Q_X|; |Q_X| > T] \leq T \cdot 4e^{-t_0} + \int_{t_0}^\infty 4e^{-t} \left(\frac{\sqrt{2}}{\sqrt{t}} + 2 \right) dt \leq 4e^{-t_0} (T + \sqrt{2} + 2) \leq 4e^{-t_0} (T + 4),$$

where $t_0 \geq 1$ is used, give, since $4\beta_\star e^{-t_0} \leq 1$, a contribution at most $\beta_\star \sqrt{N_\theta} \cdot 4e^{-t_0} (T + 4) \leq \sqrt{N_\theta} (T + 4)$. Hence $|\lambda^\top D\lambda| \leq \sqrt{N_\theta} (2T + 4) = \sqrt{N_\theta} (4\sqrt{2t_0} + 4t_0 + 4)$ uniformly over $\|\lambda\| = 1$ and over the diagonalizing basis, since the representation is basis-free in X . Combining the two branches gives $\lambda_{\max}(D) \leq \Gamma$. $\square$

A.4 Second-order score calculus and the sharp chaos constant

Write $\mathcal{L}_{\text{OU}}$ for the Ornstein–Uhlenbeck generator, which acts as multiplication by $-k$ on the k -th Hermite chaos. Gaussian integration by parts ($-\mathcal{L}_{\text{OU}} = \nabla^* \nabla$) gives, for $\phi = \sum_{k \geq 1} \phi_k$,

$$\mathbb{E} \left\| \nabla(-\mathcal{L}_{\text{OU}})^{-1} \phi \right\|^2 = \mathbb{E}[\phi(-\mathcal{L}_{\text{OU}})^{-1} \phi] = \sum_{k \geq 1} \frac{1}{k} \|\phi_k\|_{L^2}^2, \quad (79)$$

so that $\mathbb{E} \left\| \nabla(-\mathcal{L}_{\text{OU}})^{-1} \phi \right\|^2 \leq \frac{1}{2} \|\phi\|_{L^2}^2$ whenever ϕ carries only chaoses ≥ 2 . The bound fails for chaos one, and only chaoses ≥ 2 occur below.

Lemma A.5 (Score identities and the sharp second-score constant). *Fix $a = (m, C)$ and $\zeta = (u, U)$, and set $v := C^{-1/2}u$, $S := C^{-1/2}UC^{-1/2}$, $t := \|\zeta\|_a = (\|v\|^2 + \frac{1}{2} \|S\|_{\text{F}}^2)^{1/2}$, and $X = m + C^{1/2}Z$. Put*

$$q_\zeta := v^\top Z + \frac{1}{2}(Z^\top SZ - \text{Tr } S), \quad \chi_\zeta := q_\zeta^2 + \left(\frac{1}{2} \text{Tr } S^2 - \|v\|^2 - 2v^\top SZ - Z^\top S^2 Z\right).$$

Then for any ϕ of polynomial growth and $\Phi(a) := \mathbb{E}_{\rho_a}[\phi(X)]$,

$$D\Phi(a)[\zeta] = \mathbb{E}[\phi q_\zeta], \quad D^2\Phi(a)[\zeta, \zeta] = \mathbb{E}[\phi \chi_\zeta].$$

Moreover q_ζ has Hermite chaoses 1, 2 and χ_ζ has chaoses 2, 3, 4 only, so $\mathbb{E}[q_\zeta] = \mathbb{E}[\chi_\zeta] = 0$ and $\mathbb{E}[Z\chi_\zeta] = 0$; and

$$\mathbb{E}[q_\zeta^2] = t^2, \quad \|\chi_\zeta\|_{L^2} \leq \sqrt{6} t^2,$$

the constant $\sqrt{6}$ being sharp and attained by rank-one covariance directions.

Proof. Differentiation identities. With $m_s = m + su$ and $C_s = C + sU$, positive definite for small $|s|$, differentiate $\log \rho_{a_s}(\theta) = -\frac{1}{2} \log \det C_s - \frac{1}{2}(\theta - m_s)^\top C_s^{-1}(\theta - m_s) + \text{const}$ at fixed $\theta = m + C^{1/2}Z$. The entropy term contributes $-\frac{1}{2} \text{Tr } S$ at first order and $+\frac{1}{2} \text{Tr } S^2$ at second; the quadratic term contributes $v^\top Z + \frac{1}{2} Z^\top SZ$ at first order and $-\|v\|^2 - 2v^\top SZ - Z^\top S^2 Z$ at second. Hence $\partial_s \log \rho_{a_s}|_0 = q_\zeta$ and $\partial_s^2 \log \rho_{a_s}|_0 = \frac{1}{2} \text{Tr } S^2 - \|v\|^2 - 2v^\top SZ - Z^\top S^2 Z =: p_\zeta$. From $\partial_s \rho_{a_s} = \rho_{a_s} \partial_s \log \rho_{a_s}$ and $\partial_s^2 \rho_{a_s} = \rho_{a_s} ((\partial_s \log \rho_{a_s})^2 + \partial_s^2 \log \rho_{a_s})$, differentiation under the integral, licensed by polynomial growth of ϕ and locally bounded polynomial score factors, gives the two identities with $\chi_\zeta = q_\zeta^2 + p_\zeta$.

Chaos structure. Write $\ell := v^\top Z$, of chaos one, and $Q := Z^\top SZ - \text{Tr } S$, of chaos two, so that $q_\zeta = \ell + \frac{1}{2}Q$ and, since $\text{Var}(Z^\top SZ) = 2\|S\|_{\text{F}}^2$, $\mathbb{E}[q_\zeta^2] = \|v\|^2 + \frac{1}{4} \cdot 2\|S\|_{\text{F}}^2 = t^2$. The chaos-two product formula gives $Q^2 = (Q^2)_0 + (Q^2)_2 + (Q^2)_4$ with $(Q^2)_0 = 2\|S\|_{\text{F}}^2$ and $(Q^2)_2 = 4(Z^\top S^2 Z - \text{Tr } S^2) =: 4R_2$, and $\ell Q = 2v^\top SZ + G_3$ with G_3 pure chaos three, its chaos-one part being $\sum_k \mathbb{E}[\ell Q Z_k] Z_k = 2v^\top SZ$, from $\mathbb{E}[Z_i(Z_j Z_l - \delta_{jl})Z_k] = \delta_{ij}\delta_{lk} + \delta_{il}\delta_{jk}$. Writing $Z^\top S^2 Z = R_2 + \text{Tr } S^2$ in p_ζ gives $p_\zeta = -2v^\top SZ - R_2 - \frac{1}{2} \text{Tr } S^2 - \|v\|^2$. Collecting $\chi_\zeta = \ell^2 + \ell Q + \frac{1}{4}Q^2 + p_\zeta$ by chaos order, the chaos-zero part is $\|v\|^2 + \frac{1}{2}\|S\|_{\text{F}}^2 - \frac{1}{2} \text{Tr } S^2 - \|v\|^2 = 0$, the chaos-one part is $2v^\top SZ - 2v^\top SZ = 0$, and, using $\frac{1}{4}(Q^2)_2 = R_2$, there remain the mutually orthogonal parts $\chi_2 = \ell^2 - \|v\|^2$, $\chi_3 = G_3 = \ell Q - 2v^\top SZ$, and $\chi_4 = \frac{1}{4}(Q^2)_4$. In particular $\mathbb{E}[\chi_\zeta] = 0$ and $\mathbb{E}[Z\chi_\zeta] = 0$.

Sharp constant. By orthogonality $\|\chi_\zeta\|_{L^2}^2 = \mathbb{E}[\chi_2^2] + \mathbb{E}[\chi_3^2] + \mathbb{E}[\chi_4^2]$. With $\mathbf{a} := \|v\|^2$, $\mathbf{b} := \|S\|_{\text{F}}^2$, $\mathbf{c} := v^\top S^2 v$ and $d_4 := \text{Tr } S^4$, one has $\mathbb{E}[\chi_2^2] = \text{Var}((v^\top Z)^2) = 2\mathbf{a}^2$, $\mathbb{E}[\chi_3^2] = \mathbb{E}[(\ell Q)^2] - 4\|v^\top SZ\|_{L^2}^2 = 2\mathbf{a}\mathbf{b} + 4\mathbf{c}$, and $\mathbb{E}[\chi_4^2] = \frac{1}{16} \|(Q^2)_4\|_{L^2}^2 = \frac{1}{2}\mathbf{b}^2 + d_4$, whence $\|\chi_\zeta\|_{L^2}^2 = 2\mathbf{a}^2 + 2\mathbf{a}\mathbf{b} + 4\mathbf{c} + \frac{1}{2}\mathbf{b}^2 + d_4$. Since $\mathbf{c} = v^\top S^2 v \leq \|v\|^2 \|S\|_{\text{op}}^2 \leq \mathbf{a}\mathbf{b}$ and $d_4 = \sum_i \lambda_i(S)^4 \leq (\sum_i \lambda_i(S)^2)^2 = \mathbf{b}^2$,

$$\|\chi_\zeta\|_{L^2}^2 \leq 2\mathbf{a}^2 + 6\mathbf{a}\mathbf{b} + \frac{3}{2}\mathbf{b}^2 \leq 6(\mathbf{a} + \frac{\mathbf{b}}{2})^2 = 6t^4.$$

Equality holds at $v = 0$, so $\mathbf{a} = \mathbf{c} = 0$, with S of rank one, so $d_4 = \mathbf{b}^2$: then $\|\chi_\zeta\|_{L^2}^2 = \frac{1}{2}\mathbf{b}^2 + \mathbf{b}^2 = \frac{3}{2}\mathbf{b}^2 = 6(\mathbf{b}/2)^2 = 6t^4$. $\square$

A.5 The non-Gaussian modulus

Proof of Theorem 3.7. Write $x := \|v\|$ and $y := \|S\|_{\text{F}}/\sqrt{2}$, so that $t^2 = x^2 + y^2$; on $\mathcal{V}_\delta$ we have $\|U\|_{\text{op}} \leq \sqrt{2}U_\delta y$, $\|U\|_{\text{F}} \leq \sqrt{2}U_\delta y$ and $\|C\|_{\text{op}} \leq U_\delta$, and at the end we convert through $t \leq \ell_\delta^{-1} \|\zeta\|_\star$.

Mean block. Since $H_m(a) = C(g(a) + m)$ and $D^2m = 0$, $D^2H_m[\zeta, \zeta] = 2U D(g + m)[\zeta] + C D^2g[\zeta, \zeta]$. Now $g(a) + m = -\mathbb{E}[\nabla \hat{V}(X) - X]$, and the Ornstein–Uhlenbeck inverse of q_ζ , whose gradient is $\nabla \psi_1 = v + \frac{1}{2}SZ$, gives $D(g + m)[\zeta] = -\mathbb{E}[(\nabla^2 \hat{V}(X) - I)C^{1/2}(v + \frac{1}{2}SZ)]$. Splitting $\nabla^2 \hat{V} - I = (\nabla^2 \hat{V} - A(a)) + (A(a) - I)$ and using $\mathbb{E}[\nabla^2 \hat{V} - A(a)] = 0$ and $\mathbb{E}[SZ] = 0$,

$$\|D(g + m)[\zeta]\| \leq \sqrt{U_\delta} \left(\mathcal{B}_\delta \|v\| + \frac{1}{\sqrt{2}} \Sigma_\delta y \right),$$

where $\|A(a) - I\|_{\text{op}} \leq \mathcal{B}_\delta$ controls the $(A - I)v$ term and $(\mathbb{E} \|\nabla^2 \hat{V} - A\|_{\text{F}}^2)^{1/2} \leq \Sigma_\delta$ the $\frac{1}{2}(\nabla^2 \hat{V} - A)SZ$ term. For the second derivative, χ_ζ has chaoses ≥ 2 , so with $\psi_2 := (-\mathcal{L}_{\text{OU}})^{-1}\chi_\zeta$, for which $\mathbb{E} \|\nabla \psi_2\|^2 \leq \frac{1}{2} \|\chi_\zeta\|_{L^2}^2 \leq 3t^4$ by (79) and Theorem A.5, and $\mathbb{E} \nabla \psi_2 = 0$, we get $D^2g[\zeta, \zeta] = -\mathbb{E}[(\nabla^2 \hat{V}(X) - A(a))C^{1/2}\nabla \psi_2]$ and hence $\|D^2g[\zeta, \zeta]\| \leq \sqrt{3}U_\delta \bar{\Sigma}_\delta t^2$. The truncated modulus $\bar{\Sigma}_\delta = \min\{\Sigma_\delta, \beta_\star - \alpha_\star\}$ arises because $\mathbb{E} \|(\nabla^2 \hat{V} - A)C^{1/2}\nabla \psi_2\|$ may be bounded either by the pointwise operator norm $\|\nabla^2 \hat{V} - A\|_{\text{op}} \leq \beta_\star - \alpha_\star$, which is dimension-free, or by the Frobenius dispersion $(\mathbb{E} \|\nabla^2 \hat{V} - A\|_{\text{F}}^2)^{1/2} \leq \Sigma_\delta$ through Cauchy–Schwarz, whichever is smaller; no bound of the form $\|\nabla^2 \hat{V} - A\|_{\text{F}} \leq \beta_\star - \alpha_\star$ is used. Therefore

$$\|D^2H_m[\zeta, \zeta]\| \leq U_\delta^{3/2} (2\sqrt{2} \mathcal{B}_\delta xy + 2\Sigma_\delta y^2 + \sqrt{3} \bar{\Sigma}_\delta t^2),$$

and the quadratic form in (x, y) formed by the first two terms has largest eigenvalue $\Sigma_\delta + \sqrt{\Sigma_\delta^2 + 2\mathcal{B}_\delta^2}$; adding $\sqrt{3}\bar{\Sigma}_\delta t^2$ and converting t gives $K_{\text{ng},m}(\delta)$.

Covariance block. Here $H_C(a) = -C(A(a) - I)C = -CWC$ with $W := A(a) - I$, and the product rule gives

$$D^2(CWC)[\zeta, \zeta] = 2U(DW)C + 2C(DW)U + 2UWU + C(D^2W)C.$$

By the first score identity $\|DW[\zeta]\|_{\text{F}} = \|DA[\zeta]\|_{\text{F}} \leq \Sigma_\delta t$, by definition $\|W\|_{\text{op}} \leq \mathcal{B}_\delta$, and by the second score identity of Theorem A.5, $\|D^2W[\zeta, \zeta]\|_{\text{F}} = \|D^2A[\zeta, \zeta]\|_{\text{F}} \leq \sqrt{6} \Sigma_\delta t^2$. Hence

$$\|D^2H_C[\zeta, \zeta]\|_{\text{F}} \leq 4\sqrt{2}U_\delta^2 \Sigma_\delta ty + 4\mathcal{B}_\delta U_\delta^2 y^2 + \sqrt{6}U_\delta^2 \Sigma_\delta t^2 \leq U_\delta^2 ((4\sqrt{2} + \sqrt{6})\Sigma_\delta + 4\mathcal{B}_\delta) t^2,$$

using $ty \leq t^2$ and $y^2 \leq t^2$; converting t gives $K_{\text{ng},C}(\delta)$. The pair-norm bound follows from $\|D^2H[\zeta, \zeta]\|_\star^2 = \|(D^2H)_m\|^2 + \frac{1}{2} \|(D^2H)_C\|_{\text{F}}^2$. Every constant is proportional to $\mathcal{B}_\delta$, Σ_δ or $\bar{\Sigma}_\delta$, all of which vanish for Gaussian targets. $\square$

B The logarithmic-spiral construction

We construct the planar family announced in Theorem 2.5 and prove the lower bound. Throughout $\theta = (\theta_1, \theta_2) \in \mathbb{R}^2$ denotes the state, (r, ϑ) its polar coordinates, and R_ϑ the planar rotation through the angle ϑ .

The following packaging of Theorem 2.5 records the $\Theta(\kappa_\star)$ reading.

Corollary B.1 (Sharp polynomial dependence in dimension at least two). *For $N_\theta \geq 2$, the worst-case polynomial dependence of the continuous-time global-localization phase on the affine-invariant condition number is*

$$T_{\text{loc}} = \tilde{\Theta}(\kappa_\star)$$

under the general spectral initialization class, where the tilde suppresses logarithmic dependence on initialization size and localization accuracy. For the spiral family, $\kappa_\star = \Theta(\kappa)$ and $T_{\text{loc}} \geq (\kappa - 1)/200$.

B.1 The spiral potential

Fix $K \geq 16$ and set

$$h := 2\alpha, \quad c := \frac{\sqrt{K-2}}{3}, \quad \delta_K := \frac{10^{-4}}{K}, \quad D_K := \frac{18h(K-1)}{K+7}, \quad \beta := \alpha(2K+1), \quad (80)$$

so that the original-coordinate condition number is $\kappa = 2K + 1$. Define

$$s(u) := 35u^4 - 84u^5 + 70u^6 - 20u^7, \quad 0 \leq u \leq 1,$$

and the even C^3 cutoff

$$\rho(u) := \begin{cases} 1, & |u| \leq \frac{1}{2}, \\ s(2 - 2|u|), & \frac{1}{2} < |u| < 1, \\ 0, & |u| \geq 1. \end{cases} \quad (81)$$

Extend $\rho_{\delta_K}(\phi) := \rho(\phi/\delta_K)$ periodically with period π , and set

$$\bar{\rho}_{\delta_K} := \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \rho_{\delta_K}(\phi) d\phi, \quad g_K(\phi) := \frac{\rho_{\delta_K}(\phi) - \bar{\rho}_{\delta_K}}{1 - \bar{\rho}_{\delta_K}}.$$

Let f_K be the even π -periodic solution of

$$f_K'' = g_K, \quad f_K(0) = f_K'(0) = 0. \quad (82)$$

Then

$$f_K(\phi) = \frac{1}{2}\phi^2 \quad \text{for } |\phi| \leq \frac{\delta_K}{2}, \quad \|f_K'\|_\infty \leq 3\delta_K, \quad \|f_K\|_\infty \leq 2\pi\delta_K. \quad (83)$$

Define the radial cutoff

$$\chi(\ell) := \begin{cases} 0, & \ell \leq 0, \\ 35\ell^4 - 84\ell^5 + 70\ell^6 - 20\ell^7, & 0 < \ell < 1, \\ 1, & \ell \geq 1, \end{cases} \quad (84)$$

and, for a scale $r_0 > 0$, put

$$\ell := \log \frac{r}{r_0}, \quad \phi := \vartheta + c\ell. \quad (85)$$

The spiral potential is

$$V_K(r, \vartheta) := \frac{1}{2}r^2[h + D_K\chi(\ell)f_K(\phi)]. \quad (86)$$

For $r \leq r_0$ one has $V_K(\theta) = \frac{h}{2}\|\theta\|^2$. The matching of the first three derivatives of ρ and χ at their endpoints gives $V_K \in C^3(\mathbb{R}^2)$, and the π -periodicity of f_K makes V_K even.

Lemma B.2 (Global Hessian certificate). *The potential (86) satisfies*

$$0.98hI \preceq \nabla^2 V_K(\theta) \preceq h(K + 0.02)I \quad \text{for every } \theta \in \mathbb{R}^2. \quad (87)$$

Consequently $\alpha I \preceq \nabla^2 V_K \preceq \beta I$ with $\beta = \alpha(2K + 1)$. On the flat part of a spiral valley the two Hessian eigenvalues are exactly h and hK .

Proof. Put $q(\ell, \vartheta) := \chi(\ell)f_K(\vartheta + c\ell)$. For a general profile $A = A(\ell, \vartheta)$, differentiation in the orthonormal polar frame gives

$$\nabla^2 \left(\frac{1}{2} r^2 A \right) = \begin{pmatrix} A + \frac{3}{2} A_\ell + \frac{1}{2} A_{\ell\ell} & \frac{1}{2} A_\vartheta + \frac{1}{2} A_{\ell\vartheta} \\ \frac{1}{2} A_\vartheta + \frac{1}{2} A_{\ell\vartheta} & A + \frac{1}{2} A_\ell + \frac{1}{2} A_{\vartheta\vartheta} \end{pmatrix}.$$

Consequently,

$$\nabla^2 V_K - hI = \frac{D_K}{2} \chi g_K \begin{pmatrix} c^2 & c \\ c & 1 \end{pmatrix} + D_K \{ \chi B_K + E_K \}, \quad (88)$$

where all functions are evaluated at (ℓ, ϕ) and

$$B_K = \begin{pmatrix} f_K + \frac{3}{2} c f'_K & \frac{1}{2} f'_K \\ \frac{1}{2} f'_K & f_K + \frac{1}{2} c f'_K \end{pmatrix}, \quad E_K = \begin{pmatrix} (\frac{3}{2} \chi' + \frac{1}{2} \chi'') f_K + c \chi' f'_K & \frac{1}{2} \chi' f'_K \\ \frac{1}{2} \chi' f'_K & \frac{1}{2} \chi' f'_K \end{pmatrix}.$$

This identity covers both the uncut region and the transition annulus.

The normalization of the phase cutoff is explicit. Since $\int_{-1}^1 \rho(u) du = 3/2$,

$$\bar{\rho}_{\delta_K} = \frac{3\delta_K}{2\pi}, \quad -\frac{\bar{\rho}_{\delta_K}}{1 - \bar{\rho}_{\delta_K}} \leq g_K \leq 1,$$

and for $K \geq 16$,

$$(K-1) \frac{\bar{\rho}_{\delta_K}}{1 - \bar{\rho}_{\delta_K}} < 4.8 \cdot 10^{-5}. \quad (89)$$

Moreover the polynomial cutoff satisfies $0 \leq \chi \leq 1$, $\|\chi'\|_\infty = \frac{35}{16}$ and $\|\chi''\|_\infty < 7.514$. Using (83), $D_K/h \leq 18$ and $c \leq \sqrt{K}/3$, the maximum-row-sum bound in (88) gives

$$\begin{aligned} \frac{D_K}{h} \|\chi B_K + E_K\|_{\text{op}} &\leq 18 \left[\left(1 + \frac{3 \cdot 35}{2 \cdot 16} + \frac{7.514}{2} \right) 2\pi\delta_K \right. \\ &\quad \left. + \left(\frac{3}{2} c + \frac{1}{2} + c \frac{35}{16} + \frac{1}{2} \frac{35}{16} \right) 3\delta_K \right] < 0.0078. \end{aligned}$$

Every term on the right decreases when K increases, so it is enough to insert $K = 16$.

The rank-one matrix in (88) has nonzero eigenvalue $1 + c^2 = (K+7)/9$, and $\frac{D_K}{2}(1 + c^2) = h(K-1)$. Its positive part is therefore at most $h(K-1)I$, while (89) bounds its negative part by $4.8 \cdot 10^{-5}hI$. Thus, in fact,

$$0.992hI \preceq \nabla^2 V_K \preceq h(K + 0.0078)I,$$

which implies (87). On an uncut valley core $g_K = 1$ and $f_K(0) = f'_K(0) = 0$, so the two eigenvalues are exactly h and hK . Finally $0.98h = 1.96\alpha > \alpha$ and $h(K + 0.02) < \alpha(2K + 1) = \beta$. $\square$

B.2 Reference orbit and Gaussian shadowing

Set $s_K := \sqrt{K-2}$ and

$$Q_K := \frac{1}{K+7} \begin{pmatrix} (K+1)^2 & 2s_K(K+1) \\ 2s_K(K+1) & 6(K+1) \end{pmatrix}. \quad (90)$$

A principal-minor calculation gives

$$\frac{3}{2}I \prec Q_K \prec KI, \quad Q_K^{-1}e_1 = \frac{1}{K+1} \begin{pmatrix} 3 \\ -\sqrt{K-2} \end{pmatrix}. \quad (91)$$

Indeed $\text{Tr } Q_K = K + 1$, $\det Q_K = 2(K + 1)^2/(K + 7)$, and

$$\det \left(Q_K - \frac{3}{2}I \right) = \frac{2K^2 - 23K + 29}{4(K + 7)} > 0 \quad (K \geq 16),$$

while the first diagonal entry of $Q_K - \frac{3}{2}I$ is positive; the upper bound follows in the same way from the principal minors of $KI - Q_K$.

Choose m_0 with $\|m_0\| = e^2 r_0$ and polar angle ϑ_0 satisfying

$$\vartheta_0 + c \log \frac{\|m_0\|}{r_0} \equiv 0 \pmod{\pi}, \quad (92)$$

and initialize the precision by

$$P_0 := C_0^{-1} := h R_{\vartheta_0} Q_K R_{\vartheta_0}^\top. \quad (93)$$

Then (91) gives

$$\frac{1}{\beta} I \preceq C_0 \preceq \frac{1}{\alpha} I. \quad (94)$$

The point-system reference orbit is

$$r_t^{\text{ref}} := e^2 r_0 \exp \left(-\frac{3t}{K+1} \right), \quad \vartheta_t^{\text{ref}} := \vartheta_0 + \frac{\sqrt{K-2}}{K+1} t, \quad P_t^{\text{ref}} := h R_{\vartheta_t^{\text{ref}}} Q_K R_{\vartheta_t^{\text{ref}}}^\top. \quad (95)$$

Indeed, on the valley $\nabla V_K(m) = hm$, and (91) gives $\dot{r} = -\frac{3}{K+1}r$ and $r\dot{\vartheta} = \frac{\sqrt{K-2}}{K+1}r$; a direct substitution also verifies $\dot{P}^{\text{ref}} = \nabla^2 V_K(m^{\text{ref}}) - P^{\text{ref}}$.

Lemma B.3 (Gaussian shadowing certificate). *Assume*

$$r_0 \sqrt{\alpha} \geq 10^{25} K^{3/2} \sqrt{\log(eK)}. \quad (96)$$

Let (m_t, C_t) be the exact Gaussian Fisher–Rao flow (1) for V_K initialized by (92)–(93). Then, for $0 \leq t \leq K/100$,

$$-\frac{4}{K} \leq \frac{d}{dt} \log \|m_t\| \leq -\frac{2}{K}, \quad (97)$$

and in particular

$$\|m_t\| \geq e^{-0.04} \|m_0\| \quad (0 \leq t \leq K/100). \quad (98)$$

Proof. Write $P_t = C_t^{-1}$ and choose the continuous lift of the polar angle starting at ϑ_0 . Set

$$z_t := \vartheta_t + c \log \frac{r_t}{r_0}, \quad S_t := \frac{1}{h} R_{\vartheta_t}^\top P_t R_{\vartheta_t}, \quad E_t := S_t - Q_K, \quad (99)$$

where z_t is represented near 0 modulo π . Put

$$w := \begin{pmatrix} c \\ 1 \end{pmatrix}, \quad J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad d_K := \frac{D_K}{h} = \frac{18(K-1)}{K+7}.$$

On the uncut flat phase core $f_K(z) = z^2/2$, and the normalized point gradient and Hessian are

$$\begin{aligned} b_K(z) &:= \frac{1}{hr} R_{\vartheta}^\top \nabla V_K(m) = \begin{pmatrix} 1 + \frac{d_K}{2} z^2 + \frac{d_K c}{2} z \\ \frac{d_K}{2} z \end{pmatrix}, \\ H_K(z) &:= \frac{1}{h} R_{\vartheta}^\top \nabla^2 V_K(m) R_{\vartheta} \\ &= I + \frac{d_K}{2} w w^\top + d_K \begin{pmatrix} \frac{1}{2} z^2 + \frac{3}{2} c z & \frac{1}{2} z \\ \frac{1}{2} z & \frac{1}{2} z^2 + \frac{1}{2} c z \end{pmatrix}. \end{aligned} \quad (100)$$

For the exact Gaussian flow define

$$\bar{b}_t := \frac{1}{hr_t} R_{\vartheta_t}^\top \mathbb{E}[\nabla V_K(X_t)], \quad \bar{H}_t := \frac{1}{h} R_{\vartheta_t}^\top \mathbb{E}[\nabla^2 V_K(X_t)] R_{\vartheta_t}, \quad q_t := S_t^{-1} \bar{b}_t.$$

The mean and precision equations become

$$\frac{\dot{r}_t}{r_t} = -(q_t)_1, \quad \dot{\vartheta}_t = -(q_t)_2, \quad \dot{z}_t = -w^\top q_t, \quad \dot{S}_t = \bar{H}_t - S_t - (q_t)_2(S_t J - J S_t). \quad (101)$$

If the Gaussian averages in (101) are replaced by the point fields (100), then $(z, S) = (0, Q_K)$ is an equilibrium: with $\omega_K = s_K/(K+1)$,

$$Q_K^{-1} e_1 = \begin{pmatrix} 3/(K+1) \\ -\omega_K \end{pmatrix}, \quad w^\top Q_K^{-1} e_1 = 0, \quad \mathbf{H}_K(0) - Q_K + \omega_K(Q_K J - J Q_K) = 0.$$

We next give the linear and nonlinear certificates explicitly. Let

$$\mathbf{T}_K := \text{diag}(K, \sqrt{K}, 1, \sqrt{K}), \quad y := \mathbf{T}_K(z, E_{11}, E_{12}, E_{22})^\top. \quad (102)$$

For $S = Q_K + E$, let $\mathcal{F}_K(y)$ be the scaled point vector field obtained from (101) with $\bar{b} = b_K(z)$ and $\bar{H} = \mathbf{H}_K(z)$, and set $A_K := D\mathcal{F}_K(0)$ and $\mathcal{N}_K(y) := \mathcal{F}_K(y) - A_K y$. If $s = s_K = \sqrt{K-2}$, direct differentiation gives

$$A_K = \mathbf{T}_K \mathbf{B}_K \mathbf{T}_K^{-1}, \quad (103)$$

where, in the order $(z, E_{11}, E_{12}, E_{22})$,

$$\mathbf{B}_K = \begin{pmatrix} -\frac{3(K-1)}{2(K+1)} & 0 & \frac{K+7}{2(K+1)^2} & -\frac{s(K+7)}{6(K+1)^2} \\ \frac{3s(K-1)}{K+7} & -\frac{K^2+20K-17}{(K+1)(K+7)} & \frac{12s}{K+7} & -\frac{2(K-2)}{K+7} \\ \frac{3(K-1)(K+1)}{2(K+7)} & \frac{2(K-11)s}{(K+1)(K+7)} & -\frac{7K^2-10K+19}{2(K+1)(K+7)} & \frac{s(K^2-2K+9)}{2(K+1)(K+7)} \\ \frac{9s(K-1)}{K+7} & \frac{12(K-2)}{(K+1)(K+7)} & -\frac{12s}{K+7} & \frac{K-11}{K+7} \end{pmatrix}.$$

Its characteristic polynomial has the factorization

$$\det(\lambda I - A_K) = \frac{\lambda + 1}{2(K+1)^3} p_K(\lambda + 1), \quad (104)$$

where

$$p_K(\mu) = 2(K+1)^3 \mu^3 + 2(K-5)(K+1)^2 \mu^2 + 4(K-2)(K+1)^2 \mu + (K-2)(K^2 - 10K - 23).$$

For $K \geq 16$ all coefficients are positive and the cubic Routh–Hurwitz determinant is $6(K-2)(K-1)^2(K+1)^3 > 0$, so every eigenvalue of A_K has real part at most -1 . The entries of the scaled matrix obey

$$(|(A_K)_{ij}|) \preceq \begin{pmatrix} 3/2 & 0 & 2/3 & 1/4 \\ 3 & 3/2 & 12 & 2 \\ 3/2 & 1/8 & 7/2 & 1/2 \\ 9 & 3/4 & 12 & 1 \end{pmatrix},$$

so $\|A_K\|_{\text{op}} < 21$. In a complex Schur decomposition $A_K = U(D+N)U^*$, the matrix N is strictly upper triangular with $\|N\|_{\text{op}} \leq 42$ and $\|e^{Dt}\|_{\text{op}} \leq e^{-t}$; iterating Duhamel's formula terminates after three factors of N and yields

$$\|e^{A_K t}\|_{\text{op}} \leq e^{-t} \sum_{j=0}^3 \frac{(42t)^j}{j!} < 2 \cdot 10^6 e^{-t/2}. \quad (105)$$

The nonlinear estimate is also an elementary finite-dimensional calculation. Inserting the explicit inverse of the 2×2 matrix S into (100) gives, for $\|y\| \leq 10^{-12}$ and $K \geq 16$,

$$\sup_{j,\ell=1}^4 \left| \partial_{j\ell}^2(\mathcal{F}_K)_i \right| \leq (100, 2000, 500, 2000)_i. \quad (106)$$

The bounds follow term by term from $d_K \leq 18$, $c/\sqrt{K} \leq 1/3$, $Q_K \succeq \frac{3}{2}I$, and

$$E = \begin{pmatrix} y_2/\sqrt{K} & y_3 \\ y_3 & y_4/\sqrt{K} \end{pmatrix},$$

the largest denominator being controlled by $\|(Q_K + E)^{-1}\|_{\text{op}} \leq 1$. Taylor's theorem and (106) imply

$$\|\mathcal{N}_K(y)\| \leq \frac{1}{2} \sqrt{100^2 + 2000^2 + 500^2 + 2000^2} \|y\|^2 < 4000 \|y\|^2. \quad (107)$$

It remains to bound the Gaussian-versus-point error at the actual state. Define the stopping time

$$\tau := \frac{K}{100} \wedge \inf \left\{ t \geq 0 : \|y_t\| = 10^{-12} \text{ or } r_t \notin [e^{3/2}r_0, e^{5/2}r_0] \right\}. \quad (108)$$

On $[0, \tau]$, (91) and (102) give $S_t \succeq I$, hence $C_t \preceq h^{-1}I$. Put $Z_t := X_t - m_t$ and

$$\mathcal{G}_t := \left\{ \|Z_t\| \leq \frac{\delta_K r_t}{8\sqrt{K}} \right\}, \quad p_K := \sup_{t \leq \tau} \mathbb{P}(\mathcal{G}_t^c), \quad M_K := \sup_{t \leq \tau} \mathbb{E} \left[\sqrt{h} \|Z_t\| \mathbf{1}_{\mathcal{G}_t^c} \right].$$

For every point on the segment $m_t + uZ_t$, $0 \leq u \leq 1$, the logarithmic radius and angle obey, on $\mathcal{G}_t$,

$$|\Delta\vartheta| + c|\Delta\ell| \leq 2(1+c) \frac{\|Z_t\|}{r_t} < \frac{\delta_K}{4}.$$

Since $|z_t| \leq 10^{-12}/K < \delta_K/4$ and $r_t \geq e^{3/2}r_0$, the whole segment lies in the uncut flat phase core. If $G \sim \mathcal{N}(0, I_2)$, the covariance bound implies $\sqrt{h} \|Z_t\| \leq \|G\|$ under a common Gaussian representation, and the exact radial tail in dimension two gives

$$p_K \leq \exp \left(-\frac{\delta_K^2 h r_0^2}{128K} \right) \leq 10^{-50} K^{-30}, \quad M_K \leq \sqrt{2p_K} \leq \sqrt{2} 10^{-25} K^{-15}. \quad (109)$$

On the flat core, differentiating (100) in Cartesian coordinates gives

$$\text{Lip}_{\text{op}}(\nabla^2 V_K) \leq \frac{15hK}{r_t} \quad (110)$$

along these segments: $\|\partial_z \mathbf{H}_K\|_{\text{op}} \leq 12\sqrt{K}$, $\|\nabla z\| \leq 2\sqrt{K}/(5r)$, rotation of the polar frame contributes at most $2h(K+0.02)/r$, and allowing a factor two for the radius along the segment gives (110).

Let

$$\begin{aligned} \varepsilon_b(t) &:= \left\| \frac{1}{hr_t} R_{\vartheta_t}^\top (\mathbb{E}[\nabla V_K(X_t)] - \nabla V_K(m_t)) \right\|, \\ \varepsilon_H(t) &:= \left\| \frac{1}{h} R_{\vartheta_t}^\top (\mathbb{E}[\nabla^2 V_K(X_t)] - \nabla^2 V_K(m_t)) R_{\vartheta_t} \right\|_{\text{op}}. \end{aligned}$$

Put $A_0 := \sqrt{\alpha} r_0$. Splitting at $\mathcal{G}_t$, using (110) for the Taylor remainder on $\mathcal{G}_t$, the global Hessian bound on its complement, $\mathbb{E}Z_t = 0$ and $\mathbb{E}\|Z_t\|^2 \leq 2/h$, gives

$$\varepsilon_b(t) \leq 8 \frac{K}{A_0^2} + 3 \frac{K}{A_0} M_K, \quad \varepsilon_H(t) \leq 15 \frac{K}{A_0} + 3Kp_K. \quad (111)$$

The forcing $\xi_K(t)$ in the scaled transverse system is, by definition, the contribution of these actual-state Gaussian-versus-point errors. Since $\|S_t^{-1}\|_{\text{op}} \leq 1$, $\|S_t\|_{\text{op}} \leq 2K$ and $\|w\| \leq 2\sqrt{K}/5$, equations (101) and (102) imply

$$\begin{aligned} \|\xi_K(t)\| &\leq \frac{2}{5}K^{3/2}\varepsilon_b(t) + \sqrt{2K}(\varepsilon_H(t) + 4K\varepsilon_b(t)) \\ &\leq 23\frac{K^{3/2}}{A_0} + 5K^{3/2}p_K + 52\frac{K^{5/2}}{A_0^2} + 20\frac{K^{5/2}}{A_0}M_K. \end{aligned} \quad (112)$$

The radius condition (96) and (109) make the last line smaller than $3 \cdot 10^{-24}$, and in particular smaller than 10^{-22} , on $[0, \tau]$.

The exact transverse equation is

$$\dot{y}_t = A_K y_t + \mathcal{N}_K(y_t) + \xi_K(t), \quad y_0 = 0. \quad (113)$$

Variation of constants, (105), (107) and (112) yield, for $t \leq \tau$,

$$\|y_t\| \leq 2 \cdot 10^6 \int_0^t e^{-(t-s)/2} (4000 \cdot 10^{-24} + 10^{-22}) \, ds < 2 \cdot 10^{-14},$$

so the transverse component cannot cause the exit in (108).

Finally, define the point radial field $\mathcal{R}_K(y) := -e_1^\top S^{-1} b_K(z)$. Differentiating the same explicit 2×2 inverse gives

$$\sup_{\|y\| \leq 10^{-12}} \|\nabla \mathcal{R}_K(y)\| \leq \frac{20}{K}, \quad \mathcal{R}_K(0) = -\frac{3}{K+1}.$$

The Gaussian correction is at most $\varepsilon_b(t)$, so on $[0, \tau]$,

$$\left| \frac{d}{dt} \log r_t + \frac{3}{K+1} \right| \leq \frac{20}{K} \|y_t\| + \varepsilon_b(t) < \frac{1}{4K},$$

which already implies $-4/K \leq \dot{r}_t/r_t \leq -2/K$. Hence the radius cannot reach the upper barrier in (108), and before time $K/100$ it remains above $e^{2-0.04}r_0 > e^{3/2}r_0$. Therefore $\tau = K/100$, which proves (97); integration gives (98). $\square$

B.3 Intrinsic conditioning and the lower bound

Lemma B.4 (Optimizer covariance and intrinsic condition number). *Let $C_\star$ be the optimizing covariance for V_K and set $\eta_K := K \exp(-\alpha r_0^2/2)$. Under (96), $\eta_K \leq 10^{-3}$ and*

$$h(1 - \eta_K)I \preceq C_\star^{-1} \preceq h(1 + \eta_K)I. \quad (114)$$

Moreover

$$K \leq \kappa_\star(V_K) \leq 1.03K, \quad (115)$$

and the whitened initial covariance satisfies

$$\beta_\star \lambda_{0,\star} \geq \frac{1 - \eta_K}{1 + \eta_K}, \quad (116)$$

so the covariance-initialization term in Theorem 2.3 is smaller than 0.003.

Proof. The potential is exactly quadratic with Hessian hI on $\{\|\theta\| \leq r_0\}$. Since $C_\star \preceq \alpha^{-1}I$, write $X_\star = C_\star^{1/2}Z$ with $Z \sim \mathcal{N}(0, I_2)$; then $\|X_\star\| > r_0$ implies $\|Z\| > \sqrt{\alpha}r_0$, and for a standard two-dimensional Gaussian $\mathbb{P}(\|Z\| > s) = e^{-s^2/2}$. Using $\|\nabla^2 V_K - hI\|_{\text{op}} \leq hK$ and $C_\star^{-1} = \mathbb{E}[\nabla^2 V_K(X_\star)]$ gives (114).

Combining (87) with (114) gives

$$\alpha_\star \geq \frac{0.98}{1 + \eta_K}, \quad \beta_\star \leq \frac{K + 0.02}{1 - \eta_K},$$

and hence $\kappa_\star \leq 1.03K$. At the origin the Hessian is hI , whereas at a valley point it has an eigenvalue hK ; therefore $\alpha_\star \leq h\lambda_{\min}(C_\star)$ and $\beta_\star \geq hK\lambda_{\min}(C_\star)$, which gives $\kappa_\star \geq K$.

Finally (94) gives $C_0 \succeq (hK)^{-1}I$, while $C_\star^{-1} \succeq h(1 - \eta_K)I$, so $\lambda_{0,\star} \geq (1 - \eta_K)/K$; the valley Hessian and (114) also give $\beta_\star \geq K/(1 + \eta_K)$, proving (116). $\square$

Proof of Theorem 2.5. The potential is even, so the optimizing mean is $m_\star = 0$. Strong convexity in the mean and global optimality of $(0, C_\star)$ imply $\Delta(m, C) \geq \frac{\alpha}{2} \|m\|^2$ for every (m, C) . By Theorem B.3, for $0 \leq t \leq K/100$,

$$\Delta(a_t) \geq \frac{\alpha}{2} e^{-0.08} \|m_0\|^2 > \delta_{\text{loc}},$$

so $T_{\text{loc}} \geq K/100$, and $\kappa_\star \leq 1.03K$ from Theorem B.4 gives $T_{\text{loc}} \geq \kappa_\star/103$. The regularity and curvature claims are Theorem B.2, the admissibility of the initialization is (94), and the remaining assertions are Theorem B.4. $\square$

Proof of Theorem B.1. The upper bound is Theorem 2.3 and the lower bound is Theorem 2.5. For dimensions larger than two, add independent quadratic coordinates,

$$V_K^{(N_\theta)}(\theta) := V_K(\theta_1, \theta_2) + \frac{h}{2} \sum_{j=3}^{N_\theta} \theta_j^2,$$

and initialize those coordinates at equilibrium. The block-diagonal subsystem is dynamically invariant. Moreover Hadamard's determinant inequality shows that a cross-covariance block can only decrease the Gaussian determinant while leaving the separated potential expectation unchanged, hence the optimizing covariance is block diagonal. The two-dimensional slow subsystem and its condition-number comparison are therefore unchanged. For the spiral family $\kappa = 2K + 1$ and $\kappa_\star \geq K$, so $T_{\text{loc}} \geq K/100 \geq (\kappa - 1)/200$. $\square$

Remark B.5 (Hessian-Lipschitz version). Differentiating (88) once more and using $\|\chi'''\|_\infty \leq 52.5$ and $\|g'_K\|_\infty \leq 4.376/\delta_K$ gives the explicit bound

$$\text{Lip}_{\text{op}}(\nabla^2 V_K) \leq 10^8 \frac{hK^{5/2}}{r_0}.$$

Hence, for every prescribed $L_H > 0$, choosing

$$r_0 \geq \max \left\{ \frac{10^8 hK^{5/2}}{L_H}, \frac{10^{25} K^{3/2} \sqrt{\log(eK)}}{\sqrt{\alpha}} \right\}$$

preserves the lower bound while ensuring $\text{Lip}_{\text{op}}(\nabla^2 V_K) \leq L_H$. At the endpoint $L_H = 0$ the target is quadratic, so the linear lower bound is not asserted under a radius budget fixed independently of L_H .

C The bump-train construction

We construct the one-dimensional family announced in Theorem 2.12 and prove the lower bound. The construction is one-dimensional, starts with the covariance already on the curvature scale, and has a dimension-free Hessian-Lipschitz constant, so the obstruction it exhibits is the global localization stage itself rather than the covariance burn-in.

The following packaging of Theorem 2.12 records the $\Theta(\kappa^2)$ complexity reading at the globally certified stepsize.

Corollary C.1 (Sharpness of the global-localization prefactor). *For the globally fixed-step schemes (20) and (21), the factor $\kappa/\Delta t$ in Theorems 2.9 and 2.11 is sharp, up to constants independent of κ for each fixed $\gamma = \kappa\Delta t$, for a constant-factor reduction of the energy. In particular, with $\Delta t = \Theta(\kappa^{-1})$, both methods have worst-case global-localization complexity $\Theta(\kappa^2)$ in the condition-number exponent, and this remains true under the additional assumption $\|V'''\|_\infty = O(1)$.*

C.1 Scalar schemes and scope

We normalize $\alpha = 1$ and $\beta = \kappa$, and write $a = (m, c) \in \mathbb{R} \times (0, \infty)$. For $X \sim \mathcal{N}(m, c)$ put

$$b(m, c) := \mathbb{E}[V'(X)], \quad A(m, c) := \mathbb{E}[V''(X)], \quad (117)$$

so that $g(a) = -b(m, c)$. The scalar forms of (20) and (21) are, respectively,

$$m^+ = m - \Delta t c b(m, c), \quad c^+ = c \exp(\Delta t [1 - cA(m, c)]), \quad (118)$$

and

$$m^+ = m - \Delta t c b(m, c), \quad c^+ = \frac{(1 + \Delta t)c}{1 + \Delta t c A(m, c)}. \quad (119)$$

We consider a fixed globally safe scale

$$\Delta t = \frac{\gamma}{\kappa}, \quad 0 < \gamma \leq \frac{1}{2}, \quad (120)$$

with γ independent of κ . The restriction $\gamma \leq 1/2$ places the step inside the certified range of both Theorems 2.9 and 2.11; the same construction works, with changed numerical constants, for every fixed $\gamma \in (0, 2)$. The bound proves sharpness of the factor $\kappa/\Delta t$ for a constant-factor energy reduction. It does not claim that the entire product $\kappa\Delta t^{-1} \log(\Delta_0/\delta)$ is sharp for every target accuracy, and it does not apply to adaptive, relative-curvature, line-search or implicit discretizations.

C.2 A flat-top curvature bump and the bump train

Fix an even function $\varphi \in C_c^\infty(\mathbb{R})$ satisfying

$$0 \leq \varphi \leq 1, \quad \varphi(u) = 1 \text{ for } |u| \leq \frac{1}{2}, \quad \varphi(u) = 0 \text{ for } |u| \geq 1, \quad (121)$$

and set $I_\varphi := \int_{\mathbb{R}} \varphi(u) du$ and $M_\varphi := \|\varphi'\|_\infty$. Fix a numerical constant $L_H > 0$ and define

$$w_\kappa := \frac{(\kappa - 1)M_\varphi}{L_H}, \quad B_\kappa := w_\kappa I_\varphi, \quad H_\kappa := (\kappa - 1)B_\kappa, \quad (122)$$

so that

$$w_\kappa = \Theta(\kappa), \quad H_\kappa = \Theta(\kappa^2), \quad (123)$$

with implicit constants depending only on φ and L_H . The choice $w_\kappa = \Theta(\kappa)$ is what makes the Hessian-Lipschitz constant uniform:

$$\frac{\kappa - 1}{w_\kappa} \|\varphi'\|_\infty = L_H. \quad (124)$$

Choose numerical constants $T > 0$ sufficiently small and $Y > 0$ sufficiently large, depending only on φ and L_H , and set

$$N_\kappa := \left\lfloor \frac{T\kappa^2}{\gamma} \right\rfloor, \quad x_0 := \frac{Y\kappa^4}{\gamma}. \quad (125)$$

Define the positive bump centers recursively by

$$x_{j+1} := x_j - \frac{\gamma}{\kappa^2} \left[x_j + H_\kappa \left(N_\kappa - j + \frac{1}{2} \right) \right], \quad 0 \leq j < N_\kappa. \quad (126)$$

Lemma C.2 (Geometry of the bump centers). *For suitable universal choices of T and Y , and all sufficiently large κ ,*

$$x_{N_\kappa} \geq \frac{3}{4}x_0, \quad (127)$$

and

$$x_j - x_{j+1} \geq \frac{3Y}{4}\kappa^2, \quad 0 \leq j < N_\kappa. \quad (128)$$

In particular the intervals $[x_j - w_\kappa, x_j + w_\kappa]$ are pairwise disjoint, and they are disjoint from their reflections about the origin.

Proof. Summing (126) and using $x_j \leq x_0$ gives

$$x_0 - x_{N_\kappa} \leq \frac{\gamma N_\kappa}{\kappa^2}x_0 + \frac{\gamma H_\kappa}{\kappa^2} \sum_{j=0}^{N_\kappa-1} \left(N_\kappa - j + \frac{1}{2} \right) \leq T x_0 + C_\varphi \frac{T^2 \kappa^4}{\gamma},$$

where C_φ depends only on φ and L_H , because $H_\kappa \leq C_\varphi \kappa^2$ and $N_\kappa \leq T \kappa^2 / \gamma$. Since $x_0 = Y \kappa^4 / \gamma$, first choose T small and then Y large so that the right-hand side is at most $x_0/4$; this proves (127). By (126) and (127),

$$x_j - x_{j+1} \geq \frac{\gamma}{\kappa^2} x_{N_\kappa} \geq \frac{3\gamma}{4\kappa^2} \frac{Y \kappa^4}{\gamma} = \frac{3Y}{4} \kappa^2.$$

Since $w_\kappa = O(\kappa)$, the positive supports are disjoint for all large κ , and $x_{N_\kappa} \asymp \kappa^4 / \gamma \gg w_\kappa$ makes them disjoint from the reflected negative supports. $\square$

Define the potential by

$$V_\kappa''(\theta) := 1 + (\kappa - 1) \sum_{j=0}^{N_\kappa} \left[\varphi\left(\frac{\theta - x_j}{w_\kappa}\right) + \varphi\left(\frac{\theta + x_j}{w_\kappa}\right) \right], \quad (129)$$

with $V_\kappa'(0) = 0$ and $V_\kappa(0) = 0$.

Lemma C.3 (Regularity and exact condition number). *For all sufficiently large κ the function V_κ is even and belongs to $C^\infty(\mathbb{R})$, and*

$$1 \leq V_\kappa''(\theta) \leq \kappa \quad \text{for every } \theta \in \mathbb{R}, \quad (130)$$

with both bounds attained, and

$$\|V_\kappa'''\|_\infty \leq L_H. \quad (131)$$

Consequently e^{-V_κ} is integrable, the target has exact condition number κ , and its optimizer-whitened condition number also equals κ .

Proof. The reflected construction makes V_κ'' even, and the conditions at the origin then make V_κ' odd and V_κ even. By Theorem C.2 at most one summand in (129) is nonzero at any point, hence $1 \leq V_\kappa'' \leq 1 + (\kappa - 1) = \kappa$; the value 1 is attained outside the bump supports and the value κ on every flat top. Similarly $|V_\kappa'''(\theta)| \leq \frac{\kappa-1}{w_\kappa} \|\varphi'\|_\infty = L_H$. Since $V_\kappa'' \geq 1$ and $V_\kappa(0) = V_\kappa'(0) = 0$, we have $V_\kappa(\theta) \geq \theta^2/2$, which proves integrability. In one dimension optimizer whitening multiplies every Hessian value by the same positive scalar $c_\star$, so it leaves the ratio $\sup V_\kappa'' / \inf V_\kappa''$ unchanged and $\kappa_\star = \kappa$. $\square$

C.3 Gaussian averages along the train

Set

$$c_\kappa := \frac{1}{\kappa}, \quad (132)$$

and define the ideal states $a_j^\circ := (x_j, c_\kappa)$.

Lemma C.4 (Averaged curvature and score at the centers). *There exist constants $C, c > 0$ and an integer $p \geq 1$, independent of κ , such that for every $0 \leq j \leq N_\kappa$,*

$$A(x_j, c_\kappa) = \kappa + r_{j,\kappa}^A, \quad |r_{j,\kappa}^A| \leq C\kappa^p e^{-c\kappa^3}, \quad (133)$$

$$b(x_j, c_\kappa) = x_j + H_\kappa \left(N_\kappa - j + \frac{1}{2} \right) + r_{j,\kappa}^b, \quad |r_{j,\kappa}^b| \leq C\kappa^p e^{-c\kappa^3}. \quad (134)$$

The same estimates hold uniformly on the tubes

$$\mathcal{T}_{j,\kappa} := \left\{ (m, c) : |m - x_j| \leq \frac{w_\kappa}{8}, \quad |\kappa c - 1| \leq \frac{1}{8} \right\}, \quad (135)$$

in the form: for $(m, c) \in \mathcal{T}_{j,\kappa}$,

$$|A(m, c) - \kappa| \leq \varepsilon_\kappa, \quad (136)$$

$$\left| b(m, c) - x_j - H_\kappa \left(N_\kappa - j + \frac{1}{2} \right) - \kappa(m - x_j) \right| \leq \varepsilon_\kappa, \quad (137)$$

where $\varepsilon_\kappa := C\kappa^p e^{-c\kappa^3}$.

Proof. Let $X = x_j + Z/\sqrt{\kappa}$ with $Z \sim \mathcal{N}(0, 1)$. The current bump equals one whenever $|X - x_j| \leq w_\kappa/2$, and since $w_\kappa \asymp \kappa$,

$$\mathbb{P} \left(|X - x_j| > \frac{w_\kappa}{2} \right) \leq 2 \exp(-c\kappa^3).$$

Every other bump is at distance at least $c\kappa^2$ from x_j , which is at least $c\kappa^{5/2}$ Gaussian standard deviations, and summing over $N_\kappa + 1 = O(\kappa^2)$ bumps changes only the polynomial prefactor. This proves (133).

To compute the score, put $\Phi(s) := \int_{-\infty}^s \varphi(u) du$. For $\theta > 0$, away from an exponentially negligible event on which the Gaussian sample crosses the origin, integration of (129) gives

$$V'_\kappa(\theta) = \theta + (\kappa - 1)w_\kappa \sum_{k=0}^{N_\kappa} \Phi \left(\frac{\theta - x_k}{w_\kappa} \right).$$

The bumps with $k > j$ lie strictly between the origin and x_j and contribute their full mass I_φ , while those with $k < j$ contribute zero, up to the same Gaussian-tail error. For the current bump, evenness of φ gives $\Phi(s) + \Phi(-s) = I_\varphi$, and since $(X - x_j)/w_\kappa$ is symmetric, $\mathbb{E}\Phi((X - x_j)/w_\kappa) = I_\varphi/2$. Multiplying by $(\kappa - 1)w_\kappa$ proves (134).

Now take $(m, c) \in \mathcal{T}_{j,\kappa}$. The mean remains at least $3w_\kappa/8$ from the edge of the current flat top and the standard deviation is at most $\sqrt{9/(8\kappa)}$, so the probability of reaching a transition layer is still bounded by $Ce^{-c\kappa^3}$, uniformly on the tube; this proves (136). Finally $\partial_m b(m, c) = A(m, c)$, so integration in the mean variable gives the term $\kappa(m - x_j)$ with an error bounded by ε_κ , while $\partial_c b(m, c) = \frac{1}{2}\mathbb{E}[V''''_\kappa(X)]$ and V''''_κ is supported only on the transition layers, which are reached with probability $O(e^{-c\kappa^3})$ throughout the tube. Integrating in c proves (137) after increasing the polynomial prefactor in ε_κ . $\square$

C.4 Shadowing for both covariance maps

Let (m_j, c_j) denote the iterates of either (118) or (119), initialized at

$$m_0 = x_0, \quad c_0 = c_\kappa = \frac{1}{\kappa}, \quad (138)$$

and define the mean and relative-covariance errors

$$e_j := m_j - x_j, \quad q_j := \kappa c_j - 1. \quad (139)$$

Lemma C.5 (Uniform discrete shadowing). *For either covariance map there exist constants $C, c > 0$ and an integer $p \geq 1$ such that, for all sufficiently large κ and every $0 \leq j \leq N_\kappa$,*

$$|e_j| + |q_j| \leq C\kappa^p e^{-c\kappa^3}. \quad (140)$$

In particular every iterate belongs to $\mathcal{T}_{j,\kappa}$ and

$$m_j = x_j + o(1), \quad c_j = \frac{1}{\kappa}(1 + o(1)), \quad (141)$$

uniformly over $0 \leq j \leq N_\kappa$.

Proof. Write $B_j := x_j + H_\kappa(N_\kappa - j + \frac{1}{2})$, so that $x_{j+1} = x_j - \gamma B_j / \kappa^2$ by (126). On the tube, Theorem C.4 gives $b(m_j, c_j) = B_j + \kappa e_j + r_j$ with $|r_j| \leq \varepsilon_\kappa$. Since $c_j = (1 + q_j) / \kappa$ and $\Delta t = \gamma / \kappa$, the mean recursion is exactly

$$e_{j+1} = \left(1 - \frac{\gamma(1 + q_j)}{\kappa}\right) e_j - \frac{\gamma q_j}{\kappa^2} B_j - \frac{\gamma(1 + q_j)}{\kappa^2} r_j. \quad (142)$$

The center geometry gives $|B_j| \leq C\kappa^4 / \gamma$, hence, as long as $|q_j| \leq 1/8$,

$$|e_{j+1}| \leq \left(1 - \frac{7\gamma}{8\kappa}\right) |e_j| + C\kappa^2 |q_j| + C \frac{\varepsilon_\kappa}{\kappa^2}. \quad (143)$$

For the Riemannian covariance map, write $A(m_j, c_j) = \kappa + \delta_j$ with $|\delta_j| \leq \varepsilon_\kappa$; then

$$1 + q_{j+1} = (1 + q_j) \exp \left[\frac{\gamma}{\kappa} \left(1 - (1 + q_j) \left(1 + \frac{\delta_j}{\kappa} \right) \right) \right]. \quad (144)$$

For $|q_j| \leq 1/8$ and large κ the derivative of the right-hand side with respect to q_j is between zero and $1 - \gamma / (2\kappa)$, and its variation with respect to δ_j is at most $C\gamma |\delta_j| / \kappa^2$. Since the map fixes $q = 0$ when $\delta = 0$, the mean-value theorem yields

$$|q_{j+1}| \leq \left(1 - \frac{\gamma}{2\kappa}\right) |q_j| + C \frac{\varepsilon_\kappa}{\kappa^2}. \quad (145)$$

For the KL/Bregman map,

$$1 + q_{j+1} = \frac{(1 + \gamma/\kappa)(1 + q_j)}{1 + (\gamma/\kappa)(1 + q_j)(1 + \delta_j/\kappa)}, \quad (146)$$

and the same mean-value argument gives, on $|q_j| \leq 1/8$,

$$|q_{j+1}| \leq \left(1 - \frac{\gamma}{2\kappa}\right) |q_j| + C \frac{\varepsilon_\kappa}{\kappa^2}. \quad (147)$$

Starting from $q_0 = 0$, either (145) or (147) implies

$$|q_j| \leq C \frac{\varepsilon_\kappa}{\gamma\kappa} \quad (0 \leq j \leq N_\kappa). \quad (148)$$

Substituting this into (143), using $e_0 = 0$ and summing the geometric recursion gives

$$|e_j| \leq C\gamma\kappa^p e^{-c\kappa^3} \quad (149)$$

for a possibly larger integer p . Because exponential decay dominates every polynomial, (148)–(149) are eventually much smaller than $1/8$ and $w_\kappa/8$ respectively, so the tube assumptions used in the argument close by induction, proving (140). $\square$

C.5 The objective gap remains macroscopic

For $c > 0$ define the Gaussian-smoothed potential

$$F_c(m) := \mathbb{E}_{X \sim \mathcal{N}(m, c)}[V_\kappa(X)], \quad (150)$$

so that, up to a target-dependent additive constant,

$$\mathcal{E}_\kappa(m, c) = F_c(m) - \frac{1}{2} \log c. \quad (151)$$

Lemma C.6 (Constant-factor lower bound for the objective). *There exists a numerical constant $\varrho \in (0, 1)$, independent of κ , such that the iterates of either scheme satisfy*

$$\Delta(m_j, c_j) \geq \varrho \Delta(m_0, c_0), \quad 0 \leq j \leq N_\kappa, \quad (152)$$

for all sufficiently large κ .

Proof. Since V_κ is even, F_c is even and $F'_c(0) = 0$; moreover $F''_c(m) = A(m, c) \geq 1$, hence

$$F_c(m) - F_c(0) \geq \frac{1}{2} m^2. \quad (153)$$

At the matched covariance $c_\kappa = 1/\kappa$, [Theorem C.2](#) gives

$$F_{c_\kappa}(x_j) - F_{c_\kappa}(0) \geq \frac{1}{2} x_j^2 \geq \frac{1}{2} x_{N_\kappa}^2 \geq \frac{9}{32} x_0^2. \quad (154)$$

For the converse bound at initialization, integration of [\(129\)](#) gives the global estimate $|V'_\kappa(\theta)| \leq |\theta| + H_\kappa(N_\kappa + 1)$, so for $m \geq 0$,

$$F'_{c_\kappa}(m) \leq m + \sqrt{\frac{2}{\pi\kappa}} + H_\kappa(N_\kappa + 1),$$

and integrating from zero to x_0 yields

$$F_{c_\kappa}(x_0) - F_{c_\kappa}(0) \leq \frac{1}{2} x_0^2 + \left[H_\kappa(N_\kappa + 1) + \sqrt{\frac{2}{\pi\kappa}} \right] x_0 \leq C_0 x_0^2, \quad (155)$$

where C_0 is independent of κ , the last inequality following from $H_\kappa(N_\kappa + 1) = O(\kappa^4/\gamma)$ and $x_0 = Y\kappa^4/\gamma$. Combining [\(154\)](#) and [\(155\)](#),

$$F_{c_\kappa}(x_j) - F_{c_\kappa}(0) \geq \varrho_0 [F_{c_\kappa}(x_0) - F_{c_\kappa}(0)], \quad \varrho_0 := \frac{9}{32C_0} > 0. \quad (156)$$

For each fixed covariance, $m \mapsto \mathcal{E}_\kappa(m, c)$ is strictly convex and even, so the unique optimizing mean is $m_\star = 0$. Let $c_\star$ denote the optimizing covariance and set $D_\kappa := \mathcal{E}_\kappa(0, c_\kappa) - \mathcal{E}_\kappa(0, c_\star) \geq 0$. At covariance c_κ ,

$$\Delta(x_j, c_\kappa) = F_{c_\kappa}(x_j) - F_{c_\kappa}(0) + D_\kappa,$$

so after replacing ϱ_0 by $\min\{\varrho_0, 1\}$, [\(156\)](#) and $D_\kappa \geq 0$ imply

$$\Delta(x_j, c_\kappa) \geq \varrho_0 \Delta(x_0, c_\kappa). \quad (157)$$

It remains to replace the ideal states by the actual iterates. On the tube, $|\partial_m \mathcal{E}_\kappa| = |b(m, c)| \leq C\kappa^4/\gamma$, while

$$|\partial_c \mathcal{E}_\kappa(m, c)| = \frac{1}{2} \left| A(m, c) - \frac{1}{c} \right| \leq C(\varepsilon_\kappa + \kappa|q|).$$

Together with [Theorem C.5](#), the mean-value theorem gives

$$\sup_{0 \leq j \leq N_\kappa} |\mathcal{E}_\kappa(m_j, c_j) - \mathcal{E}_\kappa(x_j, c_\kappa)| = o(x_0^2). \quad (158)$$

Since [\(153\)](#) gives $\Delta(x_0, c_\kappa) \geq x_0^2/2$, the $o(x_0^2)$ error is $o(\Delta(x_0, c_\kappa))$ and can be absorbed into [\(157\)](#) after decreasing ϱ_0 by a fixed factor. This proves [\(152\)](#) with ϱ the decreased constant. $\square$

Proof of Theorem 2.12. The regularity and curvature claims follow from [Theorem C.3](#), and the trajectory statement, with $q := \varrho$, is the combination of [Theorems C.5](#) and [C.6](#). The equality $\kappa_\star = \kappa$ is part of [Theorem C.3](#), and $c_0 = 1/\kappa$ lies exactly at the lower edge of the curvature-scale band $[1/\kappa, 1]$. Finally $\Delta t = \gamma/\kappa$ converts κ^2/γ into $\kappa/\Delta t$. $\square$

D The convex-ridge local counterexample

We construct the family of [Theorem 3.15](#) and proves its assertions. The construction is a convex ridge: a one-dimensional convex tent profile evaluated along a radial coordinate of a large block of variables, tilted by a small amount s into one distinguished direction. The radial coordinate concentrates at scale $O(1)$ around a large value, which is what makes the averaged Hessian isotropic after a single scalar tuning, and the tilt is what transfers the collective radial mode into the distinguished direction and produces a large first-Hermite coupling.

Throughout, $Z \sim \mathcal{N}(0, I_{N_\theta})$, $B(z) := \nabla^2 V(z)$, and we work under the whitening conditions of [Theorem 3.1](#),

$$\alpha_\star I \preceq \nabla^2 V(z) \preceq \beta_\star I, \quad \mathbb{E}[\nabla V(Z)] = 0, \quad \mathbb{E}[\nabla^2 V(Z)] = I, \quad (159)$$

so that $\alpha_\star \leq 1 \leq \beta_\star$ and $\kappa_\star = \beta_\star/\alpha_\star$. We write $\mathcal{T}, \mathcal{T}^*, \mathcal{H}$ for the score operators of [Theorem 3.2](#) and

$$\tau_H := \sup_{\|w\|=1} \left\| \mathbb{E}[(w^\top Z)B(Z)] \right\|_F = \|\mathcal{T}^*\|_{\mathbb{R}^{N_\theta} \rightarrow \mathbb{S}(N_\theta)} = \|\mathcal{T}\|_{\mathbb{S}(N_\theta) \rightarrow \mathbb{R}^{N_\theta}}.$$

It is convenient here to use the packed form of the linearized generator (36): substituting $X \mapsto \sqrt{2}X$ identifies $\|\cdot\|_\star$ with the packed norm $\|(v, X)\|_{\text{pack}}^2 := \|v\|^2 + \|X\|_F^2$ and turns $\mathcal{L}_\star$ into

$$\mathcal{L}_\star = \begin{pmatrix} I_{N_\theta} & \mathcal{T}/\sqrt{2} \\ \mathcal{T}^*/\sqrt{2} & I + \mathcal{H}/2 \end{pmatrix} \quad \text{on } \mathbb{R}^{N_\theta} \oplus \mathbb{S}(N_\theta), \quad (160)$$

a unitary change of variables that leaves the spectrum, and hence $\gamma_\star$ and $\Lambda_\star$, unchanged.

D.1 The tent profile and the ridge potential

Let $m \in \mathbb{N}$ be large and set

$$K := (8m)^{1/4}, \quad N_\theta := m + 1, \quad \text{so that} \quad m = \frac{K^4}{8}. \quad (161)$$

Write $z = (z_1, y) \in \mathbb{R} \times \mathbb{R}^m$ and set

$$r := \|y\|, \quad \sigma(y) := \sqrt{m + r^2}, \quad \sigma_0 := \sqrt{2m} = \frac{K^2}{2}, \quad b := \sigma_0 + \frac{1}{2}. \quad (162)$$

Define a convex C^2 function $F : \mathbb{R} \rightarrow \mathbb{R}$ by

$$F''(p) := (K - |p - b|)_+, \quad F'(-\infty) = 0, \quad (163)$$

an additive constant being irrelevant. Since F'' is supported in $[b - K, b + K]$ with $\int_{\mathbb{R}} F'' = K^2$, and since $F'''(p) = -\text{sgn}(p - b)$ on $0 < |p - b| < K$ and $F''' = 0$ elsewhere,

$$0 \leq F'' \leq K, \quad 0 \leq F' \leq K^2, \quad \text{Lip}(F'') \leq 1. \quad (164)$$

For $s \geq 0$ define the tilted ridge

$$\mathcal{U}_s(z_1, y) := F(\sigma(y) + sz_1), \quad p := \sigma(y) + sz_1, \quad \mathbf{g} := \nabla p = \left(s, \frac{y}{\sigma}\right). \quad (165)$$

Since $D^2\sigma(y) = \sigma^{-1}I_m - \sigma^{-3}yy^\top$, the Hessian of $\mathcal{U}_s$ is

$$\mathcal{A}_s := D^2\mathcal{U}_s = F''(p) \mathbf{g}\mathbf{g}^\top + F'(p) \begin{pmatrix} 0 & 0 \\ 0 & D^2\sigma(y) \end{pmatrix}. \quad (166)$$

Both summands are positive semidefinite: $F'' \geq 0$, $F' \geq 0$, and $D^2\sigma \succeq 0$, having eigenvalue $1/\sigma > 0$ on $y^\perp$ and $m/\sigma^3 \geq 0$ along y .

D.2 Radial concentration

Let $Z = (Z_1, Y) \sim \mathcal{N}(0, I_1) \otimes \mathcal{N}(0, I_m)$, continue to write $r = \|Y\|$ and $\sigma = \sqrt{m + r^2}$, and set $\xi := \sigma - \sigma_0$.

Lemma D.1 (Dimension-free radial fluctuations). *For every fixed $j < \infty$ there is a numerical constant C_j with $\|\xi\|_{L^j} \leq C_j$. Moreover*

$$\mathbb{E}\xi = -\frac{\mathbb{E}\xi^2}{2\sigma_0} = O(K^{-2}). \quad (167)$$

Uniformly for $0 \leq s \leq 2K^{-1/2}$, the variable $d := \xi + sZ_1 - \frac{1}{2}$ has dimension-free sub-Gaussian tails; in particular, for every fixed $k < \infty$ there are numerical constants $c_k, C_k > 0$ with

$$\sup_{0 \leq s \leq 2K^{-1/2}} \mathbb{E}[(1 + |d|)^k \mathbf{1}_{\{|d| > K\}}] \leq C_k e^{-c_k K^2}. \quad (168)$$

Proof. The identity $\xi = (r^2 - m)/(\sigma + \sigma_0)$ and the lower bound $\sigma + \sigma_0 \geq \sigma_0 \asymp \sqrt{m}$ give $\|\xi\|_{L^j} \leq \|r^2 - m\|_{L^j} / \sigma_0$, and the centered moments of a χ_m^2 variable satisfy $\|r^2 - m\|_{L^j} \leq C_j \sqrt{m}$, for instance by writing $r^2 - m = \sum_{i=1}^m (Y_i^2 - 1)$ and applying Rosenthal's inequality. Since $\sigma^2 - \sigma_0^2 = 2\sigma_0\xi + \xi^2$ and $\mathbb{E}(\sigma^2 - \sigma_0^2) = \mathbb{E}(r^2 - m) = 0$, taking expectations gives $2\sigma_0\mathbb{E}\xi + \mathbb{E}\xi^2 = 0$, which is (167) after using $\mathbb{E}\xi^2 \leq C$ and $\sigma_0 = K^2/2$. Finally $y \mapsto \sqrt{m + \|y\|^2}$ is 1-Lipschitz, so by Gaussian concentration $\sigma - \mathbb{E}\sigma$ is sub-Gaussian with an absolute variance proxy; combined with $|\mathbb{E}\sigma - \sigma_0| = |\mathbb{E}\xi| = O(K^{-2})$, the variable ξ is uniformly sub-Gaussian, and sZ_1 is sub-Gaussian with variance proxy $s^2 \leq 4/K$. Hence d is uniformly sub-Gaussian with an absolute proxy, so $\mathbb{P}(|d| > K) \leq 2e^{-cK^2}$, and (168) follows by Cauchy-Schwarz against the polynomial weight, whose L^2 norm grows at most polynomially in K . $\square$

D.3 The averaged Hessian and the exact tuning equation

Lemma D.2 (Diagonal average). *For every $s \geq 0$,*

$$\mathbb{E}\mathcal{A}_s = \text{diag}(s^2 \overline{K}(s), \mathcal{G}(s)I_m), \quad (169)$$

where

$$\overline{K}(s) := \mathbb{E}F''(\sigma + sZ_1), \quad \mathcal{G}(s) := \frac{1}{m} \mathbb{E} \left[F''(\sigma + sZ_1) \frac{r^2}{\sigma^2} + F'(\sigma + sZ_1) \left(\frac{m}{\sigma} - \frac{r^2}{\sigma^3} \right) \right].$$

Proof. Write $\mathbb{E}\mathcal{A}_s = \mathbb{E}[F'' \mathbf{g} \mathbf{g}^\top] + \mathbb{E}[F' \text{diag}(0, D^2 \sigma)]$ using (166), with $\mathbf{g} = (s, y/\sigma)$. The (z_1, z_1) entry is $\mathbb{E}[F'' s^2] = s^2 \overline{K}(s)$. The (z_1, y_i) entry is $\mathbb{E}[F'' s y_i / \sigma]$; since F'' and σ depend on y only through r , the integrand is odd under $y_i \mapsto -y_i$, which preserves the Gaussian law, so the entry vanishes. For $i \neq j$ the (y_i, y_j) entry is $\mathbb{E}[F'' y_i y_j / \sigma^2 - F' y_i y_j / \sigma^3] = 0$ by the same reflection. The diagonal entries (y_i, y_i) are equal by exchangeability, so their common value is their average over $i = 1, \dots, m$, which by (166) and $\sum_i y_i^2 = r^2$ is exactly $\mathcal{G}(s)$. $\square$

Lemma D.3 (Uniform averaged-profile estimates). *Uniformly for $0 \leq s \leq 2K^{-1/2}$,*

$$\overline{K}(s) = K + O(1), \quad \mathbb{E}F'(\sigma + sZ_1) = \sigma_0 - \frac{K}{2} + O(1), \quad \mathcal{G}(s) = 1 - \frac{1}{K} + O(K^{-2}). \quad (170)$$

Proof. With $d = \xi + sZ_1 - \frac{1}{2}$ we have $\sigma + sZ_1 = b + d$, and on $\{|d| \leq K\}$ the definition of F gives the exact formulas

$$F''(b + d) = K - |d|, \quad F'(b + d) = \frac{K^2}{2} + Kd - \frac{1}{2}d|d|, \quad (171)$$

the second by integrating the first from b and using $F'(b) = K^2/2$, which holds because F'' is symmetric about b with total mass K^2 . By [Theorem D.1](#), $\mathbb{E}|d| = O(1)$ and $\mathbb{E}[d|d|] = O(1)$ uniformly in s , and the contribution of $\{|d| > K\}$ is exponentially small by (168), even after multiplication by any fixed polynomial in K and d . Therefore $\bar{K}(s) = \mathbb{E}[(K - |d|)\mathbf{1}_{\{|d| \leq K\}}] = K - \mathbb{E}|d| + O(e^{-cK^2}) = K + O(1)$. Also $\mathbb{E}d = \mathbb{E}\xi - \frac{1}{2} = -\frac{1}{2} + O(K^{-2})$ by (167), so (171), together with the exponentially small tail contribution, where $F' \leq K^2$ and $\mathbb{E}[K^2\mathbf{1}_{\{|d| > K\}}] = O(1)$, gives

$$\mathbb{E}F'(\sigma + sZ_1) = \frac{K^2}{2} + K\mathbb{E}d - \frac{1}{2}\mathbb{E}[d|d|] + O(1) = \frac{K^2}{2} + K\left(-\frac{1}{2} + O(K^{-2})\right) + O(1) = \sigma_0 - \frac{K}{2} + O(1),$$

using $|\mathbb{E}[d|d|]| \leq \mathbb{E}d^2 = O(1)$.

For $\mathcal{G}(s)$, decompose it as (I) + (II) - (III) with (I) = $\frac{1}{m}\mathbb{E}[F''r^2/\sigma^2]$, (II) = $\mathbb{E}[F'/\sigma]$ and (III) = $\frac{1}{m}\mathbb{E}[F'r^2/\sigma^3]$. Since $F'' \leq K$ and $r^2/\sigma^2 \leq 1$, we have $0 \leq \text{(I)} \leq K/m = O(K^{-3})$. For the second term,

$$\left\|\frac{1}{\sigma} - \frac{1}{\sigma_0}\right\|_{L^2} = \left\|\frac{\xi}{\sigma\sigma_0}\right\|_{L^2} \leq \frac{\|\xi\|_{L^2}}{\sqrt{2}m} = O(K^{-4})$$

because $\sigma \geq \sqrt{m}$, $\sigma_0 = \sqrt{2m}$ and $\|\xi\|_{L^2} = O(1)$, so with $F' \leq K^2$, (II) = $\mathbb{E}[F'/\sigma_0] + K^2 \cdot O(K^{-4}) = \mathbb{E}F'/\sigma_0 + O(K^{-2})$. Finally $r \mapsto r^2(m + r^2)^{-3/2}$ is maximized at $r^2 = 2m$ with value $2 \cdot 3^{-3/2}m^{-1/2} \leq m^{-1/2}$, so $0 \leq \text{(III)} \leq K^2m^{-3/2} = O(K^{-4})$. Combining with the second estimate of (170) and $\sigma_0 = K^2/2$,

$$\mathcal{G}(s) = \frac{\mathbb{E}F'(\sigma + sZ_1)}{\sigma_0} + O(K^{-2}) = 1 - \frac{K/2}{\sigma_0} + O(\sigma_0^{-1}) + O(K^{-2}) = 1 - \frac{1}{K} + O(K^{-2}). \quad \square$$

Lemma D.4 (Exact isotropic tuning). *For all sufficiently large K there exists $s = s_K \in (0, 2K^{-1/2})$ with*

$$s^2\bar{K}(s) = \mathcal{G}(s). \quad (172)$$

Writing $\mathfrak{a} := s^2\bar{K}(s) = \mathcal{G}(s)$ and $\alpha_\star := 1 - \mathfrak{a}$,

$$s^2 = K^{-1} + O(K^{-2}), \quad \mathfrak{a} = 1 - K^{-1} + O(K^{-2}), \quad \alpha_\star = K^{-1} + O(K^{-2}), \quad (173)$$

and $\mathbb{E}\mathcal{A}_s = \mathfrak{a}I_{N_\theta}$.

Proof. The function $\Theta_K(s) := s^2\bar{K}(s) - \mathcal{G}(s)$ is continuous on $[0, 2K^{-1/2}]$: the integrands are pointwise continuous in s , the integrand defining $\bar{K}$ is bounded by K , and the absolute value of the integrand defining $\mathcal{G}$ is bounded by $\frac{K}{m} + K^2(\frac{1}{\sigma} + \frac{r^2}{m\sigma^3}) \leq \frac{K}{m} + \frac{2K^2}{\sqrt{m}}$, so dominated convergence applies. Now $\Theta_K(0) = -\mathcal{G}(0) < 0$ for large K by (170), while at $s = 2K^{-1/2}$,

$$s^2\bar{K}(s) = \frac{4}{K}(K + O(1)) = 4 + O(K^{-1}), \quad \mathcal{G}(s) = 1 + O(K^{-1}),$$

so $\Theta_K(2K^{-1/2}) = 3 + O(K^{-1}) > 0$, and the intermediate value theorem yields a zero $s_K \in (0, 2K^{-1/2})$. At this zero (170) gives $\mathfrak{a} = \mathcal{G}(s_K) = 1 - K^{-1} + O(K^{-2})$, hence the last two estimates in (173), and in particular $0 < \alpha_\star < 1$ for large K ; dividing $s_K^2\bar{K}(s_K) = \mathfrak{a}$ by $\bar{K}(s_K) = K + O(1)$ gives $s_K^2 = (1 + O(K^{-1}))/K$. Finally (169) together with $s_K^2\bar{K}(s_K) = \mathcal{G}(s_K) = \mathfrak{a}$ gives $\mathbb{E}\mathcal{A}_s = \mathfrak{a}I$. $\square$

D.4 The potential and exact whitening

Fix the tuned value $s = s_K$. Since $|\nabla\mathcal{U}_s| = F'(p)\|\mathfrak{g}\| \leq \sqrt{2}K^2$ is bounded, the vector $\mathfrak{m} := \mathbb{E}[\nabla\mathcal{U}_s(Z)]$ is well defined, and we set

$$V(z) := \frac{\alpha_\star}{2}\|z\|^2 + \mathcal{U}_s(z) - \mathfrak{m}^\top z. \quad (174)$$

Lemma D.5 (Exact whitened equilibrium). *The potential (174) satisfies $\mathbb{E}[\nabla V(Z)] = 0$ and $\mathbb{E}[\nabla^2 V(Z)] = I_{N_\theta}$.*

Proof. Since $\mathbb{E}Z = 0$, $\mathbb{E}[\nabla V(Z)] = \alpha_\star\mathbb{E}Z + \mathbb{E}[\nabla\mathcal{U}_s(Z)] - \mathfrak{m} = 0$, and by [Theorem D.4](#), $\mathbb{E}[\nabla^2 V(Z)] = \alpha_\star I + \mathbb{E}\mathcal{A}_s = (\alpha_\star + \mathfrak{a})I = I$. $\square$

D.5 Ellipticity and a dimension-free Lipschitz Hessian

Lemma D.6 (Uniform ellipticity and condition number). *Let $L_K := \sup_z \|\mathcal{A}_s(z)\|_{\text{op}}$ and $\beta_\star := \alpha_\star + L_K$. For all sufficiently large K ,*

$$\frac{K}{2} \leq L_K \leq 3K, \quad (175)$$

so that $\alpha_\star I \preceq \nabla^2 V(z) \preceq \beta_\star I$ for every z , and

$$\kappa_\star = \frac{\beta_\star}{\alpha_\star} = \Theta(K^2). \quad (176)$$

Proof. Since $\sigma \geq \sqrt{m} = K^2/\sqrt{8}$ and $s^2 \leq 4/K$, for large K we have $\|\mathbf{g}\|^2 = s^2 + r^2/\sigma^2 \leq 1 + s^2 \leq 2$ and $\|D^2\sigma\|_{\text{op}} \leq 1/\sigma \leq \sqrt{8}/K^2$. Using (164) in (166),

$$0 \preceq \mathcal{A}_s \preceq (F'' \|\mathbf{g}\|^2 + F' \|D^2\sigma\|_{\text{op}})I \preceq (2K + \sqrt{8})I \preceq 3K I$$

for large K , which is the upper bound. For the lower bound, choose y with $r^2 = m$, so $\sigma = \sigma_0$, and $z_1 = (b - \sigma_0)/s = 1/(2s)$, so that $p = b$. Then $F''(p) = K$ and $\|\mathbf{g}\|^2 = s^2 + r^2/\sigma_0^2 = s^2 + \frac{1}{2} \geq \frac{1}{2}$, so the rank-one matrix $K\mathbf{g}\mathbf{g}^\top$ has an eigenvalue $K\|\mathbf{g}\|^2 \geq K/2$; the second summand of (166) being positive semidefinite, $L_K \geq K/2$. Since $\mathcal{A}_s \succeq 0$ the lower Hessian bound is $\alpha_\star I$, and it is attained: for any fixed y , taking z_1 sufficiently negative makes $\sigma(y) + sz_1 \leq b - K$, where $F' = F'' = 0$ and $\mathcal{A}_s = 0$, so no larger $\alpha_\star$ is admissible. Finally (173) and (175) give $\kappa_\star = (\alpha_\star + L_K)/\alpha_\star = \Theta(K^2)$. $\square$

For a trilinear form B we write $\|B\|_{\text{inj}} := \sup_{\|u\|=\|v\|=\|w\|=1} |B[u, v, w]|$.

Lemma D.7 (Dimension-free Hessian-Lipschitz bound). *The potential (174) belongs to $C^{2,1}(\mathbb{R}^{N_\theta})$ and satisfies Theorem 2.2 with the admissible constant $L_H = 64$:*

$$\|\nabla^2 V(z) - \nabla^2 V(z')\|_{\text{op}} \leq 64 \|z - z'\|, \quad z, z' \in \mathbb{R}^{N_\theta}. \quad (177)$$

Proof. The function F'' is globally Lipschitz and differentiable almost everywhere with $|F'''| \leq 1$. At every point of differentiability and for unit vectors u, v, w ,

$$\begin{aligned} D^3\mathcal{U}_s[u, v, w] &= F'''(p) \mathbf{g}[u]\mathbf{g}[v]\mathbf{g}[w] \\ &\quad + F''(p) (\mathbf{g}[u]D^2\sigma[v, w] + \mathbf{g}[v]D^2\sigma[u, w] + \mathbf{g}[w]D^2\sigma[u, v]) \\ &\quad + F'(p) D^3\sigma[u, v, w], \end{aligned} \quad (178)$$

where σ is viewed as a function of (z_1, y) independent of z_1 and the middle line collects the three product-rule terms in which exactly one differentiation falls on $\mathbf{g}\mathbf{g}^\top$. A direct differentiation gives, for $u, v, w \in \mathbb{R}^m$,

$$D^3\sigma[u, v, w] = -\frac{(u^\top v)(y^\top w) + (u^\top w)(y^\top v) + (v^\top w)(y^\top u)}{\sigma^3} + \frac{3(y^\top u)(y^\top v)(y^\top w)}{\sigma^5},$$

whence $\|D^3\sigma\|_{\text{inj}} \leq \frac{3r}{\sigma^3} + \frac{3r^3}{\sigma^5} \leq \frac{6}{\sigma^2} \leq \frac{6}{m} = \frac{48}{K^4}$. Using $\|\mathbf{g}\| \leq \sqrt{2}$, $\|D^2\sigma\|_{\text{op}} \leq \sqrt{8}/K^2$ and (164), (178) gives, almost everywhere,

$$\|D^3\mathcal{U}_s\|_{\text{inj}} \leq |F'''| \|\mathbf{g}\|^3 + 3F'' \|\mathbf{g}\| \|D^2\sigma\|_{\text{op}} + F' \|D^3\sigma\|_{\text{inj}} \leq 2\sqrt{2} + \frac{12}{K} + \frac{48}{K^2} \leq 64$$

for all sufficiently large K .

To integrate this almost-everywhere bound into a global one, fix unit vectors u, v and set $f_{u,v}(z) := u^\top \mathcal{A}_s(z)v$. Each entry of $\mathcal{A}_s$ is locally Lipschitz, since F'' is Lipschitz, F' is $C^{1,1}$, and $\sigma, \mathbf{g}, D^2\sigma$ are smooth on $\{\sigma \geq \sqrt{m}\}$, so $f_{u,v}$ is locally Lipschitz and differentiable almost everywhere with $|\nabla f_{u,v}(z)| = |D^3\mathcal{U}_s(z)[u, v, \cdot]| \leq 64$ wherever the gradient exists. For a locally Lipschitz function the global Lipschitz constant equals the essential supremum of $|\nabla f|$: mollifying, $f_{u,v} * \eta_\varepsilon$ is smooth with $|\nabla(f_{u,v} * \eta_\varepsilon)| \leq 64$ everywhere, hence 64-Lipschitz, and letting $\varepsilon \downarrow 0$ preserves the inequality. Taking the supremum over unit u, v gives $\|\mathcal{A}_s(z) - \mathcal{A}_s(z')\|_{\text{op}} \leq 64\|z - z'\|$; the quadratic and linear terms of (174) contribute the constant Hessian $\alpha_* I$, which does not affect the Hessian-Lipschitz seminorm. $\square$

D.6 A large first-Hermite coupling

Define the normalized test direction

$$X := \text{diag}\left(0, \frac{I_m}{\sqrt{m}}\right) \in \mathbb{S}(N_\theta), \quad \|X\|_F = 1. \quad (179)$$

Lemma D.8 (Codazzi conversion of the collective radial mode). *The vector $\mathcal{T}[X]$ points in the distinguished direction e_1 and*

$$\|\mathcal{T}[X]\| = \frac{s}{\sqrt{m}} \mathbb{E}\left[F''(\sigma + sZ_1) \frac{r^2}{\sigma}\right]. \quad (180)$$

Proof. The constant matrix $\alpha_* I$ has zero first-Hermite coefficient, since $\mathbb{E}[\text{Tr}(X \cdot \alpha_* I)Z] = \alpha_* \sqrt{m} \mathbb{E}Z = 0$ using $\text{Tr } X = \sqrt{m}$, so only $\mathcal{A}_s$ contributes and $\mathcal{T}[X] = \mathbb{E}[\text{Tr}(X\mathcal{A}_s)Z]$. Now

$$\text{Tr}(X\mathcal{A}_s) = \frac{1}{\sqrt{m}} \sum_{i=1}^m \partial_{ii}\mathcal{U}_s = \frac{1}{\sqrt{m}} \left(F'' \frac{r^2}{\sigma^2} + F' \left(\frac{m}{\sigma} - \frac{r^2}{\sigma^3} \right) \right)$$

is a function of (Z_1, r) alone; for each i the reflection $y_i \mapsto -y_i$ preserves the law of Z and the value of $\text{Tr}(X\mathcal{A}_s)$ while flipping Y_i , so $\mathbb{E}[\text{Tr}(X\mathcal{A}_s)Y_i] = 0$, every y -component of $\mathcal{T}[X]$ vanishes, and $\mathcal{T}[X] = \mathbb{E}[\text{Tr}(X\mathcal{A}_s)Z_1]e_1$.

Let φ_{N_θ} denote the standard Gaussian density. Since $\mathcal{A}_s$ is bounded, its entries define tempered distributions, and commutation of distributional derivatives gives $\partial_1(\mathcal{A}_s)_{ii} = \partial_i(\mathcal{A}_s)_{1i}$. Consequently

$$\mathbb{E}[Z_1(\mathcal{A}_s)_{ii}] = - \int (\mathcal{A}_s)_{ii} \partial_1 \varphi_{N_\theta} = \langle \partial_1(\mathcal{A}_s)_{ii}, \varphi_{N_\theta} \rangle = \langle \partial_i(\mathcal{A}_s)_{1i}, \varphi_{N_\theta} \rangle = - \int (\mathcal{A}_s)_{1i} \partial_i \varphi_{N_\theta} = \mathbb{E}[Y_i(\mathcal{A}_s)_{1i}],$$

these pairings being justified by inserting compactly supported cutoffs of φ_{N_θ} and letting the cutoff radius tend to infinity. From (166), $(\mathcal{A}_s)_{1i} = sF''(\sigma + sZ_1)Y_i/\sigma$, so summing over $i = 1, \dots, m$ and dividing by $\sqrt{m}$ gives (180), whose right-hand side is nonnegative. $\square$

Lemma D.9 (Size of the coupling). *There is a numerical constant C such that*

$$\|\mathcal{T}[X]\|^2 = \frac{K}{2} + O(1), \quad \tau_H^2 \geq \frac{K}{2} - C. \quad (181)$$

Proof. Since $r^2 = \sigma^2 - m$, define $q(\sigma) := r^2/\sigma = \sigma - m/\sigma$. On $[\sqrt{m}, \infty)$, $q'(\sigma) = 1 + m/\sigma^2 \leq 2$, so q is 2-Lipschitz there, and $q(\sigma_0) = \sigma_0 - m/\sigma_0 = \sigma_0/2 = K^2/4$. Using $0 \leq F'' \leq K$, $|q(\sigma) - q(\sigma_0)| \leq 2|\xi|$ and Theorem D.1,

$$\mathbb{E}[F''q(\sigma)] = q(\sigma_0) \mathbb{E}F'' + O(K \mathbb{E}|\sigma - \sigma_0|) = \overline{K}(s) \frac{K^2}{4} + O(K).$$

Because $\sqrt{m} = K^2/(2\sqrt{2})$, so that $\frac{1}{m} \cdot \frac{K^4}{16} = \frac{1}{2}$, substitution into (180) yields

$$\begin{aligned}\|\mathcal{T}[X]\|^2 &= \frac{s^2}{m} \left(\overline{K}(s) \frac{K^2}{4} + O(K) \right)^2 = \frac{s^2 \overline{K}(s)^2 K^4}{16m} + O\left(\frac{s^2 \overline{K}(s) K^3}{m} \right) + O\left(\frac{s^2 K^2}{m} \right) \\ &= \frac{s^2 \overline{K}(s)^2}{2} + O(K^{-1}) = \frac{\mathfrak{a} \overline{K}(s)}{2} + O(K^{-1}) = \frac{K}{2} + O(1),\end{aligned}$$

using $s^2 \overline{K}(s) = \mathfrak{a}$ from Theorem D.4, $\overline{K}(s) = K + O(1)$, $\mathfrak{a} = 1 + O(K^{-1})$ and $s^2 = O(K^{-1})$. Finally $\|X\|_F = 1$ and $\tau_H = \|\mathcal{T}\|$, so $\tau_H^2 \geq \|\mathcal{T}[X]\|^2 \geq K/2 - C$. $\square$

Remark D.10 (Direct cross-check). The identity (180) can also be checked without commuting mixed weak derivatives. Gaussian integration by parts in Z_1 gives $e_1^\top \mathcal{T}[X] = \frac{s}{\sqrt{m}} \mathbb{E}[F''' \frac{r^2}{\sigma^2} + F''(\frac{m}{\sigma} - \frac{r^2}{\sigma^3})]$ with F''' the almost-everywhere derivative of the Lipschitz function F'' ; for fixed Z_1 , Gaussian integration by parts in Y applied to $\Psi_{Z_1}(y) := F''(\sigma(y) + sZ_1)y/\sigma(y)$, for which $Y^\top \Psi_{Z_1}(Y) = F''r^2/\sigma$ and $\text{div } \Psi_{Z_1} = F''' \frac{r^2}{\sigma^2} + F''(\frac{m}{\sigma} - \frac{r^2}{\sigma^3})$, recovers (180) through $\mathbb{E}[Y^\top \Psi_{Z_1}(Y)] = \mathbb{E}[\text{div } \Psi_{Z_1}(Y)]$. The direct calculation verifies the identity but does not distinguish Hessian fields from general symmetric matrix fields; $\mathcal{T}[X]$ depends only on $\text{Tr}(XB)$, so for the present X any matrix field with the same yy block has the same value. What the Codazzi route additionally supplies is the polarized second-Hermite identity of Theorem A.3, which makes $\mathcal{H}$ and hence $\mathcal{L}_\star$ self-adjoint.

D.7 A nearly-null direction for the linearized generator

Set $\tau_X^2 := \|\mathcal{T}[X]\|^2$ and $h_X := \langle X, \mathcal{H}[X] \rangle_F$.

Lemma D.11 (Second-Hermite asymptotics). *For the matrix X of (179),*

$$h_X = \frac{K}{2} + O(1), \quad h_X - \tau_X^2 = O(1). \quad (182)$$

Proof. Since $XZ = (0, Y/\sqrt{m})$ and $B - I = \alpha_\star I + \mathcal{A}_s - I = \mathcal{A}_s - \mathfrak{a}I$, the polarized identity $\langle X, \mathcal{H}[X] \rangle_F = \mathbb{E}[(XZ)^\top (B(Z) - I)(XZ)]$ of Theorem A.3 gives $h_X = \frac{1}{m} \mathbb{E}[Y^\top (\mathcal{A}_s)_{yy} Y] - \mathfrak{a}$, where by (166)

$$(\mathcal{A}_s)_{yy} = F'' \frac{yy^\top}{\sigma^2} + F' \left(\frac{I_m}{\sigma} - \frac{yy^\top}{\sigma^3} \right).$$

Using $Y^\top yy^\top Y = r^4$ with $y = Y$, $Y^\top Y = r^2$ and $\sigma^2 - r^2 = m$, we get $Y^\top (\mathcal{A}_s)_{yy} Y = F'' \frac{r^4}{\sigma^2} + F' \frac{m r^2}{\sigma^3}$, so

$$h_X = \frac{1}{m} \mathbb{E} \left[F'' \frac{r^4}{\sigma^2} \right] + \mathbb{E} \left[F' \frac{r^2}{\sigma^3} \right] - \mathfrak{a}. \quad (183)$$

Let $\omega := r^2/m$, so that $\frac{r^4}{m\sigma^2} = \frac{\omega^2}{1+\omega} =: \Psi(\omega)$. Since $0 \leq \Psi'(\omega) = 1 - (1+\omega)^{-2} < 1$, the function Ψ is 1-Lipschitz with $\Psi(1) = 1/2$, and $\|\omega - 1\|_{L^2} = \|r^2 - m\|_{L^2}/m = \sqrt{2m}/m = O(K^{-2})$. Therefore

$$\frac{1}{m} \mathbb{E} \left[F'' \frac{r^4}{\sigma^2} \right] = \mathbb{E}[F'' \Psi(\omega)] = \frac{1}{2} \mathbb{E} F'' + O(K \mathbb{E} |\omega - 1|) = \frac{\overline{K}(s)}{2} + O(K^{-1}) = \frac{K}{2} + O(1).$$

For the second term of (183), $\sup_{r \geq 0} r^2(m+r^2)^{-3/2} \leq m^{-1/2}$ and $F' \leq K^2$ give $0 \leq \mathbb{E}[F' r^2/\sigma^3] \leq K^2/\sqrt{m} = 2\sqrt{2} = O(1)$. Since $\mathfrak{a} = O(1)$, this proves $h_X = K/2 + O(1)$. Finally, by Theorem D.9 and $s^2 \overline{K} = \mathfrak{a}$, $\tau_X^2 = \frac{s^2 \overline{K}^2}{2} + O(K^{-1}) = \frac{\mathfrak{a} \overline{K}}{2} + O(K^{-1}) = \frac{\mathfrak{a} K}{2} + O(1)$, so

$$h_X - \tau_X^2 = \frac{(1 - \mathfrak{a}) \overline{K}}{2} + O(1) = \frac{\alpha_\star K}{2} + O(1) = O(1)$$

by (173). $\square$

Lemma D.12 (Explicit Rayleigh direction). *Set $\Xi := X/\sqrt{2}$ and $v := -\mathcal{T}[X]/2$. Then, for $\mathcal{L}_\star$ in the packed form (160),*

$$\frac{\langle (v, \Xi), \mathcal{L}_\star(v, \Xi) \rangle_{\text{pack}}}{\|v\|^2 + \|\Xi\|_{\text{F}}^2} = \frac{2 + h_X - \tau_X^2}{2 + \tau_X^2}. \quad (184)$$

Proof. Since $\|X\|_{\text{F}} = 1$ we have $\|v\|^2 = \tau_X^2/4$ and $\|\Xi\|_{\text{F}}^2 = 1/2$. By (160) the packed quadratic form is $\|v\|^2 + 2 \langle v, (\mathcal{T}/\sqrt{2})\Xi \rangle + \|\Xi\|_{\text{F}}^2 + \frac{1}{2} \langle \Xi, \mathcal{H}[\Xi] \rangle_{\text{F}}$, the two off-diagonal contributions combining by self-adjointness. Now $(\mathcal{T}/\sqrt{2})\Xi = \mathcal{T}[X]/2$ and $\langle \Xi, \mathcal{H}[\Xi] \rangle_{\text{F}} = \frac{1}{2} \langle X, \mathcal{H}[X] \rangle_{\text{F}} = h_X/2$, so the numerator is

$$\frac{\tau_X^2}{4} + 2 \left(-\frac{\mathcal{T}[X]}{2} \right)^\top \frac{\mathcal{T}[X]}{2} + \frac{1}{2} + \frac{h_X}{4} = \frac{\tau_X^2}{4} - \frac{\tau_X^2}{2} + \frac{1}{2} + \frac{h_X}{4} = \frac{2 + h_X - \tau_X^2}{4},$$

while the denominator is $\tau_X^2/4 + 1/2 = (2 + \tau_X^2)/4$. $\square$

D.8 Smooth realization and proof of the theorem

Lemma D.13 (Smooth realization). *After increasing the numerical threshold K_0 if necessary, the potentials (174) may be chosen in $C^\infty(\mathbb{R}^{N_\theta})$ while retaining every conclusion above, including the common Hessian-Lipschitz constant $L_H = 64$.*

Proof. Let $f_K(p) := (K - |p - b|)_+$ and fix an even nonnegative $\eta \in C_c^\infty((-1, 1))$ with $\int \eta = 1$. For $\varepsilon \in (0, 1]$ set $\eta_\varepsilon(w) := \varepsilon^{-1} \eta(w/\varepsilon)$ and $f_{K,\varepsilon} := f_K * \eta_\varepsilon$, and define $F'_{K,\varepsilon}(-\infty) = 0$ with $F''_{K,\varepsilon} = f_{K,\varepsilon}$. Then $0 \leq f_{K,\varepsilon} \leq K$, $\text{Lip}(f_{K,\varepsilon}) \leq 1$, $\int_{\mathbb{R}} f_{K,\varepsilon} = K^2$ and $\|f_{K,\varepsilon} - f_K\|_{L^\infty} \leq \varepsilon$. Moreover $F'_{K,\varepsilon} = F'_K * \eta_\varepsilon$, and since F'_K has a 1-Lipschitz derivative while η_ε has zero first moment, Taylor's formula with Lipschitz remainder gives $\|F'_{K,\varepsilon} - F'_K\|_{L^\infty} \leq \varepsilon^2/2$. Consequently, uniformly for $0 \leq s \leq 2K^{-1/2}$,

$$\left| \bar{K}_\varepsilon(s) - \bar{K}(s) \right| \leq \varepsilon, \quad \left| \mathbb{E} F'_{K,\varepsilon}(\sigma + sZ_1) - \mathbb{E} F'_K(\sigma + sZ_1) \right| \leq \frac{\varepsilon^2}{2},$$

and $|\mathcal{G}_\varepsilon(s) - \mathcal{G}(s)| \leq \frac{\varepsilon}{m} + \frac{\varepsilon^2}{2}(m^{-1/2} + m^{-3/2})$, so all estimates of Theorem D.3 are unchanged. The same intermediate-value argument retunes $s = s_{K,\varepsilon}$ and restores exact whitening. Also $f_{K,\varepsilon}(b) \geq K - \varepsilon$, so the ellipticity lower bound remains of order K , and $\|f'_{K,\varepsilon}\|_{L^\infty} \leq 1$, so the proof of Theorem D.7 gives the same admissible constant $L_H = 64$. Repeating the coupling and Rayleigh calculations proves all the asserted asymptotics. $\square$

Proof of Theorem 3.15. Fix m with $K = (8m)^{1/4}$ large and let V be the potential (174), made smooth by Theorem D.13. Theorem D.5 gives the whitening conditions (159), Theorem D.6 gives Theorem 2.1 together with $\kappa_\star = \Theta(K^2)$, that is $c_\kappa K^2 \leq \kappa_\star \leq C_\kappa K^2$, and Theorem D.7 gives Theorem 2.2 with $L_H = 64$. Theorem D.9 gives $\tau_H^2 \geq K/2 - C_1$.

For the gap upper bound, $\mathcal{L}_\star$ is self-adjoint, so the Rayleigh–Ritz variational principle and Theorem D.12 give $\gamma_\star = \lambda_{\min}(\mathcal{L}_\star) \leq (2 + h_X - \tau_X^2)/(2 + \tau_X^2)$. By Theorem D.11 the numerator is at most a numerical constant $2 + C_1$, and by Theorem D.9 the denominator is at least a positive multiple of K for large K , so $\gamma_\star \leq C_1/K$, and $\kappa_\star = \Theta(K^2)$ turns this into $\gamma_\star \leq C_1'/\sqrt{\kappa_\star}$.

For the two-sided statements, $\beta_\star - 1 = \Theta(K)$ by (175) and (173), while the dimension-dependent branch of Γ in (35) is of order $\sqrt{N_\theta} \log \beta_\star = \Theta(K^2 \log K)$, so $\Gamma = \beta_\star - 1 = \Theta(K)$: the family saturates the $\beta_\star - 1$ branch. The Schur bound (77) then gives $\tau_H^2 \leq 1 + \Gamma$, which together with Theorem D.9 yields $\tau_H^2 = \Theta(K) = \Theta(\kappa_\star^{1/2})$. Likewise the lower bound $\gamma_\star \geq \max\{\alpha_\star, 1/(4 + \Gamma)\}$ of Theorem 3.3, with $\alpha_\star = \Theta(K^{-1})$ by (173) and $1/(4 + \Gamma) = \Theta(K^{-1})$, matches the Rayleigh upper bound and gives $\gamma_\star = \Theta(K^{-1}) = \Theta(\kappa_\star^{-1/2})$.

Finally, suppose a positive function $c(L_H)$ satisfied $\gamma_\star \geq c(L_H)/(1 + \log \kappa_\star)$ in every dimension under [Theorem 2.1](#) and [Theorem 2.2](#). Applying it with $L_H = 64$ along the family would give $c(64)/(1 + \log \kappa_\star) \leq C_1/K$; but $\log \kappa_\star = \Theta(\log K)$ and $K^{-1} = o((\log K)^{-1})$, a contradiction for large K . $\square$

Remark D.14 (Minimum regularity and non-circularity). The first- and second-Hermite identities use only commutation of distributional derivatives, so $V \in C^2$ with a bounded Hessian suffices; no classical third derivative is required. The tent potential (174) is $C^{2,1}$ with a globally Lipschitz Hessian, and [Theorem D.13](#) provides the C^∞ realization. The gap upper bound uses only the explicit Rayleigh direction of [Theorem D.12](#), the independently established asymptotics of h_X and τ_X , and Rayleigh–Ritz; it invokes no spectral lower bound, no third-derivative moment bound and no logarithmic coupling inequality, so the lower bound (37) may be applied afterwards to read off order-sharpness without entering the argument.

E Deterministic one-step estimates

We prove the two one-step estimates used in [Section 2](#), namely [Theorem E.4](#) for the Riemannian retraction and [Theorem E.7](#) for the KL/Bregman step, together with the relative smoothness of $\mathcal{E}_2$ that identifies the Bregman geometry of the latter. Throughout $a = (m, C)$ is a Gaussian parameter, $g := g(a)$, $A := A(a)$, $R := R(a) = C^{-1} - A$, and $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$ as in (6). Only convexity and β -smoothness of V , that is $0 \preceq \nabla^2 V \preceq \beta I$, are used; both hold under [Theorem 2.1](#).

E.1 Continuous covariance envelope

Lemma E.1 (Continuous covariance bootstrap). *Under [Theorem 2.1](#), if the initial covariance satisfies $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$, then along (1), with $\lambda_0 := \lambda_{0,\min}$ and $\lambda_{\max} := \max\{\lambda_{0,\max}, 1/\alpha\}$,*

$$C_t \succeq \ell(t)I, \quad \ell(t) := \frac{\lambda_0 e^t}{1 + \beta \lambda_0 (e^t - 1)}, \quad C_t \preceq \lambda_{\max} I.$$

In particular $C_t \succeq (2\beta)^{-1}I$ for all $t \geq \log_+(1/(\beta\lambda_0))$.

Proof. By (14) and $A(a_s) \preceq \beta I$,

$$P_t \preceq (e^{-t}\lambda_0^{-1} + \beta(1 - e^{-t}))I,$$

which gives the lower bound after inversion. For the upper bound, $A(a_t) \succeq \alpha I$ shows that no eigenvalue of C_t can cross $1/\alpha$ from below and that any eigenvalue above $1/\alpha$ is pushed downward. The curvature-scale consequence follows by requiring the decaying initial-precision term to be no larger than β . $\square$

E.2 Two scalar inequalities

Lemma E.2. *For every $w \in [-1, 1]$,*

$$\left(e^{w/2} - 1 - \frac{w}{2}\right) + \frac{1}{2}(e^{w/2} - 1)^2 \leq \frac{w^2}{2}.$$

Proof. Put $v = w/2 \in [-1/2, 1/2]$. Since $(e^v - 1) + \frac{1}{2}(e^v - 1)^2 = \frac{1}{2}(e^{2v} - 1)$, the desired inequality is equivalent, with $x = 2v = w \in [-1, 1]$, to $e^x \leq 1 + x + x^2$. Indeed

$$e^x - 1 - x = \sum_{k \geq 2} \frac{x^k}{k!} \leq x^2 \sum_{k \geq 2} \frac{|x|^{k-2}}{k!} \leq (e - 2)x^2 < x^2,$$

which proves the claim. $\square$

Lemma E.3. Let $K \geq 1$, $0 < \Delta t \leq 1/K$, $\mu \in [0, K]$, and $b := \sqrt{(1 + \Delta t)/(1 + \Delta t \mu)}$. Then

$$\mu(b - 1) - \log b + \frac{K}{2}(b - 1)^2 \leq -\frac{1}{2\Delta t}(b^2 - 1 - 2 \log b). \quad (185)$$

Proof. Because $K \leq 1/\Delta t$ and $(b - 1)^2 \geq 0$, it suffices to replace K on the left by $1/\Delta t$. The relation between b and μ is $\mu = ((1 + \Delta t)b^{-2} - 1)/\Delta t$. Multiplying the resulting inequality by Δt and moving all terms to the left, define

$$G(b) := \Delta t \mu(b - 1) - \Delta t \log b + \frac{1}{2}(b - 1)^2 + \frac{1}{2}(b^2 - 1 - 2 \log b),$$

so that $G(1) = 0$ and a direct differentiation gives

$$G'(b) = \frac{(b - 1)(2b^3 - (1 + \Delta t)(b + 2))}{b^3}.$$

Since $0 \leq \Delta t \mu \leq \Delta t K \leq 1$, we have $b \in [\sqrt{(1 + \Delta t)/2}, \sqrt{1 + \Delta t}]$. The polynomial $p(b) := 2b^3 - (1 + \Delta t)(b + 2)$ is convex on $(0, \infty)$, and at the two endpoints of this interval

$$p(\sqrt{(1 + \Delta t)/2}) = -2(1 + \Delta t) < 0, \quad p(\sqrt{1 + \Delta t}) = (1 + \Delta t)(\sqrt{1 + \Delta t} - 2) < 0,$$

using $\Delta t \leq 1$. Convexity therefore implies $p(b) < 0$ throughout the interval, so $G'(b) \geq 0$ for $b \leq 1$ and $G'(b) \leq 0$ for $b \geq 1$. Thus G attains its maximum at $b = 1$, where it vanishes, proving (185). $\square$

E.3 Retraction smoothness and the Riemannian descent estimate

Lemma E.4 (Retraction smoothness and one-step descent). Assume $0 \preceq \nabla^2 V \preceq \beta I$ and $C \preceq \Lambda I$, and set $K := \beta \Lambda \geq 1$. For a tangent vector $\zeta = (u, U)$ at $a = (m, C)$ write $W := C^{-1/2} U C^{-1/2}$. If $\|W\|_{\text{op}} \leq 1$, then

$$\mathcal{E}(\text{Retr}_a(\zeta)) \leq \mathcal{E}(a) + \langle \text{grad } \mathcal{E}(a), \zeta \rangle_a + K \|\zeta\|_a^2. \quad (186)$$

Consequently, for the update (20) with $0 < \Delta t \leq 1/K$,

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \Delta t(1 - K\Delta t)\mathbf{G}(a_n)^2,$$

and in particular $\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{\Delta t}{2}\mathbf{G}(a_n)^2$ whenever $\Delta t \leq 1/(2K)$.

Proof of Theorem E.4. Let $Z \sim \mathcal{N}(0, I)$ and introduce the coupling

$$X := m + C^{1/2}Z \sim \rho_a, \quad Y := m + u + C^{1/2}e^{W/2}Z \sim \rho_{\text{Retr}_a(\zeta)},$$

which is legitimate because $e^{W/2}$ is symmetric, so Y is Gaussian with mean $m + u$ and covariance $C^{1/2}e^W C^{1/2}$. Write $E_W := e^{W/2} - I$ and $M := C^{1/2}A C^{1/2}$. The entropy change is exact,

$$\mathcal{E}_1(\text{Retr}_a(\zeta)) - \mathcal{E}_1(a) = -\frac{1}{2} \text{Tr } W,$$

and by β -smoothness of V ,

$$\mathcal{E}_2(\text{Retr}_a(\zeta)) - \mathcal{E}_2(a) \leq \mathbb{E}[\nabla V(X)^\top (Y - X)] + \frac{\beta}{2} \mathbb{E} \|Y - X\|^2.$$

Because $Y - X = u + C^{1/2}E_W Z$, $\mathbb{E}[\nabla V(X)] = -g$, and Gaussian integration by parts gives $\mathbb{E}[Z \nabla V(X)^\top] = C^{1/2}A$, the first-order contribution equals $-g^\top u + \text{Tr}(E_W M)$. The cross term in the quadratic contribution vanishes, and $C \preceq \Lambda I$ yields

$$\frac{\beta}{2} \mathbb{E} \|Y - X\|^2 \leq \frac{K}{2} u^\top C^{-1} u + \frac{K}{2} \|E_W\|_{\text{F}}^2.$$

On the other hand, since $\text{grad } \mathcal{E}(a) = (-Cg, -CRC)$ and $C^{1/2}RC^{1/2} = I - M$,

$$\langle \text{grad } \mathcal{E}(a), \zeta \rangle_a = -g^\top u - \frac{1}{2} \text{Tr}((I - M)W).$$

Subtracting this linear term from the entropy and covariance contributions leaves

$$\Theta := \text{Tr} \left[\left(e^{W/2} - I - \frac{W}{2} \right) M \right] + \frac{K}{2} \|e^{W/2} - I\|_F^2.$$

The matrix $N_W := e^{W/2} - I - W/2$ is positive semidefinite while $0 \preceq M \preceq KI$, hence $\text{Tr}(N_W M) \leq K \text{Tr } N_W$. Diagonalizing W and applying [Theorem E.2](#) eigenvalue by eigenvalue gives

$$\Theta \leq K \sum_i \left[e^{w_i/2} - 1 - \frac{w_i}{2} + \frac{1}{2}(e^{w_i/2} - 1)^2 \right] \leq \frac{K}{2} \|W\|_F^2.$$

Combining the mean and covariance estimates and using $\|\zeta\|_a^2 = u^\top C^{-1}u + \frac{1}{2} \|W\|_F^2$ proves [\(186\)](#).

For the deterministic step take $\zeta = -\Delta t \text{grad } \mathcal{E}(a) = (\Delta t Cg, \Delta t CRC)$, so that

$$W = \Delta t C^{1/2}RC^{1/2} = \Delta t(I - C^{1/2}AC^{1/2}).$$

The eigenvalues of the matrix in parentheses lie in $[1 - K, 1]$, so $\|W\|_{\text{op}} \leq \Delta t K \leq 1$. Moreover $\langle \text{grad } \mathcal{E}(a), \zeta \rangle_a = -\Delta t G(a)^2$ and $\|\zeta\|_a^2 = \Delta t^2 G(a)^2$, and substitution into [\(186\)](#) gives

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \Delta t(1 - K\Delta t)G(a_n)^2.$$

The final assertion is the case $\Delta t \leq 1/(2K)$. □

E.4 The KL/Bregman step: variational characterization

Lemma E.5. *The update [\(21\)](#) is the forward-backward proximal step*

$$a_{n+1} = \arg \min_{a'=(m',C')} \left\{ \Delta t \langle \nabla \mathcal{E}_2(a_n), a' - a_n \rangle + \Delta t \mathcal{E}_1(a') + \text{KL}(\rho_{a'} \parallel \rho_{a_n}) \right\}, \quad (187)$$

which linearizes the potential $\mathcal{E}_2$ explicitly and treats the entropy $\mathcal{E}_1$ implicitly, and its minimizer is

$$m_{n+1} = m_n + \Delta t C_n g(a_n), \quad C_{n+1} = (1 + \Delta t)(C_n^{-1} + \Delta t A(a_n))^{-1}.$$

Proof. The Gaussian KL splits as

$$\text{KL}(\rho_{a'} \parallel \rho_{a_n}) = \frac{1}{2}(m_n - m')^\top C_n^{-1}(m_n - m') + \frac{1}{2}[\text{Tr}(C_n^{-1}C') - N_\theta + \log \det C_n - \log \det C'],$$

so the objective in [\(187\)](#) decouples in m' and C' . The m' -part is $-\Delta t g(a_n)^\top (m' - m_n) + \frac{1}{2}(m' - m_n)^\top C_n^{-1}(m' - m_n)$, minimized at $m' - m_n = \Delta t C_n g(a_n)$. The C' -part, dropping C' -independent terms and using $\nabla_C \mathcal{E}_2 = \frac{1}{2}A$ and $\mathcal{E}_1(C') = -\frac{1}{2} \log \det C'$, is

$$\frac{\Delta t}{2} \text{Tr}(A(a_n)C') + \frac{1}{2} \text{Tr}(C_n^{-1}C') - \frac{1+\Delta t}{2} \log \det C' = \frac{1}{2} \text{Tr}((C_n^{-1} + \Delta t A(a_n))C') - \frac{1+\Delta t}{2} \log \det C',$$

whose stationarity condition $\frac{1}{2}(C_n^{-1} + \Delta t A(a_n)) = \frac{1+\Delta t}{2}(C')^{-1}$ gives the stated covariance. Both objectives are strictly convex in their arguments, since $C_n^{-1} \succ 0$ and $-\log \det$ is convex, so the stationary points are the minimizers. □

E.5 Relative smoothness of the potential part

Proposition E.6. *Under Theorem 2.1, let $a_n = (m_n, C_n)$ satisfy $\lambda_{\min} I \preceq C_n \preceq \lambda_{\max} I$. Then $\mathcal{E}_2$ is L -smooth with respect to the local Bregman divergence $D(\cdot, a_n) = \text{KL}(\rho \cdot \|\rho_{a_n})$ of (22): for every $a = (m, C)$ with $\lambda_{\min} I \preceq C \preceq \lambda_{\max} I$,*

$$\mathcal{E}_2(a) \leq \mathcal{E}_2(a_n) + \langle \nabla \mathcal{E}_2(a_n), a - a_n \rangle + L D(a, a_n), \quad L = \beta \lambda_{\max}.$$

Proof. We consider the displacement from $a_n = (m_n, C_n)$ to $a = (m, C)$ and denote the direction $u = m - m_n$ and $H = B - B_n$, where $M := C_n^{-1/2} C C_n^{-1/2}$, $B_n := C_n^{1/2}$ and $B := C_n^{1/2} M^{1/2}$, so that $B B^\top = C$. Under this formulation we write $\mathcal{E}_2(m, B) := \mathbb{E}_{\xi \sim \mathcal{N}(0, I)}[V(m + B\xi)]$ and denote the trajectory

$$\varphi(t) = \mathbb{E}_{\xi \sim \mathcal{N}(0, I)}[V(m_n + tu + (B_n + tH)\xi)].$$

Writing $X_t = m_n + tu + (B_n + tH)\xi$,

$$\varphi'(t) = \mathbb{E}[\nabla V(X_t)^\top (u + H\xi)], \quad \varphi''(t) = \mathbb{E}[(u + H\xi)^\top \nabla^2 V(X_t)(u + H\xi)],$$

and $\nabla^2 V \preceq \beta I$ gives $\varphi''(t) \leq \beta(\|u\|^2 + \|H\|_F^2)$. As Taylor's expansion gives $\varphi(1) = \varphi(0) + \varphi'(0) + \int_0^1 (1-s)\varphi''(s)ds$,

$$\varphi(1) - \varphi(0) - \varphi'(0) \leq \frac{\beta}{2}(\|u\|^2 + \|H\|_F^2),$$

which is equivalent to

$$\mathcal{E}_2(a) - \mathcal{E}_2(a_n) - \langle \nabla_m \mathcal{E}_2(a_n), u \rangle - \langle \nabla_B \mathcal{E}_2(a_n), H \rangle \leq \frac{\beta}{2}(\|u\|^2 + \|H\|_F^2).$$

Denote the gradient of $\mathcal{E}_2$ with respect to C by

$$G_n := \nabla_C \mathcal{E}_2(a_n) = \frac{1}{2} \mathbb{E}_{\rho_{a_n}}[\nabla^2 V(\theta)] = \frac{1}{2} A(a_n),$$

where $G_n \succeq 0$ because V is convex. Since $\nabla_B \mathcal{E}_2(a_n) = \nabla_C \mathcal{E}_2(a_n) \cdot 2B_n = 2G_n B_n$,

$$\begin{aligned} \langle \nabla_B \mathcal{E}_2(a_n), H \rangle &= \text{Tr}((2G_n B_n)^\top H) = \text{Tr}(G_n (B_n H^\top + H B_n)) \\ &\leq \text{Tr}(G_n (B_n H^\top + H B_n + H H^\top)) = \langle \nabla_C \mathcal{E}_2(a_n), C - C_n \rangle, \end{aligned}$$

the inequality coming from $G_n \succeq 0$ and $H H^\top \succeq 0$. We therefore obtain

$$\mathcal{E}_2(a) \leq \mathcal{E}_2(a_n) + \langle \nabla \mathcal{E}_2(a_n), a - a_n \rangle + \frac{\beta}{2}(\|u\|^2 + \|H\|_F^2).$$

It remains to connect $\frac{\beta}{2}(\|u\|^2 + \|H\|_F^2)$ to the divergence D . For the mean part,

$$\frac{\beta}{2} \|u\|^2 \leq \beta \lambda_{\max} \cdot \frac{1}{2} u^\top C_n^{-1} u \leq \beta \lambda_{\max} D^m(m, m_n).$$

For the covariance part, $D^C(C, C_n) = \frac{1}{2}(\text{Tr } M - \log \det M - N_\theta)$, and the eigenvalues $\eta_1, \dots, \eta_{N_\theta}$ of M satisfy $\eta_i \in [\lambda_{\min}/\lambda_{\max}, \lambda_{\max}/\lambda_{\min}]$. This gives

$$D^C(C, C_n) = \frac{1}{2} \sum_{i=1}^{N_\theta} (\eta_i - \log \eta_i - 1) \geq \frac{1}{2} \sum_{i=1}^{N_\theta} (\sqrt{\eta_i} - 1)^2 = \frac{1}{2} \|M^{1/2} - I\|_F^2 \geq \frac{1}{2\lambda_{\max}} \|H\|_F^2,$$

where the first inequality comes from $\log x \leq x - 1$ and the second from $H = B - B_n = C_n^{1/2}(M^{1/2} - I)$ together with $C_n \preceq \lambda_{\max} I$. These collectively give $\frac{\beta}{2}(\|u\|^2 + \|H\|_F^2) \leq L D(a, a_n)$ with $L = \beta \lambda_{\max}$, which completes the proof. $\square$

E.6 Forward-KL descent and the Wasserstein-gradient lower bound

Proposition E.7 (Forward-KL descent and Wasserstein-gradient control). *Assume $0 \preceq \nabla^2 V \preceq \beta I$ and $LI \preceq C_n \preceq \Lambda I$, put $K := \beta \Lambda \geq 1$, and let $0 < \Delta t \leq 1/K$. Then the update (21) satisfies*

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t} \text{KL}(\rho_{a_{n+1}} \| \rho_{a_n}),$$

and

$$\text{KL}(\rho_{a_{n+1}} \| \rho_{a_n}) \geq \frac{L \Delta t^2}{4(1 + \Delta t K)^2} \left\| \text{grad}^{W_2} \mathcal{E}(a_n) \right\|_{W_2}^2.$$

Proof of Theorem E.7. Set

$$M := C_n^{1/2} A(a_n) C_n^{1/2}, \quad B := \sqrt{1 + \Delta t} (I + \Delta t M)^{-1/2},$$

so that $0 \preceq M \preceq KI$, B is a symmetric function of M , and $C_{n+1} = C_n^{1/2} B^2 C_n^{1/2}$ by Theorem E.5. With $\xi \sim \mathcal{N}(0, I)$, couple

$$X := m_n + C_n^{1/2} \xi \sim \rho_{a_n}, \quad Y := m_n + \Delta t C_n g + C_n^{1/2} B \xi \sim \rho_{a_{n+1}}.$$

The entropy change is $\mathcal{E}_1(a_{n+1}) - \mathcal{E}_1(a_n) = -\log \det B$. Writing $E := B - I$, the β -smoothness estimate of the proof of Theorem E.4 applies verbatim with E in place of $e^{W/2} - I$, since both are symmetric functions of M and the computation uses only $V(Y) \leq V(X) + \nabla V(X)^\top (Y - X) + \frac{\beta}{2} \|Y - X\|^2$, $\mathbb{E}[\nabla V(X)] = -g$, the Price identity $\mathbb{E}[\xi \nabla V(X)^\top] = C_n^{1/2} A$, and $C_n \preceq \Lambda I$. It gives

$$\mathcal{E}_2(a_{n+1}) - \mathcal{E}_2(a_n) \leq -\Delta t g^\top C_n g + \text{Tr}(EM) + \frac{K \Delta t^2}{2} g^\top C_n g + \frac{K}{2} \|E\|_F^2.$$

Because $K \Delta t \leq 1$, the mean contribution is at most $-\frac{\Delta t}{2} g^\top C_n g$, which is exactly $-\Delta t^{-1}$ times the mean part $\frac{\Delta t^2}{2} g^\top C_n g$ of $\text{KL}(\rho_{a_{n+1}} \| \rho_{a_n})$. For the covariance part, let μ_i be the eigenvalues of M and $b_i = \sqrt{(1 + \Delta t)/(1 + \Delta t \mu_i)}$ the aligned eigenvalues of B ; adding the entropy change, the covariance contribution is

$$\sum_i \left[\mu_i (b_i - 1) - \log b_i + \frac{K}{2} (b_i - 1)^2 \right],$$

which by Theorem E.3 is at most $-\frac{1}{2\Delta t} \sum_i (b_i^2 - 1 - 2 \log b_i)$. Since $C_n^{-1} C_{n+1}$ has eigenvalues b_i^2 , this is exactly $-\Delta t^{-1}$ times the covariance part $\frac{1}{2} \sum_i (b_i^2 - 1 - \log b_i^2)$ of $\text{KL}(\rho_{a_{n+1}} \| \rho_{a_n})$. Adding the mean and covariance parts proves the descent inequality.

For the lower bound, the mean part of the forward divergence is

$$\frac{1}{2} (m_{n+1} - m_n)^\top C_n^{-1} (m_{n+1} - m_n) = \frac{\Delta t^2}{2} g^\top C_n g \geq \frac{L \Delta t^2}{2} \|g\|^2.$$

For the covariance part, set $S := C_n^{1/2} R C_n^{1/2} = I - M$ and

$$Q := C_n^{1/2} C_{n+1}^{-1} C_n^{1/2} = \frac{I + \Delta t M}{1 + \Delta t}.$$

For $q > 0$ one has

$$q^{-1} - 1 + \log q \geq \frac{(q - 1)^2}{2 \max\{q, 1\}^2}, \tag{188}$$

since for $q \geq 1$ the left side is $\int_1^q (t - 1) t^{-2} dt \geq (q - 1)^2 / (2q^2)$, while for $q \leq 1$ it is $\int_q^1 (1 - t) t^{-2} dt \geq (1 - q)^2 / 2$. The eigenvalues of Q satisfy

$$q_i - 1 = -\frac{\Delta t}{1 + \Delta t} s_i, \quad \max\{q_i, 1\} \leq \frac{1 + \Delta t K}{1 + \Delta t},$$

with $s_i = 1 - \mu_i$ the eigenvalues of S , so applying (188) eigenvalue by eigenvalue gives

$$\text{KL}_C(\rho_{a_{n+1}} \| \rho_{a_n}) = \frac{1}{2} \sum_i (q_i^{-1} - 1 + \log q_i) \geq \frac{\Delta t^2}{4(1 + \Delta t K)^2} \|S\|_F^2.$$

Finally $\|S\|_F^2 = \text{Tr}(C_n R C_n R) = \text{Tr}(C_n [R C_n R]) \geq L \text{Tr}(R C_n R)$, because $R C_n R \succeq 0$ and $C_n \succeq L I$. Adding the mean and covariance parts gives

$$\text{KL}(\rho_{a_{n+1}} \| \rho_{a_n}) \geq \frac{L \Delta t^2}{4(1 + \Delta t K)^2} (\|g\|^2 + \text{Tr}(R C_n R)) = \frac{L \Delta t^2}{4(1 + \Delta t K)^2} \|\text{grad}^{W_2} \mathcal{E}(a_n)\|_{W_2}^2,$$

which is the stated bound. $\square$

E.7 Warm-start estimates

Lemma E.8 (Wasserstein covariance bootstrap). *Under Theorem 2.1 and $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$, along (24) one has $C_t \preceq \lambda_{\max} I$ and*

$$C_t \succeq \ell_{\text{BW}}(t) I, \quad \ell_{\text{BW}}(t) := \frac{1}{\beta} + \left(\lambda_0 - \frac{1}{\beta} \right) e^{-2\beta t}.$$

Consequently, for $c \in (0, 1)$, the time $T_{\text{BW}}(c) := \frac{1}{2\beta} \log \frac{1-\beta\lambda_0}{1-c}$ if $\lambda_0 < c/\beta$, and $T_{\text{BW}}(c) := 0$ otherwise, satisfies $C_{T_{\text{BW}}(c)} \succeq \frac{c}{\beta} I$. For fixed c the warm-up time is $O(1/\beta)$, with no logarithmic dependence on λ_0 .

Discretely, given $a = (m, C)$ with $C \preceq \lambda_{\max} I$ and a transport stepsize $0 < \eta \leq 1/\beta$, the Bures–Wasserstein forward–backward step (25) satisfies $C^+ \succeq \eta I$, $C^+ \preceq \lambda_{\max} I$, and $\mathcal{E}(m^+, C^+) \leq \mathcal{E}(m, C)$.

Proof. For an eigenvalue μ_i of C_t with normalized eigenvector u_i , $\dot{\mu}_i = 2(1 - \mu_i h_i)$ with $h_i = u_i^\top A(a_t) u_i \in [\alpha, \beta]$, hence $\dot{\mu}_i \geq 2(1 - \beta \mu_i)$, and scalar comparison with $\dot{\ell}_{\text{BW}} = 2(1 - \beta \ell_{\text{BW}})$, $\ell_{\text{BW}}(0) = \lambda_0$, gives $\mu_i \geq \ell_{\text{BW}}$. If $\mu_i \geq 1/\alpha$ then $\dot{\mu}_i \leq 2(1 - \alpha \mu_i) \leq 0$, which gives the upper bound. Finally $\ell_{\text{BW}}(t) \geq c/\beta$ is equivalent to $e^{-2\beta t} \leq (1 - c)/(1 - \beta \lambda_0)$.

For the discrete step, $\psi_\eta(x) := \frac{1}{2}(x + 2\eta + \sqrt{x(x + 4\eta)})$ is increasing with $\psi_\eta(0) = \eta$ and (25) applies it to the eigenvalues of $\tilde{C} \succeq 0$, so $C^+ \succeq \eta I$. Since $0 \preceq I - \eta A(a) \preceq (1 - \eta \alpha) I$ and $\lambda_{\max} \geq 1/\alpha$, $\tilde{C} \preceq (1 - \eta \alpha)^2 \lambda_{\max} I \preceq (1 - \eta/\lambda_{\max})^2 \lambda_{\max} I$, and the scalar identity $\psi_\eta((1 - \eta/L)^2 L) = L$ at $L = \lambda_{\max}$ with monotonicity gives $C^+ \preceq \lambda_{\max} I$. The step is the Bures–Wasserstein forward–backward step for the splitting (6) with the entropy treated backward, so under $\eta \leq 1/\beta$ the forward–backward descent inequality of Diao et al. [7] gives the energy decrease. $\square$

Lemma E.9 (Curvature-scale covariance rescue). *Under Theorem 2.1, Theorem 2.14 gives $\frac{1}{\beta} I \preceq C_0 \preceq \frac{1}{\alpha} I$.*

Proof. We have $\alpha I \preceq H_{\text{in}} \preceq \beta I$, and the matrix inverse is operator antitone on $\mathbb{S}_{++}(N_\theta)$, so $\frac{1}{\beta} I \preceq H_{\text{in}}^{-1} \preceq \frac{1}{\alpha} I$. $\square$

E.8 Deferred statements of the local theory

Lemma E.10 (Entropy-enhanced coercivity). *Write $B := C^{1/2}$ with eigenvalues $s_1, \dots, s_{N_\theta}$ and let $\phi_{\alpha_\star}(s) := s - 1 - \log s + \frac{\alpha_\star}{2}(s - 1)^2$ be as in Section 3.2. Then*

$$\Delta(m, C) \geq \frac{\alpha_\star}{2} \|m\|^2 + \sum_{i=1}^{N_\theta} \phi_{\alpha_\star}(s_i). \quad (189)$$

For a Gaussian target this is an equality, and $\Delta_G(m, C) = \frac{1}{2} \|m\|^2 + \sum_i \varphi(c_i)$ with $\varphi(c) := \frac{1}{2}(c - 1 - \log c)$ and c_i the eigenvalues of C .

Proof. With $X = m + BZ$ we have $\Delta = \mathbb{E}[\widehat{V}(m + BZ) - \widehat{V}(Z)] - \log \det B$. By $\alpha_\star$ -strong convexity, $\widehat{V}(m + BZ) - \widehat{V}(Z) \geq \nabla \widehat{V}(Z)^\top (m + (B - I)Z) + \frac{\alpha_\star}{2} \|m + (B - I)Z\|^2$. Taking expectations, $\mathbb{E}[\nabla \widehat{V}(Z)]^\top m = 0$, $\mathbb{E}[\nabla \widehat{V}(Z)^\top (B - I)Z] = \text{Tr}(B - I)$ by Gaussian integration by parts, and $\mathbb{E} \|m + (B - I)Z\|^2 = \|m\|^2 + \|B - I\|_F^2$. Hence $\Delta \geq \frac{\alpha_\star}{2} \|m\|^2 + \text{Tr}(B - I) - \log \det B + \frac{\alpha_\star}{2} \|B - I\|_F^2$, which diagonalizes to (189). For a Gaussian target the convexity step is an equality and $\phi_1(s) = \frac{1}{2}(s^2 - 1 - \log s^2) = \varphi(c)$ with $c = s^2$. $\square$

Proposition E.11 (Exact Gaussian Riemannian map). *For a Gaussian target the update (20) in whitened coordinates reduces to $m_{n+1} = (I - \Delta t C_n)m_n$ and, in the eigenbasis of C_n , to $c^+ = c e^{\Delta t(1-c)}$, with scalar factors $r_m^R(c) = (1 - \Delta t c)^2$ and*

$$r_C^R(c) = \left(\frac{c e^{\Delta t(1-c)} - 1}{c - 1} \right)^2, \quad r_C^R(1) = (1 - \Delta t)^2.$$

For $0 < \Delta t < 2$ and $q \in (0, 1)$ with $q \geq (1 - \Delta t)^2$, let $I_{\Delta t}^R(q) = [\ell, u]$ be the connected component containing 1 of $\{c > 0 : r_m^R(c) \leq q, r_C^R(c) \leq q\}$. If in addition

$$\Delta t \leq \min \left\{ 1, \frac{\log u}{u - 1} \right\}$$

and $\text{spec}(C_n) \subseteq I_{\Delta t}^R(q)$, then $\|a_{n+k} - a_\star\|_\star^2 \leq q^k \|a_n - a_\star\|_\star^2$ for all $k \geq 0$.

Proof. For a Gaussian target $F_C(a) = C - C^2 = -C(C - I)$, so $C^{-1/2}F_C C^{-1/2} = I - C$ and the retraction gives $C^+ = C^{1/2}e^{\Delta t(I-C)}C^{1/2} = C e^{\Delta t(I-C)}$, since C and $I - C$ commute; in the eigenbasis this is $c^+ = c e^{\Delta t(1-c)}$, and expanding at $c = 1$ gives $r_C^R(1) = (1 - \Delta t)^2$. On $I_{\Delta t}^R(q)$ both factors are at most q ; the first stepsize condition prevents a point below 1 from overshooting above 1 and the second prevents points in $(1, u]$ from crossing below 1, so $c \mapsto c e^{\Delta t(1-c)}$ keeps $[\ell, u]$ forward invariant and the per-step contraction propagates. $\square$

Lemma E.12 (Pointwise one-step contraction). *Fix $\bullet \in \{\text{R}, \text{KL}\}$ and a level $\delta > 0$ satisfying the radius condition $K_{\text{disc}}(\delta) \mathfrak{b}_{\star, \bullet}(\delta) \sqrt{\delta} \leq \gamma_\bullet$ of Theorem 3.12. If $0 < \Delta t \leq 2/(4 + \Gamma)$, then every $a \in \mathcal{U}_\delta$ satisfies*

$$\|T_\bullet(a) - a_\star\|_{\star, \bullet} \leq \left(1 - \frac{1}{2} \Delta t \gamma_\bullet \right) \|a - a_\star\|_{\star, \bullet}.$$

No energy-descent hypothesis is used.

Proof. With $\xi := a - a_\star$ and $a \in \mathcal{U}_\delta \subseteq \mathcal{V}_\delta$, the integral Taylor identity of the proof of Theorem 3.12 holds at the single point a . Its linear part obeys $\|M_\bullet \xi\|_{\star, \bullet} \leq (1 - \Delta t \gamma_\bullet) \|\xi\|_{\star, \bullet}$ by Theorem 3.4, whose only stepsize requirement is $\Delta t \leq 2/(4 + \Gamma)$, and the segment $a_\star + s\xi$ lies in $\mathcal{V}_\delta$, so the remainder is at most $\frac{1}{2} \Delta t K_{\text{disc}}(\delta) \|\xi\|_{\star, \bullet}^2 \leq \frac{1}{2} \Delta t \gamma_\bullet \|\xi\|_{\star, \bullet}$ by the radius condition and $\|\xi\|_{\star, \bullet} \leq \mathfrak{b}_{\star, \bullet}(\delta) \sqrt{\delta}$. $\square$

E.9 The computable entry gates

We prove Theorem 3.14. The two gates are the two basic inequalities of Section 2 read in the direction of a stopping test.

Proof of Theorem 3.14. Write $R(a) = C^{-1} - A(a)$ for the precision residual, so that the Bures–Wasserstein residual of the statement is

$$\mathcal{R}_{\text{BW}}(a)^2 = \|g(a)\|^2 + \text{Tr}(R(a)C R(a)) = \left\| \text{grad}^{W_2} \mathcal{E}(a) \right\|_{W_2}^2,$$

which is exactly the left-hand side of the Wasserstein Polyak–Łojasiewicz inequality (12). That inequality therefore reads $\mathcal{R}_{\text{BW}}(a)^2 \geq 2\alpha \Delta(a)$, and consequently $\mathcal{R}_{\text{BW}}(a_n)^2 \leq 2\alpha \delta_{\text{loc}}$ forces

$\Delta(a_n) \leq \delta_{\text{loc}}$. Every quantity entering $\mathcal{R}_{\text{BW}}(a_n)$ is an expectation under ρ_{a_n} together with C_n and α , so the test uses no information about $a_\star$.

For the intrinsic test, suppose $C_n \succeq \ell_n I$. The comparison (13) gives $\mathsf{G}(a_n)^2 \geq \alpha \ell_n \Delta(a_n)$, so $\mathsf{G}(a_n)^2 \leq \alpha \ell_n \delta_{\text{loc}}$ again forces $\Delta(a_n) \leq \delta_{\text{loc}}$. The first inequality in (13) also gives $\mathcal{R}_{\text{BW}}(a_n)^2 \leq 2\mathsf{G}(a_n)^2/\ell_n$, so a triggered Fisher–Rao gate triggers the Bures–Wasserstein gate as well, which is the asserted comparison between the two tests.

Finally, both deterministic schemes decrease the energy at the stepsizes of Theorems 2.9 and 2.11, and the flow decreases it by (11). Hence $\Delta(a_k) \leq \Delta(a_n) \leq \delta_{\text{loc}}$ for every $k \geq n$, so once either test triggers the iterate remains in the certified sublevel set permanently. $\square$

Proposition E.13 (Exact Gaussian spectral gates). *Let the target be Gaussian with precision H , so that $C_\star = H^{-1}$, and put $S_n := H^{1/2}C_nH^{1/2}$, computable from C_n and H without the target mean. Then, for $0 < \rho \leq 2$, the condition $\lambda_{\min}(S_n) \geq \rho/2$ certifies instantaneous rate $\geq \rho$ for the flow and is forward invariant, each covariance eigenvalue obeying $\dot{s} = s(1-s)$ and moving monotonically toward 1. For the KL scheme, $\text{spec}(S_n) \subseteq I_{\Delta t}^{\text{KL}}(q)$ certifies the per-step contraction $\|a_{n+1} - a_\star\|_{\star, \Delta t}^2 \leq q \|a_n - a_\star\|_{\star, \Delta t}^2$ and is forward invariant.*

Proof. S_n is the whitened covariance $\hat{C}_n$. For the flow, $q_c = q_G \geq 2\lambda_{\min}(\hat{C}_n) = 2\lambda_{\min}(S_n)$ by Theorem 3.6, and for $\rho \leq 2$ the threshold $\rho/2$ lies below 1, so the scalar dynamics $\dot{s} = s(1-s)$ preserves it. The KL statement is Theorem 3.11 in whitened coordinates. $\square$

F Proofs for the stochastic theory

We prove the statements of Section 4 deferred there. Throughout we use the notation of that section: $g(a) = -\mathbb{E}_{\rho_a}[\nabla V]$, $A(a) = \mathbb{E}_{\rho_a}[\nabla^2 V]$, $R(a) = C^{-1} - A(a)$, the intrinsic fluctuation Ψ of Theorem 4.3, and the minibatch estimators (46).

F.1 The rescued initializer

Lemma F.1 (Rescue energy for nonquadratic targets). *Under Theorem 2.1, the quadratic rescue of Theorem 2.14 gives $\Delta(a_0)$ deterministic and finite. Under the additional Theorem 2.2, writing $r_{\text{in}} := \|m_{\text{in}} - m_\star\|$,*

$$\|m_0 - m_\star\| \leq \frac{L_H}{2\alpha} \left(r_{\text{in}}^2 + \frac{N_\theta}{\alpha} \right), \quad \|C_0 - C_\star\|_{\text{F}} \leq \frac{\sqrt{N_\theta} L_H}{\alpha^2} \left(r_{\text{in}} + \sqrt{N_\theta/\alpha} \right),$$

and hence

$$\Delta(a_0) \leq \frac{\beta L_H^2}{8\alpha^2} \left(r_{\text{in}}^2 + \frac{N_\theta}{\alpha} \right)^2 + \frac{\beta^2 N_\theta L_H^2}{4\alpha^4} \left(r_{\text{in}} + \sqrt{N_\theta/\alpha} \right)^2,$$

which is quadratic in L_H and vanishes when $L_H = 0$.

Proof of Theorem F.1. Write $d := m_{\text{in}} - m_\star$ and $\bar{H} := \int_0^1 \nabla^2 V(m_\star + td) dt$, so that $\nabla V(m_{\text{in}}) = \nabla V(m_\star) + \bar{H}d$ by the integral Taylor form. Note that $m_\star$ is the optimal variational mean, not the mode of the target: the stationarity condition is $\mathbb{E}_{\rho_{a_\star}}[\nabla V] = 0$, which does not force $\nabla V(m_\star) = 0$. We first bound $\nabla V(m_\star)$. With $X = m_\star + C_\star^{1/2}Z$ and Taylor expansion, $\nabla V(X) = \nabla V(m_\star) + \nabla^2 V(m_\star)C_\star^{1/2}Z + r(X)$ with $\|r(X)\| \leq \frac{L_H}{2} \|C_\star^{1/2}Z\|^2$. Taking $\mathbb{E}_{\rho_{a_\star}}$, the linear term vanishes and $\mathbb{E}_{\rho_{a_\star}}[\nabla V] = 0$, so

$$\|\nabla V(m_\star)\| \leq \frac{L_H}{2} \mathbb{E} \|C_\star^{1/2}Z\|^2 = \frac{L_H}{2} \text{Tr } C_\star \leq \frac{L_H N_\theta}{2\alpha}. \quad (190)$$

For the mean, since $C_0^{-1} = H_{\text{in}}$,

$$m_0 - m_\star = d - H_{\text{in}}^{-1}(\nabla V(m_\star) + \bar{H}d) = H_{\text{in}}^{-1}(H_{\text{in}} - \bar{H})d - H_{\text{in}}^{-1}\nabla V(m_\star).$$

Hessian-Lipschitzness gives $\|H_{\text{in}} - \bar{H}\| \leq \int_0^1 L_H(1-t)r_{\text{in}}dt = \frac{L_H}{2}r_{\text{in}}$, and $\|H_{\text{in}}^{-1}\| \leq 1/\alpha$, so with (190), $\|m_0 - m_\star\| \leq \frac{1}{\alpha} \cdot \frac{L_H}{2}r_{\text{in}}^2 + \frac{1}{\alpha} \cdot \frac{L_H N_\theta}{2\alpha} = \frac{L_H}{2\alpha}(r_{\text{in}}^2 + \frac{N_\theta}{\alpha})$. For the covariance, with $C_\star^{-1} = A(a_\star) = \mathbb{E}_{\rho_{a_\star}}[\nabla^2 V]$,

$$\|H_{\text{in}} - C_\star^{-1}\| \leq \mathbb{E}_{\rho_{a_\star}} \|\nabla^2 V(m_{\text{in}}) - \nabla^2 V(X)\| \leq L_H \mathbb{E}_{\rho_{a_\star}} \|m_{\text{in}} - X\| \leq L_H(r_{\text{in}} + \sqrt{N_\theta/\alpha}),$$

using $\mathbb{E}_{\rho_{a_\star}} \|m_{\text{in}} - X\| \leq r_{\text{in}} + \sqrt{\text{Tr } C_\star}$ and $\text{Tr } C_\star \leq N_\theta/\alpha$. From $C_0 - C_\star = H_{\text{in}}^{-1}(C_\star^{-1} - H_{\text{in}})C_\star$ and $\|X\|_{\text{F}} \leq \sqrt{N_\theta} \|X\|_{\text{op}}$, $\|C_0 - C_\star\|_{\text{F}} \leq \frac{\sqrt{N_\theta} L_H}{\alpha^2}(r_{\text{in}} + \sqrt{N_\theta/\alpha})$. It remains to bound the energy gap. Since $a_\star$ is stationary, the gap is second order in the parameter error. Parameterize $f(m, B) := \mathbb{E}_Z[V(m + BZ)] - \log \det B$, so $f(m, B) = \mathcal{E}(m, BB^\top) + \text{const}$ and $\Delta(a_0) = f(m_0, B_0) - f(m_\star, B_\star)$ with $B_0 := C_0^{1/2}$, $B_\star := C_\star^{1/2}$. Along the affine path $m_s = m_\star + s\delta m$, $B_s = B_\star + s\delta B$ with $\delta m = m_0 - m_\star$ and $\delta B = B_0 - B_\star$ symmetric, the first derivative vanishes at $s = 0$: $\partial_m f = \mathbb{E}[\nabla V] = 0$ at $a_\star$, and by Stein's identity $\frac{d}{ds} \mathbb{E}[V(m_\star + B_s Z)]|_0 = \text{Tr}(\delta B \mathbb{E}[\nabla^2 V(X)]B_\star) = \text{Tr}(\delta B B_\star^{-1}) = \frac{d}{ds} \log \det B_s|_0$. For the second derivatives,

$$\frac{d^2}{ds^2} \mathbb{E}[V(m_s + B_s Z)] = \mathbb{E}[(\delta m + \delta B Z)^\top \nabla^2 V(X_s)(\delta m + \delta B Z)] \leq \beta(\|\delta m\|^2 + \|\delta B\|_{\text{F}}^2),$$

the cross term vanishing by $\mathbb{E}Z = 0$, and with $M_s := B_s^{-1}\delta B$, $\frac{d^2}{ds^2}[-\log \det B_s] = \text{Tr}(M_s^2) \leq \|M_s\|_{\text{F}}^2 \leq \beta \|\delta B\|_{\text{F}}^2$, using $\text{Tr}(M^2) \leq \|M\|_{\text{F}}^2$ and $\|B_s^{-1}\|_{\text{op}} \leq \sqrt{\beta}$, both C_0 and $C_\star$ lying on the band $\succeq \frac{1}{\beta}I$. Taylor with vanishing first derivative gives $\Delta(a_0) \leq \frac{\beta}{2} \|m_0 - m_\star\|^2 + \beta \|C_0^{1/2} - C_\star^{1/2}\|_{\text{F}}^2$, and the square root is Frobenius-Lipschitz on the band, eigenvalue divided differences of $\sqrt{\cdot}$ on $[\frac{1}{\beta}, \infty)$ being at most $\frac{\sqrt{\beta}}{2}$, so $\|C_0^{1/2} - C_\star^{1/2}\|_{\text{F}} \leq \frac{\sqrt{\beta}}{2} \|C_0 - C_\star\|_{\text{F}}$. Substituting the two parameter bounds gives the stated estimate; every term carries L_H^2 , so $\Delta(a_0) = 0$ when $L_H = 0$. $\square$

F.2 The global one-step estimates

Lemma F.2 (Self-bounding estimate for the intrinsic fluctuation). *Under Theorem 2.1, if $C \preceq \Lambda I$ then*

$$\Psi(a) \leq 2\beta\Lambda N_\theta + \beta\Lambda \|C^{1/2}R(a)C^{1/2}\|_{\text{F}}^2 \leq 2\beta\Lambda N_\theta + 2\beta\Lambda G(a)^2. \quad (191)$$

Proof. Write $B_X := C^{1/2}\nabla^2 V(X)C^{1/2}$ and $S_A := C^{1/2}A(a)C^{1/2} = \mathbb{E}[B_X]$. First, $\Psi(a) = \mathbb{E} \text{Tr}(B_X^2) - \text{Tr}(S_A^2) \leq \mathbb{E} \text{Tr}(B_X^2)$. Second, $0 \preceq B_X \preceq \beta C \preceq \beta\Lambda I$ almost surely, and $0 \preceq M \preceq LI$ implies $M^2 \preceq LM$, so $B_X^2 \preceq \beta\Lambda B_X$ and $\mathbb{E} \text{Tr}(B_X^2) \leq \beta\Lambda \text{Tr}(S_A)$. Third, $s \leq 2 + (s-1)^2$ for every real s , so $\text{Tr}(S_A) \leq 2N_\theta + \|S_A - I\|_{\text{F}}^2$; since $S_A - I = -C^{1/2}R(a)C^{1/2}$, this gives the first inequality, and $\|C^{1/2}R(a)C^{1/2}\|_{\text{F}}^2 \leq 2G(a)^2$ gives the second. $\square$

Lemma F.3 (Hessian-Lipschitz control of the intrinsic fluctuation). *Under Theorem 2.2, the anisotropic bound $\Psi(a) \leq L_H^2 \text{Tr}(C^2) \text{Tr } C \leq L_H^2 \|C\|_{\text{op}} (\text{Tr } C)^2$ holds. On the band $C \preceq \frac{1}{\alpha}I$,*

$$\Psi(a) \leq \Psi_H := \frac{N_\theta^2 L_H^2}{\alpha^3} = N_\theta^2 \bar{L}_H^2. \quad (192)$$

Proof. Write $X = m + C^{1/2}Z$ and $F(Z) := C^{1/2}\nabla^2 V(m + C^{1/2}Z)C^{1/2}$. Gaussian Poincaré applied componentwise gives $\Psi(a) = \mathbb{E} \|F(Z) - \mathbb{E}F(Z)\|_{\text{F}}^2 \leq \mathbb{E} \|D_Z F(Z)\|_{\text{HS}}^2$. For a unit vector u , $D_Z F(Z)[u] = C^{1/2}M_u C^{1/2}$ with $M_u := D(\nabla^2 V)(X)[C^{1/2}u]$ symmetric and $\|M_u\|_{\text{op}} \leq L_H(u^\top C u)^{1/2}$. In the eigenbasis of C with eigenvalues c_i , Cauchy-Schwarz on the weights gives $\|C^{1/2}M_u C^{1/2}\|_{\text{F}}^2 = \sum_{ij} c_i c_j (M_u)_{ij}^2 \leq \text{Tr}(C^2 M_u^2) \leq \|M_u\|_{\text{op}}^2 \text{Tr}(C^2)$, so $\|D_Z F(Z)[u]\|_{\text{F}}^2 \leq L_H^2 (u^\top C u) \text{Tr}(C^2)$; summing over an orthonormal basis gives $\|D_Z F(Z)\|_{\text{HS}}^2 \leq L_H^2 \text{Tr } C \text{Tr}(C^2)$, and $\text{Tr}(C^2) \leq \|C\|_{\text{op}} \text{Tr } C$ gives the coarser form. On $C \preceq \frac{1}{\alpha}I$ one has $\text{Tr}(C^2) \leq N_\theta/\alpha^2$ and $\text{Tr } C \leq N_\theta/\alpha$. $\square$

Lemma F.4 (The KL resolvent update is an exponential retraction). Fix $a = (m, C)$ with $\frac{1}{\beta}I \preceq C \preceq \frac{1}{\alpha}I$ and $\alpha I \preceq \bar{H} \preceq \beta I$, and set $P := C^{1/2}\bar{H}C^{1/2}$, so that $\frac{1}{\kappa}I \preceq P \preceq \kappa I$. Write $\bar{S} := I - P$ and define the logarithmic tangent

$$W(P) := \log(1 + \Delta t) I - \log(I + \Delta t P). \quad (193)$$

Then the KL covariance update $C^+ = (1 + \Delta t)(C^{-1} + \Delta t \bar{H})^{-1}$ satisfies

$$C^+ = C^{1/2}e^{W(P)}C^{1/2}, \quad \text{equivalently} \quad a^+ = \text{Retr}_a(\Delta t C \bar{b}, C^{1/2}W(P)C^{1/2}). \quad (194)$$

Moreover, with $\mathcal{R}(P) := W(P) - \Delta t \bar{S}$,

$$\|W(P)\|_{\text{op}} \leq \Delta t \kappa, \quad \|W(P)\|_{\text{F}} \leq \Delta t \|\bar{S}\|_{\text{F}}, \quad \|\mathcal{R}(P)\|_{\text{F}} \leq \kappa \Delta t^2 \|\bar{S}\|_{\text{F}}, \quad (195)$$

so for $\Delta t \leq 1/(8\kappa)$ the tangent $\zeta := (\Delta t C \bar{b}, C^{1/2}W(P)C^{1/2})$ satisfies $\|W(P)\|_{\text{op}} \leq \frac{1}{8} < 1$ and hence the scope condition of (53).

Proof. Since $C^{-1} + \Delta t \bar{H} = C^{-1/2}(I + \Delta t P)C^{-1/2}$, we have $C^+ = C^{1/2}[(1 + \Delta t)(I + \Delta t P)^{-1}]C^{1/2}$, and, $\log(1 + \Delta t)I$ and $\log(I + \Delta t P)$ being commuting functions of P , $e^{W(P)} = (1 + \Delta t)(I + \Delta t P)^{-1}$; the retraction form follows from $\text{Retr}_a(u, U) = (m + u, C^{1/2} \exp(C^{-1/2}UC^{-1/2})C^{1/2})$ with $U = C^{1/2}W(P)C^{1/2}$. The band $\frac{1}{\kappa}I \preceq P \preceq \kappa I$ is $\alpha C \preceq P \preceq \beta C$ combined with $\frac{1}{\beta}I \preceq C \preceq \frac{1}{\alpha}I$. Each eigenvalue $p \in [\frac{1}{\kappa}, \kappa]$ of P contributes $w(p) = \log(1 + \Delta t) - \log(1 + \Delta t p) = \int_p^1 \frac{\Delta t}{1 + \Delta t r} dr$, and $0 \leq \frac{\Delta t}{1 + \Delta t r} \leq \Delta t$ for $r \geq 0$ gives $|w(p)| \leq \Delta t |1 - p|$, whence the first two bounds of (195). For the remainder, $w(p) - \Delta t(1 - p) = -\Delta t^2 \int_p^1 \frac{r}{1 + \Delta t r} dr$ and $0 \leq \frac{r}{1 + \Delta t r} \leq r \leq \kappa$ give $|w(p) - \Delta t(1 - p)| \leq \kappa \Delta t^2 |1 - p|$; summing over eigenvalues gives the third. $\square$

Lemma F.5 (Riemannian stochastic one-step descent). Assume Theorem 2.1 and $\frac{1}{2\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$. For one step of (48) with $0 < \Delta t \leq 1/\kappa$,

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \Delta t \left(1 - \kappa \Delta t \left(1 + \frac{2}{B}\right)\right) G(a_n)^2 + \frac{3\kappa \Delta t^2}{2B} \Psi(a_n). \quad (196)$$

Proof of Theorem F.5. By Theorem 4.2(1) the band $C_n \preceq \frac{1}{\alpha}I$ holds pathwise, so the scope condition $\|C_n^{-1/2}UC_n^{-1/2}\|_{\text{op}} \leq 1$ of the retraction-smoothness estimate (53) holds pathwise for the tangent $\zeta_n = -\Delta t \widehat{\text{grad}} \mathcal{E}(a_n)$ because $\Delta t \kappa \leq 1$, and (53) with $\Lambda = \frac{1}{\alpha}$, hence $L = 2\beta\Lambda = 2\kappa$, gives pathwise

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) + \langle \text{grad } \mathcal{E}(a_n), \zeta_n \rangle_{a_n} + \kappa \Delta t^2 \left\| \widehat{\text{grad}} \mathcal{E}(a_n) \right\|_{a_n}^2.$$

Taking $\mathbb{E}[\cdot \mid \mathcal{F}_n]$, conditional unbiasedness (47) gives $\mathbb{E}[\langle \text{grad } \mathcal{E}(a_n), \widehat{\text{grad}} \mathcal{E}(a_n) \rangle_{a_n} \mid \mathcal{F}_n] = G(a_n)^2$, and the intrinsic variance bound (51) gives $\mathbb{E}[\left\| \widehat{\text{grad}} \mathcal{E}(a_n) \right\|_{a_n}^2 \mid \mathcal{F}_n] \leq (1 + \frac{2}{B})G(a_n)^2 + \frac{3}{2B}\Psi(a_n)$. Collecting terms gives (196). $\square$

Lemma F.6 (KL stochastic one-step descent). Assume Theorem 2.1 and $\frac{1}{\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$. For one step of (49) with $0 < \Delta t \leq 1/(8\kappa)$,

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \frac{\Delta t}{2} G(a_n)^2 + \frac{7\kappa \Delta t^2}{4B} \Psi(a_n). \quad (197)$$

Proof of Theorem F.6. Suppress the index n and condition on $\mathcal{F}_n$. Write $A := A(a)$, $R := C^{-1} - A$, $S := C^{1/2}RC^{1/2}$, $P_0 := C^{1/2}AC^{1/2} = I - S$, and $D := P - P_0$, so that $\bar{S} = S - D$ with $\mathbb{E}[D \mid \mathcal{F}] = 0$ and $\mathbb{E}[\|D\|_{\text{F}}^2 \mid \mathcal{F}] = \Psi(a)/B$ by Theorem 4.3 and minibatch independence. Set

$$M := g(a)^\top C g(a), \quad V := \|S\|_{\text{F}}^2, \quad \sigma^2 := \frac{\Psi(a)}{B}, \quad \text{so that} \quad G(a)^2 = M + \frac{1}{2}V.$$

On the band $C \preceq \frac{1}{\alpha}I$ the retraction-smoothness estimate (53) applies with $\Lambda = \frac{1}{\alpha}$ and $L = 2\kappa$, and by Theorem F.4 the tangent $\zeta = (\Delta t C \bar{b}, C^{1/2}W(P)C^{1/2})$ meets its scope condition, so pathwise $\mathcal{E}(a^+) \leq \mathcal{E}(a) + \langle \text{grad } \mathcal{E}(a), \zeta \rangle_a + \kappa \|\zeta\|_a^2$. The first-order term evaluates to $\langle \text{grad } \mathcal{E}(a), \zeta \rangle_a = -\Delta t g^\top C \bar{b} - \frac{1}{2} \text{Tr}(S W(P))$: the mean block uses $\text{grad}_m \mathcal{E} = -Cg$ in the metric $v^\top C^{-1}v$, and the covariance block uses $\text{grad}_C \mathcal{E} = -CRC$ in the metric $\frac{1}{2} \text{Tr}(C^{-1} \cdot C^{-1} \cdot)$, giving $\frac{1}{2} \text{Tr}(C^{-1}(-CRC)C^{-1}C^{1/2}W(P)C^{1/2}) = -\frac{1}{2} \text{Tr}(S W(P))$. Likewise $\|\zeta\|_a^2 = \Delta t^2 \bar{b}^\top C \bar{b} + \frac{1}{2} \|W(P)\|_F^2$. Hence, pathwise,

$$\mathcal{E}(a^+) - \mathcal{E}(a) \leq -\Delta t g^\top C \bar{b} - \frac{1}{2} \text{Tr}(S W(P)) + \kappa \Delta t^2 \bar{b}^\top C \bar{b} + \frac{\kappa}{2} \|W(P)\|_F^2.$$

Write $\bar{b} = g + \varepsilon_b$ with $\mathbb{E}[\varepsilon_b | \mathcal{F}] = 0$. By (51) and independence, $\mathbb{E}[\varepsilon_b^\top C \varepsilon_b | \mathcal{F}] \leq (V + \Psi(a))/B = \frac{V}{B} + \sigma^2$, so $\mathbb{E}[\bar{b}^\top C \bar{b} | \mathcal{F}] \leq M + \frac{V}{B} + \sigma^2$ and

$$\mathbb{E}[-\Delta t g^\top C \bar{b} + \kappa \Delta t^2 \bar{b}^\top C \bar{b} | \mathcal{F}] \leq -\Delta t(1 - \kappa \Delta t)M + \frac{\kappa \Delta t^2}{B}V + \kappa \Delta t^2 \sigma^2. \quad (198)$$

For the covariance drift, with $W(P) = \Delta t \bar{S} + \mathcal{R}(P)$ we have $-\frac{1}{2} \text{Tr}(S W(P)) = -\frac{\Delta t}{2} \text{Tr}(S \bar{S}) - \frac{1}{2} \text{Tr}(S \mathcal{R}(P))$, and $\mathbb{E}[\bar{S} | \mathcal{F}] = S$ gives $\mathbb{E}[\text{Tr}(S \bar{S}) | \mathcal{F}] = V$. For the remainder, Cauchy–Schwarz, (195), $\mathbb{E}[\|\bar{S}\|_F^2 | \mathcal{F}] = V + \sigma^2$, Jensen, and $x\sqrt{x^2 + y^2} \leq x^2 + \frac{1}{2}y^2$ with $x = \sqrt{V}$, $y = \sigma$ give $|\mathbb{E}[\text{Tr}(S \mathcal{R}(P)) | \mathcal{F}]| \leq \kappa \Delta t^2(V + \frac{1}{2}\sigma^2)$, so

$$\mathbb{E}[-\frac{1}{2} \text{Tr}(S W(P)) | \mathcal{F}] \leq -\frac{\Delta t}{2}V + \frac{\kappa \Delta t^2}{2}V + \frac{\kappa \Delta t^2}{4}\sigma^2. \quad (199)$$

Likewise, from $\|W(P)\|_F \leq \Delta t \|\bar{S}\|_F$ we get $\mathbb{E}[\|W(P)\|_F^2 | \mathcal{F}] \leq \Delta t^2(V + \sigma^2)$, so

$$\frac{\kappa}{2} \mathbb{E}[\|W(P)\|_F^2 | \mathcal{F}] \leq \frac{\kappa \Delta t^2}{2}V + \frac{\kappa \Delta t^2}{2}\sigma^2. \quad (200)$$

Adding (198), (199) and (200) to Δ_n ,

$$\mathbb{E}[\Delta_{n+1} | \mathcal{F}] \leq \Delta_n - \Delta t(1 - \kappa \Delta t)M - \Delta t \left[\frac{1}{2} - \kappa \Delta t(1 + \frac{1}{B}) \right] V + \frac{7}{4} \kappa \Delta t^2 \sigma^2.$$

For $\Delta t \leq 1/(8\kappa)$ one has $1 - \kappa \Delta t \geq \frac{7}{8} \geq \frac{1}{2}$ and, since $B \geq 1$, $\frac{1}{2} - \kappa \Delta t(1 + \frac{1}{B}) \geq \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. Thus $\mathbb{E}[\Delta_{n+1} | \mathcal{F}] \leq \Delta_n - \frac{\Delta t}{2}M - \frac{\Delta t}{4}V + \frac{7\kappa \Delta t^2}{4B}\Psi(a)$, which is (197) since $G(a)^2 = M + \frac{1}{2}V$. $\square$

F.3 The global theorems

Proof of Theorem 4.4. Theorem 4.2(1) gives the band $\frac{1}{2\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ pathwise, so the one-step estimate (196) and the gradient domination $G(a_n)^2 \geq \frac{1}{2\kappa}\Delta_n$ of (52) are available at every step. For (1), at $\Delta t = \frac{1}{2\kappa(1+2/B)}$ the bracket in (196) equals $\frac{1}{2}$, so with $\Psi \leq \Psi_H = N_\theta^2 \bar{L}_H^2$ from Theorem F.3, $\mathbb{E}[\Delta_{n+1} | \mathcal{F}_n] \leq (1 - \frac{\Delta t}{4\kappa})\Delta_n + \frac{3\kappa \Delta t^2}{2B}\Psi_H$. Unrolling, the floor is $\frac{3\kappa \Delta t^2 \Psi_H / (2B)}{\Delta t / (4\kappa)} = \frac{6\kappa^2 \Delta t}{B}\Psi_H = \frac{3\kappa}{B+2}\Psi_H$ and the contraction is $1 - \frac{\Delta t}{4\kappa} = 1 - \frac{1}{8\kappa^2(1+2/B)}$. For (2), on the band $C_n \preceq \frac{1}{\alpha}I$ we have $\beta\Lambda = \kappa$, so Theorem F.2 gives $\frac{3}{2}\Psi(a_n) \leq 3\kappa N_\theta + 3\kappa G(a_n)^2$. Substituting into (196) and collecting the G^2 terms,

$$\mathbb{E}[\Delta_{n+1} | \mathcal{F}_n] \leq \Delta_n - \Delta t(1 - \kappa \Delta t \rho_B)G(a_n)^2 + \frac{3\kappa^2 \Delta t^2 N_\theta}{B},$$

since $\kappa \Delta t^2(1 + \frac{2}{B}) + \frac{3\kappa \Delta t^2}{2B} \cdot 2\kappa = \kappa \Delta t^2 \rho_B$ with $\rho_B = 1 + \frac{2+3\kappa}{B}$. At $\Delta t = \frac{1}{2\kappa \rho_B}$ the bracket is $\frac{1}{2}$, so $\mathbb{E}[\Delta_{n+1} | \mathcal{F}_n] \leq (1 - \frac{\Delta t}{4\kappa})\Delta_n + \frac{3\kappa^2 \Delta t^2 N_\theta}{B}$. Unrolling, the floor is $\frac{3\kappa^2 \Delta t^2 N_\theta / B}{\Delta t / (4\kappa)} = \frac{12\kappa^3 \Delta t N_\theta}{B} = \frac{6\kappa^2 N_\theta}{B \rho_B} = \frac{6\kappa^2 N_\theta}{B+2+3\kappa}$ and the contraction is $1 - \frac{1}{8\kappa^2 \rho_B}$. Since $B + 2 + 3\kappa \geq 3\kappa$, the floor is at most $2\kappa N_\theta$. $\square$

Proof of Theorem 4.5. Theorem 4.2(2) gives the band $\frac{1}{\beta}I \preceq C_n \preceq \frac{1}{\alpha}I$ pathwise for every $\Delta t > 0$, and (52) gives $G(a_n)^2 \geq \frac{1}{\kappa}\Delta_n$. For (1), by Theorem F.3, $\Psi \leq \Psi_H = N_\theta^2 \bar{L}_H^2$; substituting into (197) gives $\mathbb{E}[\Delta_{n+1} | \mathcal{F}_n] \leq (1 - \frac{\Delta t}{2\kappa})\Delta_n + \frac{7\kappa\Delta t^2}{4B}\Psi_H$. At $\Delta t = \frac{1}{8\kappa}$ the contraction is $1 - \frac{1}{16\kappa^2}$ and the floor is $\frac{7\kappa\Delta t^2\Psi_H/(4B)}{\Delta t/(2\kappa)} = \frac{7\kappa}{16B}\Psi_H$. For (2), we have $\beta\Lambda = \kappa$ on the band, so Theorem F.2 gives $\Psi(a_n) \leq 2\kappa N_\theta + 2\kappa G(a_n)^2$, and (197) becomes

$$\mathbb{E}[\Delta_{n+1} | \mathcal{F}_n] \leq \Delta_n - \frac{\Delta t}{2} \left(1 - \frac{7\kappa^2\Delta t}{B}\right) G(a_n)^2 + \frac{7\kappa^2\Delta t^2}{2B} N_\theta.$$

The choice $\Delta t \leq \frac{B}{14\kappa^2}$ makes $7\kappa^2\Delta t/B \leq \frac{1}{2}$, so the bracket is at least $\frac{1}{2}$, and gradient domination gives $\mathbb{E}[\Delta_{n+1} | \mathcal{F}_n] \leq (1 - \frac{\Delta t}{4\kappa})\Delta_n + \frac{7\kappa^2\Delta t^2}{2B} N_\theta$. Unrolling, the floor is $\frac{7\kappa^2\Delta t^2 N_\theta/(2B)}{\Delta t/(4\kappa)} = \frac{14\kappa^3\Delta t N_\theta}{B}$, which equals κN_θ when $\Delta t = \frac{B}{14\kappa^2}$, and the contraction is then $1 - \frac{B}{56\kappa^3}$. $\square$

F.4 Local variance and the matrix-exponential perturbation

Lemma F.7 (Local variance decomposition). *Assume Theorems 2.1 and 2.2, work in whitened coordinates, and let $a = (m, C)$ with $\frac{1}{2}I \preceq C \preceq \frac{3}{2}I$. Then for a single sample $X \sim \rho_a$,*

$$\mathbb{E}[\|\varepsilon_b\|^2 + \frac{1}{2}\|\varepsilon_H\|_F^2] \leq V_0 + V_1 \|a - a_\star\|_\star^2, \quad V_0 := 3N_\theta^2 L_{H,\star}^2, \quad V_1 := 24 + 6N_\theta L_{H,\star}^2, \quad (201)$$

and by conditional independence the minibatch noise obeys $\mathbb{E}[\|\bar{\varepsilon}_b\|^2 + \frac{1}{2}\|\bar{\varepsilon}_H\|_F^2 | \mathcal{F}] \leq B^{-1}(V_0 + V_1 \|a - a_\star\|_\star^2)$. At the optimizer, $\mathbb{E}[\|\varepsilon_b(a_\star)\|^2 + \frac{1}{2}\|\varepsilon_H(a_\star)\|_F^2] \leq \sigma_\star^2 := \frac{3}{2}N_\theta^2 L_{H,\star}^2$, which is 0 for Gaussian targets.

Proof of Theorem F.7. Write $H(x) := \nabla^2 \hat{V}(x)$, so $\varepsilon_H = H(X) - A(a)$ and $A(a) = \mathbb{E}_{\rho_a}[H(X)]$. Gaussian Poincaré applied componentwise to $X \mapsto H_{ij}(X)$ with $X \sim \mathcal{N}(m, C)$ gives $\text{Var}(H_{ij}(X)) \leq \|C\|_{\text{op}} \mathbb{E} \|\nabla H_{ij}(X)\|^2$, so $\mathbb{E} \|\varepsilon_H\|_F^2 \leq \|C\|_{\text{op}} \mathbb{E} \sum_{i,j,k} (\partial_k H_{ij}(X))^2$. For each k , $\sum_{i,j} (\partial_k H_{ij}(x))^2 = \|DH(x)[e_k]\|_F^2 \leq N_\theta \|DH(x)[e_k]\|_{\text{op}}^2 \leq N_\theta L_{H,\star}^2$ by (55); summing over the N_θ coordinates k and using $\|C\|_{\text{op}} \leq \frac{3}{2}$,

$$\mathbb{E} \|\varepsilon_H\|_F^2 \leq \frac{3}{2} N_\theta^2 L_{H,\star}^2. \quad (202)$$

Let $F(x) := -\nabla \hat{V}(x) + C^{-1}(x - m)$, so $\varepsilon_b = F(X) - \mathbb{E}F(X)$ and $DF(x) = C^{-1} - H(x)$ is symmetric. Gaussian Poincaré componentwise and summed gives $\mathbb{E} \|\varepsilon_b\|^2 \leq \|C\|_{\text{op}} \mathbb{E} \|C^{-1} - H(X)\|_F^2$, and since $\mathbb{E}H(X) = A(a)$ the orthogonal decomposition $C^{-1} - H(X) = (C^{-1} - A(a)) - \varepsilon_H$ has vanishing cross expectation, so

$$\mathbb{E} \|\varepsilon_b\|^2 \leq \frac{3}{2} \left(\|C^{-1} - A(a)\|_F^2 + \mathbb{E} \|\varepsilon_H\|_F^2 \right). \quad (203)$$

Adding $\frac{1}{2}\mathbb{E} \|\varepsilon_H\|_F^2$ to (203),

$$\mathbb{E} \left[\|\varepsilon_b\|^2 + \frac{1}{2} \|\varepsilon_H\|_F^2 \right] \leq 2 \mathbb{E} \|\varepsilon_H\|_F^2 + \frac{3}{2} \|C^{-1} - A(a)\|_F^2,$$

and by (202) the first term is at most $3N_\theta^2 L_{H,\star}^2 = V_0$. For the second, $\|C^{-1} - A(a)\|_F^2 \leq 2\|C^{-1} - I\|_F^2 + 2\|A(a) - I\|_F^2$; since $C^{-1} - I = -C^{-1}(C - I)$ and $\|C^{-1}\|_{\text{op}} \leq 2$, the first piece is at most $8\|C - I\|_F^2$. For the second piece, $A(a) - I = \mathbb{E}_Z[H(m + C^{1/2}Z) - H(Z)]$ because $A(a_\star) = \mathbb{E}_Z H(Z) = I$, so by Jensen and (55),

$$\|A(a) - I\|_F^2 \leq N_\theta L_{H,\star}^2 \mathbb{E} \|m + (C^{1/2} - I)Z\|^2 = N_\theta L_{H,\star}^2 (\|m\|^2 + \|C^{1/2} - I\|_F^2),$$

the cross term vanishing by $\mathbb{E}Z = 0$, and $\|C^{1/2} - I\|_F^2 \leq \|C - I\|_F^2$ eigenvalue-wise. Hence $\frac{3}{2}\|C^{-1} - A(a)\|_F^2 \leq 12\|C - I\|_F^2 + 3N_\theta L_{H,\star}^2 (\|m\|^2 + \|C - I\|_F^2)$. Converting through $\|a - a_\star\|_\star^2 =$

$\|m\|^2 + \frac{1}{2}\|C - I\|_F^2$ gives $\|C - I\|_F^2 \leq 2\|a - a_\star\|_\star^2$ and $\|m\|^2 + \|C - I\|_F^2 \leq 2\|a - a_\star\|_\star^2$, so the state-dependent part is at most $(24 + 6N_\theta L_{H,\star}^2)\|a - a_\star\|_\star^2 = V_1\|a - a_\star\|_\star^2$. The minibatch bound is $\mathbb{E}\|\bar{\varepsilon}\|^2 = B^{-1}\mathbb{E}\|\varepsilon\|^2$ for centered independent averages. At $a_\star$ we have $C = I$ and $C^{-1} - A(a_\star) = 0$, so (202) tightens to $\mathbb{E}\|\varepsilon_H\|_F^2 \leq N_\theta^2 L_{H,\star}^2$ and (203) to $\mathbb{E}\|\varepsilon_b\|^2 \leq N_\theta^2 L_{H,\star}^2$, giving $\sigma_\star^2 \leq \frac{3}{2}N_\theta^2 L_{H,\star}^2$. $\square$

The only nonlinear ingredient of the local perturbation analysis is the following pair of elementary bounds on the matrix exponential.

Lemma F.8 (Exponential perturbation bounds). *Let $S, \hat{S} \in \mathbb{S}(N_\theta)$ with $S \preceq I$ and $\hat{S} \preceq I$, put $\Xi := \hat{S} - S$, and let $0 < \Delta t \leq 1$. Then*

$$\begin{aligned} \text{(i)} \quad & \left\| e^{\Delta t \hat{S}} - e^{\Delta t S} \right\|_F \leq e \Delta t \|\Xi\|_F, \\ \text{(ii)} \quad & \left\| e^{\Delta t \hat{S}} - e^{\Delta t S} - D \exp(\Delta t S) [\Delta t \Xi] \right\|_F \leq \frac{e}{2} \Delta t^2 \|\Xi\|_{\text{op}} \|\Xi\|_F. \end{aligned}$$

Proof. For symmetric $M = Q \text{diag}(\lambda_i) Q^\top$ the Fréchet derivative acts in the eigenbasis by the Loewner divided-difference matrix, $(Q^\top D \exp(M) [E] Q)_{ij} = \Phi_{ij} (Q^\top E Q)_{ij}$ with $\Phi_{ij} = (e^{\lambda_i} - e^{\lambda_j})/(\lambda_i - \lambda_j)$, equal to e^{λ_i} when $\lambda_i = \lambda_j$; since $\Phi_{ij} \leq e^{\lambda_{\max}(M)}$, one has $\|D \exp(M) [E]\|_F \leq e^{\lambda_{\max}(M)} \|E\|_F$. The Duhamel identity $e^Y - e^X = \int_0^1 \int_0^1 e^{uM_s} (Y - X) e^{(1-u)M_s} du ds$ with $M_s = X + s(Y - X)$, taken at $X = \Delta t S$ and $Y = \Delta t \hat{S}$ so that $M_s \preceq \Delta t I$ by convexity, together with $\|e^{uM_s}\|_{\text{op}} \leq e^{u\Delta t}$, gives $\|e^{\Delta t \hat{S}} - e^{\Delta t S}\|_F \leq e^{\Delta t} \Delta t \|\Xi\|_F \leq e \Delta t \|\Xi\|_F$, which is (i).

For (ii), Taylor's theorem with integral remainder gives $e^{\Delta t \hat{S}} - e^{\Delta t S} - D \exp(\Delta t S) [\Delta t \Xi] = \int_0^1 (1-s) \Phi''(s) ds$ with $\Phi(s) := \exp(\Delta t S + s \Delta t \Xi)$ and $\Phi''(s) = D^2 \exp(M_s) [\Delta t \Xi, \Delta t \Xi]$, $M_s \preceq \Delta t I$. The second Fréchet derivative admits the simplex representation $D^2 \exp(M) [E, E] = 2 \int_{u_0+u_1+u_2=1, u_i \geq 0} e^{u_0 M} E e^{u_1 M} E e^{u_2 M}$, whose Frobenius norm is bounded, using $\|e^{u_i M}\|_{\text{op}} \leq e^{u_i \lambda_{\max}(M)}$ and $\|ABC\|_F \leq \|A\|_{\text{op}} \|B\|_{\text{op}} \|C\|_F$, by $2 \cdot \frac{1}{2} \cdot e^{\lambda_{\max}(M)} \|E\|_{\text{op}} \|E\|_F$, the factor $\frac{1}{2}$ being the volume of the simplex. Hence $\|\Phi''(s)\|_F \leq e^{\Delta t} \Delta t^2 \|\Xi\|_{\text{op}} \|\Xi\|_F$, and $\int_0^1 (1-s) ds = \frac{1}{2}$ with $e^{\Delta t} \leq e$ gives (ii). $\square$

Lemma F.9 (One-step stochastic perturbation, Riemannian). *Assume Theorems 2.1 and 2.2, work in whitened coordinates, and let $a = (m, C)$ with $\frac{1}{2}I \preceq C \preceq \frac{3}{2}I$ and $\|T_R(a) - a_\star\|_\star \leq 1$, where T_R is the deterministic map of (20) and $\hat{a}^+$ one stochastic step of (48) with batch B and $0 < \Delta t \leq 1$. Then*

$$\mathbb{E}[\|\hat{a}^+ - a_\star\|_\star^2 | \mathcal{F}] \leq \|T_R(a) - a_\star\|_\star^2 + \frac{64 \Delta t^2}{B} (V_0 + V_1 \|a - a_\star\|_\star^2).$$

Proof of Theorem F.9. Let $a^+ := T_R(a)$ and $\hat{a}^+ - a^+ = (\delta_m, \delta_C)$. Expanding the whitened square,

$$\mathbb{E}[\|\hat{a}^+ - a_\star\|_\star^2 | \mathcal{F}] = \|a^+ - a_\star\|_\star^2 + 2 \left\langle a^+ - a_\star, \mathbb{E}[\hat{a}^+ - a^+ | \mathcal{F}] \right\rangle_\star + \mathbb{E}[\|\hat{a}^+ - a^+\|_\star^2 | \mathcal{F}], \quad (204)$$

and abbreviate $Q := \mathbb{E}[\|\bar{\varepsilon}_b\|^2 + \frac{1}{2}\|\bar{\varepsilon}_H\|_F^2 | \mathcal{F}]$. For the second moment, the mean components differ by $\delta_m = \Delta t C \bar{\varepsilon}_b$, so $\mathbb{E}\|\delta_m\|^2 \leq \frac{9}{4} \Delta t^2 \mathbb{E}\|\bar{\varepsilon}_b\|^2$. The covariance blocks are $C^{1/2} e^{\Delta t S} C^{1/2}$ and $C^{1/2} e^{\Delta t \hat{S}} C^{1/2}$ with $S = I - C^{1/2} A(a) C^{1/2} \preceq I$, $\hat{S} = I - C^{1/2} \bar{H} C^{1/2} \preceq I$, and $\hat{S} - S = -C^{1/2} \bar{\varepsilon}_H C^{1/2} =: \Xi_S$. By Theorem F.8(i) and $\|\Xi_S\|_F \leq \frac{3}{2} \|\bar{\varepsilon}_H\|_F$,

$$\|\delta_C\|_F \leq \|C\|_{\text{op}} \left\| e^{\Delta t \hat{S}} - e^{\Delta t S} \right\|_F \leq \frac{3}{2} \cdot e \Delta t \cdot \frac{3}{2} \|\bar{\varepsilon}_H\|_F = \frac{9e}{4} \Delta t \|\bar{\varepsilon}_H\|_F.$$

Therefore $\mathbb{E} \|\hat{a}^+ - a^+\|_\star^2 \leq \frac{9}{4} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_b\|^2 + \frac{81e^2}{32} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_H\|_F^2 \leq \frac{81e^2}{16} \Delta t^2 Q$, using $\frac{9}{4} \leq \frac{81e^2}{16}$. The mean bias vanishes, $\mathbb{E}[\delta_m | \mathcal{F}] = 0$. For the covariance, $e^{\Delta t \hat{S}} - e^{\Delta t S} = D \exp(\Delta t S) [\Delta t \Xi_S] + \mathcal{R}$ with the first term centered, so $\mathbb{E}[\delta_C | \mathcal{F}] = C^{1/2} \mathbb{E}[\mathcal{R} | \mathcal{F}] C^{1/2}$ and, by [Theorem F.8\(ii\)](#) with $\|\Xi_S\|_{\text{op}} \|\Xi_S\|_F \leq \frac{9}{4} \|\bar{\varepsilon}_H\|_F^2$,

$$\|\mathbb{E}[\delta_C | \mathcal{F}]\|_F \leq \frac{3}{2} \cdot \frac{e}{2} \Delta t^2 \cdot \frac{9}{4} \mathbb{E} \|\bar{\varepsilon}_H\|_F^2 = \frac{27e}{16} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_H\|_F^2.$$

Since the mean bias is zero, $\|\mathbb{E}[\hat{a}^+ - a^+ | \mathcal{F}]\|_\star = \frac{1}{\sqrt{2}} \|\mathbb{E}[\delta_C | \mathcal{F}]\|_F$, so with $\|a^+ - a_\star\|_\star \leq 1$, Cauchy–Schwarz, and $\mathbb{E} \|\bar{\varepsilon}_H\|_F^2 \leq 2Q$, the cross term of (204) is at most $\frac{27e}{4\sqrt{2}} \Delta t^2 Q$. Adding the two contributions, the coefficient of $\Delta t^2 Q$ is $\frac{81e^2}{16} + \frac{27e}{4\sqrt{2}} < 64$, numerically about 50.4. Bounding $Q \leq B^{-1}(V_0 + V_1 \|a - a_\star\|_\star^2)$ by [Theorem F.7](#) gives the claim. $\square$

Lemma F.10 (One-step stochastic perturbation, KL). *Under the hypotheses of [Theorem F.9](#) but for the KL maps (21) and (49) and the weighted norm $\|\cdot\|_{\star, \Delta t}$, with $\|T_{\text{KL}}(a) - a_\star\|_{\star, \Delta t} \leq 1$,*

$$\mathbb{E} \left[\left\| \hat{a}^+ - a_\star \right\|_{\star, \Delta t}^2 \mid \mathcal{F} \right] \leq \|T_{\text{KL}}(a) - a_\star\|_{\star, \Delta t}^2 + \frac{68 \Delta t^2}{B} (V_0 + V_1 \|a - a_\star\|_\star^2).$$

Proof of [Theorem F.10](#). Let $a^+ := T_{\text{KL}}(a)$, $P := C^{-1} + \Delta t A(a)$ and $\hat{P} := C^{-1} + \Delta t \bar{H} = P + \Delta t \bar{\varepsilon}_H$, so that $C^+ = (1 + \Delta t)P^{-1}$, $\hat{C}^+ = (1 + \Delta t)\hat{P}^{-1}$ and $\delta_C = (1 + \Delta t)(\hat{P}^{-1} - P^{-1})$. Since $A(a), \bar{H} \succeq 0$ we have $P \succeq C^{-1}$ and $\hat{P} \succeq C^{-1}$, so $\|P^{-1}\|_{\text{op}}, \|\hat{P}^{-1}\|_{\text{op}} \leq \|C\|_{\text{op}} \leq \frac{3}{2}$ almost surely. The expansion (204) holds with $\|\cdot\|_\star$ replaced by $\|\cdot\|_{\star, \Delta t}$, whose covariance weight is $\frac{1+\Delta t}{2}$. For the second moment, as before $\mathbb{E} \|\delta_m\|^2 \leq \frac{9}{4} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_b\|^2$. The first-order resolvent identity $\hat{P}^{-1} - P^{-1} = -P^{-1}(\Delta t \bar{\varepsilon}_H) \hat{P}^{-1}$ gives $\|\hat{P}^{-1} - P^{-1}\|_F \leq \frac{9}{4} \Delta t \|\bar{\varepsilon}_H\|_F$, hence $\|\delta_C\|_F \leq \frac{9}{2} \Delta t \|\bar{\varepsilon}_H\|_F$. As $\frac{1+\Delta t}{2} \leq 1$, we get $\mathbb{E} \|\hat{a}^+ - a^+\|_{\star, \Delta t}^2 \leq \frac{9}{4} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_b\|^2 + \frac{81}{4} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_H\|_F^2 \leq \frac{81}{2} \Delta t^2 Q$. For the bias, the second-order resolvent identity

$$\hat{P}^{-1} - P^{-1} = -P^{-1}(\Delta t \bar{\varepsilon}_H)P^{-1} + P^{-1}(\Delta t \bar{\varepsilon}_H)P^{-1}(\Delta t \bar{\varepsilon}_H)\hat{P}^{-1}$$

has centered first term, so the conditional bias is the second term, and $\|\mathbb{E}[\hat{P}^{-1} - P^{-1} | \mathcal{F}]\|_F \leq (\frac{3}{2})^3 \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_H\|_F^2 = \frac{27}{8} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_H\|_F^2$. Thus $\|\mathbb{E}[\delta_C | \mathcal{F}]\|_F \leq \frac{27}{4} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_H\|_F^2$ and, the mean bias being zero, $\|\mathbb{E}[\hat{a}^+ - a^+ | \mathcal{F}]\|_{\star, \Delta t} \leq \frac{27}{4} \Delta t^2 \mathbb{E} \|\bar{\varepsilon}_H\|_F^2$; with $\|a^+ - a_\star\|_{\star, \Delta t} \leq 1$ the cross term is at most $27 \Delta t^2 Q$. Adding the two contributions, the coefficient of $\Delta t^2 Q$ is $\frac{81}{2} + 27 = 67.5 < 68$. Finally $\|a - a_\star\|_\star^2 \leq \|a - a_\star\|_{\star, \Delta t}^2$, so $Q \leq B^{-1}(V_0 + V_1 \|a - a_\star\|_\star^2)$ from [Theorem F.7](#) gives the stated bound. $\square$

F.5 Killed-process contraction and finite-horizon containment

Proof of [Theorem 4.7](#). Write $r_n := \|a_n - a_\star\|_\star$. On $\{n < \tau_R\}$ we have $a_n \in \mathcal{U}_{\delta_R}$, so $\frac{1}{2}I \preceq C_n \preceq \frac{3}{2}I$ and $r_n \leq \mathbf{b}_{\star, R}(\delta_R) \sqrt{\delta_R} \leq 1$ by [Theorem 4.6](#). The pointwise one-step contraction of [Theorem E.12](#), which needs only $\Delta t \leq \frac{2}{4+\Gamma}$ and no energy-descent bound, gives $\|T_R(a_n) - a_\star\|_\star \leq (1 - \mu_R \Delta t) r_n \leq r_n \leq 1$, since $\frac{1}{2} \gamma_R = \mu_R$. Hence [Theorem F.9](#) applies and

$$\mathbb{E}[r_{n+1}^2 | \mathcal{F}_n] \leq (1 - \mu_R \Delta t)^2 r_n^2 + \frac{64 \Delta t^2}{B} (V_0 + V_1 r_n^2).$$

Because $\Delta t \leq \frac{2}{4+\Gamma}$ we have $\mu_R \Delta t = \frac{\Delta t}{2(4+\Gamma)} \leq \frac{1}{(4+\Gamma)^2} \leq \frac{1}{16} \leq \frac{1}{2}$, so $(1 - \mu_R \Delta t)^2 \leq 1 - \frac{3}{2} \mu_R \Delta t$ from $x^2 \leq \frac{1}{2}x$ for $x \leq \frac{1}{2}$. The stepsize bound $\Delta t \leq \frac{B}{256 A_R V_1}$ gives $\frac{64 \Delta t^2}{B} V_1 \leq \frac{1}{2} \mu_R \Delta t$, hence

$$\mathbb{E}[r_{n+1}^2 | \mathcal{F}_n] \leq (1 - \mu_R \Delta t) r_n^2 + \eta_R, \quad \eta_R := \frac{64 \Delta t^2}{B} V_0 = \frac{192 \Delta t^2 N_\theta^2 L_{H, \star}^2}{B}.$$

Set $M_n := \mathbb{E}[r_n^2 \mathbf{1}_{\{\tau_R > n\}}]$. Since $\{\tau_R > n+1\} \subseteq \{\tau_R > n\} \in \mathcal{F}_n$,

$$M_{n+1} \leq \mathbb{E}[\mathbf{1}_{\{\tau_R > n\}} \mathbb{E}[r_{n+1}^2 \mid \mathcal{F}_n]] \leq (1 - \mu_R \Delta t) M_n + \eta_R,$$

and iterating from $M_0 = r_0^2$ with $\sum_{j \geq 0} (1 - \mu_R \Delta t)^j = (\mu_R \Delta t)^{-1}$ gives $M_N \leq (1 - \mu_R \Delta t)^N r_0^2 + \eta_R / (\mu_R \Delta t)$, which is the stated bound since $\eta_R / (\mu_R \Delta t) = 384 \Delta t (4 + \Gamma) N_\theta^2 L_{H,*}^2 / B$. $\square$

Proof of Theorem 4.8. Identical to the previous proof, replacing $\|\cdot\|_*$ by $\|\cdot\|_{*,\Delta t}$, γ_R by γ_{KL} so that μ_R becomes μ_{KL} , Theorem F.9 by Theorem F.10 with the coefficient 64 replaced by 68, and η_R by $\eta_{KL} = \frac{68 \Delta t^2}{B} V_0 = \frac{204 \Delta t^2 N_\theta^2 L_{H,*}^2}{B}$. The step condition $\Delta t \leq \frac{B}{272 A_{KL} V_1}$ gives $\frac{68 \Delta t^2}{B} V_1 \leq \frac{1}{2} \mu_{KL} \Delta t$, and $\mu_{KL} \Delta t \leq \frac{1}{2}$ holds as before, so $\mathbb{E}[r_{n+1}^2 \mid \mathcal{F}_n] \leq (1 - \mu_{KL} \Delta t) r_n^2 + \eta_{KL}$ with $r_n = \|a_n - a_*\|_{*,\Delta t}$; the non-exit recursion gives $M_N \leq (1 - \mu_{KL} \Delta t)^N r_0^2 + \eta_{KL} / (\mu_{KL} \Delta t)$, which is the stated bound. $\square$

Proof of Theorem 4.9. Since $\{a : \|a - a_*\|_{*,\bullet} \leq r_\bullet\} \subseteq \mathcal{U}_{\delta_\bullet}$, on $\{n < \tau_{r_\bullet}\}$ the one-step recursion $\mathbb{E}[r_{n+1}^2 \mid \mathcal{F}_n] \leq q_\bullet r_n^2 + \eta_\bullet$ established in the two preceding proofs holds, with $q_\bullet := 1 - \mu_\bullet \Delta t$ and $r_n := \|a_n - a_*\|_{*,\bullet}$. Define the process frozen at exit, $R_n := r_{n \wedge \tau_{r_\bullet}}$. On $\{\tau_{r_\bullet} > n\}$ one has $R_{n+1} = r_{n+1}$ and, since $q_\bullet \leq 1$, $\mathbb{E}[R_{n+1}^2 \mid \mathcal{F}_n] \leq q_\bullet r_n^2 + \eta_\bullet \leq R_n^2 + \eta_\bullet$; on $\{\tau_{r_\bullet} \leq n\}$ one has $R_{n+1} = R_n$. In either case $\mathbb{E}[R_{n+1}^2 \mid \mathcal{F}_n] \leq R_n^2 + \eta_\bullet$, so iterating gives $\mathbb{E} R_N^2 \leq r_0^2 + N \eta_\bullet$. On $\{\tau_{r_\bullet} \leq N\}$ the frozen value satisfies $R_N = r_{\tau_{r_\bullet}} > r_\bullet$, so Markov's inequality gives $r_\bullet^2 \mathbb{P}(\tau_{r_\bullet} \leq N) \leq \mathbb{E} R_N^2 \leq r_0^2 + N \eta_\bullet$, which is the claim. Splitting the failure budget equally between the two terms gives the displayed sufficient conditions. $\square$

Freezing the process at the exit time removes the factor $(\mu_\bullet \Delta t)^{-1}$ that a per-exit-time union bound would incur; that denominator remains genuine in the stationary killed-moment floor $\eta_\bullet / (\mu_\bullet \Delta t)$ of Theorems 4.7 and 4.8, which sums repeated noise within the surviving process, but not in the exit probability. At the certified stepsize $\Delta t = \Theta((4 + \Gamma)^{-1})$, where also $\mu_\bullet = \Theta((4 + \Gamma)^{-1})$, the resulting batch requirement is smaller by a factor of order $(4 + \Gamma)^2$ than the condition $B \gtrsim N \Delta t N_\theta^2 L_{H,*}^2 / (\epsilon' \mu_\bullet r_\bullet^2)$ that the union-bound argument would demand.

Proof of Theorem 4.10. By Markov's inequality applied to $\Delta_{N_1} \geq 0$ and $\mathcal{U}_{\delta_{\text{in}}} = \{\Delta \leq \delta_{\text{in}}\}$,

$$\mathbb{P}(a_{N_1} \notin \mathcal{U}_{\delta_{\text{in}}}) = \mathbb{P}(\Delta_{N_1} > \delta_{\text{in}}) \leq \frac{\mathbb{E} \Delta_{N_1}}{\delta_{\text{in}}} \leq \epsilon_{\text{ent}}.$$

On E_{in} the norm conversion (39) gives $\|a_{N_1} - a_*\|_{*,\bullet} \leq \mathbf{b}_{*,\bullet}(\delta_{\text{in}}) \sqrt{\delta_{\text{in}}} =: r_0$, and the deep-entry choice makes $r_0^2 \leq \frac{\epsilon'}{2} r_\bullet^2 \leq r_\bullet^2$, so a_{N_1} lies in the outer ball. Since $E_{\text{in}} \in \mathcal{F}_{N_1}$, applying Theorem 4.9 to the process $(a_{N_1+n})_{n \geq 0}$ on E_{in} , the deep-entry condition controls the initial-condition term by $\epsilon'/2$ and the batch condition controls the $N_2 \eta_\bullet$ term by $\epsilon'/2$, giving $\mathbb{P}(\{\tau^{(N_1)} \leq N_2\} \cap E_{\text{in}}) \leq \epsilon'$. Hence $\mathbb{P}(E_{\text{in}} \cap \{\tau^{(N_1)} > N_2\}) \geq (1 - \epsilon_{\text{ent}}) - \epsilon'$.

For the second moment write $\rho_n := \|a_{N_1+n} - a_*\|_{*,\bullet}$. On $\{n < \tau^{(N_1)}\} \cap E_{\text{in}}$ the outer ball contains a_{N_1+n} , so the one-step recursion $\mathbb{E}[\rho_{n+1}^2 \mid \mathcal{F}_{N_1+n}] \leq q_\bullet \rho_n^2 + \eta_\bullet$ holds there; setting $M_n := \mathbb{E}[\rho_n^2 \mathbf{1}_{E_{\text{in}}} \mathbf{1}_{\{\tau^{(N_1)} > n\}}]$ and using $\{\tau^{(N_1)} > n\} \cap E_{\text{in}} \in \mathcal{F}_{N_1+n}$ gives $M_{n+1} \leq q_\bullet M_n + \eta_\bullet$, so $M_{N_2} \leq q_\bullet^{N_2} r_0^2 + \eta_\bullet / (\mu_\bullet \Delta t)$, and $\eta_\bullet / (\mu_\bullet \Delta t)$ equals the displayed F_R resp. F_{KL} . $\square$

F.6 The decreasing schedule

Proof of Theorem 4.11. Since $n + n_0 \geq n_0 \geq 64\kappa^2$, we have $\Delta t_n = 8\kappa / (n + n_0) \leq 1 / (8\kappa)$, and the covariance bands persist along the varying schedule. For the KL scheme this follows at each step from the induction of Theorem 4.2(2), which allows every positive stepsize. For the Riemannian scheme, suppose $\frac{1}{2\beta} I \preceq C_n \preceq \frac{1}{\alpha} I$; since $\Delta t_n \kappa \leq 1$ and $\Delta t_n \leq 1$, the upper-band argument gives $C_{n+1}^{-1} \succeq (1 - \Delta t_n) C_n^{-1} + \Delta t_n \overline{H}_n \succeq \alpha I$, and the lower envelope gives

$$C_{n+1}^{-1} \preceq e^{-\Delta t_n} [C_n^{-1} + (e - 1) \Delta t_n \overline{H}_n] \preceq \beta e^{-\Delta t_n} [2 + (e - 1) \Delta t_n] I \preceq 2\beta I,$$

because $e^{-h}[2 + (e - 1)h] \leq 2$ for $0 \leq h \leq 1$. The rescued initializer lies in both bands, so induction proves their persistence and the constant-stepsize one-step estimates apply at each n .

For the Riemannian scheme, [Theorem F.5](#) with $\Delta t_n \leq 1/(8\kappa)$ has bracket $1 - \kappa\Delta t_n(1 + 2/B) \geq 1 - \frac{3}{8} \geq \frac{1}{2}$, so the drift is at least $\frac{\Delta t_n}{2}G^2$, and the Riemannian gradient domination $G^2 \geq \frac{1}{2\kappa}\Delta$ of (52) makes it at least $\frac{\Delta t_n}{4\kappa}\Delta_n$, with floor coefficient $\frac{3}{2} \leq c_S$. For the KL scheme, [Theorem F.6](#) gives drift $\frac{\Delta t_n}{2}G^2 \geq \frac{\Delta t_n}{2\kappa}\Delta_n \geq \frac{\Delta t_n}{4\kappa}\Delta_n$ and floor coefficient $\frac{7}{4} = c_S$. In both cases, with $\Psi \leq \Psi_H$ from [Theorem F.3](#),

$$\mathbb{E}\Delta_{n+1} \leq \left(1 - \frac{\Delta t_n}{4\kappa}\right)\mathbb{E}\Delta_n + \frac{c_S\kappa\Delta t_n^2}{B}\Psi_H. \quad (205)$$

Write $t := n + n_0$, so that $\frac{\Delta t_n}{4\kappa} = \frac{2}{t}$ and $\frac{c_S\kappa\Delta t_n^2}{B}\Psi_H = \frac{S}{t^2}$ with $S := 64c_S\kappa^3\Psi_H/B$, and (205) becomes $\mathbb{E}\Delta_{n+1} \leq (1 - \frac{2}{t})\mathbb{E}\Delta_n + \frac{S}{t^2}$. We show $\mathbb{E}\Delta_n \leq K/t$ by induction with $K = \max\{S, n_0\Delta(a_0)\}$. For $n = 0$, $\mathbb{E}\Delta_0 = \Delta(a_0) \leq K/n_0$. Assuming $\mathbb{E}\Delta_n \leq K/t$,

$$\mathbb{E}\Delta_{n+1} \leq \left(1 - \frac{2}{t}\right)\frac{K}{t} + \frac{S}{t^2} = \frac{K}{t} - \frac{2K - S}{t^2} \leq \frac{K}{t} - \frac{K}{t^2} = \frac{K(t - 1)}{t^2} \leq \frac{K}{t + 1},$$

using $2K - S \geq K$ and $(t - 1)(t + 1) \leq t^2$. Since $t + 1 = (n + 1) + n_0$, the induction closes. $\square$

F.7 Gaussian targets in the local phase

Corollary F.11 (Gaussian targets in the local phase). *Let V be as in [Theorem 4.12](#), so that $L_{H,\star} = 0$ and $V_0 = \sigma_\star^2 = \eta_\bullet = 0$, and adopt the stepsize hypotheses of [Theorem 4.7](#) resp. [Theorem 4.8](#). For a Gaussian target $\Gamma = 0$, so the first branch reads $\Delta t \leq \frac{1}{2}$, while the second branch persists: $L_{H,\star} = 0$ leaves $V_1 = 24$, so the hypotheses are $0 < \Delta t \leq \min\{\frac{1}{2}, B/(256 A_R V_1)\}$ with $A_R = 4$, and $0 < \Delta t \leq \min\{\frac{1}{2}, B/(272 A_{KL} V_1)\}$ respectively. Only the additive floors, not the state-dependent stepsize restriction, are removed by Gaussianity. Three cases are distinguished by the whitened covariance initialization.*

- (a) *For generic $C_0 \neq I$, the additive floors of [Theorems 4.7](#) and [4.8](#) vanish and the exit bound of [Theorem 4.9](#) loses its horizon-growing term,*

$$\mathbb{E}[\|a_N - a_\star\|_{\star,\bullet}^2; \tau_\bullet > N] \leq (1 - \mu_\bullet\Delta t)^N \|a_0 - a_\star\|_{\star,\bullet}^2, \quad \mathbb{P}(\tau_\bullet \leq N) \leq \frac{\|a_0 - a_\star\|_{\star,\bullet}^2}{r_\bullet^2},$$

both uniformly in N . These remain mean-square and in-probability statements: the mean-score noise $\varepsilon_b = (C^{-1/2} - C^{1/2})Z$ is nonzero whenever $C \neq I$, the state-dependent constant $V_1 = 24$ survives because V_1 does not contain $L_{H,\star}$, and the exit probability is not zero.

- (b) *For matched covariance $C_0 = I$, that is $C_0 = C_\star$, with $m_0 \neq 0$, one has $\varepsilon_b = 0$ and $\varepsilon_H = 0$ pathwise, both covariance updates fix $C_n \equiv I$, and the mean update is deterministic, $m_{n+1} = (1 - \Delta t)m_n$ for both schemes. The iterate is pathwise deterministic and contracts to $a_\star$ with no noise, so no exit control is needed. Zero-noise pathwise evolution therefore requires only $C_0 = C_\star$, not $a_0 = a_\star$.*

- (c) *Under quadratic rescue the initializer is $a_0 = a_\star$ by [Theorem 2.15](#), the special case of (b) with $m_0 = 0$; then $a_n \equiv a_\star$ and no local phase is needed.*

Proof. For a quadratic potential $\nabla^2\hat{V} \equiv C_\star^{1/2}HC_\star^{1/2}$ is constant, so $L_{H,\star} = 0$, and [Theorem F.7](#) gives $V_0 = 3N_\theta^2L_{H,\star}^2 = 0$ and $\sigma_\star^2 = \frac{3}{2}N_\theta^2L_{H,\star}^2 = 0$, whence $\eta_\bullet = 0$. Setting $\eta_\bullet = 0$ in [Theorems 4.7](#) to [4.9](#) gives the two displays of (a). For $\hat{V}(z) = \frac{1}{2}\|z\|^2$ and $X = m + C^{1/2}Z$ the whitened single-sample mean noise is $\varepsilon_b = -C^{1/2}Z + C^{-1/2}Z = (C^{-1/2} - C^{1/2})Z$, nonzero for $C \neq I$, so no pathwise or zero-exit conclusion follows. For (b), $C_0 = I$ gives $\varepsilon_b = 0$ and, since $\nabla^2\hat{V} \equiv I$, $\bar{H}_n = I = A(a)$ and $\varepsilon_H = 0$ pathwise; the Riemannian update returns $C_1 = \exp(\Delta t \cdot 0) = I$

and the KL update $C_1 = (1 + \Delta t)(I + \Delta t I)^{-1} = I$, so $C_n \equiv I$, and then $\bar{b}_n = g(a_n) = -m_n$ gives $m_{n+1} = (1 - \Delta t)m_n$. For (c), [Theorem 2.15](#) gives $a_0 = a_\star = (0, I)$, the $m_0 = 0$ instance of (b). $\square$

Remark F.12 (Conservative constants). The explicit constants entering the local floors, namely $V_0 = 3N_\theta^2 L_{H,\star}^2$ and $V_1 = 24 + 6N_\theta L_{H,\star}^2$ of [Theorem F.7](#), the aggregate coefficients 64 and 68 of [Theorems F.9](#) and [F.10](#), the exponential remainder constant $e/2$ of [Theorem F.8](#), the stepsize denominators 256 and 272, and the floor coefficients 192, 204, 384, 408, are conservative and not optimized. Only their orders in $(\Delta t, N_\theta, L_{H,\star}, B, A_\bullet)$ enter the conclusions of [Section 4](#). The same applies to the global constants: the retraction-smoothness constant $L = 2\beta\Lambda$ may be replaced by $2(e - 2)\beta\Lambda \approx 1.437\beta\Lambda$, which rescales the stepsizes and floors of [Theorems 4.4](#) and [4.5](#) by the factor $e - 2$ without affecting any order, and the dimension coefficient $2\beta\Lambda$ of [Theorem F.2](#) may be reduced to at most $\frac{5}{4}\beta\Lambda$ by retaining the centering term discarded in its first step.

G Details for the non-log-concave scheme

We prove [Theorem 2.21](#). Throughout, [Theorem 2.20](#) is in force, the stepsize satisfies $0 < \Delta t \leq \frac{1}{2\beta\lambda_+}$, and we write

$$\mathcal{E}(m, C) = \Phi(C) + \mathcal{E}_2(m, C) + \text{const}, \quad \Phi(C) := -\frac{1}{2} \log \det C, \quad \mathcal{E}_2(m, C) := \mathbb{E}_{\mathcal{N}(m, C)}[V(\theta)].$$

For two states $a = (m, C)$ and $a' = (m', C')$ in $\mathcal{A}'$ we set

$$\begin{aligned} D_m(a', a) &:= \frac{1}{2} \|m' - m\|_{C^{-1}}^2, \\ D_C^+(C', C) &:= \text{KL}(\mathcal{N}(0, C') \parallel \mathcal{N}(0, C)), \\ D_C^-(C, C') &:= \text{KL}(\mathcal{N}(0, C) \parallel \mathcal{N}(0, C')). \end{aligned}$$

The forward divergence D_C^+ carries the smoothness remainder and appears in the proximal model, while the reverse divergence D_C^- is what remains in the descent certificate, and the two must be kept apart. With the trace pairing on symmetric matrices,

$$D_C^+(C', C) = D_\Phi(C', C) := \Phi(C') - \Phi(C) - \text{Tr}(\nabla\Phi(C)(C' - C)), \quad \nabla\Phi(C) = -\frac{1}{2}C^{-1}, \quad (206)$$

so the forward covariance divergence is the log-determinant Bregman divergence in its first argument.

Lemma G.1. *Let $C, C' \in [\lambda_- I, \lambda_+ I]$, let $b = \mathbb{E}_{\mathcal{N}(m, C)}[\nabla_\theta V(\theta)]$ and $A = \mathbb{E}_{\mathcal{N}(m, C)}[\nabla_\theta \nabla_\theta V(\theta)]$. Then*

$$\mathcal{E}_2(m', C') \leq \mathcal{E}_2(m, C) + b^\top(m' - m) + \frac{1}{2} \text{Tr}(A(C' - C)) + 2\beta\lambda_+ \left\{ D_m(a', a) + D_C^+(C', C) \right\}. \quad (207)$$

Proof. Set $u = m' - m$, $M = C^{-1/2}C'C^{-1/2}$, $B_0 = C^{1/2}$, $B' = C^{1/2}M^{1/2}$ and $H = B' - B_0$, so that $B'B'^\top = C^{1/2}MC^{1/2} = C'$. We couple

$$X = m + B_0 Z, \quad Y = m' + B' Z, \quad Z \sim \mathcal{N}(0, I),$$

so that $X \sim \mathcal{N}(m, C)$ and $Y \sim \mathcal{N}(m', C')$. Since $\|\nabla^2 V\|_{\text{op}} \leq \beta$, Taylor expansion at X gives

$$\begin{aligned} \mathbb{E}V(Y) &\leq \mathbb{E}V(X) + \mathbb{E}\left[\nabla V(X)^\top(u + HZ)\right] + \frac{\beta}{2} \mathbb{E}\|u + HZ\|^2 \\ &= \mathbb{E}V(X) + b^\top u + \mathbb{E}\left[\nabla V(X)^\top HZ\right] + \frac{\beta}{2} \left(\|u\|^2 + \|H\|_{\text{F}}^2\right). \end{aligned}$$

By Stein's lemma, $\mathbb{E}[\nabla V(X)Z^\top] = AB_0$, hence $\mathbb{E}[\nabla V(X)^\top HZ] = \text{Tr}(H^\top AB_0)$. Expanding

$$C' - C = B_0 H^\top + H B_0^\top + H H^\top$$

and using the symmetry of A and B_0 gives

$$\text{Tr}(H^\top AB_0) = \frac{1}{2} \text{Tr}(A(C' - C)) - \frac{1}{2} \text{Tr}(A H H^\top).$$

Since $\|A\|_{\text{op}} \leq \beta$, the term $-\frac{1}{2} \text{Tr}(A H H^\top)$ and the Taylor remainder together are bounded above by $\frac{\beta}{2} \|u\|^2 + \beta \|H\|_{\text{F}}^2$. It remains to convert both into the split divergence. The covariance box gives $\|u\|^2 \leq 2\lambda_+ D_m(a', a)$, so $\frac{\beta}{2} \|u\|^2 \leq \beta \lambda_+ D_m(a', a)$. For the covariance part, $\|H\|_{\text{F}}^2 = \|C^{1/2}(M^{1/2} - I)\|_{\text{F}}^2 \leq \lambda_+ \|M^{1/2} - I\|_{\text{F}}^2$, while the scalar inequality

$$x - 1 - \log x \geq (\sqrt{x} - 1)^2, \quad x > 0,$$

applied to the eigenvalues of M gives

$$\|M^{1/2} - I\|_{\text{F}}^2 \leq \sum_{i=1}^{N_\theta} \{\lambda_i(M) - 1 - \log \lambda_i(M)\} = 2D_C^+(C', C),$$

so that $\beta \|H\|_{\text{F}}^2 \leq 2\beta \lambda_+ D_C^+(C', C)$. Substituting the two bounds proves (207). $\square$

We next identify (30) as a constrained proximal step. At $a_n = (m_n, C_n) \in \mathcal{A}'$ define

$$\mathcal{Q}_n(m, C) := \Phi(C) + b_n^\top (m - m_n) + \frac{1}{2} \text{Tr}(A_n(C - C_n)) + \frac{1}{\Delta t} \left\{ D_m((m, C), a_n) + D_C^+(C, C_n) \right\}. \quad (208)$$

The mean and covariance variables separate in (208). The mean minimizer is $m_n - \Delta t C_n b_n$. For the covariance, put $\sigma := 1 + \Delta t^{-1}$ and recall $\tilde{C}_{n+1} = (1 + \Delta t)(C_n^{-1} + \Delta t A_n)^{-1}$; by (206) the covariance-dependent part of $\mathcal{Q}_n$ equals, up to an additive constant,

$$\sigma \Phi(C) + \frac{1}{2} \text{Tr} \left((A_n + \Delta t^{-1} C_n^{-1}) C \right) = \sigma D_\Phi(C, \tilde{C}_{n+1}) + \text{const}, \quad (209)$$

since $\sigma \tilde{C}_{n+1}^{-1} = \Delta t^{-1} C_n^{-1} + A_n$ matches the linear coefficient. The matrix $\tilde{C}_{n+1}$ is well defined and positive definite because

$$C_n^{-1} + \Delta t A_n \succeq (\lambda_+^{-1} - \Delta t \beta) I \succeq (2\lambda_+)^{-1} I$$

under $\Delta t \beta \lambda_+ \leq 1/2$.

The Bregman projection associated with (209) is exactly spectral clipping. Write $\tilde{C}_{n+1} = Q \text{diag}(\tilde{c}_i) Q^\top$ with columns q_i of Q , and $C^+ = Q \text{diag}(c_i^+) Q^\top$ with $c_i^+ = \min\{\lambda_+, \max\{\lambda_-, \tilde{c}_i\}\}$. Then

$$G := \nabla_C D_\Phi(C^+, \tilde{C}_{n+1}) = \frac{1}{2} \left(\tilde{C}_{n+1}^{-1} - (C^+)^{-1} \right) = Q \text{diag}(g_i) Q^\top,$$

where $g_i > 0$ only when $c_i^+ = \lambda_-$, $g_i < 0$ only when $c_i^+ = \lambda_+$, and $g_i = 0$ otherwise. Hence every $C \in [\lambda_- I, \lambda_+ I]$ satisfies

$$\text{Tr} \left(G(C - C^+) \right) = \sum_{i=1}^{N_\theta} g_i \left\{ q_i^\top C q_i - c_i^+ \right\} \geq 0,$$

which is the variational inequality for minimizing $D_\Phi(\cdot, \tilde{C}_{n+1})$ over the covariance box; strict convexity in the first argument gives uniqueness. Therefore $C^+ = \text{Clip}_{[\lambda_-, \lambda_+]}(\tilde{C}_{n+1}) = C_{n+1}$, and $a_{n+1} = (m_{n+1}, C_{n+1})$ produced by (30) is the unique minimizer of (208) over $\mathcal{A}'$. In particular the iterates remain feasible.

From the mean minimizer, with $u_n = m_{n+1} - m_n = -\Delta t C_n b_n$,

$$b_n^\top u_n = -\frac{1}{\Delta t} \|u_n\|_{C_n^{-1}}^2 = -\frac{2}{\Delta t} D_m(a_{n+1}, a_n). \quad (210)$$

From the covariance minimizer, constrained first-order optimality gives, for every $C \in [\lambda_- I, \lambda_+ I]$,

$$\text{Tr} \left(\left\{ \nabla \Phi(C_{n+1}) + \frac{1}{2} A_n + \frac{1}{\Delta t} (\nabla \Phi(C_{n+1}) - \nabla \Phi(C_n)) \right\} (C - C_{n+1}) \right) \geq 0. \quad (211)$$

We take $C = C_n$ and write $\Delta C_n = C_{n+1} - C_n$. The Bregman identities

$$\begin{aligned} \text{Tr}(\nabla \Phi(C_{n+1}) \Delta C_n) &= \Phi(C_{n+1}) - \Phi(C_n) + D_C^-(C_n, C_{n+1}), \\ \text{Tr}(\{\nabla \Phi(C_{n+1}) - \nabla \Phi(C_n)\} \Delta C_n) &= D_C^-(C_n, C_{n+1}) + D_C^+(C_{n+1}, C_n) \end{aligned}$$

then turn (211) into

$$\Phi(C_{n+1}) - \Phi(C_n) + \frac{1}{2} \text{Tr}(A_n \Delta C_n) \leq -\left(1 + \frac{1}{\Delta t}\right) D_C^-(C_n, C_{n+1}) - \frac{1}{\Delta t} D_C^+(C_{n+1}, C_n). \quad (212)$$

This is the constrained log-determinant three-point inequality; it is where the clipping is absorbed, since (211) is an inequality rather than an equation precisely when the update is pinned at one of the two clip levels.

We now apply [Theorem G.1](#) with $a = a_n$ and $a' = a_{n+1}$ and combine it with (210) and (212). Writing $L = 2\beta\lambda_+$,

$$\begin{aligned} \mathcal{E}(a_{n+1}) - \mathcal{E}(a_n) &= \Phi(C_{n+1}) - \Phi(C_n) + \mathcal{E}_2(a_{n+1}) - \mathcal{E}_2(a_n) \\ &\leq -\left(\frac{2}{\Delta t} - L\right) D_m(a_{n+1}, a_n) - \frac{1}{\Delta t} D_C^-(C_n, C_{n+1}) - \left(\frac{1}{\Delta t} - L\right) D_C^+(C_{n+1}, C_n). \end{aligned}$$

Because $L\Delta t \leq 1$, the last term is nonpositive and $2/\Delta t - L \geq 1/\Delta t$, hence

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t} \left\{ \frac{1}{2} \|m_{n+1} - m_n\|_{C_n^{-1}}^2 + D_C^-(C_n, C_{n+1}) \right\} = \mathcal{E}(a_n) - \frac{1}{\Delta t} \mathbf{S}_n,$$

which is (32). Summing over $0 \leq n < N$ and bounding the minimum by the average gives (33), and [Theorem 2.21](#) is proved.

The quantity $\mathbf{S}_n$ vanishes if and only if $(m_{n+1}, C_{n+1}) = (m_n, C_n)$. At such a fixed point, mean optimality gives $b_n = 0$, while (211) reduces to

$$\text{Tr} \left(\frac{1}{2} (A_n - C_n^{-1}) (C - C_n) \right) \geq 0 \quad \text{for every } C \in [\lambda_- I, \lambda_+ I],$$

which are exactly the first-order conditions for $\mathcal{E}$ on the covariance box. Thus $\mathbf{S}_n$ is a constrained fixed-point residual and not an unconstrained Fisher–Rao gradient norm, as recorded in the closing scope paragraph of [Section 2](#)’s non-log-concave subsection.

Finally we record the mean-block identity behind [Theorem G.2](#). The full successive reverse KL divergence is

$$\text{KL}(\mathcal{N}(m_n, C_n) \parallel \mathcal{N}(m_{n+1}, C_{n+1})) = D_C^-(C_n, C_{n+1}) + \frac{1}{2} \|m_{n+1} - m_n\|_{C_{n+1}^{-1}}^2,$$

whose mean block is measured in C_{n+1}^{-1} rather than in C_n^{-1} . Feasibility gives $C_{n+1}^{-1} \preceq \lambda_-^{-1} I \preceq (\lambda_+/\lambda_-) C_n^{-1}$, which is the origin of the factor λ_+/λ_- in the following corollary.

Corollary G.2. *Under the assumptions of [Theorem 2.21](#), the full successive-Gaussian reverse KL divergence satisfies*

$$\min_{0 \leq n < N} \text{KL}(\mathcal{N}(m_n, C_n) \parallel \mathcal{N}(m_{n+1}, C_{n+1})) \leq \frac{\lambda_+}{\lambda_-} \frac{\Delta t}{N} \{\mathcal{E}(a_0) - \mathcal{E}(a_N)\}.$$

Proof. The reverse KL divergence between successive Gaussians splits as

$$\text{KL}(\mathcal{N}(m_n, C_n) \parallel \mathcal{N}(m_{n+1}, C_{n+1})) = \text{KL}(\mathcal{N}(0, C_n) \parallel \mathcal{N}(0, C_{n+1})) + \frac{1}{2} \|m_{n+1} - m_n\|_{C_{n+1}^{-1}}^2,$$

and the mean block is measured in C_{n+1}^{-1} rather than C_n^{-1} . Feasibility gives

$$C_{n+1}^{-1} \preceq \lambda_-^{-1} I \preceq \frac{\lambda_+}{\lambda_-} C_n^{-1},$$

so the full divergence is at most $(\lambda_+/\lambda_-)\mathbb{S}_n$, and (33) completes the proof. $\square$

H Representation-theoretic classification

We record the structural form of Theorem 5.1. Throughout, $N_\theta \geq 2$, the base point is $o = (0, I) \in \mathcal{A}$, and we identify $T_o\mathcal{A}$ with $\mathbb{R}^{N_\theta} \times \mathbb{S}(N_\theta)$.

Since $\Phi_{(C^{1/2}, m)}(0, I) = (m, C)$, the affine group acts transitively on $\mathcal{A}$, and $\Phi_{(A, b)}(0, I) = (b, AA^\top) = (0, I)$ exactly when $b = 0$ and $A \in O(N_\theta)$. An affine invariant metric is therefore determined by an inner product on $T_o\mathcal{A}$ that is invariant under the isotropy representation

$$Q \cdot (u, X) = (Qu, QXQ^\top), \quad Q \in O(N_\theta), \quad (213)$$

and conversely every such inner product extends to an affine invariant metric by transport. At o the isotropy representation splits as

$$T_o\mathcal{A} = \mathbb{R}^{N_\theta} \oplus \mathbb{S}_0(N_\theta) \oplus \mathbb{R}I, \quad \mathbb{S}_0(N_\theta) := \{X \in \mathbb{S}(N_\theta) : \text{Tr } X = 0\}, \quad (214)$$

into the mean block, the traceless covariance block and the trace covariance block, which are pairwise inequivalent irreducible representations of $O(N_\theta)$. Inequivalence is visible on the sign element $Q = -I$, which acts as -1 on $\mathbb{R}^{N_\theta}$ and trivially on $\mathbb{S}(N_\theta)$, and on dimensions, since $\dim \mathbb{S}_0(N_\theta) = N_\theta(N_\theta + 1)/2 - 1 \geq 2 > 1 = \dim \mathbb{R}I$. Schur's lemma then leaves a single scalar weight on each summand and no invariant pairing between distinct summands, so every invariant quadratic form on $T_o\mathcal{A}$ is

$$\eta \|u\|^2 + \omega \text{Tr}(X_0^2) + \tau'(\text{Tr } X)^2 \quad (215)$$

for scalars η, ω, τ' , where $X = X_0 + \frac{\text{Tr } X}{N_\theta} I$. We verify this by an explicit computation with equivariant operators, which also fixes the normalization of the trace weight.

Lemma H.1. *Let $N_\theta \geq 2$ and let $\mathcal{M} : \mathbb{S}(N_\theta) \rightarrow \mathbb{S}(N_\theta)$ be linear with $\mathcal{M}(QXQ^\top) = Q\mathcal{M}(X)Q^\top$ for every $Q \in O(N_\theta)$. Then there are scalars ω, τ with $\mathcal{M}(X) = \omega X + \tau(\text{Tr } X)I$. Similarly, if $M \in \mathbb{S}(N_\theta)$ satisfies $Q^\top MQ = M$ for every $Q \in O(N_\theta)$, then $M = \eta I$ for a scalar η .*

Proof. Write $E_{ij} = \frac{1}{2}(e_i e_j^\top + e_j e_i^\top)$ for the standard basis of $\mathbb{S}(N_\theta)$ and expand $\mathcal{M}(E_{ij}) = \sum_{k \leq l} c_{kl}^{ij} E_{kl}$. Taking $Q = \text{diag}(\epsilon_1, \dots, \epsilon_{N_\theta})$ with $\epsilon_i \in \{\pm 1\}$ gives $QE_{ij}Q^\top = \epsilon_i \epsilon_j E_{ij}$, so equivariance forces $\epsilon_k \epsilon_l c_{kl}^{ij} = \epsilon_i \epsilon_j c_{kl}^{ij}$ for every choice of signs, hence $c_{kl}^{ij} = 0$ unless $\epsilon_k \epsilon_l$ and $\epsilon_i \epsilon_j$ agree identically in the signs. For $i \neq j$ this holds only when $\{k, l\} = \{i, j\}$, so $\mathcal{M}(E_{ij}) = \omega_{ij} E_{ij}$; for $i = j$ it holds only when $k = l$, so $\mathcal{M}(E_{ii})$ is diagonal, $\mathcal{M}(E_{ii}) = \sum_k c_k^i E_{kk}$.

Taking Q to be a permutation matrix and using $QE_{ij}Q^\top = E_{\sigma(i)\sigma(j)}$ gives $\omega_{ij} = \omega$ for all $i \neq j$, and $c_k^i = \mu$, $c_k^i = \nu$ for all $k \neq i$, with ω, μ, ν independent of the indices. Hence $\mathcal{M}(E_{ii}) = (\mu - \nu)E_{ii} + \nu I$.

It remains to identify ω with $\mu - \nu$. Let Q be the rotation by $\pi/4$ in the plane spanned by e_1, e_2 and the identity on its orthogonal complement, which exists because $N_\theta \geq 2$. Then $Q(E_{11} - E_{22})Q^\top = 2E_{12}$, while $\mathcal{M}(E_{11} - E_{22}) = (\mu - \nu)(E_{11} - E_{22})$ since the two copies of νI cancel. Equivariance gives $\mathcal{M}(2E_{12}) = (\mu - \nu) \cdot 2E_{12}$, and comparing with $\mathcal{M}(E_{12}) = \omega E_{12}$

yields $\omega = \mu - \nu$. Setting $\tau = \nu$, the map $X \mapsto \omega X + \tau(\text{Tr } X)I$ agrees with $\mathcal{M}$ on every E_{ij} , hence everywhere.

For the second claim, the same sign matrices force M diagonal and the permutation matrices force its diagonal entries to be equal. $\square$

Representation-theoretic proof of Theorem 5.1. Let g be an affine invariant metric and let $b := g_o$ be the induced $O(N_\theta)$ -invariant inner product on $T_o\mathcal{A}$.

The mean-covariance cross term vanishes. Indeed, put $L(u, X) := b((u, 0), (0, X))$. Invariance under the sign element $Q = -I$ of (213) gives $L(u, X) = L(-u, X) = -L(u, X)$, hence $L \equiv 0$; this is the representation-theoretic form of the sign-element argument used in the proof of Theorem 5.1.

On the mean block, write $b((u, 0), (v, 0)) = u^\top M v$ with $M \in \mathbb{S}(N_\theta)$. Invariance reads $Q^\top M Q = M$ for every $Q \in O(N_\theta)$, so $M = \eta I$ by Theorem H.1, and $b((u, 0), (v, 0)) = \eta u^\top v$.

On the covariance block, write $b((0, X), (0, Y)) = \text{Tr}(\mathcal{M}(X)Y)$ for the unique linear $\mathcal{M} : \mathbb{S}(N_\theta) \rightarrow \mathbb{S}(N_\theta)$ representing b in the trace pairing, which is nondegenerate on $\mathbb{S}(N_\theta)$. Invariance of b and of the trace pairing give $\mathcal{M}(QXQ^\top) = Q\mathcal{M}(X)Q^\top$, so $\mathcal{M}(X) = \omega X + \tau(\text{Tr } X)I$ by Theorem H.1 and

$$b((0, X), (0, Y)) = \omega \text{Tr}(XY) + \tau \text{Tr}(X) \text{Tr}(Y). \quad (216)$$

In the splitting (214) this reads $\omega \text{Tr}(X_0 Y_0) + \frac{\omega + N_\theta \tau}{N_\theta} \text{Tr}(X) \text{Tr}(Y)$, which is (215) with $\tau' = (\omega + N_\theta \tau)/N_\theta$ and confirms that the three summands are pairwise b -orthogonal.

Positivity is read off blockwise. On the mean block, $b((u, 0), (u, 0)) = \eta \|u\|^2 > 0$ requires $\eta > 0$. On the traceless block, taking X traceless with $\text{Tr}(X^2) = 1$ gives $b((0, X), (0, X)) = \omega$, so $\omega > 0$. On the trace block, taking $X = sI$ gives $b((0, X), (0, X)) = N_\theta(\omega + N_\theta \tau)s^2$, so $\omega + N_\theta \tau > 0$, that is, $\tau > -\omega/N_\theta$. Conversely, under these three conditions b is positive definite on each summand and the summands are orthogonal, so b is an inner product.

Finally we transport b to an arbitrary $a = (m, C)$. The map $\Phi_{(C^{1/2}, m)}$ carries o to a and, by (58), carries $(u, X) \in T_o\mathcal{A}$ to $(C^{1/2}u, C^{1/2}XC^{1/2})$; equivalently a tangent vector at a pulls back to $(C^{-1/2}u, C^{-1/2}XC^{-1/2})$. Substituting into b gives

$$g_a((u, X), (v, Y)) = \eta u^\top C^{-1}v + \omega \text{Tr}(C^{-1}XC^{-1}Y) + \tau \text{Tr}(C^{-1}X) \text{Tr}(C^{-1}Y),$$

which is (60). The triple (η, ω, τ) is determined by b through the three displayed evaluations, and multiplying it by $c > 0$ multiplies b , hence g , by c . $\square$

For $N_\theta = 1$ the splitting (214) has no traceless summand and $\mathbb{S}(1) = \mathbb{R}I$, so Theorem H.1 degenerates: every linear map on $\mathbb{S}(1)$ is equivariant, only $\omega + \tau$ is determined, and the positivity conditions become $\eta > 0$ and $\omega + \tau > 0$, as recorded in Theorem 5.2.

I Experimental protocols

All paths below are relative to the repository root, and all numerical kernels use float64 arrays. The objective reported throughout is the Gaussian variational objective, up to an additive target-dependent constant,

$$\mathcal{E}(m, C) = -\frac{1}{2} \log \det C + \mathbb{E}_{\Theta \sim \mathcal{N}(m, C)} V(\Theta),$$

and $\Delta(a) = \mathcal{E}(a) - \mathcal{E}(a_*)$ is the energy gap against the optimizer of the same experiment. Where deterministic quadrature is used, NumPy's K -point Gauss-Hermite nodes (z_j, w_j) are rescaled by $\mathbb{E}f(Z) = \pi^{-1/2} \sum_{j \leq K} w_j f(\sqrt{2}z_j)$ for $Z \sim \mathcal{N}(0, 1)$. Writing $G_n = \mathbb{E}_{\rho_{a_n}} \nabla V$ and $A_n = \mathbb{E}_{\rho_{a_n}} \nabla^2 V$, the two Fisher-Rao steps are (20) and (21) in the equivalent form $m_{n+1} = m_n - \Delta t C_n G_n$ with $C_{n+1}^R = C_n^{1/2} \exp[\Delta t \{I - C_n^{1/2} A_n C_n^{1/2}\}] C_n^{1/2}$ and $C_{n+1}^{\text{KL}} = (1 + \Delta t)(C_n^{-1} + \Delta t A_n)^{-1}$.

Unless stated otherwise a run is evolved for its fixed iteration budget, and a time to tolerance is a diagnostic computed from the saved trajectory rather than an early-stopping rule. The PDF files displayed in [Section 6](#) are byte-for-byte copies of the corresponding files in `reports/assets/figs`; the script `reports/make_report_assets.py` performs only post-processing and never reruns a dynamics experiment.

Each experiment below belongs to a named group whose configurations live under `configs/<group>`, whose runners live under `scripts/<group>`, and whose kernels and plot selectors live under `src/<group>`. The lineage paragraphs name the group, its configuration and runner files, and, for each selected panel, the CSV it reads, the builder function that consumes it, and the resulting file in `reports/assets/figs`.

I.1 Covariance bootstrap

Let $H = \text{diag}(\nu_1, \dots, \nu_{N_\theta})$ with $(\nu_i)_{i=1}^{N_\theta}$ log-spaced in $[1, \kappa]$. The Gaussian target is $V(\theta) = \frac{1}{2}\theta^\top H\theta$ with $(m_\star, C_\star) = (0, H^{-1})$, and the non-Gaussian target used only in the transport experiment is $V(\theta) = \frac{1}{2}\theta^\top H\theta + \gamma \sum_{i=1}^{N_\theta} \log \cosh(\theta_i)$ with $\gamma = 1$. For the latter $m_\star = 0$ and each diagonal entry of $C_\star$ solves

$$c_i^{-1} = \nu_i + \gamma \mathbb{E}_{Z \sim \mathcal{N}(0, c_i)} \text{sech}^2 Z, \quad (217)$$

computed by fixed-point iteration with tolerance 10^{-13} , at most 500 iterations, and 80-point Gauss–Hermite quadrature. Thus $\alpha = 1$, while $\beta = \kappa$ for the Gaussian target and $\beta = \kappa + 1$ for the log cosh target. Every run starts from $m_0 = m_\star + 3\sqrt{\text{diag}(C_\star)}$ and $C_0 = \lambda_{0,\min}I$, and the Gaussian scaling grid is

$$N_\theta \in \{10, 20\}, \quad \kappa \in \{10, 100, 1000\}, \quad \lambda_{0,\min} \in \{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\},$$

for both covariance maps, with $\Delta t = 0.5$ and 200 steps. The burn-in time is the first n such that $\lambda_{\min}(C_n) \geq (2\beta)^{-1}$. The left panel of [Figure 1](#) selects KL, $N_\theta = 20$, $\kappa = 1000$, and $\lambda_{0,\min} = 10^{-10}$; its three ordinates are $\lambda_{\min}(C_n)$, the exact Gaussian energy gap, and $\|C_n^{1/2}G_n\|$, and its vertical markers are the covariance threshold and the first saved iterate with energy gap at most 10^{-3} .

For the transport grid, the Gaussian and log cosh targets both have $N_\theta = 20$ and $\kappa = 50$, and the grid is

$$\lambda_{0,\min} \in \{10^{-8}, 10^{-6}, 10^{-4}\}, \quad c \in \{0.25, 0.5, 1\}, \quad \Delta t = 0.5,$$

with both Fisher–Rao tails and a 6000-step budget. When $\lambda_{0,\min} < c/\beta$, one Bures step of size $\eta = c/\beta$ is applied first,

$$\begin{aligned} m^+ &= m - \eta G, & \tilde{C} &= (I - \eta A)C(I - \eta A), \\ C^+ &= \frac{1}{2} \left\{ \tilde{C} + 2\eta I + [\tilde{C}(\tilde{C} + 4\eta I)]^{1/2} \right\}. \end{aligned}$$

The right panel of [Figure 1](#) selects $c = 0.5$ and plots, for every target, tail, and $\lambda_{0,\min}$, the first iteration whose gap is at most 10^{-1} , together with $\lambda_{\min}(C)$ after the first step.

Data and figure lineage. The group is `natural_gradient_covariance_bootstrap`, with configuration `covariance_bootstrap.yaml` and runner `run_experiments.py`. The left panel reads `outputs/natural_gradient_covariance_bootstrap/results_long.csv` through `plotting.py::fig_three_stage` into `fig_covboot_three_stage.pdf`; the right panel reads `outputs/natural_gradient_covariance_bootstrap/wasserstein_bootstrap_summary.csv` through `plotting.py::fig_wasserstein_bootstrap` into `fig_covboot_wasserstein_bootstrap.pdf`.

I.2 Deterministic discretization grid

All three targets have $N_\theta = 2$, use $\lambda \in \{0.01, 0.1, 1\}$, and are

$$\begin{aligned} V_G(\theta) &= \frac{1}{2} \theta^\top \text{diag}(1, \lambda) \theta, \\ V_q(\theta) &= \frac{(\sqrt{\lambda} \theta_1 - \theta_2)^2}{20} + \frac{\theta_2^4}{20}, \\ V_s(\theta) &= \frac{(\sqrt{\lambda} \theta_1 - \theta_2)^2}{20} + \frac{\delta}{2} \|\theta\|^2 + \gamma \log \cosh \theta_2, \quad (\delta, \gamma) = (0.05, 1). \end{aligned}$$

The Gaussian initialization is $m_0 = (10, 10)$, $C_0 = \text{diag}(0.5, 2)$; the other two use $m_0 = (10, 10)$, $C_0 = 4I$. Gaussian and quartic expectations are analytic through order four, and the smooth target uses 80-point one-dimensional Gauss–Hermite quadrature for the θ_2 -marginal. The Gaussian optimum is analytic; the other two optima are computed in Cholesky coordinates by four deterministic L-BFGS-B starts with a 5000-iteration budget, gradient tolerance 10^{-10} , and function tolerance 10^{-16} .

The common grid is $\Delta t \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10\}$ with $T = 15$ and $N = \lceil T/\Delta t \rceil$, and both schemes use exact population G_n and A_n . A Radau solution of (1), with relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and 400 output times, is used only for the terminal discretization-accuracy diagnostic. Simulation stops early only on a nonfinite update or $\lambda_{\min}(C_n) \leq 10^{-12}$. A completed trajectory is called stable when it is SPD-feasible, ends below its initial energy, and never exceeds 10^3 times its initial energy gap; monotonicity allows an absolute per-step tolerance of 10^{-10} ; and accuracy means $\|m_N - m(T)\| + \|C_N - C(T)\|_F / \|C(T)\|_F \leq 10^{-2}$.

The left panel of Figure 2 uses only V_s , all three values of λ , and the matched stable steps $\Delta t \in \{0.05, 0.1, 0.5, 1\}$, plotting the energy gap against $n\Delta t$. The right panel retains every stable run that reaches gap 10^{-6} , and for each target, method, and Δt averages the hitting iteration and the hitting time $n\Delta t$ over the eligible values of λ ; its top row is iteration count and its bottom row is physical flow time, not measured wall time.

Data and figure lineage. The group is `natural_gradient_discretization_stepsize`, with configuration `stepsize_grid.yaml` and runner `run_stepsize_grid.py`; the target, method, optimizer, ODE, and classification code sits in its `src` tree. Both selected panels read `outputs/natural_gradient_discretization_stepsize/results_long.csv` and/or `outputs/natural_gradient_discretization_stepsize/summary.csv`, through the builders `scripts/natural_gradient_discretization_stepsize/plot_results.py::fig_convergence_speed_grid` and `plot_results.py::fig_iterations_vs_stepsize`, into `fig_convergence_speed_smooth_logconcave.pdf` and `fig_iterations_vs_stepsize.pdf`.

I.3 Sticking-the-landing variance and constant-step floors

Here $A = \text{diag}(a_1, \dots, a_{N_\theta})$ with (a_i) log-spaced in $[1, \kappa]$, and the targets are $V_G(\theta) = \frac{1}{2} \theta^\top A \theta$ and $V_{lc}(\theta) = \frac{1}{2} \theta^\top A \theta + \tau \sum_{i=1}^{N_\theta} \log \cosh(\theta_i)$. The grid is $N_\theta \in \{2, 8, 32\}$ and $\kappa \in \{10, 100, 10^4\}$ for V_G , and $N_\theta \in \{2, 8, 32\}$, $\kappa \in \{10, 100\}$, $\tau \in \{0.1, 1\}$ for V_{lc} . For V_{lc} one has $m_\star = 0$ and $C_\star$ diagonal with $(C_\star)_{ii}$ solving (217) with ν_i replaced by a_i and γ by τ , under the same quadrature, tolerance, and iteration cap.

The estimator-level experiment compares the baseline score $b_{\text{base}}(\theta) = \nabla \log \rho_{\text{post}}(\theta)$ with the sticking-the-landing estimator $b_{\text{STL}}(\theta) = b_{\text{base}}(\theta) + C^{-1}(\theta - m)$. At each target configuration the four states plotted on the left of Figure 3 have $C = C_\star$ and $m = m_\star + \rho \sqrt{\text{diag}(C_\star)}$ with $\rho \in \{0, 0.25, 1, 4\}$; the full estimator grid additionally includes $(m_\star, 0.25C_\star)$ and $(m_\star, 4C_\star)$. For every state and each seed $s \in \{0, \dots, 19\}$, $M = 100000$ Gaussian samples are drawn in one chunk, the configured chunk cap being 200000. The reported mean-block Fisher–Rao variance is $\text{Var}_{\text{FR}}(b) = \mathbb{E}[(b - \mathbb{E}b)^\top C(b - \mathbb{E}b)]$. The left panel plots, against ρ , the median over the 20 seeds

of $\text{Var}_{\text{FR}}(b_{\text{STL}})$ divided by $\text{Var}_{\text{FR}}(b_{\text{base}})$, separately for all 12 non-Gaussian configurations; for display on a logarithmic axis the plotting code adds 10^{-3} to ρ and clips ratios below 10^{-16} .

The algorithm-level experiment compares $\{\text{R}, \text{R-STL}, \text{KL}, \text{KL-STL}\}$. At every step it draws B samples from the current Gaussian, averages the corresponding mean estimator, and averages the diagonal pointwise log-target Hessian for the covariance step, the two mean estimators within a covariance scheme using the same standard-normal stream. The settings are $\Delta t = 0.1$, $B \in \{1, 4, 16\}$, $m_0 = m_\star + 3\sqrt{\text{diag}(C_\star)}$, and $C_0 = 0.5C_\star$, with 2000 steps for V_G and 3000 for V_c . Gaussian energy gaps are analytic, and non-Gaussian gaps use 40-point Gauss–Hermite quadrature and are therefore independent of the mini-batch used for the update. An emergency eigenvalue guard clips values below 10^{-12} ; the runner counts such clips in a cell diagnostic printed during execution, but the surviving summary CSV does not preserve that count.

The replicate labels are $0, \dots, 19$. For the algorithm grid, one batched random stream advances these 20 leading-array entries, and its cell seed is the first 15 hexadecimal digits of `SHA256(20260623|kind|d|kappa|tau|B)`, interpreted as an integer; the method name is deliberately absent, so all four paired methods receive the same standard-normal sequence. For the estimator grid, the generator seed is the replicate label itself.

For each run the empirical floor is the median gap over the final 20 percent of the full, undecimated trajectory. The right panel of Figure 3 takes the median of these per-run medians over every target configuration and seed, separately by target kind, method, and B , and clips nonpositive display values to 10^{-16} .

Data and figure lineage. The group is `natural_gradient_stl_variance`, with configuration `stl_variance.yaml` and the two runners `run_estimator_variance.py` and `run_algorithm_grid.py`. The left panel reads `outputs/natural_gradient_stl_variance/estimator_variance.csv` through `plotting.py::fig_variance_by_distance` with `kind=log_cosh` into `fig_stl_variance_by_distance_to_optimum.pdf`; the right panel reads `outputs/natural_gradient_stl_variance/tail_noise_floor.csv` through `plotting.py::fig_batchsize_effect` into `fig_stl_batchsize_effect.pdf`.

I.4 Trace and traceless affine modes

At a state (m, C) , set $A_C = \mathbb{E}_{\rho_a} \nabla^2 V$ and $B = C^{1/2} A_C C^{1/2}$. Both experiments run

$$m^+ = m - \Delta t C \mathbb{E}_{\rho_a} \nabla V, \quad C^+ = C^{1/2} \exp \left[\frac{\Delta t}{2\omega} \left\{ -B + \frac{\omega + \tau \text{Tr } B}{\omega + N_\theta \tau} I \right\} \right] C^{1/2}. \quad (218)$$

For the Gaussian calibration target $\rho_{\text{post}} = \mathcal{N}(0, I_{N_\theta})$ expectations are exact, so $B = C$, the mean step is $m^+ = m - \Delta t C m$, and the covariance step diagonalizes as $\lambda_i^+ = \lambda_i \exp[\frac{\Delta t}{2\omega} \{-\lambda_i + q_\tau(C)\}]$ with $q_\tau(C) = (\omega + \tau \text{Tr } C)/(\omega + N_\theta \tau)$. Its grid is $N_\theta \in \{2, 5, 10\}$, $\omega \in \{0.125, 0.25, 0.5, 1, 2\}$, and $\tau \in \{-\omega/(2N_\theta), 0, \omega/(2N_\theta)\}$, with $\Delta t = 0.02$, $T = 20$, and metrics saved every five steps.

The non-Gaussian target is

$$V_\rho(\theta) = \frac{1}{2} \|\theta\|^2 + \frac{\rho}{M} \sum_{\ell=1}^M \log \cosh(a_\ell^\top \theta), \quad N_\theta = 5, \quad \rho = 5, \quad M = 4N_\theta = 20,$$

where the unit vectors a_ℓ are normalized Gaussian draws from NumPy seed 123. The expectations in (218) use one fixed scrambled Sobol design, transformed coordinatewise by the Gaussian inverse CDF, with $K = 4096$ and seed 2026, and the same points are reused at every state and for every (ω, τ) and initialization. The reference optimum uses an independent scrambled Sobol design with $K_{\text{ref}} = 8192$ and seed 2027, followed by Cholesky-coordinate L-BFGS-B with a 2000-iteration budget, gradient tolerance 10^{-6} , and function tolerance 10^{-15} . The dynamics grid is $\omega \in \{0.25, 0.5, 1\}$ and $\tau \in \{-\omega/(2N_\theta), 0, \omega/(2N_\theta)\}$, with $\Delta t = 0.005$, $T = 40$, and metrics saved every ten steps. There are no random replicates: once the seeds are fixed, the quasi-Monte Carlo design is a single common deterministic one.

The five Gaussian initializations are

$$\begin{aligned}
\text{mean: } m_0 &= \mathbf{31}/\sqrt{N_\theta}, \quad C_0 = I, \\
\text{volume-high/low: } m_0 &= 0, \quad C_0 = 4I \text{ or } I/4, \\
\text{shape: } m_0 &= 0, \quad C_0 = \text{diag}(e^2, e^{-2}, 1, \dots, 1), \\
\text{mixed: } m_0 &= \mathbf{21}/\sqrt{N_\theta}, \quad C_0 = 2 \text{diag}(e^{1.5}, e^{-1.5}, 1, \dots, 1),
\end{aligned}$$

and for the non-Gaussian target the same formulas are applied in the $(m_\star, C_\star)$ -whitened coordinates, so that for instance the shape covariance is $C_\star^{1/2} \text{diag}(e^2, e^{-2}, 1, \dots, 1) C_\star^{1/2}$.

Both panels of [Figure 4](#) show $T(\tau)/T(0)$ for every ω and initialization, which is the ratio predicted mode by mode in [Theorem 5.4](#). The Gaussian panel selects $N_\theta = 5$ and the first saved time at which the exact Gaussian KL, normalized by its initial value, is at most 10^{-4} ; the non-Gaussian panel uses the fixed quasi-Monte Carlo objective gap, normalized by its initial gap, at tolerance 10^{-2} . A cell is left blank if either hitting time is not finite within the horizon.

Data and figure lineage. The group is `omega_tau_modes`, with configurations `gaussian_target.yaml` and `logconcave_target.yaml`. The report builder reads the legacy committed locations `outputs/gaussian_grid/summary.csv` and `outputs/logconcave_grid/summary.csv` through `reports/make_report_assets.py::_fig_tau_speedup`, producing `fig_ot_gaussian_tau_speedup.pdf` and `fig_ot_logconcave_tau_speedup.pdf`. To reproduce these exact locations with the current runners, their `-outdir` arguments must override the nested `output_dir` values in the YAML files.

I.5 Non-log-concave boundary

All selected runs are centered, scalar, and deterministic, and use 160-point Gauss–Hermite quadrature for the double well [\(26\)](#), whose Hessian $V_R''(\theta) = -1 + 2 \tanh^2(\theta/R)$ lies in $[-1, 1]$. For $\Theta \sim \mathcal{N}(0, c)$ we write $A_R(c) = \mathbb{E}V_R''(\Theta)$, and the plotted squared Fisher–Rao gradient is $(1 - cA_R(c))^2$.

The Riemannian cascade runs the map [\(28\)](#) of [Theorem 2.18](#) with $R \in \{20, 50, 100, 300, 1000\}$, $c_0 = 1$, $\Delta t = 1$, and a 20-step cap, stopping before the next update if $c_n > R^2/4$, if $\log c$ would exceed 700, or on nonfinite arithmetic. The first panel of [Figure 5](#) plots every recorded iterate for all five values of R .

The pole experiment runs the unprojected resolvent [\(29\)](#) of [Theorem 2.19](#) with $R = 1000$, $\Delta t = 1$, $c_0 = 1 - \varepsilon$, $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$, and a two-step cap. The middle panel plots the squared gradient after the first step against ε , together with the reference $4\varepsilon^{-2}$. At $R = 1000$ the sufficient condition $R_\varepsilon^2 \geq 2\varepsilon^{-2}$ used in the proof of [Theorem 2.19](#) holds only for $\varepsilon \in \{10^{-1}, 10^{-2}\}$; the measured-to-reference ratios are 0.90 and 0.99 there, remain 0.99 and 0.89 at the uncertified points 10^{-3} and 10^{-4} , and collapse to 0.39 and 0.014 at 10^{-5} and 10^{-6} , where the fixed truncation radius saturates the pole. The small- ε bend of the middle panel is this fixed- R saturation and not a failure of the ε^{-2} law along the proposition’s admissible sequences $R_\varepsilon \rightarrow \infty$.

The clipped experiment runs the scalar form of [\(30\)](#),

$$c_{n+1} = \text{clip}_{[0.5, 2]} \frac{(1 + \Delta t)c_n}{1 + \Delta t c_n A_R(c_n)}, \quad c_0 = 1, \quad \beta = 1.$$

The report panel selects the primary safety fraction 0.5, hence $L_{\text{clip}} = 2\beta\lambda_+ = 4$ and $\Delta t = 0.5/L_{\text{clip}} = 0.125$, and runs 300 steps for each of the five values of R . The full output also contains safety fractions 0.25 and 0.9, together with the deliberately outside-theorem step $\Delta t = 1/(\beta\lambda_+) = 0.5$; the selected panel excludes those rows. At step n the code records the centered covariance term of the split stationarity measure [\(31\)](#),

$$D_n = \text{KL}(\mathcal{N}(0, c_n) \parallel \mathcal{N}(0, c_{n+1})), \quad D_{\min}(N) = \min_{0 \leq n < N} D_n, \quad B_N = \frac{\Delta t}{N} \{\mathcal{E}(0, c_0) - \mathcal{E}(0, c_N)\},$$

and the right panel plots $D_{\min}(N)$ against the envelope B_N of [Theorem 2.21](#). In the recorded primary runs the upper clip first activates at step 2 for every R ; the step-2 displacement is nonzero, while the next and subsequent pinned steps have $D_n = 0$.

Data and figure lineage. The group is `natural_gradient_nonconvex_instability`, with configuration `nonconvex_instability.yaml`. Writing outputs for `outputs/natural_gradient_nonconvex_instability`, the cascade panel reads `outputs/results_long.csv` through `plotting.py::fig_riemannian_cascade`, the pole panel reads `outputs/kl_pole_summary.csv` through `plotting.py::fig_kl_pole`, and the clipped panel reads `outputs/clipped_kl_stationarity.csv` through `plotting.py::fig_clipped_kl_stationarity`, restricted to `dt_rule=theorem_safe`. The assets are, respectively, `fig_nonconvex_riemannian_cascade.pdf`, `fig_nonconvex_kl_pole.pdf`, and `fig_nonconvex_clipped_kl_stationarity.pdf`.

I.6 Provenance limits of the committed snapshot

The scientific parameters above are recoverable from the configurations, source, and raw columns used by the selected figures, but some run-level provenance files that the runners are designed to write are not committed: `outputs/natural_gradient_covariance_bootstrap` lacks `run_metadata.json`; `outputs/natural_gradient_stl_variance` lacks `algorithm_results_long.csv`, `target_metadata.json`, `run_metadata.json`, `estimator_target_metadata.json`, and `estimator_run_metadata.json`; and the Gaussian affine-mode runner does not write a run-metadata file. The exact Python, NumPy, Torch, CUDA, and hardware versions of those original runs therefore cannot be reconstructed from this snapshot. The estimator-level STL CSV records the Torch/CUDA backend but not the GPU model, the algorithm-level backend is not recorded in the surviving summary files, and `requirements.txt` and `requirements-gpu.txt` give lower dependency bounds rather than a locked environment. These omissions do not change the selected data-to-figure mappings, since the two STL panels need only the surviving estimator and tail CSVs and the covariance panels need only the surviving long and transport-summary CSVs. The target-level metadata of the covariance-bootstrap group is a deterministic function of the committed configuration and has been regenerated from it (`target_metadata.json`); the run-level metadata of the original runs is not reconstructed. The asset builder `reports/make_report_assets.py` treats the GPU-regenerable trajectory file `algorithm_results_long.csv` and the algorithm run metadata as optional and builds every figure and table used by this manuscript from the committed summary files, so both `-only stl_variance` and `-only covariance_bootstrap` run to completion from a clean checkout; the supplementary algorithm-gap panels, which this manuscript does not use, require rerunning `scripts/natural_gradient_stl_variance/run_algorithm_grid.py` on a CUDA host. No missing value has been guessed here. General WFR experiments under `outputs/wfr_gradient_flow` are not used in any figure or claim in this manuscript.

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